

Supporting Information:

Data-Driven Design Guidelines for TADF

Emitters from a High-Throughput Screening of

747 Molecules

Jean-Pierre Tchapet Njafa*,* Elvira Vanelle Kameni Tcheuffa, Aissatou Maghame
Foumkpou, and Serge Guy Nana Engo

*Department of Physics, Faculty of Science, University of Yaounde I, P.O. Box 812, Yaounde,
Cameroon*

E-mail: jean-pierre.tchapet@facsciences-uy1.cm

This Supporting Information contains:

- Table S1: Architectural classification and performance statistics for TADF emitters
- Table S2: Experimental validation data for representative molecules
- Table S4: Multi-reference TADF validation - xTB vs TD-DFT comparison
- Table S5: Spin-orbit coupling matrix elements for 17 representative molecules
- Figure S5: Correlation analysis of SOC with structural descriptors
- Additional computational details and analysis parameters

Table S1: Molecular architecture and classification of the studied TADF emitters. This table provides the SMILES representation of each molecule, along with its structural classification.

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
4CzTPN	<chem>N#Cc1c(-n2c3cccc3c3cccc32)c(-n2c3cccc3c3cccc32)c(C#N)c(-n2c3cccc3c3cccc32)c1-n1c2cccc2c2cccc21</chem>	4	4	Multi-D/A	788.914	13.818
DMAC-DPS	<chem>CC1(C)c2cccc2N(c2ccc(S(=O)(=O)c3ccc(N4c5cccc5C(C)(C)c5cccc54)cc3)cc2)c2cccc21</chem>	2	0	2D-0A	632.829	10.738
PSPCz	<chem>C1=CC=C(C=C1)S(=O)(=O)C2=CC=CC(=C2)N3C4=CC=CC=C4C5=CC=CC=C53</chem>	1	0	1D-0A	383.472	5.617
Px2BP	<chem>C1=CC=C2C(=C1)N(C3=CC=CC=C3O2)C4=CC=C(C=C4)C(=O)C5=CC=C(C=C5)N6C7=CC=CC=C7O</chem>	4	0	Multi-D/A	544.610	10.069
CzS2	<chem>C8=CC=CC=C86C1=CC=C2C(=C1)C3=CC=CC=C3N2C4=CC=C(C=C4)S(=O)(=O)C5=CC=C(C=C5)N6C7=CC=CC=C7</chem>	2	0	2D-0A	548.667	8.714
2TCz-DPS	<chem>C8=CC=CC=C86CC(C)(C)C1=CC2=C(C=C1)N(C3=C2C=C(C=C3)C(C)(C)C)C4=CC=C(C=C4)S(=O)(=O)C5=CC=C(C=C5)N6C7=C(C=C(C=C7)C(C)(C)C)C8=C6C=CC(=C8)C(C)(C)C</chem>	2	0	2D-0A	773.099	13.904
TDBA-DI	<chem>CC(C)(C)c1ccc2c(c1)B1c3cc(C(C)(C)C)ccc3O3cc(-n4c5cccc5c5c6c(c7cccc7n6-c6cccc6)c6c(c7cccc7n6-c6cccc6)c54)cc(c31)O2</chem>	3	0	Multi-D/A	877.897	14.300
OPDPO	<chem>O=CC1=CCC(c2cccc2N2c3cccc3Sc3cccc32)(P(=O)(c2cccc2)c2cccc2)C=C1</chem>	2	0	2D-0A	581.677	8.916

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
2CzTPEPCz	<chem>C(=C(c1ccc(-n2c3ccccc3c3cccc32)cc1)c1ccc(-n2c3ccccc3c3ccccc32)cc1)c1ccc(-c2ccc(-n3c4ccccc4c4ccccc43)cc2)cc1</chem>	3	0	Multi-D/A	828.031	16.234
TBRb	<chem>CC(C)(C)c1ccc(-c2c3ccc(C(C)(C)C)cc3c(-c3ccccc3)c3c(-c4ccc(C(C)(C)C)cc4)c4cc(C(C)(C)C)cc4c(-c4ccccc4)c23)cc1</chem>	0	0	Unknown	757.118	17.004
TBPe	<chem>CC(C)(C)c1cc2cc(C(C)(C)C)cc3c4cc(C(C)(C)C)cc5cc(C(C)(C)C)cc(c(c1)c23)c54</chem>	0	0	Unknown	476.748	10.927
2CzPN	<chem>N#Cc1cc(-c2ccccc3c2[nH]c2ccccc23)c(-c2ccccc3c2[nH]c2ccccc23)cc1C#N</chem>	2	4	Multi-D/A	458.524	8.033
3CzFTFP	<chem>Fc1nc(F)c(F)c(C2(c3ccc4c(cc3)c3ccccc3n4-c3ccccc3)c3ccccc3-c3ccccc32)c1F</chem>	1	0	1D-0A	556.562	9.098
DDPhCz-DCPP	<chem>N#Cc1nc2c3ccc(-n4c5ccc(-c6ccccc6)cc5c5cc(-c6ccccc6)ccc54)cc3c3cc(-n4c5ccc(-c6ccccc6)cc5c5cc(-c6ccccc6)ccc54)ccc3c2nc1C#N</chem>	2	3	Multi-D/A	915.072	16.542
ACRXTN	<chem>CC1(C)c2ccccc2N(c2ccc3c(=O)c4ccccc4oc3c2)c2ccccc21</chem>	1	0	1D-0A	403.481	7.055
TXO-TPA	<chem>O=C1c2ccccc2S(=O)(=O)c2ccc(-c3ccc(N(c4ccccc4)c4ccc(cc4)cc3)cc21</chem>	1	0	1D-0A	487.580	7.201
PCTXO	<chem>O=C1c2ccccc2S(=O)(=O)c2ccc(-c3ccc(N(c4ccc(Br)cc4)c4ccc(Br)cc4)cc3)cc21</chem>	1	0	1D-0A	645.372	8.726

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
IP-PPI	<chem>c1ccc(-n2c(-c3ccc(-c4ccc(-c5cn6cccc6n5)cc4)cc3)nc3c4cccc4c4cccc4c32)cc1</chem>	0	0	Unknown	562.676	9.981
DMDHNP-DPS	<chem>CC1(C)c2cnccc2Nc2ccncc21.O=S(=O)(c1cccc1)c1cccc1c1ccc(-c2nnc(-c3ccc(-c4cc</chem>	0	0	Unknown	429.545	5.379
4NTAZ-PPI	<chem>c(-c5nc6c7cccc7c7cccc7c6n5-c5cccc5)cc4)cc3)n2-c2cccc3cccc23)cc1c1ccc(-c2nnc(-c3ccc(-c4cc</chem>	0	0	Unknown	715.860	12.734
TAZ-PPI	<chem>c(-c5nc6c7cccc7c7cccc7c6n5-c5cccc5)cc4)cc3)n2-c2cccc2)cc1Cc1cc(N2c3cccc3Sc3cccc3</chem>	0	0	Unknown	665.800	11.581
PTZ-mPYR	<chem>2)ccc1-c1cc(-c2ccc(N3c4ccc4Sc4cccc43)cc2C)ncn1N#Cc1ccc(-n2c3cccc3c3cc(-c4ccc5c(c4)c4cccc4n5-c4</chem>	4	1	Multi-D/A	654.864	12.296
pCNBCzoCF3	<chem>ccc(C#N)cc4C(F)(F)F)ccc32)c(C(F)(F)F)c1CC1(C)OB(c2ccc(C(=C(c3ccc</chem>	2	4	Multi-D/A	670.616	11.329
PTZTPE-2	<chem>cc3)c3cccc3)c3cccc3)cc2)OC1(C)CN#Cc1c(-n2c3cccc3c3cccc32)c(C#N)c(-n2c3cccc3c3cccc32)c(-n2c3cccc3c3ccc</chem>	0	0	Unknown	458.410	6.993
4CzIPN	<chem>cc32)c1-n1c2cccc2c2cccc21CC1(C)c2cccc2N(c2ccc(-c3</chem>	4	4	Multi-D/A	788.914	13.818
DMAC-TRZ	<chem>nc(-c4cccc4)nc(-c4cccc4)n3)cc2)c2cccc21</chem>	1	1	D-A	516.648	8.982

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
TCzTrz	<chem>c1ccc(-c2nc(-c3ccccc3)nc(-c3cc(-n4c5ccccc5c5ccccc54)c(-n4c5ccccc5c5ccccc54)c3)n2)cc1</chem>	3	1	Multi-D/A	804.957	14.164
δ -2CbPN	<chem>N#Cc1cc(-n2c3ccccc3c3nccc32)c(-n2c3ccccc3c3nccc32)cc1C#N</chem>	0	4	Multi-D/A	460.500	6.414
CzDCbTrz	<chem>c1ccc(-c2nc(-c3ccccc3)nc(-c3cc(-n4c5ccccc5c5ncccc54)c(-n4c5ccccc5c5ncccc54)c3)n2)cc1</chem>	1	1	D-A	806.933	12.954
α -2CbPN	<chem>N#Cc1cc(-n2c3ccccc3c3cccn32)c(-n2c3ccccc3c3ccncc32)cc1C#N</chem>	0	4	Multi-D/A	460.500	6.414
CN-QP	<chem>N#Cc1ccc2nc(-c3ccc(N4c5cccc50c5ccccc54)cc3)c(-c3ccc(N4c5ccccc50c5ccccc54)c3)nc2c1</chem>	4	3	Multi-D/A	669.744	11.986
TFM-QP	<chem>FC(F)(F)c1ccc2nc(-c3ccc(N4c5ccccc50c5ccccc54)cc3)c(-c3ccc(N4c5ccccc50c5ccccc54)cc3)nc2c1</chem>	4	1	Multi-D/A	712.731	13.134
NDN	<chem>c1ccc2c(c1)c1cccn1n2-c1ccc2sc3ccc(-n4c5ccccc5c5ccncc54)cc3c2c1</chem>	0	0	Unknown	516.629	9.039
mCP	<chem>c1cc(-n2c3ccccc3c3ccccc32)cc(-n2c3ccccc3c3ccccc32)c1</chem>	2	0	2D-0A	408.504	7.881
ZDZ	<chem>c1ccc2c(c1)c1ccccc1n2-c1ccc2sc3ccc(-n4c5ccccc5c5ccccc54)cc3c2c1</chem>	2	0	2D-0A	514.653	10.249

Continued on next page

Table S1 – Continued from previous page

[illegible]

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
SpiroAC-TRZ	<chem>c1ccc(-c2nc(-c3ccccc3)nc(-c3ccc(N4c5ccccc5C5(c6ccc(cc6-c6ccccc65)c5ccccc54)c(c3)n2)cc1CC1(C)c2ccccc2Nc2cccc(C(=O)c3ccccc3)c21.c1ccc2c(c1)sc1ccccc12</chem>	1	1	D-A	638.774	11.019
DBT-BZ-DMAC	<chem>CC1(C)c2ccccc2Nc2cccc(C(=O)c3ccccc3)c21.c1ccc2c(c1)sc1ccccc12</chem>	0	0	Unknown	497.663	9.355
DQP	<chem>c1ccc2nc3c(nc2c1)c1nc2cccc2nc1c1nc2cccc2nc31N#Cc1nc2c3ccc(-c4ccc(N(c5ccccc5)c5ccccc5)cc4)cc3c3cc(-c4ccc(N(c5ccccc5)c5ccccc5)cc4)ccc3c2nc1C#N</chem>	0	6	Multi-D/A	384.402	4.976
TPA-DCPP	<chem>c1ccc2c(c1)0c1ccccc1N2c1ccc(-c2nnc(-c3ccc(N4c5ccccc50c5ccccc54)cc3)o2)cc1CC1(C)c2ccccc2N(c2ccc(-c3nc(-c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)nc(-c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)n3)cc2)c2ccccc21C0c1ccc(N(c2ccc(OC)cc2)c2ccc(-c3ccc4c(c3)C(=O)c3ccc(-c5ccc(N(c6ccc(OC)cc6)c6ccc(OC)cc6)cc5)cc3C4=O)c2)cc1N#Cc1ccc2c(c1)C1(c3cc(C#N)ccc3-2)c2ccccc2N(c2ccccc2)c2ccccc21N#Cc1nc2c3ccc(N4c5ccccc5S</chem>	2	3	Multi-D/A	766.908	13.953
2PXZ-OXD	<chem>c1ccc2c(c1)0c1ccccc1N2c1ccc(-c2nnc(-c3ccc(N4c5ccccc50c5ccccc54)cc3)o2)cc1CC1(C)c2ccccc2N(c2ccc(-c3nc(-c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)nc(-c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)n3)cc2)c2ccccc21C0c1ccc(N(c2ccc(OC)cc2)c2ccc(-c3ccc4c(c3)C(=O)c3ccc(-c5ccc(N(c6ccc(OC)cc6)c6ccc(OC)cc6)cc5)cc3C4=O)c2)cc1N#Cc1ccc2c(c1)C1(c3cc(C#N)ccc3-2)c2ccccc2N(c2ccccc2)c2ccccc21N#Cc1nc2c3ccc(N4c5ccccc5S</chem>	4	0	Multi-D/A	584.635	10.555
BmPyPhB	<chem>c1ccc2c(c1)0c1ccccc1N2c1ccc(-c2nnc(-c3ccc(N4c5ccccc50c5ccccc54)cc3)o2)cc1CC1(C)c2ccccc2N(c2ccc(-c3nc(-c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)nc(-c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)n3)cc2)c2ccccc21C0c1ccc(N(c2ccc(OC)cc2)c2ccc(-c3ccc4c(c3)C(=O)c3ccc(-c5ccc(N(c6ccc(OC)cc6)c6ccc(OC)cc6)cc5)cc3C4=O)c2)cc1N#Cc1ccc2c(c1)C1(c3cc(C#N)ccc3-2)c2ccccc2N(c2ccccc2)c2ccccc21N#Cc1nc2c3ccc(N4c5ccccc5S</chem>	3	1	Multi-D/A	931.200	17.200
MPPA	<chem>c1ccc2c(c1)0c1ccccc1N2c1ccc(-c2nnc(-c3ccc(N4c5ccccc50c5ccccc54)cc3)o2)cc1CC1(C)c2ccccc2N(c2ccc(-c3nc(-c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)nc(-c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)n3)cc2)c2ccccc21C0c1ccc(N(c2ccc(OC)cc2)c2ccc(-c3ccc4c(c3)C(=O)c3ccc(-c5ccc(N(c6ccc(OC)cc6)c6ccc(OC)cc6)cc5)cc3C4=O)c2)cc1N#Cc1ccc2c(c1)C1(c3cc(C#N)ccc3-2)c2ccccc2N(c2ccccc2)c2ccccc21N#Cc1nc2c3ccc(N4c5ccccc5S</chem>	2	0	2D-0A	814.938	12.770
ACRFLCN	<chem>c1ccc2c(c1)0c1ccccc1N2c1ccc(-c2nnc(-c3ccc(N4c5ccccc50c5ccccc54)cc3)o2)cc1CC1(C)c2ccccc2N(c2ccc(-c3nc(-c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)nc(-c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)n3)cc2)c2ccccc21C0c1ccc(N(c2ccc(OC)cc2)c2ccc(-c3ccc4c(c3)C(=O)c3ccc(-c5ccc(N(c6ccc(OC)cc6)c6ccc(OC)cc6)cc5)cc3C4=O)c2)cc1N#Cc1ccc2c(c1)C1(c3cc(C#N)ccc3-2)c2ccccc2N(c2ccccc2)c2ccccc21N#Cc1nc2c3ccc(N4c5ccccc5S</chem>	1	4	Multi-D/A	457.536	7.576
PTZ-DCPP	<chem>c1ccc2c(c1)0c1ccccc1N2c1ccc(-c2nnc(-c3ccc(N4c5ccccc50c5ccccc54)cc3)o2)cc1CC1(C)c2ccccc2N(c2ccc(-c3nc(-c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)nc(-c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)n3)cc2)c2ccccc21C0c1ccc(N(c2ccc(OC)cc2)c2ccc(-c3ccc4c(c3)C(=O)c3ccc(-c5ccc(N(c6ccc(OC)cc6)c6ccc(OC)cc6)cc5)cc3C4=O)c2)cc1N#Cc1ccc2c(c1)C1(c3cc(C#N)ccc3-2)c2ccccc2N(c2ccccc2)c2ccccc21N#Cc1nc2c3ccc(N4c5ccccc5S</chem>	4	3	Multi-D/A	674.814	11.548

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
BBA-Pyr	<chem>c1ccc(-c2ccc(N(c3ccc(-c4cccc4)cc3)c3ccc(-c4ccc(-c5cc(-c6cccc6)nc(-c6cccc6)n5)cc4)cc3)cc2)cc1</chem>	1	1	D-A	703.889	13.948
PhICz-Pyr	<chem>c1ccc(-c2ccc3c(c2)c2cc(-c4cccc4)cc4c5cc(-c6ccc(-c7cc(-c8cccc8)nc(-c8cccc8)n7)cc6)ccc5n3c24)cc1</chem>	2	1	D-A-D	699.857	13.629
PhCz-Pyr	<chem>c1ccc(-c2ccc3c(c2)c2cc(-c4cccc4)ccc2n3-c2ccc(-c3ccc(-c4cc(-c5cccc5)nc(-c5cccc5)n4)cc3)cc2)cc1</chem>	1	1	D-A	701.873	13.576
PDA-DP	<chem>CC1(C)c2cc(-c3cccc3)ccc2N(c2ccc(S(=O)(=O)c3ccc(N4c5ccc(-c6cccc6)cc5C(C)(C)c5cc(-c6cccc6)ccc54)cc3)cc2)c2ccc(-c3cccc3)cc21O=P(c1cccc1)(c1cccc1)c1</chem>	2	0	2D-0A	937.221	17.406
9CzFDBFDPO	<chem>cccc2c1oc1c(P(=O)(c3cccc3)c3cccc3)cc(-c3ccc(C4(c5ccc(-n6c7cccc7c7cccc76)cc5)c5cccc5-c5cccc54)c3)cc12</chem>	1	0	1D-0A	1050.147	15.992
9PhCzFDBFDPO	<chem>0=P(c1cccc1)(c1cccc1)c1cccc2c1oc1c(P(=O)(c3cccc3)c3cccc3)cc(-c3ccc(C4(c5ccc(-c6ccc(-n7c8cccc8c8cccc87)cc6)cc5)c5cccc5-c5cccc54)cc3)cc12</chem>	1	0	1D-0A	1126.245	17.659
9PhCzFDBFSPO	<chem>0=P(c1cccc1)(c1cccc1)c1cccc2c1oc1ccc(-c3ccc(C4(c5ccc(-c6ccc(-n7c8cccc8c8cccc87)cc6)cc5)c5cccc5-c5cccc54)cc3)cc12</chem>	1	0	1D-0A	926.068	16.020

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
9CzFDBFSPO	<chem>O=P(c1ccccc1)(c1ccccc1)c1cccc2c1oc1ccc(-c3ccc(C4(c5ccc(-n6c7ccccc7c7ccccc76)cc5)c5ccccc5-c5ccccc54)c3)cc12</chem>	1	0	1D-0A	849.970	14.353
9ArFDBFxPO	<chem>O=P(c1ccccc1)(c1ccccc1)c1cccc2c1oc1c(P(=O)(c3ccccc3)c3ccccc3)cc(-c3ccc(C4(c5ccc(-n6c7ccccc7c7ccccc76)cc5)c5ccccc5-c5ccccc54)c3)cc12</chem>	1	0	1D-0A	1050.147	15.992
12BTCzTPN	<chem>N#Cc1c(-n2c3ccccc3c3ccccc32)c(-n2c3ccccc3c3ccc4c5ccccc5sc4c32)c(C#N)c(-n2c3ccccc3c3ccccc32)c1-n1c2ccccc2c2ccc3c4ccccc4sc3c21Cc1ccccc1-c1cc2nc3c4ccc(N5c6ccccc60c6ccccc65)cc4c4cc(N5c6ccccc60c6ccccc65)cc4c3nc2cc1-c1ccccc1C</chem>	4	4	Multi-D/A	1001.212	18.554
oDTBPZ-DPXZ	<chem>CC1(C)c2ccccc2N(c2ccc3c(c2)c2cc(N4c5ccccc5C(C)(C)c5ccccc54)ccc2c2nc4cc(N5c6ccccc6C(C)(C)c6ccccc65)c(C#N)cc4nc32)c2ccccc21</chem>	4	2	Multi-D/A	822.968	16.191
3DMAC-BP-CN	<chem>CC1(C)c2ccccc2N(c2ccc3c(c2)c2cc(N4c5ccccc5C(C)(C)c5ccccc54)ccc2c2nc4cc(N5c6ccccc6C(C)(C)c6ccccc65)c(C#N)cc4nc32)c2ccccc21</chem>	3	4	Multi-D/A	927.168	17.288
3DMAC-BP-Br	<chem>N#Cc1ccc2c(c1)c1cc(C#N)cc1c1nc3cc(-c4ccc(N(c5ccccc5)c5ccccc5)cc4)c(-c4ccc(N(c5ccccc5)c5ccccc5)cc4)c3nc21</chem>	3	2	Multi-D/A	981.054	18.179
TPA-PZCN	<chem>N#Cc1ccc2c(c1)c1cc(C#N)cc1c1nc3cc(-c4ccc(N(c5ccccc5)c5ccccc5)cc4)c(-c4ccc(N(c5ccccc5)c5ccccc5)cc4)c3nc21</chem>	2	6	Multi-D/A	816.968	15.106

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
1PXZ-BP	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc2c(c1)c1cccc1c1nc3cccc3nc21</chem>	2	2	2D-2A	461.524	8.665
PTZ-DBTO2	<chem>O=S1(=O)c2cccc2-c2cc(N3c4cccc4Sc4cccc43)ccc21</chem>	2	0	2D-0A	413.523	6.434
BACQ	<chem>CC1(C)c2c(ccc3cccc23)-n2c3cccc3c(=O)c3cccc1c32CC1(C)c2cc(-n3c4cccc4c4cccc43)ccc2N(c2ccc3sc4cccc4c(=O)c3c2)c2ccc(-n3c4cccc4c4cccc43)cc21CC1(C)c2cc(-n3c4cccc4c4cccc43)ccc2N(c2ccc3c(c2)S(=O)(=O)c2cccc2S3(=O)=O)c2ccc(-n3c4cccc4c4cccc43)cc21CC1(C)c2cc(-n3c4cccc4c4cccc43)ccc2N(c2ccc3c(c2)-c2cccc2S3(=O)=O)c2ccc(-n3c4cccc4c4cccc43)cc21CC1(C)c2cccc2N(c2cc(-c3cc(-c4ccc(-c5cccc5)cc4)nc(-c4cccc4)n3)cc(N3c4cccc4C(C)(C)c4cccc43)c2)c2cccc21C0c1ccc2c(c1)c1cccc1n2-c1ccc(-c2c(C#N)c(C)nc(C)c2C#N)cc1</chem>	1	0	1D-0A	361.444	5.936
2Cz-DMAC-TXO	<chem>CC1(C)c2cc(-n3c4cccc4c4cccc43)ccc2N(c2ccc3sc4cccc4c(=O)c3c2)c2ccc(-n3c4cccc4c4cccc43)cc21CC1(C)c2cc(-n3c4cccc4c4cccc43)ccc2N(c2ccc3c(c2)S(=O)(=O)c2cccc2S3(=O)=O)c2ccc(-n3c4cccc4c4cccc43)cc21CC1(C)c2cc(-n3c4cccc4c4cccc43)ccc2N(c2ccc3c(c2)-c2cccc2S3(=O)=O)c2ccc(-n3c4cccc4c4cccc43)cc21CC1(C)c2cccc2N(c2cc(-c3cc(-c4ccc(-c5cccc5)cc4)nc(-c4cccc4)n3)cc(N3c4cccc4C(C)(C)c4cccc43)c2)c2cccc21C0c1ccc2c(c1)c1cccc1n2-c1ccc(-c2c(C#N)c(C)nc(C)c2C#N)cc1</chem>	3	0	Multi-D/A	749.939	13.718
2Cz-DMAC-TTR	<chem>(=O)(=O)c2cccc2S3(=O)=O)c2ccc(-n3c4cccc4c4cccc43)cc21CC1(C)c2cc(-n3c4cccc4c4cccc43)ccc2N(c2ccc3c(c2)-c2cccc2S3(=O)=O)c2ccc(-n3c4cccc4c4cccc43)cc21CC1(C)c2cccc2N(c2cc(-c3cc(-c4ccc(-c5cccc5)cc4)nc(-c4cccc4)n3)cc(N3c4cccc4C(C)(C)c4cccc43)c2)c2cccc21C0c1ccc2c(c1)c1cccc1n2-c1ccc(-c2c(C#N)c(C)nc(C)c2C#N)cc1</chem>	3	0	Multi-D/A	817.992	11.969
2Cz-DMAC-BTB	<chem>CC1(C)c2cc(-n3c4cccc4c4cccc43)ccc2N(c2ccc3c(c2)-c2cccc2S3(=O)=O)c2ccc(-n3c4cccc4c4cccc43)cc21CC1(C)c2cccc2N(c2cc(-c3cc(-c4ccc(-c5cccc5)cc4)nc(-c4cccc4)n3)cc(N3c4cccc4C(C)(C)c4cccc43)c2)c2cccc21C0c1ccc2c(c1)c1cccc1n2-c1ccc(-c2c(C#N)c(C)nc(C)c2C#N)cc1</chem>	3	0	Multi-D/A	753.927	12.803
DMAC-BPP	<chem>(=O)(=O)c2cccc2S3(=O)=O)c2ccc(-n3c4cccc4c4cccc43)cc21CC1(C)c2cccc2N(c2cc(-c3cc(-c4ccc(-c5cccc5)cc4)nc(-c4cccc4)n3)cc(N3c4cccc4C(C)(C)c4cccc43)c2)c2cccc21C0c1ccc2c(c1)c1cccc1n2-c1ccc(-c2c(C#N)c(C)nc(C)c2C#N)cc1</chem>	2	1	D-A-D	799.034	15.363
Me-MOC	<chem>C0c1ccc2c(c1)c1cccc1n2-c1ccc(-c2c(C#N)c(C)nc(C)c2C#N)cc1</chem>	1	2	A-D-A	428.495	6.215
Me-DOC	<chem>C0c1ccc2c(c1)c1cc(OC)ccc1n2-c1ccc(-c2c(C#N)c(C)nc(C)c2C#N)cc1</chem>	1	2	A-D-A	458.521	6.223
3DCPO	<chem>O=P(c1cccc1)(c1ccc2c(c1)c1cccc1n2-c1cccc1c1ccc2c(c1)c1cccc1n2-c1cccc1</chem>	2	0	2D-0A	608.681	9.520

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
TDBA-TPDICz	<chem>CC(C)(C)c1ccc2c(c1)B1c3cc(C(C)(C)C)ccc3Oc3cc(-n4c5cccc5c5c(-c6cccc6)c6c(c(-c7cccc7)c54)c4cccc4n6-c4cccc4)cc(c31)O2</chem>	2	0	2D-0A	864.898	14.537
TRZ-pIC	<chem>CC(C)(C)c1ccc2c(c1)B1c3cc(C(C)(C)C)ccc3Oc3cc(-n4c5cccc5c5c(-c6cccc6)c6c(c(-c7cccc7)c54)c4cccc4n6-c4cccc4)cc(c31)O2</chem>	2	0	2D-0A	864.898	14.537
SBF-PXZ	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc2c(c1)C1(c3cccc3-c3ccc31)c1cc(N3c4cccc40c4cccc43)ccc1-2</chem>	4	0	Multi-D/A	678.791	13.181
PtOEP	<chem>CCC1=C(CC)c2cc3[nH]c(cc4nc(cc5[nH]c(cc1n2)c(CC)c5C)C(CC)=C4CC)c(CC)c3CC.[Pt+2]</chem>	0	0	Unknown	729.870	10.023
PxCYP	<chem>N#Cc1ccc(-c2ccc(N3c4cccc40c4cccc43)cc2)nc1-c1ccc(N2c3cccc30c3cccc32)cc1</chem>	4	1	Multi-D/A	618.696	11.438
AcCYP	<chem>CC1(C)c2cccc2N(c2ccc(-c3ccc(C#N)c(-c4ccc(N5c6cccc6C(C)(C)c6cccc65)cc4)n3)cc2)c2cccc21</chem>	2	1	D-A-D	670.860	12.505
BCN	<chem>CC1(C)c2cccc2N2c3ccc4cc(C#N)ccc4c3C(C)(C)c3cccc1c32</chem>	1	2	A-D-A	400.525	7.460
DPPSPF	<chem>O=S(=O)(c1cccc1)c1ccc(-c2ccc3c(c2)C(c2cccc2)(c2cccc2)c2cccc2-3)cc1</chem>	0	0	Unknown	534.680	8.550

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
Me2ACBO	<chem>CC(C)(c1ccc2c(c1)B1c3ccc(N4c5ccccc5C(C)(C)c5ccccc54)cc30c3cccc(c31)O2)c1ccc2c(c1)B1c3ccc(N4c5ccccc5C(C)(C)c5ccccc54)cc30c3cccc(c31)O2</chem>	2	0	2D-0A	994.809	13.686
F2AcBO	<chem>CC1(C)c2ccccc2N(c2ccc3c(c2)Oc2ccccc4c2B3c2cc(C3(c5ccc6c(c5)B5c7ccc(N8c9ccccc9C(C)(C)c9ccccc98)cc70c7cccc(c75)O6)c5ccccc5-c5ccc(cc53)ccc204)c2ccccc21O=S(=O)(c1ccccc1)c1ccc(-c2ccc3c(c2)C2(c4ccccc4-c4cccc42)c2ccccc2-3)cc1</chem>	2	0	2D-0A	1116.935	15.723
PSPSF	<chem>O=C1c2ccccc2S(=O)(=O)c2ccc(-c3ccc4c2ccccc4-c4cccc42)c2ccccc2-3)cc1</chem>	0	0	Unknown	532.664	8.530
TXO-P-Si	<chem>O=C1c2ccccc2S(=O)(=O)c2ccc(-c3ccc([Si](c4ccccc4)(c4ccccc4)c4ccccc4)cc3)cc21</chem>	0	0	Unknown	578.765	5.108
CzB-FMPIM	<chem>FC(F)(F)c1ccc(-n2c(-c3ccc(-c4ccc(-n5c6ccccc6c6ccccc65)cc4)cc3)nc(-c3ccccc3)c2-c2ccccc2)cc1</chem>	1	0	1D-0A	681.761	12.656
CzB-FMPPI	<chem>FC(F)(F)c1ccc(-n2c(-c3ccc(-c4ccc(-n5c6ccccc6c6ccccc65)cc4)cc3)nc3c4ccccc4c4ccccc4c32)cc1</chem>	1	0	1D-0A	679.745	12.782
BACF	<chem>CC1(C)c2ccccc2N(c2ccccc2)c2ccc3cc(F)ccc3c21</chem>	1	0	1D-0A	353.440	7.088
BACH	<chem>CC1(C)c2ccccc2N(c2ccccc2)c2ccc3ccccc3c21</chem>	1	0	1D-0A	335.450	6.949

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
PzTDBA	<chem>CC(C)(C)c1ccc2c(c1)B1c3cc(C(C)(C)C)ccc3Oc3cc(N4c5ccccc5N(c5cc6c7c(c5)Oc5ccc(C(C)(C)C)cc5B7c5cc(C(C)(C)C)ccc5O6)c5cccc54)cc(c31)O2</chem>	2	0	2D-0A	942.818	13.585
3TCPM	<chem>O=C(c1cccc1)c1ccc(-n2c3ccccc3c3cc(-c4ccc(N(c5cccc5)c5cccc5)cc4)ccc32)cc1</chem>	2	0	2D-0A	590.726	11.151
TPA-PPO	<chem>c1ccc(-c2nc(-c3ccc(-c4ccc(N(c5cccc5)c5cccc5)cc4)cn3)oc2-c2cccc2)cc1</chem>	1	0	1D-0A	541.654	10.207
CzPh-PPO	<chem>c1ccc(-c2nc(-c3ccc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)oc2-c2cccc2)cc1</chem>	1	0	1D-0A	539.638	9.835
m-CF3PIBI	<chem>FC(F)(F)c1ccc(-n2c(-c3ccc(-c4ccc(-c5ccc(-c6nc7cccc7n6-c6cccc6)cc5)cc4)cc3)nc3c4cccc4c4cccc4c32)c1</chem>	0	0	Unknown	782.869	14.358
DPTPEA	<chem>c1ccc(C(=C(c2cccc2)c2ccc(N(c3cccc3)c3cccc3)cc2)c2cccc2)cc1</chem>	1	0	1D-0A	499.657	10.164
PICNAnCz	<chem>N#Cc1ccc(-n2c(-c3ccc(-c4c5cccc5c(Br)c5cccc45)cc3)nc3c4cccc4c4cccc4c32)c1</chem>	0	2	0D-2A	650.579	11.606
DPA-DPQ	<chem>c1ccc(-c2cc3ncc(N(c4cccc4)c4cccc4)nc3cc2-c2cccc2)cc1</chem>	0	1	0D-1A	449.557	8.434
PhQLPXZ	<chem>c1ccc(-c2cc(-c3cccc3)c3c(N4c5cccc5Oc5cccc54)cc3n2)cc1</chem>	2	0	2D-0A	462.552	9.144

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
2DPA-DPQ	<chem>c1ccc(-c2cc3nc(N(c4cccc4) c4cccc4) c(N(c4cccc4) c4cccc4) nc3cc2-c2cccc2) cc1</chem>	0	1	0D-1A	616.768	11.903
FDQPXZ	<chem>Fc1cc2nc(-c3cccc3) c(-c3cccc3) nc2cc1N1c2cccc2O c2cccc21</chem>	2	1	D-A-D	481.530	8.678
DTRZSFX	<chem>c1ccc(-c2nc(-c3cccc3) nc(-c3ccc4c(c3) C3(c5cccc5O c5cccc53) c3cc(-c5nc(-c6cccc6) nc(-c6cccc6) n5) ccc3-4) n2) cc1</chem>	0	2	0D-2A	794.918	12.527
E-o-DHMPA	<chem>Oc1cccc1-c1c2cccc2c(-c2cccc2O) c2cccc12</chem>	0	0	Unknown	390.482	6.312
Z-o-DHMPA	<chem>Oc1cccc1-c1c2cccc2c(-c2cccc2O) c2cccc12</chem>	0	0	Unknown	390.482	6.312
p-DHMPA	<chem>Oc1ccc(-c2c3cccc3c(-c3cc(CO) cc3) c3cccc23) cc1</chem> <chem>O=C(c1ccc(N2c3cccc3C3(c4cccc4Sc4cccc43) c3cccc32) cc1</chem>	0	0	Unknown	390.482	6.312
STAC-BP-STAC	<chem>O=C(c1ccc(N2c3cccc3C3(c4cccc4Sc4cccc43) c3cccc32) cc1</chem> <chem>O=C(c1ccc(N2c3cccc3C3(c4cccc4O c4cccc43) c3cccc32) cc1</chem>	2	0	2D-0A	905.160	16.179
SXAC-BP-SXAC	<chem>O=C(c1ccc(N2c3cccc3C3(c4cccc4O c4cccc43) c3cccc32) cc1</chem>	2	0	2D-0A	873.024	15.461
m-DHMPA	<chem>Oc1cccc(-c2c3cccc3c(-c3cccc(CO) c3) c3cccc23) c1</chem>	0	0	Unknown	390.482	6.312

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
SFAC-BP-SFAC	<chem>O=C(c1ccc(N2c3ccccc3C3(c4ccccc4-c4ccccc43)c3ccccc32)cc1)c1ccc(N2c3ccccc3C3(c4ccccc4-c4ccccc43)c3ccccc32)cc1</chem>	2	0	2D-0A	841.026	15.210
PBTPA	<chem>CCN1c2ccccc2Sc2cc(-c3ccc(-c4ccc(N(c5ccccc5)c5ccccc5)cc4)c4nsnc34)ccc21</chem>	2	0	2D-0A	604.804	11.118
HDP	<chem>O=C1c2ccccc2C(=O)c2nc3c(c(c(0)c4ccccc43)nc21)CC(C)(C)c1ccc(N(c2ccc(-c3cc4nc5c6ccc(C#N)cc6c6cc(C#N)ccc6c5nc4cc3-c3cccn3)cc2)c2ccc(C(C)(C)C)cc2)cc1</chem>	0	1	0D-1A	326.311	3.264
PY-TPA	<chem>CC(C)(C)c1ccc(N(c2ccc(-c3cc4nc5c6ccc(C#N)cc6c6cc(C#N)ccc6c5nc4cc3-c3cccn3)cc2)c2ccc(C(C)(C)C)cc2)cc1</chem>	1	6	Multi-D/A	762.961	13.627
CN-TPA	<chem>CC(C)(C)c1ccc(N(c2ccc(-c3cc4nc5c6ccc(C#N)cc6c6cc(C#N)ccc6c5nc4cc3C#N)cc2)c2ccc(C(C)(C)C)cc2)cc1</chem>	1	8	Multi-D/A	710.885	12.436
DMIC-TRZ	<chem>CC1(C)CC(c2cccc(-c3nc(-c4ccccc4)nc(-c4ccccc4)n3)c2)C=c2[nH]c3cc4c(cc3c21)-c1ccccc1C=4</chem>	0	1	0D-1A	592.746	8.409
NyDPAc	<chem>CC1(C)c2ccccc2Nc2cccc(-c3ccc(-c4ccc5nc(-c6ccc(-c7cccc8c7C(C)(C)c7ccccc7N8)c6)ccc5n4)cc3)c21</chem>	0	0	Unknown	696.898	13.064
oCBP	<chem>c1ccc(-n2c3ccccc3c3ccccc32)c(-c2ccccc2-n2c3ccccc3c3ccccc32)c1</chem>	2	0	2D-0A	484.602	9.548
PABP	<chem>Cc1cc(-c2c3ccccc3c(-c3cccc3)c3ccccc23)ccc1-c1ccc(-c2c3ccccc3c(-c3ccccc3)c3ccccc23)cc1C</chem>	0	0	Unknown	686.898	15.251

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
DHID-DPS	<chem>CCn1c2cccc2c2c1c1cccc1n2-c1ccc(S(=O)(=O)c2ccc(-n3c4cccc4c4c3c3cccc3n4CC)cc2)cc1CC(C)(C)c1ccc(-n2c(-c3ccc(-c4c(F)c(F)c(-c5ccc(-c6nc7c8cccc8c8cccc8c7n6-c6ccc(C(C)(C)C)cc6)cc5)c(F)c4F)cc3)nc3c4cccc4c4cccc4c32)cc1Cc1cc(C)c(B(c2c(C)cc(C)cc2C)n2c3cccc3c3cc(-c4ccc5c(c4)c4cccc4n5-c4cccc4)ccc32)c(C)c1Cc1cc(C)c(B(c2ccc(-n3c4cccc4c4cc(-n5c6cccc6c6ccc5)ccc43)cc2)c2c(C)cc(C)cc2C)c(C)c1Cc1cc(C)c(B(c2ccc(-n3c4cccc4c4cc(N5c6cccc60c6ccc5)ccc43)cc2)c2c(C)cc(C)cc2C)c(C)c1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1cccc1-c1nc(-c2cccc2)nc(-c2ccc(-c3nc(-c4cccc4)nc(-c4cccc4-n4c5ccc(C(C)(C)C)cc5c5cc(C(C)(C)C)ccc54)n3)cc2)n1CCCCn1c2cccc2c2c1c1cccc1n2-c1ccc2c(c1)-c1cc(-n3c4cccc4c4c3c3cccc3n4CCCC)ccc1S2(=O)=O</chem>	0	0	Unknown	682.849	10.663
4FBTPI	<chem>c7c8cccc8c8cccc8c7n6-c6ccc(C(C)(C)C)cc6)cc5)c(F)c4F)cc3)nc3c4cccc4c4cccc4c32)cc1Cc1cc(C)c(B(c2c(C)cc(C)cc2C)n2c3cccc3c3cc(-c4ccc5c(c4)c4cccc4n5-c4cccc4)ccc32)c(C)c1Cc1cc(C)c(B(c2ccc(-n3c4cccc4c4cc(-n5c6cccc6c6ccc5)ccc43)cc2)c2c(C)cc(C)cc2C)c(C)c1Cc1cc(C)c(B(c2ccc(-n3c4cccc4c4cc(N5c6cccc60c6ccc5)ccc43)cc2)c2c(C)cc(C)cc2C)c(C)c1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1cccc1-c1nc(-c2cccc2)nc(-c2ccc(-c3nc(-c4cccc4)nc(-c4cccc4-n4c5ccc(C(C)(C)C)cc5c5cc(C(C)(C)C)ccc54)n3)cc2)n1CCCCn1c2cccc2c2c1c1cccc1n2-c1ccc2c(c1)-c1cc(-n3c4cccc4c4c3c3cccc3n4CCCC)ccc1S2(=O)=O</chem>	0	0	Unknown	999.168	18.797
BCzBMes	<chem>CCn1c2cccc2c2c1c1cccc1n2-c1ccc(S(=O)(=O)c2ccc(-n3c4cccc4c4c3c3cccc3n4CC)cc2)cc1CC(C)(C)c1ccc(-n2c(-c3ccc(-c4c(F)c(F)c(-c5ccc(-c6nc7c8cccc8c8cccc8c7n6-c6ccc(C(C)(C)C)cc6)cc5)c(F)c4F)cc3)nc3c4cccc4c4cccc4c32)cc1Cc1cc(C)c(B(c2c(C)cc(C)cc2C)n2c3cccc3c3cc(-c4ccc5c(c4)c4cccc4n5-c4cccc4)ccc32)c(C)c1Cc1cc(C)c(B(c2ccc(-n3c4cccc4c4cc(-n5c6cccc6c6ccc5)ccc43)cc2)c2c(C)cc(C)cc2C)c(C)c1Cc1cc(C)c(B(c2ccc(-n3c4cccc4c4cc(N5c6cccc60c6ccc5)ccc43)cc2)c2c(C)cc(C)cc2C)c(C)c1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1cccc1-c1nc(-c2cccc2)nc(-c2ccc(-c3nc(-c4cccc4)nc(-c4cccc4-n4c5ccc(C(C)(C)C)cc5c5cc(C(C)(C)C)ccc54)n3)cc2)n1CCCCn1c2cccc2c2c1c1cccc1n2-c1ccc2c(c1)-c1cc(-n3c4cccc4c4c3c3cccc3n4CCCC)ccc1S2(=O)=O</chem>	2	0	2D-0A	656.682	11.063
CCDMB	<chem>c7c8cccc8c8cccc8c7n6-c6ccc(C(C)(C)C)cc6)cc5)c(F)c4F)cc3)nc3c4cccc4c4cccc4c32)cc1Cc1cc(C)c(B(c2c(C)cc(C)cc2C)n2c3cccc3c3cc(-c4ccc5c(c4)c4cccc4n5-c4cccc4)ccc32)c(C)c1Cc1cc(C)c(B(c2ccc(-n3c4cccc4c4cc(-n5c6cccc6c6ccc5)ccc43)cc2)c2c(C)cc(C)cc2C)c(C)c1Cc1cc(C)c(B(c2ccc(-n3c4cccc4c4cc(N5c6cccc60c6ccc5)ccc43)cc2)c2c(C)cc(C)cc2C)c(C)c1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1cccc1-c1nc(-c2cccc2)nc(-c2ccc(-c3nc(-c4cccc4)nc(-c4cccc4-n4c5ccc(C(C)(C)C)cc5c5cc(C(C)(C)C)ccc54)n3)cc2)n1CCCCn1c2cccc2c2c1c1cccc1n2-c1ccc2c(c1)-c1cc(-n3c4cccc4c4c3c3cccc3n4CCCC)ccc1S2(=O)=O</chem>	2	0	2D-0A	656.682	10.248
PCDMB	<chem>c7c8cccc8c8cccc8c7n6-c6ccc(C(C)(C)C)cc6)cc5)c(F)c4F)cc3)nc3c4cccc4c4cccc4c32)cc1Cc1cc(C)c(B(c2c(C)cc(C)cc2C)n2c3cccc3c3cc(-c4ccc5c(c4)c4cccc4n5-c4cccc4)ccc32)c(C)c1Cc1cc(C)c(B(c2ccc(-n3c4cccc4c4cc(-n5c6cccc6c6ccc5)ccc43)cc2)c2c(C)cc(C)cc2C)c(C)c1Cc1cc(C)c(B(c2ccc(-n3c4cccc4c4cc(N5c6cccc60c6ccc5)ccc43)cc2)c2c(C)cc(C)cc2C)c(C)c1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1cccc1-c1nc(-c2cccc2)nc(-c2ccc(-c3nc(-c4cccc4)nc(-c4cccc4-n4c5ccc(C(C)(C)C)cc5c5cc(C(C)(C)C)ccc54)n3)cc2)n1CCCCn1c2cccc2c2c1c1cccc1n2-c1ccc2c(c1)-c1cc(-n3c4cccc4c4c3c3cccc3n4CCCC)ccc1S2(=O)=O</chem>	3	0	Multi-D/A	672.681	10.726
BTrztCz	<chem>c7c8cccc8c8cccc8c7n6-c6ccc(C(C)(C)C)cc6)cc5)c(F)c4F)cc3)nc3c4cccc4c4cccc4c32)cc1Cc1cc(C)c(B(c2c(C)cc(C)cc2C)n2c3cccc3c3cc(-c4ccc5c(c4)c4cccc4n5-c4cccc4)ccc32)c(C)c1Cc1cc(C)c(B(c2ccc(-n3c4cccc4c4cc(-n5c6cccc6c6ccc5)ccc43)cc2)c2c(C)cc(C)cc2C)c(C)c1Cc1cc(C)c(B(c2ccc(-n3c4cccc4c4cc(N5c6cccc60c6ccc5)ccc43)cc2)c2c(C)cc(C)cc2C)c(C)c1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1cccc1-c1nc(-c2cccc2)nc(-c2ccc(-c3nc(-c4cccc4)nc(-c4cccc4-n4c5ccc(C(C)(C)C)cc5c5cc(C(C)(C)C)ccc54)n3)cc2)n1CCCCn1c2cccc2c2c1c1cccc1n2-c1ccc2c(c1)-c1cc(-n3c4cccc4c4c3c3cccc3n4CCCC)ccc1S2(=O)=O</chem>	2	2	2D-2A	1095.452	19.443
DHID-DBS	<chem>CCn1c2cccc2c2c1c1cccc1n2-c1ccc2c(c1)-c1cc(-n3c4cccc4c4c3c3cccc3n4CCCC)ccc1S2(=O)=O</chem>	0	0	Unknown	736.941	12.204

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
CZ-DPS-DMAC	<chem>CC1(C)c2cccc2N(c2ccc(S(=O)(=O)c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)cc2)c2cccc21Cc1ccc(N(c2ccc(C)cc2)c2cccc2C2(c3cccc3N(c3ccc(C)cc3)c3ccc(C)cc3)CCCC2)cc</chem>	3	0	Multi-D/A	755.943	12.823
XT-T	<chem>CC1(C)c2cccc2N(c2nc(-c3cccc3)nc(-c3cccc3)n2)c2cc3c4cccc4n(-c4cccc4)c3cc1ccc(-c2cc(-c3cccc3)cc(-c3c4cccc4c(-c4ccc(-c5nc6c7cccc8ccc9cccc(c9c87)c6n5-c5cccc5)cc4)c4cccc34)c2)cc1N#Cc1c(-c2ccc(-n3c4cccc4c4cccc43)cc2)cncc1-c1ccc(-n2c3cccc3c3cccc32)cc1C=Cc1ccc(C0c2ccc3c(c2)c2cccc2n3-c2ccc(-n3c4cccc4c4cc(OCc5ccc(C=C)cc5)ccc43)c2C#N)cc1c1ccc(-c2cc(-c3cccc3)cc(-c3c4cccc4c(-c4ccc(-c5nc6c7cccc7c7cccc7c6n5-c5cccc5)cc4)c4cccc34)c2)cc</chem>	2	0	2D-0A	626.888	13.110
3DPTIA	<chem>CC1(C)c2cccc2N(c2nc(-c3cccc3)nc(-c3cccc3)n2)c2cc3c4cccc4n(-c4cccc4)c3cc1ccc(-c2cc(-c3cccc3)cc(-c3c4cccc4c(-c4ccc(-c5nc6c7cccc8ccc9cccc(c9c87)c6n5-c5cccc5)cc4)c4cccc34)c2)cc1</chem>	1	1	D-A	605.745	10.412
PyIAntPh	<chem>CC1(C)c2cccc2N(c2nc(-c3cccc3)nc(-c3cccc3)n2)c2cc3c4cccc4n(-c4cccc4)c3cc1ccc(-c2cc(-c3cccc3)cc(-c3c4cccc4c(-c4ccc(-c5nc6c7cccc8ccc9cccc(c9c87)c6n5-c5cccc5)cc4)c4cccc34)c2)cc1</chem>	0	0	Unknown	798.989	16.564
2CzP-INN	<chem>CC1(C)c2cccc2N(c2nc(-c3cccc3)nc(-c3cccc3)n2)c2cc3c4cccc4n(-c4cccc4)c3cc1ccc(-c2cc(-c3cccc3)cc(-c3c4cccc4c(-c4ccc(-c5nc6c7cccc8ccc9cccc(c9c87)c6n5-c5cccc5)cc4)c4cccc34)c2)cc1</chem>	2	1	D-A-D	586.698	10.481
VBNO	<chem>CC1(C)c2cccc2N(c2nc(-c3cccc3)nc(-c3cccc3)n2)c2cc3c4cccc4n(-c4cccc4)c3cc1ccc(-c2cc(-c3cccc3)cc(-c3c4cccc4c(-c4ccc(-c5nc6c7cccc8ccc9cccc(c9c87)c6n5-c5cccc5)cc4)c4cccc34)c2)cc1</chem>	2	2	2D-2A	697.838	12.196
PIAntPh	<chem>CC1(C)c2cccc2N(c2nc(-c3cccc3)nc(-c3cccc3)n2)c2cc3c4cccc4n(-c4cccc4)c3cc1ccc(-c2cc(-c3cccc3)cc(-c3c4cccc4c(-c4ccc(-c5nc6c7cccc8ccc9cccc(c9c87)c6n5-c5cccc5)cc4)c4cccc34)c2)cc1</chem>	0	0	Unknown	774.967	15.973
BCZ-DPS-AD	<chem>CC1(C)c2cccc2N(c2cc(N3c4cccc4C(C)(C)c4cccc43)cc(S(=O)(=O)c3ccc4c(c3)c3ccc3n4-c3cccc3)c2)c2cccc21</chem>	3	0	Multi-D/A	798.024	13.835

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
2BTPA-INN	<chem>N#Cc1c(-c2ccc(N(c3ccc(-c4cccc4)cc3)c3ccc(-c4cccc4)cc3)cc2)cncc1-c1ccc(N(c2ccc(-c3cccc3)cc2)c2ccc(-c3cccc3)cc2)cc1</chem>	2	1	D-A-D	895.122	17.895
SpiroAc	<chem>c1ccc2c(c1)Nc1cccc1C21c2cccc2-c2cccc21</chem>	0	0	Unknown	331.418	6.107
BuPCzBCO	<chem>Cc1ccc2c3cc(Br)c(C)cc3n(-c3ccc(C(C)(C)C)cc3)c2c1</chem>	1	0	1D-0A	406.367	7.461
AQ-S-CZ	<chem>O=C1c2cccc2C(=O)c2cc(Sc3ccc(-n4c5cccc5c5cccc54)cc3)ccc21</chem>	1	0	1D-0A	481.576	7.710
DBP-3TPA	<chem>c1ccc(N(c2cccc2)c2ccc(-c3ccc4nc5c6ccc(-c7ccc(N(c8cccc8)c8cccc8)cc7)cc6c6cc(-c7ccc(N(c8cccc8)c8cccc8)cc7)ccc6c5nc4c3)cc2)cc1</chem>	3	2	Multi-D/A	1010.257	20.500
DCzIPN	<chem>N#Cc1cc(C#N)c(-n2c3cccc3c3cccc32)cc1-n1c2cccc2c2cccc21</chem>	2	4	Multi-D/A	458.524	7.624
DMeCzIPN	<chem>Cc1cccc2c3cccc3n(-c3cc(-n4c5cccc5c5ccc(C)c54)c(C#N)cc3C#N)c12</chem>	2	4	Multi-D/A	486.578	8.241
SAFDPA	<chem>c1ccc(N(c2cccc2)c2ccc3c(c2)-c2cccc2C32c3cccc3N(c3cccc3)c3cccc32)cc1</chem>	2	0	2D-0A	574.727	11.303
t-DABNA	<chem>CC(C)(C)c1ccc(N2c3ccc(C(C)(C)C)cc3B3c4cc(C(C)(C)C)ccc4N(c4ccc(C(C)(C)C)cc4)c4cccc2c43)cc1</chem>	2	0	2D-0A	644.756	10.959

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
3-AcBPtCN	<chem>CC1(C)c2cccc2N(c2ccc3c(c2)c2cc(C#N)ccc2c2nc4cc(C#N)c(C#N)cc4nc32)c2cccc21</chem>	1	8	Multi-D/A	562.636	8.814
3-AcBPdCN	<chem>CC1(C)c2cccc2N(c2ccc3c(c2)c2cccc2c2nc4cc(C#N)c(C#N)cc4nc32)c2cccc21</chem>	1	6	Multi-D/A	537.626	8.942
11-AcBPdCN	<chem>CC1(C)c2cccc2N(c2ccc3nc4c5ccc(C#N)cc5c5cc(C#N)ccc5c4nc3c2)c2cccc21</chem>	1	6	Multi-D/A	537.626	8.942
PID-3	<chem>c1ccc(-n2c(-c3ccc(-c4cccs4)s3)nc3c4cccc4c4cccc4c32)cc1</chem>	0	0	Unknown	458.611	8.789
3-ACBPmCN	<chem>CC1(C)c2cccc2N(c2ccc3c(c2)c2cc(C#N)ccc2c2nc4cccc4nc32)c2cccc21</chem>	1	4	Multi-D/A	512.616	9.070
3-DTPA-BBI	<chem>O=c1c2ccc(-c3ccc(N(c4cccc4)c4cccc4)cc3)c3c(-c4ccc(N(c5cccc5)c5cccc5)cc4)ccc(c32)c2nc3cccc3n12</chem>	2	0	2D-0A	756.909	13.865
IABTCN	<chem>CC1(C)c2cccc2-c2cc3c(cc21)N(c1cccc1)c1ccc(-c2ccc(C#N)c4nsnc24)cc1C3(c1ccc1)c1cccc1</chem>	1	2	A-D-A	684.868	11.702
BTZPP	<chem>CC(C)(C)c1ccc(-n2c(-c3ccc(-c4cccc5nsnc45)cc3)cc3c2cc(-c2ccc(-c4cccc5nsnc45)cc2)n3-c2ccc(C(C)(C)C)cc2)cc1</chem>	0	0	Unknown	791.062	13.694
DPimdPPA	<chem>N#C/C(=C/c1ccc(-c2nc(-c3cccc3)c(-c3cccc3)[nH]2)c1)c1cccc1</chem>	0	1	0D-1A	423.519	7.475
BCzPh	<chem>c1ccc(-n2c3cccc3c3cc(-c4ccc5c(c4)c4cccc4n5-c4ccc4)ccc32)cc1</chem>	2	0	2D-0A	484.602	9.548

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
TBPM2CN	<chem>N#Cc1ccc(-c2ccc3c(c2)c2cc(-c4ccc(C#N)cc4)ccc2c2c3nc(-c3ccc(-c4ccc(N(c5ccccc5)c5ccccc5)cc4)cc3)n2-c2ccccc2)cc1</chem>	1	4	Multi-D/A	815.980	15.213
TPM2CN	<chem>N#Cc1ccc(-c2ccc3c(c2)c2cc(-c4ccc(C#N)cc4)ccc2c2c3nc(-c3ccc(N(c4ccccc4)c4ccccc4)cc3)n2-c2ccccc2)cc1</chem>	1	4	Multi-D/A	739.882	13.546
PT-BZ-DMAC	<chem>CC1(C)c2ccccc2N(c2ccc(C(=O)c3ccc(Sc4ccccc4)cc3)cc2)c2ccccc21</chem>	1	0	1D-0A	497.663	9.178
PS-BZ-DMAC	<chem>CC1(C)c2ccccc2N(c2ccc(C(=O)c3ccc(S(=O)(=O)c4ccccc4)cc3)cc2)c2ccccc21</chem>	1	0	1D-0A	529.661	7.860
2PCzBN-tPh	<chem>CC(C)(C)c1ccc(-c2cc(-n3c4ccc(-c5ccccc5)cc4c4cc(-c5ccccc5)ccc43)c(C#N)c(-n3c4ccc(-c5ccccc5)cc4c4cc(-c5ccccc5)ccc43)c2)cc1</chem>	2	2	2D-2A	870.112	17.385
2PCzBN	<chem>N#Cc1c(-n2c3ccc(-c4ccccc4)cc3c3cc(-c4ccccc4)ccc32)cccc1-n1c2ccc(-c3ccccc3)c2c2cc(-c3ccccc3)ccc21</chem>	2	2	2D-2A	737.906	14.420
TPA-2	<chem>O=C1c2ccccc2C(=O)c2c1ccc1nc(-c3ccc(N(c4ccccc4)c4ccccc4)cc3)c(-c3ccc(N(c4ccccc4)c4ccccc4)cc3)nc21</chem>	2	1	D-A-D	746.870	12.679
36DTPAFM	<chem>N#CC(C#N)=C1c2ccc(-c3ccc(N(c4ccccc4)c4ccccc4)cc3)c2-c2cc(-c3ccc(N(c4ccccc4)c4ccccc4)cc3)ccc21</chem>	2	2	2D-2A	714.872	13.790
AQTC	<chem>N#Cc1cc2nc3c(nc2c(C#N)c1C#N)-c1cccc2cccc-3c12</chem>	0	7	Multi-D/A	329.322	4.045

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
3TPAFM	<chem>N#CC(C#N)=C1c2cccc2-c2cc(-c3ccc(N(c4cccc4)c4ccc4)cc3)ccc21</chem>	1	2	A-D-A	471.563	8.653
BBPA	<chem>CCCC0c1cccc1-c1c2cccc2c(-c2cccc20CCCC)c2cccc12</chem>	0	0	Unknown	474.644	9.685
DMPA	<chem>C0c1ccc(-c2c3cccc3c(-c3ccc(OC)cc3OC)c3cccc23)c(OC)c1</chem>	0	0	Unknown	450.534	7.361
PXZ-NAI	<chem>O=C1c2cccc3c(N4c5cccc5Oc5cccc54)ccc(c23)C(=O)N1c1cccc1</chem>	2	0	2D-0A	454.485	7.216
PTZ-NAI	<chem>O=C1c2cccc3c(N4c5cccc5Sc5cccc54)ccc(c23)C(=O)N1c1cccc1</chem>	2	0	2D-0A	470.553	7.575
AI-4Cz	<chem>O=C1c2c(c(-n3c4cccc4c4cccc43)c(-n3c4cccc4c4cccc43)c(-n3c4cccc4c4cccc43)c2-n2c3cccc3c3cccc32)C(=O)N1c1cccc1</chem>	4	0	Multi-D/A	884.011	14.876
DAcIPN	<chem>CC1(C)c2cccc2N(c2cc(N3c4cccc4C(C)(C)c4cccc43)c(C#N)cc2C#N)c2cccc21</chem>	2	4	Multi-D/A	542.686	9.648
TCz1	<chem>CCCC(CC)Cn1c2ccc(-n3c4cccc4c4cccc43)cc2c2cc(-n3c4cccc4c4cccc43)ccc21</chem>	3	0	Multi-D/A	609.817	12.205
MBF	<chem>F[B-]1(F)OC(c2cccc2)=Cc2cc(-c3ccc(N4c5cccc5Oc5cccc54)cc3)nc[n+]21</chem>	2	1	D-A-D	503.317	7.363
MPA	<chem>O=C(Cc1cc(-c2ccc(N3c4cccc4Oc4cccc43)cc2)ncn1)c1cccc1</chem>	2	1	D-A-D	455.517	7.145
3,3-NHC	<chem>O=c1oc2cccc2c(OC)c1C(c1cc2cccc2c1)c1c(OC)c2cccc2oc1=O</chem>	0	0	Unknown	462.457	5.644

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
3CP-DPS-DMAC	<chem>CC1(C)c2cccc2N(c2ccc(S(=O)(=O)c3ccc4c(c3)c3cccc3n4-c3cccc3)cc2)c2cccc21</chem>	2	0	2D-0A	590.748	9.726
3CP-DPS-PXZ	<chem>O=S(=O)(c1ccc(N2c3cccc3O2)c3cccc32)cc1)c1ccc2c(c1)c1cccc1n2-c1cccc1</chem>	3	0	Multi-D/A	564.666	9.192
3CzAnN	<chem>c1ccc(-n2c3cccc3c3cc(-c4c5cccc5c(-c5ccc6cccc6c5)cc4)ccc32)cc1</chem>	1	0	1D-0A	545.685	11.577
DPQ-DMAC	<chem>CC1(C)c2cccc2N(c2ccc3nc(-c4ccccn4)c(-c4ccccn4)nc3c2)c2cccc21</chem>	1	1	D-A	491.598	7.863
DPAA	<chem>c1ccc(-c2c3cccc3c(-c3ccc3)ccc23)cc1</chem>	0	0	Unknown	330.430	7.327
t3Cz-SO	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc2c(c1)c1cc(-n3c4ccc(C(C)(C)C)c4c4cc(C(C)(C)C)ccc43)ccc1n2-c1ccc(S(=O)(=O)c2cccc2)cc1</chem>	3	0	Multi-D/A	938.294	17.001
Cz-SO	<chem>O=S(=O)(c1cccc1)c1ccc(-n2c3cccc3c3cccc32)cc1</chem>	1	0	1D-0A	383.472	5.617
DPQ-PXZ	<chem>c1ccc(-c2nc3ccc(N4c5cccc5O4c5cccc54)cc3nc2-c2ccccn2)nc1</chem>	2	1	D-A-D	465.516	7.329
pCzAnN	<chem>c1ccc2cc(-c3c4cccc4c(-c4ccc(-n5c6cccc6c6cccc65)cc4)c4cccc34)ccc2c1</chem>	1	0	1D-0A	545.685	11.577
3-BHH	<chem>O=c1oc2cccc2c(O)c1C(c1cc(Br)s1)c1c(O)c2cccc2oc1=O</chem>	0	0	Unknown	497.322	5.315

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
DPP-PXZ	<chem>c1ccc(-c2nc3cc(N4c5ccccc5Oc5ccccc54)cnc3nc2-c2ccccn2)nc1</chem>	1	0	1D-0A	466.504	6.724
APPT-PXZ	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc2c3c(cccc13)-c1nc3c4cccnc4c4ncccc4c3nc1-2</chem>	2	1	D-A-D	537.582	9.102
HITPE	<chem>0c1ccc(-c2ccc(C(=C(c3cccc3)c3cccc3)cc2)cc1-c1nc(-c2cccc2)c(-c2cccc2)n1-c1cccc1C=Cc1ccc(COc2cccc3c2cccc2n3-c2cc(-n3c4cccc4c4c(OCc5ccc(C=C)cc5)cccc43)c(C#N)c(-n3c4cccc4c4c(OCc5ccc(C=C)cc5)cccc43)c2)cc1</chem>	0	0	Unknown	718.900	13.253
DV-3CzCN	<chem>0=S(=O)(c1ccc(-c2ccc3c(c2)C2(c4cccc4-c4cccc42)c2cccc2-3)cc1)c1ccc(-c2ccc3c(c2)C2(c4cccc4-c4cccc42)c2cccc2-3)cc1</chem>	3	2	Multi-D/A	995.195	17.516
SF-DPSO	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c2nc(-c3cccc3)c(-c3cccc3)nc12</chem>	0	0	Unknown	847.051	14.540
BDQ-tBuCz	<chem>CC1(C)c2cccc2N(c2ccc(N3c4cccc4C(C)(C)c4cccc43)c3nc(-c4cccc4)c(-c4cccc4)nc23)c2cccc21</chem>	2	1	D-A-D	837.168	16.348
BDQDMAC	<chem>c1ccc(-c2nc3c(-c4ccc(-n5c6cccc6c6cccc65)cc4)ccc(-c4ccc(-n5c6cccc6c6cccc65)cc4)c3nc2-c2cccc2)cc1</chem>	2	1	D-A-D	696.898	13.182
BDQPCz	<chem>c1ccc(-c2nc3c(-c4ccc(-n5c6cccc6c6cccc65)cc4)ccc(-c4ccc(-n5c6cccc6c6cccc65)cc4)c3nc2-c2cccc2)cc1</chem>	2	1	D-A-D	764.932	14.492

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
4-SCZ	<chem>c1ccc2c(c1)-c1ccccc1C21c2cccc2-c2c1ccc1[nH]c3cccc</chem>	1	0	1D-0A	405.500	7.665
Py-TPA	<chem>c3c21CC(C)(C)c1ccc(N(c2ccc(-c3cc4nc5c6ccc(-c7ccncc7)nc6c6nc(-c7ccncc7)ccc6c5nc4cc3C#N)cc2)c2ccc(C(C)(C)C)cc2)cc1CC(C)(C)c1ccc(-n2c(-c3ccc(-c4ccc(N5c6cccc6C(c6ccc6)(c6cccc6)c6cccc65)c4)cc3)nc3c4cccc4c4cccc4c32)cc1N#Cc1ccc(-c2cc(C#N)cc(-c3ccc(C#N)cc3)c2N2c3cccc30c3cccc32)cc1N#Cc1ccc(-c2cc(N3c4cccc40c4cccc43)cc(-c3ccc(C#N)cc3)c2C#N)cc1CCCC(C)Cn1c2cc(C#N)ccc2c2ccc(C#Cc3ccc4ccc5cccc6ccc3c4c56)cc21</chem>	1	4	Multi-D/A	817.013	13.607
DPACPhTPI	<chem>c1ccc2c(c1)0c1ccccc1N2c1cc2c3cccc3c3ccc4ccc5c6ccc6c1c1c5c4c3c21O=C1c2cccc2C(=O)c2cc(N(c3ccc4c(c3)C(=O)c3cccc3C4=O)c3ccc4c(c3)C(=O)c3cccc3C4=O)ccc21O=C(c1ccc(N2c3cccc30c3ccc32)cc1)c1ccc2c(c1)c1ccc1n2-c1cccc(-n2c3cccc3c3cccc32)c1</chem>	1	0	1D-0A	834.079	16.129
oPTBC	<chem>c1ccc2c(c1)0c1ccccc1N2c1cc2c3cccc3c3ccc4ccc5c6ccc6c1c1c5c4c3c21O=C1c2cccc2C(=O)c2cc(N(c3ccc4c(c3)C(=O)c3cccc3C4=O)c3ccc4c(c3)C(=O)c3cccc3C4=O)ccc21O=C(c1ccc(N2c3cccc30c3ccc32)cc1)c1ccc2c(c1)c1ccc1n2-c1cccc(-n2c3cccc3c3cccc32)c1</chem>	2	6	Multi-D/A	486.534	8.211
mPTBC	<chem>c1ccc2c(c1)0c1ccccc1N2c1cc2c3cccc3c3ccc4ccc5c6ccc6c1c1c5c4c3c21O=C1c2cccc2C(=O)c2cc(N(c3ccc4c(c3)C(=O)c3cccc3C4=O)c3ccc4c(c3)C(=O)c3cccc3C4=O)ccc21O=C(c1ccc(N2c3cccc30c3ccc32)cc1)c1ccc2c(c1)c1ccc1n2-c1cccc(-n2c3cccc3c3cccc32)c1</chem>	2	6	Multi-D/A	486.534	8.211
P3C	<chem>c1ccc2c(c1)0c1ccccc1N2c1cc2c3cccc3c3ccc4ccc5c6ccc6c1c1c5c4c3c21O=C1c2cccc2C(=O)c2cc(N(c3ccc4c(c3)C(=O)c3cccc3C4=O)c3ccc4c(c3)C(=O)c3cccc3C4=O)ccc21O=C(c1ccc(N2c3cccc30c3ccc32)cc1)c1ccc2c(c1)c1ccc1n2-c1cccc(-n2c3cccc3c3cccc32)c1</chem>	1	2	A-D-A	528.699	10.180
DBCP	<chem>c1ccc2c(c1)0c1ccccc1N2c1cc2c3cccc3c3ccc4ccc5c6ccc6c1c1c5c4c3c21O=C1c2cccc2C(=O)c2cc(N(c3ccc4c(c3)C(=O)c3cccc3C4=O)c3ccc4c(c3)C(=O)c3cccc3C4=O)ccc21O=C(c1ccc(N2c3cccc30c3ccc32)cc1)c1ccc2c(c1)c1ccc1n2-c1cccc(-n2c3cccc3c3cccc32)c1</chem>	2	0	2D-0A	531.614	11.648
NAQ3	<chem>c1ccc2c(c1)0c1ccccc1N2c1cc2c3cccc3c3ccc4ccc5c6ccc6c1c1c5c4c3c21O=C1c2cccc2C(=O)c2cc(N(c3ccc4c(c3)C(=O)c3cccc3C4=O)c3ccc4c(c3)C(=O)c3cccc3C4=O)ccc21O=C(c1ccc(N2c3cccc30c3ccc32)cc1)c1ccc2c(c1)c1ccc1n2-c1cccc(-n2c3cccc3c3cccc32)c1</chem>	1	0	1D-0A	635.631	7.483
mCP-BP-DMAC	<chem>c1ccc2c(c1)0c1ccccc1N2c1cc2c3cccc3c3ccc4ccc5c6ccc6c1c1c5c4c3c21O=C1c2cccc2C(=O)c2cc(N(c3ccc4c(c3)C(=O)c3cccc3C4=O)c3ccc4c(c3)C(=O)c3cccc3C4=O)ccc21O=C(c1ccc(N2c3cccc30c3ccc32)cc1)c1ccc2c(c1)c1ccc1n2-c1cccc(-n2c3cccc3c3cccc32)c1</chem>	4	0	Multi-D/A	693.806	12.687

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
BFCZPZ3	<chem>c1ccc2c(c1)oc1ccc3c4ccccc4n(-c4cnc(-n5c6ccccc6c6cc7oc8ccccc8c7c65)cn4)c3c12</chem>	2	0	2D-0A	590.642	10.470
BTCZPZ1	<chem>c1ccc2c(c1)sc1c2ccc2c3cccc3n(-c3cnc(-n4c5ccccc5c5ccc6c7ccccc7sc6c54)cn3)c21</chem>	2	0	2D-0A	622.778	11.407
BFCZPZ2	<chem>c1ccc2c(c1)oc1c2ccc2c1c1cccc1n2-c1cnc(-n2c3ccccc3c3c4oc5ccccc5c4ccc32)cn1</chem>	2	0	2D-0A	590.642	10.470
CZ9CZPZ	<chem>c1ccc2c(c1)c1ccccc1n2-c1cnc(-n2c3ccccc3c3cc(-n4c5ccccc5c5ccccc54)ccc32)cn1</chem>	4	0	Multi-D/A	740.870	12.865
BFCZPZ1	<chem>c1ccc2c(c1)oc1c2ccc2c3cccc3n(-c3cnc(-n4c5ccccc5c5ccc6c7ccccc7oc6c54)cn3)c21</chem>	2	0	2D-0A	590.642	10.470
TPBPPI-PY	<chem>c1ccc(-c2cc(-c3ccc(-c4nc5c6ccccc6c6ccccc6c5n4-c4cc(-c5cccn5)cc4)cc3)c(-c3ccccc3)c(-c3ccccc3)c2-c2ccccc2)cc1</chem>	0	0	Unknown	828.031	16.396
TPBPPI-PBI	<chem>c1ccc(-c2cc(-c3ccc(-c4nc5c6ccccc6c6ccccc6c5n4-c4cc(-c5ccc(-c6nc7ccccc7n6-c6ccccc6)cc5)cc4)cc3)c(-c3ccccc3)c(-c3ccccc3)c2-c2ccccc2)cc1</chem>	0	0	Unknown	1019.264	20.007
PTZ-2PTO	<chem>O=S1(=O)c2ccccc2N(c2ccccc2)c2cc(N3c4ccccc4Sc4ccccc43)ccc21</chem>	4	0	Multi-D/A	504.636	8.237

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
4F-m- ν -DABNA	<chem>Cc1cccc(N(c2cccc(C)c2)c2cc3c4c(c2)N(c2c(F)cccc2F)c2cc5c(cc2B4c2ccc(C)cc2N3c2cccc(C)c2)B2c3ccc(C)cc3N(c3cccc(C)c3)c3cc(N(c4cccc(C)c4)c4cccc(C)c4)cc(c32)N5c2c(F)cccc2F)c1Cc1cccc(N(c2cccc(C)c2)c2cc3c4c(c2)N(c2cccc2)c2cc5c(cc2B4c2ccc(C)cc2N3c2cccc(C)c2)B2c3ccc(C)cc3N(c3cccc(C)c3)c3cc(N(c4cccc(C)c4)c4cccc(C)c4)cc(c32)N5c2cccc2)c1</chem>	6	0	Multi-D/A	1 281.132	19.815
m- ν -DABNA	<chem>Cc1cccc(N(c2cccc(C)c2)c2cc3c4c(c2)N(c2cccc2)c2cc5c(cc2B4c2ccc(C)cc2N3c2cccc(C)c2)B2c3ccc(C)cc3N(c3cccc(C)c3)c3cc(N(c4cccc(C)c4)c4cccc(C)c4)cc(c32)N5c2cccc2)c1</chem>	6	0	Multi-D/A	1 209.172	19.259
TPA-PyI	<chem>Brc1cccc(-n2c(-c3ccc(-c4ccc(N(c5cccc5)c5cccc5)cc4)cc3)nc(-c3cccc3)c2-c2cccc2)c1Fc1cccc(F)c1N1c2cc3c(cc2B2c4cccc4N(c4cccc4)c4cc(N(c5cccc5)c5cccc5)cc1c42)B1c2cccc2N(c2cccc2)c2cc(N(c4cccc4)c4cccc4)cc(c21)N3c1c(F)cccc1Fc1ccc(-c2nc(-c3cccc3)nc(-c3cccc(-c4cc(-n5c6cccc6c6cccc65)ccc4-c4ccc(-n5c6cccc6c6cccc65)cc4)c3)n2)cc1c1ccc(-n2c(-c3ccc(-c4cc(-n5c6cccc6c6cccc65)ccc4-c4ccc(-n5c6cccc6c6cccc65)cc4)cc3)nc3cccc32)cc1</chem>	1	0	1D-0A	694.676	12.773
4F- ν -DABNA	<chem>Cc1cccc(N(c2cccc(C)c2)c2cc3c4c(c2)N(c2cccc2)c2cc5c(cc2B4c2ccc(C)cc2N3c2cccc(C)c2)B2c3ccc(C)cc3N(c3cccc(C)c3)c3cc(N(c4cccc(C)c4)c4cccc(C)c4)cc(c32)N5c2cccc2)c1</chem>	6	0	Multi-D/A	1 168.916	17.348
CBPTZ	<chem>c1ccc(-c2nc(-c3cccc3)nc(-c3cccc(-c4cc(-n5c6cccc6c6cccc65)ccc4-c4ccc(-n5c6cccc6c6cccc65)cc4)c3)n2)cc1</chem>	2	1	D-A-D	791.958	14.401
CBPBM	<chem>c1ccc(-c2nc(-c3cccc3)nc(-c3cccc(-c4cc(-n5c6cccc6c6cccc65)ccc4-c4ccc(-n5c6cccc6c6cccc65)cc4)cc3)nc3cccc32)cc1</chem>	2	0	2D-0A	752.921	14.221

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
CBPPY	<chem>c1cncc(-c2cc(-n3c4cccc4c4cccc43)ccc2-c2ccc(-n3c4cccc4c4cccc43)cc2)c1</chem>	2	0	2D-0A	561.688	10.610
N-B	<chem>Cc1cc(C)c(B(c2cccc2)c2c(C)cc(C)cc2C)c(C)c1</chem>	0	0	Unknown	326.292	4.053
DPACpBr	<chem>Cc1cc(N2c3cccc3C(c3cccc3)(c3cccc3)c3cccc32)cc(C)c1Br</chem>	1	0	1D-0A	516.482	9.232
DPCN	<chem>CC1(C)c2cccc2N(c2ccc(-c3ccc(-c4nc5c6cccc6c6cccc6c5n4-c4ccc(C#N)cc4)cc3)c2)c2cccc21</chem>	1	2	A-D-A	678.839	12.647
BZC-PXZ	<chem>O=C(CC(=O)c1cccc10)c1ccc(N2c3cccc30c3cccc32)cc1</chem>	2	0	2D-0A	421.452	6.423
CzBN	<chem>N#Cc1c(-n2c3cccc3c3cccc32)cccc1-n1c2cccc2c2cccc21</chem>	2	2	2D-2A	433.514	7.752
2CzBN	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1cccc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1C#N</chem>	2	2	2D-2A	657.946	12.942
TPA-DPPI	<chem>c1ccc(-c2nc3c4cccc4c4ccc4c3n2-c2ccc(-n3c(-c4ccc(-c5ccc(N(c6cccc6)c6cccc6)cc5)cc4)nc4c5cccc5c5cccc5c43)cc2)cc1</chem>	1	0	1D-0A	906.105	17.448
ANQDC-PSTA	<chem>N#Cc1cc2nc3c(nc2cc1C#N)-c1ccc(N2c4ccc(-c5cccc5)cc4C4(c5cccc5-c5cccc54)c4ccc5c(sc6cccc65)c42)c2ccc-3c12</chem>	1	5	Multi-D/A	815.961	14.355

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
pDPAPQ-PXZ	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc2c3c(cccc13)-c1nc3cc(-c4ccncc4)c(-c4ccncc4)cc3nc</chem>	2	1	D-A-D	589.658	10.130
m-CZ-DPS-DMAC	<chem>CC1(C)c2cccc2N(c2cccc(S(=O)(=O)c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)c2)c2cccc21</chem>	3	0	Multi-D/A	755.943	12.823
pDPBPZ-PXZ	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc2c(c1)c1cccc1c1nc3cc(-c4ccncc4)c(-c4ccncc4)cc3nc</chem>	2	2	2D-2A	615.696	10.789
DDPhCz-DPPZ	<chem>c1ccc(-c2ccc3c(c2)c2cc(-c4cccc4)ccc2n3-c2cc3nc4c5ccnc5c5ncccc5c4nc3cc2-n2c3ccc(-c4cccc4)cc3c3cc(-c4cccc4)ccc32)cc1</chem>	2	2	2D-2A	917.088	16.742
DCz-DPPZ	<chem>c1cnc2c(c1)c1nc3cc(-n4c5cccc5c5cccc54)c(-n4c5cccc5c5cccc54)cc3nc1c1cccn</chem>	2	2	2D-2A	612.696	10.074
Oxd-bCZ	<chem>c12c1ccc2c(c1)c1cccc1n2-c1ccc(-c2ccc(-c3nnc(-c4ccc(-c5ccc(-n6c7cccc7c7cccc76)cc5)cc4)o3)cc2)cc1</chem>	2	0	2D-0A	704.833	12.932
Fene	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc2cccc(N3c4cccc40c4cccc43)c2n1</chem>	3	0	Multi-D/A	491.550	9.386
Fens	<chem>c1ccc2c(c1)Sc1cccc1N2c1ccc2cccc(N3c4cccc4Sc4cccc43)c2n1</chem>	3	0	Multi-D/A	523.686	10.104
Yad	<chem>CC1(C)c2cccc2N(c2ccc3ccc(N4c5cccc5C(C)(C)c5cccc54)c3n2)c2cccc21</chem>	1	0	1D-0A	543.714	10.453

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
DTPCZPYTZ	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-c2ccc3ccc4c(-c5ccc(-c6nnc(-c7cccc7)n6-c6cccc6)cc5)cc5ccc2c3c54)cc1</chem>	1	0	1D-0A	851.110	16.525
TPADSO2	<chem>O=S1(=O)c2cccc2S(=O)(=O)c2cc(-c3ccc(N(c4cccc4)c4cccc4)cc3)ccc21</chem>	1	0	1D-0A	523.635	6.802
B4PyPPM	<chem>c1ccc(-c2nc(-c3cc(-c4ccnc4)cc(-c4ccncc4)c3)cc(-c3cc(-c4ccncc4)cc(-c4ccncc4)c3)n2)cc1</chem>	0	1	0D-1A	616.728	9.726
mAP	<chem>CC1(C)c2cccc2N(c2cccc(N3c4cccc4C(C)(C)c4cccc43)c2)c2cccc21</chem>	2	0	2D-0A	492.666	9.905
DBBPZ-DPXZ	<chem>c1ccc(-c2cc3nc4c5ccc(N6c7cccc70c7cccc76)cc5c5cc(N6c7cccc70c7cccc76)ccc5c4nc3cc2-c2cccc2)cc1</chem>	4	2	Multi-D/A	794.914	15.574
DPIBPZ-DPXZ	<chem>c1ccc2c(c1)0c1cccc1N2c1cc2c(c1)c1cc(N3c4cccc40c4cccc43)ccc1c1nc3cc(-c4cncnc4)c(-c4cncnc4)cc3nc21</chem>	4	4	Multi-D/A	798.866	13.154
m-DTPACO	<chem>O=C(c1ccc(N(c2cccc2)c2ccc2)cc1)c1cccc(C(=O)c2ccc(N(c3cccc3)c3cccc3)cc2)c1</chem>	2	0	2D-0A	620.752	11.088
mDPBPZ-DPXZ	<chem>c1cncc(-c2cc3nc4c5ccc(N6c7cccc70c7cccc76)cc5c5cc(N6c7cccc70c7cccc76)ccc5c4nc3cc2-c2ccnc2)c1</chem>	4	2	Multi-D/A	796.890	14.364
p-DTPACO	<chem>O=C(c1ccc(C(=O)c2ccc(N(c3cccc3)c3cccc3)cc2)cc1)c1ccc(N(c2cccc2)c2cccc2)cc1</chem>	2	0	2D-0A	620.752	11.088

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
2TDBA-SBA	<chem>CC(C)(C)c1ccc2c(c1)B1c3ccc(C(C)(C)C)ccc3Oc3ccc(N4c5ccccc5C5(c6cccc64)c4cccc4N(c4cc6c7c(c4)Oc4ccc(C(C)(C)C)cc4B7c4cc(C(C)(C)C)ccc4O6)c4cccc45)cc(c31)O2</chem>	2	0	2D-0A	1107.025	16.281
TPA-APm	<chem>c1ccc(N(c2cccc2)c2ccc(-c3cnc4nc5c(nc4n3)-c3cccc4ccc-5c34)cc2)cc1</chem>	1	0	1D-0A	499.577	8.357
TPA-AP	<chem>c1ccc(N(c2cccc2)c2ccc(-c3ccc4nc5c(nc4c3)-c3cccc4ccc-5c34)cc2)cc1</chem>	1	1	D-A	497.601	9.567
TDBA-SBA	<chem>CC(C)(C)c1ccc(N2c3cccc3C3(c4cccc42)c2cccc2N(c2c4c5c(c2)Oc2ccc(C(C)(C)C)cc2B5c2cc(C(C)(C)C)ccc2O4)c2cccc23)cc1</chem>	2	0	2D-0A	858.935	14.256
TPA-APy	<chem>c1ccc(N(c2cccc2)c2ccc(-c3cnc4nc5c(nc4c3)-c3cccc4ccc-5c34)cc2)cc1</chem>	1	0	1D-0A	498.589	8.962
TPD4PA	<chem>c1ccc(N(c2cccc2)c2cc3c4c(c2)N(c2cccc2)c2cc5c6c(c2B4c2cccc2N3c2cccc2)Oc2cccc2B6c2cccc2O5)cc1</chem>	3	0	Multi-D/A	779.517	9.967
o-Ac2BBP	<chem>CC1(C)c2cccc2N(c2ccc(C(=O)c3cccc3C(=O)c3ccc(N4c5cccc5C(C)(C)c5cccc54)cc3)cc2)c2cccc21</chem>	2	0	2D-0A	700.882	12.367
TPA-PyF2	<chem>Cc1ccc(N(c2ccc(C3(c4ccnc4)c4cc(C(C)(C)C)ccc4-c4cc(C(C)(C)C)cc43)cc2)c2ccc(C3(c4ccnc4)c4cc(C(C)(C)C)ccc4-c4ccc(C(C)(C)C)cc43)cc2)cc1</chem>	1	0	1D-0A	966.370	18.171

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
o-Px2BBP	<chem>O=C(c1ccc(N2c3cccc30c3ccccc32)cc1)c1cccc1C(=O)c1ccc(N2c3cccc30c3cccc32)cc1</chem>	4	0	Multi-D/A	648.718	11.300
D-BP-PXZ	<chem>O=C(c1ccc(N2c3cccc30c3ccccc32)cc1)c1cccc1-c1cccc1C(=O)c1ccc(N2c3cccc30c3cccc32)cc1</chem>	4	0	Multi-D/A	724.816	12.967
BTD-PXZ	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc2nsnc2c1Cc1cc(-n2c3ccc(C(C)(C)C)c3c3cc(C(C)(C)C)ccc32)cc(C)C1B1c2cccc2B(c2c(C)cc(-n3c4ccc(C(C)(C)C)cc4c4cc(C(C)(C)C)ccc43)cc2C)c2ccccc21CC(C)(C)N1C(=O)c2cc(-n3c4cccc4c4cccc43)ccc2S1(=O)=O</chem>	2	0	2D-0A	317.373	5.267
tBuCzDBA	<chem>Cc1cc(-n2c3ccc(C(C)(C)C)c3c3cc(C(C)(C)C)ccc32)cc(C)C1B1c2cccc2B(c2c(C)cc(-n3c4ccc(C(C)(C)C)cc4c4cc(C(C)(C)C)ccc43)cc2C)c2ccccc21CC(C)(C)N1C(=O)c2cc(-n3c4cccc4c4cccc43)ccc2S1(=O)=O</chem>	2	0	2D-0A	938.962	13.650
5-Cz-Sac	<chem>CC(C)(C)N1C(=O)c2cc(-n3c4cccc4c4cccc43)ccc2S1(=O)=O</chem>	1	0	1D-0A	404.491	4.727
6-Cz-Sac	<chem>CC(C)(C)N1C(=O)c2ccc(-n3c4cccc4c4cccc43)cc2S1(=O)=O</chem>	1	0	1D-0A	404.491	4.727
DiCz-Sac	<chem>CC(C)(C)N1C(=O)c2cc(-n3c4cccc4c4cccc43)c(-n3c4cccc4c4cccc43)cc2S1(=O)=O</chem>	2	0	2D-0A	569.686	7.824
2CM	<chem>CCCCCCCCn1c2cccc2c2cc(Nc3nc(Nc4ccc5c(c4)c4cccc4n5CCCCCCCC)nc(OC)n3)ccc21Cc1cc(C)c(B(c2cccc2-n2c3cccc3c3cccc32)c2c(C)cc(C)cc2C)c(C)c1</chem>	2	1	D-A-D	695.956	12.304
CzoB	<chem>Cc1cc(C)c(B(c2cccc2-n2c3cccc3c3cccc32)c2c(C)cc(C)cc2C)c(C)c1</chem>	1	0	1D-0A	491.487	7.150
CzBroB	<chem>Cc1cc(C)c(B(c2ccc(Br)cc2-n2c3cccc3c3cccc32)c2c(C)cc(C)cc2C)c(C)c1</chem>	1	0	1D-0A	570.383	7.913

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
PxCN	<chem>N#Cc1c(-c2ccc(N3c4cccc40c4cccc43)cc2)cncc1-c1ccc(N2c3cccc30c3cccc32)cc1CC1(C)c2cccc2N(c2ccc(-c3cncc(-c4ccc(N5c6cccc6C(C)(C)c6cccc65)cc4)c3C#N)c2)c2cccc21Cc1cc(C)c(B(c2ccc(C#N)cc2-n2c3cccc3c3cccc32)c2c(C)cc(C)cc2C)c(C)c1N#Cc1nc2c3ccc(-c4ccc(N(c5cccc5)c5cccc5)cc4)nc3c3nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)ccc3c2nc1C#N0=C1c2ccc(N3c4cccc4C4(c5cccc5-c5cccc54)c4cccc43)ccc21CC1(C)c2cccc2N(c2ccc3c(c2)-c2cc(N4c5cccc5C(C)(C)c5cccc54)ccc2C3=O)c2cccc21CC1(C)c2cccc2N(c2ccc3c4c(cccc24)C(=O)N(c2cccc2)C3=O)c2cccc21CC(C)(C)c1ccc(-n2c(-c3ccc4ccc5cccc6ccc3c4c56)nc(-c3cccc3)c2-c2cccc2)cc1c1ccc2c(c1)c1cccc1n2-c1cc(-c2ccc(-c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)c3nsnc23)cc(-n2c3cccc3c3cccc32)c1</chem>	4	1	Multi-D/A	618.696	11.438
AcCN	<chem>cnc(-c4ccc(N5c6cccc6C(C)(C)c6cccc65)cc4)c3C#N)c2)c2cccc21Cc1cc(C)c(B(c2ccc(C#N)cc2-n2c3cccc3c3cccc32)c2c(C)cc(C)cc2C)c(C)c1N#Cc1nc2c3ccc(-c4ccc(N(c5cccc5)c5cccc5)cc4)nc3c3nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)ccc3c2nc1C#N0=C1c2ccc(N3c4cccc4C4(c5cccc5-c5cccc54)c4cccc43)ccc21CC1(C)c2cccc2N(c2ccc3c(c2)-c2cc(N4c5cccc5C(C)(C)c5cccc54)ccc2C3=O)c2cccc21CC1(C)c2cccc2N(c2ccc3c4c(cccc24)C(=O)N(c2cccc2)C3=O)c2cccc21CC(C)(C)c1ccc(-n2c(-c3ccc4ccc5cccc6ccc3c4c56)nc(-c3cccc3)c2-c2cccc2)cc1c1ccc2c(c1)c1cccc1n2-c1cc(-c2ccc(-c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)c3nsnc23)cc(-n2c3cccc3c3cccc32)c1</chem>	2	1	D-A-D	670.860	12.505
CzCNoB	<chem>-n2c3cccc3c3cccc32)c2c(C)cc(C)cc2C)c(C)c1N#Cc1nc2c3ccc(-c4ccc(N(c5cccc5)c5cccc5)cc4)nc3c3nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)ccc3c2nc1C#N0=C1c2ccc(N3c4cccc4C4(c5cccc5-c5cccc54)c4cccc43)ccc21CC1(C)c2cccc2N(c2ccc3c(c2)-c2cc(N4c5cccc5C(C)(C)c5cccc54)ccc2C3=O)c2cccc21CC1(C)c2cccc2N(c2ccc3c4c(cccc24)C(=O)N(c2cccc2)C3=O)c2cccc21CC(C)(C)c1ccc(-n2c(-c3ccc4ccc5cccc6ccc3c4c56)nc(-c3cccc3)c2-c2cccc2)cc1c1ccc2c(c1)c1cccc1n2-c1cc(-c2ccc(-c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)c3nsnc23)cc(-n2c3cccc3c3cccc32)c1</chem>	1	2	A-D-A	516.497	7.022
CNPP-TPA	<chem>cnc(-c4ccc(N(c5cccc5)c5cccc5)cc4)nc3c3nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)ccc3c2nc1C#N0=C1c2ccc(N3c4cccc4C4(c5cccc5-c5cccc54)c4cccc43)ccc21CC1(C)c2cccc2N(c2ccc3c(c2)-c2cc(N4c5cccc5C(C)(C)c5cccc54)ccc2C3=O)c2cccc21CC1(C)c2cccc2N(c2ccc3c4c(cccc24)C(=O)N(c2cccc2)C3=O)c2cccc21CC(C)(C)c1ccc(-n2c(-c3ccc4ccc5cccc6ccc3c4c56)nc(-c3cccc3)c2-c2cccc2)cc1c1ccc2c(c1)c1cccc1n2-c1cc(-c2ccc(-c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)c3nsnc23)cc(-n2c3cccc3c3cccc32)c1</chem>	2	3	Multi-D/A	768.884	12.743
SPAC-FO	<chem>3)cc2-c2cc(N3c4cccc4C4(c5cccc5-c5cccc54)c4cccc43)ccc21CC1(C)c2cccc2N(c2ccc3c(c2)-c2cc(N4c5cccc5C(C)(C)c5cccc54)ccc2C3=O)c2cccc21CC1(C)c2cccc2N(c2ccc3c4c(cccc24)C(=O)N(c2cccc2)C3=O)c2cccc21CC(C)(C)c1ccc(-n2c(-c3ccc4ccc5cccc6ccc3c4c56)nc(-c3cccc3)c2-c2cccc2)cc1c1ccc2c(c1)c1cccc1n2-c1cc(-c2ccc(-c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)c3nsnc23)cc(-n2c3cccc3c3cccc32)c1</chem>	2	0	2D-0A	839.010	15.191
DMAC-FO	<chem>2)-c2cc(N4c5cccc5C(C)(C)c5cccc54)ccc2C3=O)c2cccc21CC1(C)c2cccc2N(c2ccc3c4c(cccc24)C(=O)N(c2cccc2)C3=O)c2cccc21CC(C)(C)c1ccc(-n2c(-c3ccc4ccc5cccc6ccc3c4c56)nc(-c3cccc3)c2-c2cccc2)cc1c1ccc2c(c1)c1cccc1n2-c1cc(-c2ccc(-c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)c3nsnc23)cc(-n2c3cccc3c3cccc32)c1</chem>	2	0	2D-0A	594.758	11.116
6AcBIQ	<chem>CC1(C)c2cccc2N(c2ccc3c4c(cccc24)C(=O)N(c2cccc2)C3=O)c2cccc21CC(C)(C)c1ccc(-n2c(-c3ccc4ccc5cccc6ccc3c4c56)nc(-c3cccc3)c2-c2cccc2)cc1c1ccc2c(c1)c1cccc1n2-c1cc(-c2ccc(-c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)c3nsnc23)cc(-n2c3cccc3c3cccc32)c1</chem>	1	0	1D-0A	480.567	7.750
PtB	<chem>4ccc5cccc6ccc3c4c56)nc(-c3cccc3)c2-c2cccc2)cc1c1ccc2c(c1)c1cccc1n2-c1cc(-c2ccc(-c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)c3nsnc23)cc(-n2c3cccc3c3cccc32)c1</chem>	0	0	Unknown	552.721	11.068
BCPBT	<chem>5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)c3nsnc23)cc(-n2c3cccc3c3cccc32)c1</chem>	4	0	Multi-D/A	949.155	17.414

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
InDM	<chem>CN(C)c1ccc(-n2c(-c3ccccc3)nc(-c3ccccc3)c2-c2ccccc2)cc1</chem>	0	0	Unknown	415.540	6.939
PnB	<chem>CCCCc1ccc(-n2c(-c3ccc4ccc5cccc6ccc3c4c56)nc(-c3ccc(cc3)c2-c2ccccc2)cc1c1ccc(-c2ccc(-c3c4ccccc4c(-c4c5ccccc5c(-c5ccc(-c6ccccc6)cc5)c5ccccc45)c4ccc(cc34)cc2)cc1</chem>	0	0	Unknown	552.721	11.113
BANE	<chem>CCCCc1ccc(-n2c(-c3ccc4ccc5cccc6ccc3c4c56)nc(-c3ccc(cc3)c2-c2ccccc2)cc1c1ccc(-c2ccc(-c3c4ccccc4c(-c4c5ccccc5c(-c5ccc(-c6ccccc6)cc5)c5ccccc45)c4ccc(cc34)cc2)cc1</chem>	0	0	Unknown	658.844	14.634
CBM	<chem>CCCCn1c2ccc(C=C(C#N)C#N)c2c2cc(C=C(C#N)C#N)ccc21N#Cc1ccc(-c2cccc(-c3ccc(C#N)c(-n4c5ccccc5c5ccccc54)c3)c2)cc1-n1c2ccccc2c2ccc21</chem>	1	4	Multi-D/A	375.435	5.456
DCzBNPh-1	<chem>CCCCn1c2ccc(C=C(C#N)C#N)c2c2cc(C=C(C#N)C#N)ccc21N#Cc1ccc(-c2cccc(-c3ccc(C#N)c(-n4c5ccccc5c5ccccc54)c3)c2)cc1-n1c2ccccc2c2ccc21</chem>	2	4	Multi-D/A	610.720	10.958
MCZ-P2-DTM	<chem>CCCCn1c2ccc(C=C(C#N)C#N)c2c2cc(C=C(C#N)C#N)ccc21N#Cc1ccc(-c2cccc(-c3ccc(C#N)c(-n4c5ccccc5c5ccccc54)c3)c2)cc1-n1c2ccccc2c2ccc21</chem>	1	0	1D-0A	666.802	11.129
DBTO-PTZ	<chem>CCCCn1c2ccc(C=C(C#N)C#N)c2c2cc(C=C(C#N)C#N)ccc21N#Cc1ccc(-c2cccc(-c3ccc(C#N)c(-n4c5ccccc5c5ccccc54)c3)c2)cc1-n1c2ccccc2c2ccc21</chem>	4	0	Multi-D/A	578.649	9.651
α -DMAC-DBP	<chem>CCCCn1c2ccc(C=C(C#N)C#N)c2c2cc(C=C(C#N)C#N)ccc21N#Cc1ccc(-c2cccc(-c3ccc(C#N)c(-n4c5ccccc5c5ccccc54)c3)c2)cc1-n1c2ccccc2c2ccc21</chem>	1	2	A-D-A	487.606	9.198
α -2DMAC-DBP	<chem>CCCCn1c2ccc(C=C(C#N)C#N)c2c2cc(C=C(C#N)C#N)ccc21N#Cc1ccc(-c2cccc(-c3ccc(C#N)c(-n4c5ccccc5c5ccccc54)c3)c2)cc1-n1c2ccccc2c2ccc21</chem>	2	2	2D-2A	694.882	13.308
β -DI-DBP	<chem>CCCCn1c2ccc(C=C(C#N)C#N)c2c2cc(C=C(C#N)C#N)ccc21N#Cc1ccc(-c2cccc(-c3ccc(C#N)c(-n4c5ccccc5c5ccccc54)c3)c2)cc1-n1c2ccccc2c2ccc21</chem>	3	2	Multi-D/A	775.915	14.381

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
DIC-TRZ	<chem>c1ccc(-c2nc(-c3ccccc3)nc(-n3c4ccccc4c4ccc5c6ccccc6n(-c6ccccc6)c5c43)n2)cc1CC(C)(C)c1ccc(N(c2ccc(-c3cc4c(cc3C#N)=NC3c5ccc(P(=O)(c6ccccc6)c6ccccc6)nc5-c5nc(P(=O)(c6ccccc6)c6ccc6)ccc5C3N=4)cc2)c2ccc(C(C)(C)C)cc2)cc1CC1(C)c2ccccc2N(c2ccc3c(c2)c2cc(N4c5ccccc5C(C)(C)c5ccccc54)ccc2n3-c2ccc3c(c2)S(=O)(=O)c2ccccc2S3(=O)=O)c2ccccc21Cc1ccc2c(c1)c1cc(C)ccc1n2</chem>	2	1	D-A-D	563.664	9.400
PO-TPA	<chem>-c1cccc(-n2c3ccc(C)cc3c3c(c(C)ccc32)c1P(=O)(c1ccccc1)c1ccccc1O=C1c2ccccc2-c2nc3cc(-c4ccc(N(c5ccccc5)c5ccccc5)cc4)c(-c4ccc(N(c5ccccc5)c5ccccc5)cc4)cc3nc21</chem>	1	2	A-D-A	1065.211	13.075
DMAC-Cz-TTR	<chem>N#CC(C#N)=Cc1ccc(-c2ccc3c4ccccc5ccc2c3c45)cc1F</chem>	3	0	Multi-D/A	860.073	12.981
DCzPO	<chem>N#CC(C#N)=Cc1c(F)cc(-c2ccc3ccc4ccccc5ccc2c3c45)cc1F</chem>	2	0	2D-0A	664.789	10.754
IQ-oTPA	<chem>O=S(=O)(c1cccn1)c1cccc(N2c3ccccc3Oc3ccccc32)c1CC(C)c1cccc(C(C)C)c1N1C(=O)c2cc(N3c4ccccc4C(C)(C)c4ccccc43)cc3cc(N4c5ccccc5C(C)(C)c5ccccc54)cc(c23)C1=O</chem>	2	1	D-A-D	718.860	13.115
2C	<chem>N#CC(C#N)=Cc1ccc(-c2ccc3c4ccccc5ccc2c3c45)cc1F</chem>	0	2	0D-2A	372.402	6.821
C-H	<chem>N#CC(C#N)=Cc1c(F)cc(-c2ccc3ccc4ccccc5ccc2c3c45)cc1F</chem>	0	2	0D-2A	390.392	6.960
PSPP	<chem>O=S(=O)(c1cccn1)c1cccc(N2c3ccccc3Oc3ccccc32)c1CC(C)c1cccc(C(C)C)c1N1C(=O)c2cc(N3c4ccccc4C(C)(C)c4ccccc43)cc3cc(N4c5ccccc5C(C)(C)c5ccccc54)cc(c23)C1=O</chem>	2	0	2D-0A	400.459	5.490
2DN	<chem>O=S(=O)(c1cccn1)c1cccc(N2c3ccccc3Oc3ccccc32)c1CC(C)c1cccc(C(C)C)c1N1C(=O)c2cc(N3c4ccccc4C(C)(C)c4ccccc43)cc3cc(N4c5ccccc5C(C)(C)c5ccccc54)cc(c23)C1=O</chem>	2	0	2D-0A	772.005	14.105

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
PSPBP	<chem>O=S(=O)(c1cccnc1)c1cc(N2c3cccc3Oc3cccc32)cc(N2c3cccc3Oc3cccc32)c1CC(C)c1cccc(C(C)C)c1N1C(=O)c2cc(N3c4cccc4Oc4cccc43)cc3cc(N4c5cccc5Oc5cccc54)cc(c23)C1=O</chem>	4	0	Multi-D/A	581.653	9.065
2PN	<chem>c1ccc(-c2ccc3c(c2)c2cc(-c4cccc4)ccc2n3-c2cc(-n3c4ccc(-c5cccc5)cc4c4cc(-c5cccc5)ccc43)ncn2)cc1</chem>	4	0	Multi-D/A	719.841	13.038
pCbz-PYR	<chem>c1ccc(-c2nc(-c3cccc3)n(-c3ccc(-c4c5cccc5c(-c5ccc(-c6nc7c8cccc8c8cccc8c7n6-c6cccc6)cc5)c5cccc45)cc3)c2-c2cccc2)cc1</chem>	2	1	D-A-D	714.872	13.339
PPI-An-NPIM	<chem>Fc1cc(N2c3cccc3Sc3cccc32)c(F)cc1N1c2cccc2Sc2cccc21C(=Cc1ccc(-n2c3cccc3c3cccc32)cc1)c1ccc(C(=C(c2ccccc2)c2cccc2)c2cccc2)cc1</chem>	0	0	Unknown	917.128	17.826
2PS-2FPh	<chem>C(=Cc1ccc(-n2c3cccc3c3cccc32)cc1)c1ccc(C(=C(c2ccccc2)c2cccc2)c2cccc2)cc1</chem>	4	0	Multi-D/A	508.618	9.834
9CzTPE	<chem>O=P(c1cccc1)(c1cccc1)c1ccc([Si](c2cccc2)(c2cccc2)c2cccc2)cc1</chem>	1	0	1D-0A	599.777	11.961
2CzTPE	<chem>c1ccc(-n2c(-c3ccc(-c4cc(-c5cccc(-c6cccnc6)c5)cc(-c5cccc(-c6cccnc6)c5)c4)cc3)nc3c4cccc4c4cccc4c32)c1</chem>	1	0	1D-0A	627.831	12.659
FTAT-FBO	<chem>O=P(c1cccc1)(c1cccc1)c1ccc([Si](c2cccc2)(c2cccc2)c2cccc2)cc1</chem>	0	0	Unknown	536.687	4.703
DmPy-PPI	<chem>c1ccc(-n2c(-c3ccc(-c4cc(-c5cccc(-c6cccnc6)c5)cc(-c5cccc(-c6cccnc6)c5)c4)cc3)nc3c4cccc4c4cccc4c32)c1</chem>	0	0	Unknown	752.921	14.124

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
mCP-BmPy	<chem>CC(C)(C)c1ccc2c(c1)C(c1cc(-c3cccn3)cc(-c3cccn3)c1)(c1ccc3c(c1)c1ccccc1n3-c1cccc(-n3c4cccc4c4cccc43)c1)c1cc(C(C)(C)C)ccc1-2</chem>	2	0	2D-0A	915.197	16.963
mCP-Py	<chem>CC(C)(C)c1ccc2c(c1)C(c1ccn3-c1cccc(-n3c4cccc4c4cccc43)c1)c1cc(C(C)(C)C)cc1-2</chem>	2	0	2D-0A	762.013	14.234
mCP-Ph	<chem>CC(C)(C)c1ccc2c(c1)C(c1cccc1)(c1ccc3c(c1)c1ccccc1n3-c1cccc(-n3c4cccc4c4cccc43)c1)c1cc(C(C)(C)C)cc1-2</chem>	2	0	2D-0A	761.025	14.839
DMTAC	<chem>CC1ccc(-c2c(-c3cccc3)c(-c3cccc3)c3c(N)cc4c5cccc5c(N)cc4c3c2-c2ccc(C)c(C)c2)cc1C</chem>	0	0	Unknown	618.824	12.212
IPTAC	<chem>CC1ccc(-c2c(-c3ccc(C(C)C)cc3)c(-c3ccc(C(C)C)cc3)c3c(c(N)cc4c5cccc5c(N)cc43)c2-c2ccc(C)cc2)cc1</chem>	0	0	Unknown	674.932	13.842
TBTAC	<chem>CC1cccc1-c1c(-c2ccc(C(C)(C)C)cc2)c(-c2ccc(C(C)(C)C)cc2)c2c(c(N)cc3c4cccc4c(N)cc32)c1-c1ccccc1C</chem>	0	0	Unknown	702.986	14.190
p2Cz2Trz	<chem>c1ccc(-c2nc(-c3cccc3)nc(-c3cc(-n4c5cccc5c5cccc54)c(-c4nc(-c5cccc5)nc(-c5cccc5)n4)cc3-n3c4cccc4c4cccc43)n2)cc1</chem>	2	2	2D-2A	871.020	14.253

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
m2Cz2Trz	<chem>c1ccc(-c2nc(-c3ccccc3)nc(-c3cc(-c4nc(-c5ccccc5)nc(-c5ccccc5)n4)c(-n4c5ccccc5c5ccccc54)cc3-n3c4ccccc4c4ccccc43)n2)cc1</chem>	2	2	2D-2A	871.020	14.253
2,4-CzPyCl3	<chem>C1c1nc(-n2c3ccccc3c3ccccc32)c(C1)c(-n2c3ccccc3c3ccccc32)c1C1</chem>	2	0	2D-0A	512.827	9.236
DMAC-PCN	<chem>CC1(C)c2ccccc2N(c2ccc(-c3c(C#N)c(-c4ccccc4)nc(-c4ccccc4)c3C#N)cc2)c2ccccc21</chem>	1	2	A-D-A	564.692	9.935
2PXZP	<chem>Cn1cnc2c(-c3ccc(N4c5ccccc50c5ccccc54)cc3)nc(-c3ccc(N4c5ccccc50c5ccccc54)cc3)nc21</chem>	4	1	Multi-D/A	648.726	10.848
2,4,6-CzPyCl2	<chem>C1c1c(-n2c3ccccc3c3ccccc32)nc(-n2c3ccccc3c3ccccc32)c(C1)c1-n1c2ccccc2c2ccccc21</chem>	3	0	Multi-D/A	643.577	11.680
1PXZP	<chem>Cn1cnc2c(-c3ccc(N4c5ccccc50c5ccccc54)cc3)ncnc21</chem>	2	1	D-A-D	391.434	5.606
3,6-T2C	<chem>N#Cc1ccc(-c2nc3c4ccc(-c5ccc(N(c6ccccc6)c6ccccc6)cc5)cc4c4cc(-c5ccc(N(c6ccccc6)c6ccccc6)cc5)ccc4c3n2-c2ccc(C#N)cc2)cc1</chem>	2	4	Multi-D/A	907.093	17.016
3,6-TC	<chem>N#Cc1ccc(-c2nc3c4ccc(-c5ccc(N(c6ccccc6)c6ccccc6)cc5)cc4c4cc(-c5ccc(N(c6ccccc6)c6ccccc6)cc5)ccc4c3n2-c2ccccc2)cc1</chem>	2	2	2D-2A	882.083	17.144
Ph-CF3	<chem>c1ccc2c(c1)c1ccccc1n2-c1ccc(Nc2ccc(-n3c4ccccc4c4ccccc43)cc2)cc1</chem>	2	0	2D-0A	499.617	9.624

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
arylene	<chem>c1ccc2c(c1)c1ccccc1n2-c1ccc(Nc2ccc(-n3c4ccccc4c4cccc43)cc2)cc1</chem>	2	0	2D-0A	499.617	9.624
TS-1	<chem>O=S(=O)(c1ccccc1)c1ccc(-n2c3ccccc3c3ccccc32)cc1Cc1ccc2c(c1)c1cc(C)ccc1n2</chem>	1	0	1D-0A	383.472	5.617
3,6-MeCzOXD	<chem>-c1ccccc1-c1nnc(-c2ccccc2-n2c3ccc(C)cc3c3cc(C)ccc32)O1</chem>	2	0	2D-0A	608.745	10.831
p-PO15NCzDPA	<chem>O=P(c1ccccc1)(c1ccccc1)c1ccc(-c2c3ccccc3c(-c3ccc(-n4c5cccn5c5cccn54)cc3)c3ccccc23)cc1Cc1ccc2c3ccc(C)cc3n(-c3ccc3-c3nnc(-c4ccccc4-n4c5cc(C)ccc5c5ccc(C)cc54)O3)c2c1</chem>	0	0	Unknown	697.778	10.853
2,7-MeCzOXD	<chem>N#Cc1ccccc1-c1ccc(-n2c3ccc(-c4ccccc4)cc3c3cc(-c4cccc4)ccc32)cc1c1ccc(-n2c3ccccc3c3ccccc32)c(-c2nnc(-c3ccccc3-n3c4ccccc4c4cccc43)O2)c1</chem>	2	0	2D-0A	608.745	10.831
BPCN-Cz2Ph	<chem>N#Cc1ccccc1-c1ccc(-n2c3ccc3c3cc(-c4ccc5c(c4)c4cccc4n5-c4ccccc4)ccc32)cc1CCCCCN1C(=O)C(c2ccc(-c3ccc(-n4c5ccccc5c5ccccc54)c3)cc2)=C(c2ccc(-c3ccc(-n4c5ccccc5c5ccccc54)cc3)cc2)C1=O</chem>	1	2	A-D-A	496.613	9.656
o-CzOXD	<chem>O=P(c1ccccc1)(c1ccccc1)c1cccc2c1oc1c(P(=O)(c3ccccc3)c3ccccc3)cccc12</chem>	2	0	2D-0A	552.637	9.598
BPCN-2Cz	<chem>O=P(c1ccccc1)(c1ccccc1)c1cccc2c1oc1c(P(=O)(c3ccccc3)c3ccccc3)cccc12</chem>	2	2	2D-2A	585.710	11.086
Cbz-MI	<chem>O=P(c1ccccc1)(c1ccccc1)c1cccc2c1oc1c(P(=O)(c3ccccc3)c3ccccc3)cccc12</chem>	2	0	2D-0A	816.017	14.075
DBFDPO	<chem>O=P(c1ccccc1)(c1ccccc1)c1cccc2c1oc1c(P(=O)(c3ccccc3)c3ccccc3)cccc12</chem>	0	0	Unknown	568.549	6.865

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
XCBP	<chem>Cc1cc(-n2c3ccccc3c3ccccc32)c(C)cc1-c1ccc(C(=O)c2ccc2)cc1</chem>	1	0	1D-0A	451.569	8.299
Cbz-PYR	<chem>c1ccc2c(c1)c1ccccc1n2-c1cc(-n2c3ccccc3c3ccccc32)nc</chem>	2	1	D-A-D	410.480	6.671
tCbz-mPYRs	<chem>n1Cc1nc(-n2c3ccc(C(C)(C)C)cc3c3c(C(C)(C)C)cccc32)c(C)c(-n2c3ccc(C(C)(C)C)cc3c3c(C(C)(C)C)cccc32)n1</chem>	2	1	D-A-D	695.033	12.891
PCBP	<chem>O=C(c1cccc1)c1ccc(-c2ccc(-n3c4cccc4c4cccc43)cc2)cc1</chem>	1	0	1D-0A	423.515	7.682
MeCZ-TRZ	<chem>Cc1ccc2c(c1)c1ccccc1n2-c1ccc(-c2nc(-c3ccc(-n4c5ccc5c5cc(C)ccc54)c(C)c3)nc(-c3ccc(-n4c5cccc5c5cc(C)ccc54)c(C)c3)n2)cc1C</chem>	3	1	Multi-D/A	889.119	16.014
3ca	<chem>COc1ccc(-c2cc(-c3ccc(C(F)(F)F)cc3)c(-c3nc4cccc4[nH]3)s2)cc1</chem>	0	0	Unknown	450.485	7.653
2MeCZ-TRZ	<chem>Cc1ccc2c(c1)c1cc(C)ccc1n2-c1ccc(-c2nc(-c3ccc(-n4c5ccc(C)cc5c5cc(C)ccc54)c(C)c3)nc(-c3ccc(-n4c5ccc(C)cc5c5cc(C)ccc54)c(C)c3)n2)cc1C</chem>	3	1	Multi-D/A	931.200	16.940
CZ-TRZ	<chem>Cc1cc(-c2nc(-c3ccc(-n4c5cccc5c5cccc54)c(C)c3)nc(-c3ccc(-n4c5cccc5c5cccc54)c(C)c3)n2)ccc1-n1c2ccc2c2cccc21</chem>	3	1	Multi-D/A	847.038	15.089
DMAC-AQ	<chem>CC1(C)c2cccc2N(c2ccc3c(c2)C(=O)c2cccc2C3=O)c2ccc21</chem>	1	0	1D-0A	415.492	6.571

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
DTPTCzDP-CN	<chem>CC(C)(C)c1nc(-c2ccc(-n3c4ccc(N(c5ccccc5)c5ccccc5)c4c4cc(N(c5ccccc5)c5ccccc5)ccc43)c(C#N)c2)nc(C(C)(C)C)n1</chem>	3	3	Multi-D/A	794.019	14.042
mP2MPC	<chem>Cc1cc(-c2ccc(-n3c4ccccc4c4ccccc43)c2)c(C)cc1-c1ccc(-c2nc3c4ccccc4c4ccccc4c3n2-c2ccccc2)c1</chem>	1	0	1D-0A	715.900	14.047
XPhos	<chem>CC(C)C1=CC(C(C)C)=C(c2ccc(cc2)C(C(C)C)(P(C2CCCC2)C2CCCC2)C1c1ccc2c(c1)Oc1cccc1N2c1ccc(-c2nnc(-c3ccc(N4c5ccccc50c5ccccc54)cc3)c(-c3ccc(N4c5ccccc50c5ccccc54)cc3)n2)cc1</chem>	0	0	Unknown	478.745	10.624
TPXZ-as-TAZ	<chem>N#Cc1cccc2c1sc1ccc(-c3nc(-c4ccccc4)nc(-c4ccccc4)n3)cc12</chem>	6	0	Multi-D/A	852.954	15.599
2Trz6CNDBT	<chem>c1ccc(N(c2ccccc2)c2ccc(-c3ccc(-c4nc5c6ccccc6c6cccc6c5n4-c4cccc(-c5ccc6ccc7cccc8ccc5c6c78)c4)cc3)cc2)cc1</chem>	0	3	Multi-D/A	440.531	7.112
TPAPyPI	<chem>c1ccc(N(c2ccccc2)c2ccc(-c3ccc(-c4nc5c6ccccc6c6cccc6c5n4-c4cccc(-c5ccc6ccc7cccc8ccc5c6c78)c4)cc3)cc2)cc1</chem>	1	0	1D-0A	814.004	16.700
TPACzPI	<chem>c1ccc(N(c2ccccc2)c2ccc(-c3ccc(-c4nc5c6ccccc6c6cccc6c5n4-c4cccc(-c5ccc6c(c5)c5ccccc5n6-c5ccccc5)c4)c3)cc2)cc1</chem>	2	0	2D-0A	855.057	16.900
NA-TNA	<chem>CC(C)(C)c1ccc(N2C(=O)c3ccc4c(/C=C/c5ccc6cc(N(c7ccccc7)c7ccccc7)ccc6c5)ccc(c34)C2=O)cc1</chem>	1	0	1D-0A	648.806	11.731

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
CzPrSBI	<chem>O=C1c2cccc2S(=O)(=O)N1CC</chem>	1	0	1D-0A	390.464	4.029
m-CzCNDMT	<chem>Cn1c2cccc2c2cccc21</chem> <chem>Cc1cc(-c2cccc(-n3c4cccc4</chem> <chem>c4cc(C#N)ccc43)c2)c(C)cc1</chem> <chem>-c1cccc(-n2c3cccc3c3cc(C</chem> <chem>#N)ccc32)c1</chem>	2	4	Multi-D/A	638.774	11.575
DTXO-TPA2	<chem>Cc1cc(N(c2cccc2)c2cccc3c</chem> <chem>cccc23)ccc1-c1ccc(N(c2ccc</chem> <chem>cc2)c2cccc3cccc23)cc1C</chem> <chem>c1ccc2c(-c3ccc(-n4c5cccc</chem> <chem>5c5cccc54)cc3)nc(-c3ccc(</chem> <chem>-n4c5cccc5c5cccc54)cc3)</chem> <chem>nc2c1</chem>	2	0	2D-0A	616.808	13.216
24PQ-Cz	<chem>O=S(=O)(c1ccc(N2c3cccc3S</chem> <chem>c3cccc32)cc1)c1ccc(-n2c3</chem> <chem>cccc3c3cccc32)cc1</chem>	3	0	Multi-D/A	580.734	9.551
2CzMC	<chem>C0c1cc(=O)oc2cc3c(cc12)c1</chem> <chem>cccc1n3-c1cccc1</chem> <chem>FC(F)(F)c1ccc(-n2c(-c3ccc</chem> <chem>(-c4ccc(-n5c(-c6cccc6)nc</chem> <chem>(-c6cccc6)c5-c5cccc5)cc</chem> <chem>4)cc3)nc3c4cccc4c4cccc4</chem> <chem>c32)cc1</chem>	1	0	1D-0A	341.366	4.899
PPI-PIM-2	<chem>C0c1cc(=O)oc2c1ccc1c2c2cc</chem> <chem>ccc2n1-c1cccc1</chem> <chem>FC(F)(F)c1ccc(-n2c(-c3ccc</chem> <chem>(-c4ccc(-c5nc(-c6cccc6)c</chem> <chem>(-c6cccc6)n5-c5cccc5)cc</chem> <chem>4)cc3)nc3c4cccc4c4cccc4</chem> <chem>c32)cc1</chem>	0	0	Unknown	808.907	14.871
4CzMC	<chem>C0c1cc(=O)oc2c1ccc1c2c2cc</chem> <chem>ccc2n1-c1cccc1</chem> <chem>FC(F)(F)c1ccc(-n2c(-c3ccc</chem> <chem>(-c4ccc(-c5nc(-c6cccc6)c</chem> <chem>(-c6cccc6)n5-c5cccc5)cc</chem> <chem>4)cc3)nc3c4cccc4c4cccc4</chem> <chem>c32)cc1</chem>	1	0	1D-0A	341.366	4.899
PPI-PIM-1	<chem>c1ccc(N(c2cccc2)c2cccc2</chem> <chem>-n2c3cccc3c3cccc32)cc1</chem>	2	0	2D-0A	410.520	8.254

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
SFX-bPy	<chem>c1ccc2c(c1)Oc1ccc(-c3ccc4cccc4n3)cc1C21c2cccc2-c2cccc21</chem>	0	0	Unknown	459.548	8.371
t-BuCz2BP	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(C(=O)c2ccc(-n3c4ccc(C(C)(C)C)cc4c4cc(C(C)(C)C)ccc43)cc2)cc1</chem>	2	0	2D-0A	737.044	14.302
SAF-2CzB	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-c2ccc3c(c2)C2(c4cc(-c5ccc(-n6c7ccc(C(C)(C)C)cc7c7cc(C(C)(C)C)ccc76)cc5)ccc4-3)c3cccc3N(c3cccc3)c3cccc32)cc1</chem>	3	0	Multi-D/A	1114.534	22.551
F-2CzB	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-c2ccc3c(c2)C(C)(C)C2cc(-c4ccc(-n5c6ccc(C(C)(C)C)cc6c6cc(C(C)(C)C)ccc65)cc4)ccc2-3)cc1</chem>	2	0	2D-0A	901.295	18.711
SAF-CzB	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-c2ccc3c(c2)C2(c4cccc4-3)c3cccc3N(c3cccc3)c3cccc32)cc1</chem>	2	0	2D-0A	761.025	15.192
BPy-TP2	<chem>c1ccc(-c2ccc(-c3ccc4c5ccc(-c6ccc(-c7cccn7)nc6)cc5c5cccc5c4c3)cn2)nc1</chem>	0	0	Unknown	536.638	9.394
Cz-AQ	<chem>O=C1c2cccc2C(=O)c2cc(-n3c4cccc4c4cccc43)ccc21</chem>	1	0	1D-0A	373.411	5.559

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
-ph-pCzAnBzt	<chem>Cc1cc(-c2c3ccccc3c(-c3ccc(C#N)cc3)c3ccccc23)c(C)cc1-c1ccc(-n2c3ccccc3c3ccccc32)cc1</chem>	1	2	A-D-A	624.787	12.580
PyI	<chem>c1cc2ccc3cccc4c5[nH]cnc5c(c1)c2c34C=Cc1ccc(COc2cccc3c2c2ccc2n3CCCCCOc2cccc3c2c2cccc2n3-c2cccc(-n3c4ccccc4c4c(OCc7ccc(C=C)cc7)cccc65)cc43)c2C#N)cc1</chem>	0	0	Unknown	242.281	4.460
DVCz-2CzCN	<chem>c1ccc(-c2c3ccccc3c(-c3ccc(-n4c5ccccc5c5c6sc7ccccc7c6ccc54)cc3)c3ccccc23)cc1c1ccc(-n2c3ccccc3c3cc(-c4ccc5c(c4)c4cccc4n5-c4cccc4)ccc32)cc1CC1(C)c2cccc2N(c2ccc(-c3cc(-c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)ncn3)cc2)c2cccc21</chem>	4	2	Multi-D/A	1 228.550	21.691
ATDBT	<chem>c1ccc(-c2c3ccccc3c(-c3ccc(-n4c5ccccc5c5c6sc7ccccc7c6ccc54)cc3)c3ccccc23)cc1c1ccc(-n2c3ccccc3c3cc(-c4ccc5c(c4)c4cccc4n5-c4cccc4)ccc32)cc1</chem>	1	0	1D-0A	601.774	12.792
DPhBCz	<chem>CC1(C)c2cccc2N(c2ccc(-c3cc(-c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)ncn3)cc2)c2cccc21</chem>	2	0	2D-0A	484.602	9.548
AcHPM	<chem>CC1(C)c2cccc2N(c2ccc(C(=O)c3ccccc3)cc2)c2cccc21c1ccc(-c2cccc(-c3nc(-c4ccc(-c5ccccc5)c4)nc(-c4ccc(-c5ccccc5)c4)n3)c2)cc1Br</chem>	2	1	D-A-D	646.838	12.029
PhCO-DMAC	<chem>CC1(C)c2cccc2N(c2ccc(C(=O)c3ccccc3)cc2)c2cccc21c1ccc(-c2cccc(-c3nc(-c4ccc(-c5ccccc5)c4)nc(-c4ccc(-c5ccccc5)c4)n3)c2)cc1</chem>	1	0	1D-0A	389.498	7.027
Diphenylsulfane	<chem>Br</chem>	0	1	0D-1A	537.666	9.874
2BrCPT	<chem>n2-c1ccc(-c2nc(-c3ccccc3)nc(-c3ccccc3)n2)cc1c1ccc(-n2c(-c3ccc(-c4cc(-n5c6ccccc6c6ccccc65)cc(-n5c6ccccc6c6ccccc65)c4)cc3)nc3ccccc32)cc1</chem>	1	1	D-A	632.359	9.495
DCbz-Ph-Bz	<chem>n5c6ccccc6c6ccccc65)cc(-n5c6ccccc6c6ccccc65)c4)cc3)nc3ccccc32)cc1</chem>	2	0	2D-0A	676.823	12.554

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
DDPA-Ph-Bz	<chem>c1ccc(N(c2ccccc2)c2cc(-c3ccc(-c4nc5ccccc5n4-c4cccc4)cc3)cc(N(c3ccccc3)c3ccccc3)c2)cc1</chem>	2	0	2D-0A	680.855	13.299
IDAC-MCO	<chem>Cc1cc2cc(-n3c4ccccc4c4cc5c(cc43)C(C)(C)c3ccccc3N5c3ccccc3)ccc2c(=O)o1</chem>	2	0	2D-0A	532.643	9.308
diarylbiindole	<chem>c1ccc(-c2c(-c3ccccc3)c3cc4cccc5cc(-c6ccccc6)c(c2-c2ccccc2)c3c45)cc1</chem>	0	0	Unknown	506.648	11.252
CNDSB	<chem>N#CC(=Cc1ccccc1)c1ccc(C(C#N)=Cc2ccccc2)cc1</chem> <chem>Cc1cc(C)c(CC#Cc2ccc(N(C)C)cc2)c(B(c2c(C)c(C)cc(C)c2CC#Cc2ccc(N(C)C)cc2)c1C</chem>	0	2	0D-2A	332.406	5.815
TMAB	<chem>c1cncc(CC(c2nc3ccccc3[nH]2)N(C(Cc2cccn2)c2nc3cccc3[nH]2)C(Cc2cccn2)c2nc3ccccc3[nH]2)c1</chem>	0	0	Unknown	840.021	9.959
MOF	<chem>N#Cc1cc(-c2nc(-c3ccccc3)n c(-c3ccccc3)n2)ccc1-n1c2cccc2c2cc3c4ccccc4n4c5cccc5c(c21)c34</chem>	0	0	Unknown	680.820	8.054
DICzTrz	<chem>O=S1(=O)c2ccccc2N(c2ccccc2)c2cc(N3c4ccccc4c4ccccc43)ccc21</chem>	3	3	Multi-D/A	662.756	10.991
PXZ2PTO	<chem>O=C1c2ccccc2C2(c3cc(N4c5cccc5C(c5ccccc5)(c5ccccc5)c5ccccc54)ccc31)c1ccccc1-c1ccccc12</chem>	4	0	Multi-D/A	488.568	7.878
SPBP-DPAC		1	0	1D-0A	675.831	11.760

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
PX-TRZ	<chem>Cc1cc(C)c(B(c2ccc(-c3nc(C1)nc(-c4ccccc4)n3)cc2)c2c(C)cc(C)cc2C)c(C)c1Cc1ccc(N(c2ccc(C)cc2)c2ccc2C2(c3ccccc3N(c3ccc(C)cc3)c3ccc(C)cc3)CCCC2)cc1</chem>	0	1	0D-1A	515.897	6.226
C2MPI	<chem>Cc1ccc(N(c2ccc(C)cc2)c2ccc2C2(c3ccccc3N(c3ccc(C)cc3)c3ccc(C)cc3)CCCC2)cc1</chem>	2	0	2D-0A	626.888	13.110
p-PCzTPD	<chem>Cc1ccc(N(c2ccc(C)cc2)c2ccc2C2(c3ccccc3N(c3ccc(C)cc3)c3ccc(C)cc3)CCCC2)cc1</chem>	2	0	2D-0A	747.964	13.233
m-PCzTPD	<chem>Cc1ccc(N(c2ccc(C)cc2)c2ccc2C2(c3ccccc3N(c3ccc(C)cc3)c3ccc(C)cc3)CCCC2)cc1</chem>	2	0	2D-0A	747.964	13.233
o-PCzTPD	<chem>Cc1ccc(N(c2ccc(C)cc2)c2ccc2C2(c3ccccc3N(c3ccc(C)cc3)c3ccc(C)cc3)CCCC2)cc1</chem>	2	0	2D-0A	747.964	13.233
CPBzFC	<chem>Cc1ccc(N(c2ccc(C)cc2)c2ccc2C2(c3ccccc3N(c3ccc(C)cc3)c3ccc(C)cc3)CCCC2)cc1</chem>	2	0	2D-0A	706.874	12.791
SiCz2Py2	<chem>Cc1ccc(N(c2ccc(C)cc2)c2ccc2C2(c3ccccc3N(c3ccc(C)cc3)c3ccc(C)cc3)CCCC2)cc1</chem>	2	0	2D-0A	821.072	11.382
SiCz1Py3	<chem>Cc1ccc(N(c2ccc(C)cc2)c2ccc2C2(c3ccccc3N(c3ccc(C)cc3)c3ccc(C)cc3)CCCC2)cc1</chem>	1	0	1D-0A	732.963	9.347

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
SiCz3Py1	<chem>c1cncc(-c2ccc([Si](c3ccc(-n4c5ccccc5c5ccccc54)cc3)(c3ccc(-n4c5ccccc5c5ccccc54)cc3)c3ccc(-n4c5ccccc5c5ccccc54)cc3)cc2)c1</chem>	3	0	Multi-D/A	909.181	13.417
POBBPE	<chem>CC(C)(C)c1ccc(OC2ccc(C(C)(C)C)cc2P(=O)(c2ccccc2)c2ccccc2)cc1</chem>	0	0	Unknown	482.604	7.713
CzppT	<chem>c1ccc(-c2c(-c3ccccc3)c(-c3ccc(-c4cccc5c4[nH]c4cccc45)cc3)c(-c3ccccc3)c(-c3ccccc3)c2-c2ccc(-c3cccc4c3[nH]c3ccccc34)cc2)cc1</chem>	2	0	2D-0A	867.068	17.082
SBDBQ-PXZ	<chem>c1ccc(-c2nc3ccc(N4c5ccccc5OC5ccccc54)cc3nc2-c2ccc(c2)cc1</chem>	2	1	D-A-D	463.540	8.539
DBQ-3PXZ	<chem>c1ccc2c(c1)OC1CCCC1N2c1ccc(-c2nc3ccc(N4c5ccccc5OC5ccccc54)cc3nc2-c2ccc(N3c4ccccc4OC4ccccc43)cc2)cc1</chem>	6	1	Multi-D/A	825.928	15.690
P4CzCN-BCz	<chem>C=Cc1ccc(COC2ccc(C(C)(C)C3ccc(OC)cc3)cc2)cc1</chem>	0	0	Unknown	344.454	5.940
NAP-1,5-DPA	<chem>COc1ccc(N(c2ccc(OC)cc2)c2cccc3c(N(c4ccc(OC)cc4)c4ccc(OC)cc4)ccccc23)cc1</chem>	2	0	2D-0A	582.700	9.814
3DPA3CN	<chem>N#Cc1c(-c2ccc(N(c3ccccc3)c3ccccc3)cc2)c(C#N)c(-c2ccc(N(c3ccccc3)c3ccccc3)cc2)c(C#N)c1-c1ccc(N(c2cccc2)c2ccccc2)cc1</chem>	3	6	Multi-D/A	883.071	16.712
TPA-AN-NA	<chem>c1ccc(N(c2ccccc2)c2ccc(-c3c4ccccc4c(-c4ccc5ccccc5c4)c4ccccc34)cc2)cc1</chem>	1	0	1D-0A	547.701	11.950

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
OCzBN	<chem>C0c1ccc2c(c1)c1cccc1n2-c1cc(-n2c3cccc3c3cc(OC)cc32)c(C#N)c(-n2c3cccc3c3cc(OC)ccc32)c1</chem>	3	2	Multi-D/A	688.787	10.875
BBCz	<chem>c1ccc2c(c1)[nH]c1cc3[nH]c4cccc4c3cc12C0c1c(C#N)c(C)cc2c([Se][Se]c3c(C)nc(N4CCCC4)c4c(OC)c(C#N)c(C)cc34)c(C)nc(N3CCCC3)c12</chem>	2	0	2D-0A	256.308	4.956
IQ-Se	<chem>Cc1cccc1P(CP(CP(c1cccc1C)c1cccc1C)c1cccc1C)N#Cc1cccc(-c2nc(-c3cccc([Si](c4cccc4)(c4cccc4)c4cccc4)c3)nc(-n3c4cccc4c4cccc43)n2)c1</chem>	0	4	Multi-D/A	746.676	5.032
dTolmp	<chem>Cc1cccc1P(CP(CP(c1cccc1C)c1cccc1C)c1cccc1C)N#Cc1cccc(-c2nc(-c3cccc([Si](c4cccc4)(c4cccc4)c4cccc4)c3)nc(-n3c4cccc4c4cccc43)n2)c1</chem>	0	0	Unknown	562.614	8.208
SiTCNCz	<chem>Cc1cc(P(c2cccc2)c2cccc2)c(P(c2cccc2)c2cccc2)cc1C</chem>	1	3	Multi-D/A	681.875	7.552
dpmb	<chem>N#Cc1c(-n2c3cccc3c3cccc32)cc(-n2c3cccc3c3cccc32)cc1-n1c2cccc2c2cccc21O=C(c1cccc1)c1ccc(-n2c3cccc3c3cc(-c4ccc(N(c5cccc5)c5cccc5)cc4)ccc32)cc1</chem>	0	0	Unknown	474.524	5.820
3CzBN	<chem>c1ccc2c(c1)c1cccc1n2-c1cncc(-c2cncc(-n3c4cccc4c4cccc43)n2)c1</chem>	3	2	Multi-D/A	598.709	10.850
TCB	<chem>c1ccc2c(c1)c1cccc1n2-c1cncc(-c2cncc(-n3c4cccc4c4cccc43)n2)c1</chem>	2	0	2D-0A	590.726	11.151
CPzPyC	<chem>c1ccc2c(c1)c1cccc1n2-c1cncc(-c2cncc(-n3c4cccc4c4cccc43)n2)c1</chem>	2	0	2D-0A	487.566	7.733
CPzPzC	<chem>c1ccc2c(c1)c1cccc1n2-c1cncc(-c2cncc(-n3c4cccc4c4cccc43)n2)n1</chem>	2	0	2D-0A	488.554	7.128

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
CPzPC	<chem>c1cc(-c2cncc(-n3c4ccccc4c4ccccc43)n2)cc(-n2c3ccccc3c3ccccc32)c1</chem>	2	0	2D-0A	486.578	8.338
CBIQD	<chem>O=C1c2cccc3cccc(c23)C(=O)N1c1nc2ccc(Cl)cc2s1</chem>	0	0	Unknown	364.813	4.904
PXZ-QL	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc2ccccc2n1</chem>	1	0	1D-0A	310.356	5.810
SBA	<chem>c1ccc2c(c1)Nc1cccc1C21c2cccc2Nc2cccc21O=C(c1ccc(-n2c3ccccc3c3c4c(c5ccccc5n4-c4ccccc4)c4c(c5ccccc5n4-c4ccccc4)c32)cc1)c1ccc(-n2c3ccccc3c3c4c(c5ccccc5n4-c4ccccc4)c4c(c5ccccc5n4-c4ccccc4)c32)cc1CCCCN(CCCC)c1ccc(C=Nc2cc(C#N)c(C#N)cc2N=Cc2ccc(N(CCC)CCC)cc2O)c(O)c1Cc1cc(C)c(B(c2ccccc2-n2c3ccccc3c3ccccc32)c2c(C)cc(C)cc2C)c(C)c1Cc1cc(C)c(B(c2ccc(C(F)(F)F)cc2-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c2c(C)cc(C)cc2C)c(C)c1</chem>	0	0	Unknown	346.433	6.183
TATP-BP	<chem>c1ccc2c(c1)Nc1cccc1C21c2cccc2Nc2cccc21O=C(c1ccc(-n2c3ccccc3c3c4c(c5ccccc5n4-c4ccccc4)c4c(c5ccccc5n4-c4ccccc4)c32)cc1)c1ccc(-n2c3ccccc3c3c4c(c5ccccc5n4-c4ccccc4)c4c(c5ccccc5n4-c4ccccc4)c32)cc1CCCCN(CCCC)c1ccc(C=Nc2cc(C#N)c(C#N)cc2N=Cc2ccc(N(CCC)CCC)cc2O)c(O)c1Cc1cc(C)c(B(c2ccccc2-n2c3ccccc3c3ccccc32)c2c(C)cc(C)cc2C)c(C)c1Cc1cc(C)c(B(c2ccc(C(F)(F)F)cc2-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c2c(C)cc(C)cc2C)c(C)c1</chem>	6	0	Multi-D/A	1 173.392	21.500
Pd-1	<chem>CCCCN(CCCC)c1ccc(C=Nc2cc(C#N)c(C#N)cc2N=Cc2ccc(N(CCC)CCC)cc2O)c(O)c1Cc1cc(C)c(B(c2ccccc2-n2c3ccccc3c3ccccc32)c2c(C)cc(C)cc2C)c(C)c1Cc1cc(C)c(B(c2ccc(C(F)(F)F)cc2-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c2c(C)cc(C)cc2C)c(C)c1</chem>	0	4	Multi-D/A	620.842	9.156
C3F7	<chem>c1ccc2c(c1)Nc1cccc1C21c2cccc2Nc2cccc21O=C(c1ccc(-n2c3ccccc3c3c4c(c5ccccc5n4-c4ccccc4)c4c(c5ccccc5n4-c4ccccc4)c32)cc1)c1ccc(-n2c3ccccc3c3c4c(c5ccccc5n4-c4ccccc4)c4c(c5ccccc5n4-c4ccccc4)c32)cc1CCCCN(CCCC)c1ccc(C=Nc2cc(C#N)c(C#N)cc2N=Cc2ccc(N(CCC)CCC)cc2O)c(O)c1Cc1cc(C)c(B(c2ccccc2-n2c3ccccc3c3ccccc32)c2c(C)cc(C)cc2C)c(C)c1Cc1cc(C)c(B(c2ccc(C(F)(F)F)cc2-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c2c(C)cc(C)cc2C)c(C)c1</chem>	1	0	1D-0A	491.487	7.150
BuCzCF3oB	<chem>F)cc2-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c2c(C)cc(C)cc2C)c(C)c1</chem>	1	0	1D-0A	671.700	10.764
phenazasiline	<chem>Brcc1ccc2nc3c4ccc(Br)cc4c4ccccc4c3nc2c1</chem>	0	2	0D-2A	438.122	6.614
D-A-D	<chem>Brcc1ccc2nc3c4ccc(Br)cc4c4ccccc4c3nc2c1</chem>	0	2	0D-2A	438.122	6.614

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
pzpy	<chem>c1ccc(-c2nc3cccc3n2-c2cc</chem> <chem>c(-c3c4cccc4c(-c4ccc(-n5</chem> <chem>c(-c6cccc6)nc6cccc65)cc</chem> <chem>4)c4cccc34)cc2)cc1</chem>	0	0	Unknown	714.872	13.339
m-PO-ABN	<chem>N#Cc1ccc(-c2c3cccc3c(-c3</chem> <chem>cccc(-c4nc5c6cccc6c6cccc</chem> <chem>c6c5o4)c3)c3cccc23)cc1</chem>	0	2	0D-2A	572.667	11.313
PO-ABN	<chem>N#Cc1ccc(-c2c3cccc3c(-c3</chem> <chem>cccc(-c4nc5c6cccc6c6cccc</chem> <chem>c6c5o4)c3)c3cccc23)cc1</chem>	0	2	0D-2A	572.667	11.313
p-PO-ABN	<chem>N#Cc1ccc(-c2c3cccc3c(-c3</chem> <chem>ccc(-c4nc5c6cccc6c6cccc</chem> <chem>6c5o4)cc3)c3cccc23)cc1</chem>	0	2	0D-2A	572.667	11.313
teracridan	<chem>c1ccc2c(c1)Cc1c(cccc1-c1c</chem> <chem>cc3c(c1-c1cccc4c1Cc1cccc</chem> <chem>1N4)Cc1cccc1N3)N2</chem> <chem>CC(C)(C)c1ccc2c(c1)c1cc(C</chem> <chem>(C)(C)C)ccc1n2-c1ccc(-c2c</chem>	0	0	Unknown	539.682	9.964
PTPC	<chem>cc(-c3cc(-c4cccc4)c(-c4c</chem> <chem>cc(-c5nc6c7cccc7c7cccc7</chem> <chem>c6n5-c5cccc5)cc4)cc3-c3c</chem> <chem>cccc3)cc2)cc1</chem>	1	0	1D-0A	1028.356	21.026
TPTI	<chem>O=c1[nH]c2ccsc2c2cc3c(=O)</chem> <chem>[nH]c4ccsc4c3cc12</chem> <chem>c1ccc(-c2nc(-c3cccc3)nc(</chem> <chem>-c3ccc(-n4c5ccc(-c6ccc7sc</chem>	0	0	Unknown	324.386	3.799
BDBTCz-Trz	<chem>8cccc8c7c6)cc5c5cc(-c6cc</chem> <chem>c7sc8cccc8c7c6)ccc54)cc3</chem> <chem>)n2)cc1</chem> <chem>c1ccc(-c2nc(-c3cccc3)nc(</chem> <chem>-c3ccc(-n4c5ccc(-c6ccc7oc</chem>	1	1	D-A	839.061	16.039
BDBFCz-Trz	<chem>8cccc8c7c6)cc5c5cc(-c6cc</chem> <chem>c7oc8cccc8c7c6)ccc54)cc3</chem> <chem>)n2)cc1</chem>	1	1	D-A	806.925	15.102

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
HAP-3MF	<chem>Cc1cc(C2=NC3=NC(c4ccc(F)c(C(C)c4)=NC4=NC(c5ccc(F)c(CC1(C)c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc2c(c1)-c1cccc1C21c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc(N(c2cccc2)c2cccc2-c2c3cccc3c(-c3ccc4ccc5c(-c6cccc6N(c6cccc6)c6cccc6)ccc6ccc3c4c65)c3cccc23)cc1c1ccc(N(c2cccc2)c2ccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7ccc(N(c8cccc8)c8cccc8)cc7)ccc7ccc4c5c76)c4cccc34)cc2)cc1N#Cc1ccc(-c2ccc3c(c2)c2cc(-c4ccc(C#N)cc4)ccc2c2nc4cc(-c5ccc(N(c6cccc6)c6cccc6)cc5)c(-c5ccc(N(c6cccc6)c6cccc6)cc5)cc4nc32)cc1c1ccc(N(c2cccc2)c2cccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7cccc(N(c8cccc8)c8cccc8)c7)ccc7ccc4c5c76)c4cccc34)c2)cc1CC1(C)C(=O)c2cc(-c3ccc(N4c5cccc50c5cccc54)cc3)c(-c3ccc(N4c5cccc50c5cccc54)cc3)cc2C1=O</chem>	0	0	Unknown	497.484	5.080
FlCz	<chem>Cc1cc(C2=NC3=NC(c4ccc(F)c(C(C)c4)=NC4=NC(c5ccc(F)c(CC1(C)c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc2c(c1)-c1cccc1C21c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc(N(c2cccc2)c2cccc2-c2c3cccc3c(-c3ccc4ccc5c(-c6cccc6N(c6cccc6)c6cccc6)ccc6ccc3c4c65)c3cccc23)cc1c1ccc(N(c2cccc2)c2ccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7ccc(N(c8cccc8)c8cccc8)cc7)ccc7ccc4c5c76)c4cccc34)cc2)cc1N#Cc1ccc(-c2ccc3c(c2)c2cc(-c4ccc(C#N)cc4)ccc2c2nc4cc(-c5ccc(N(c6cccc6)c6cccc6)cc5)c(-c5ccc(N(c6cccc6)c6cccc6)cc5)cc4nc32)cc1c1ccc(N(c2cccc2)c2cccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7cccc(N(c8cccc8)c8cccc8)c7)ccc7ccc4c5c76)c4cccc34)c2)cc1CC1(C)C(=O)c2cc(-c3ccc(N4c5cccc50c5cccc54)cc3)c(-c3ccc(N4c5cccc50c5cccc54)cc3)cc2C1=O</chem>	1	1	D-A	513.644	9.214
SFCz	<chem>Cc1cc(C2=NC3=NC(c4ccc(F)c(C(C)c4)=NC4=NC(c5ccc(F)c(CC1(C)c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc2c(c1)-c1cccc1C21c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc(N(c2cccc2)c2cccc2-c2c3cccc3c(-c3ccc4ccc5c(-c6cccc6N(c6cccc6)c6cccc6)ccc6ccc3c4c65)c3cccc23)cc1c1ccc(N(c2cccc2)c2ccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7ccc(N(c8cccc8)c8cccc8)cc7)ccc7ccc4c5c76)c4cccc34)cc2)cc1N#Cc1ccc(-c2ccc3c(c2)c2cc(-c4ccc(C#N)cc4)ccc2c2nc4cc(-c5ccc(N(c6cccc6)c6cccc6)cc5)c(-c5ccc(N(c6cccc6)c6cccc6)cc5)cc4nc32)cc1c1ccc(N(c2cccc2)c2cccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7cccc(N(c8cccc8)c8cccc8)c7)ccc7ccc4c5c76)c4cccc34)c2)cc1CC1(C)C(=O)c2cc(-c3ccc(N4c5cccc50c5cccc54)cc3)c(-c3ccc(N4c5cccc50c5cccc54)cc3)cc2C1=O</chem>	1	1	D-A	635.770	11.251
o-TPA-AP-TPA	<chem>Cc1cc(C2=NC3=NC(c4ccc(F)c(C(C)c4)=NC4=NC(c5ccc(F)c(CC1(C)c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc2c(c1)-c1cccc1C21c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc(N(c2cccc2)c2cccc2-c2c3cccc3c(-c3ccc4ccc5c(-c6cccc6N(c6cccc6)c6cccc6)ccc6ccc3c4c65)c3cccc23)cc1c1ccc(N(c2cccc2)c2ccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7ccc(N(c8cccc8)c8cccc8)cc7)ccc7ccc4c5c76)c4cccc34)cc2)cc1N#Cc1ccc(-c2ccc3c(c2)c2cc(-c4ccc(C#N)cc4)ccc2c2nc4cc(-c5ccc(N(c6cccc6)c6cccc6)cc5)c(-c5ccc(N(c6cccc6)c6cccc6)cc5)cc4nc32)cc1c1ccc(N(c2cccc2)c2cccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7cccc(N(c8cccc8)c8cccc8)c7)ccc7ccc4c5c76)c4cccc34)c2)cc1CC1(C)C(=O)c2cc(-c3ccc(N4c5cccc50c5cccc54)cc3)c(-c3ccc(N4c5cccc50c5cccc54)cc3)cc2C1=O</chem>	2	0	2D-0A	865.092	18.831
p-TPA-AP-TPA	<chem>Cc1cc(C2=NC3=NC(c4ccc(F)c(C(C)c4)=NC4=NC(c5ccc(F)c(CC1(C)c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc2c(c1)-c1cccc1C21c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc(N(c2cccc2)c2cccc2-c2c3cccc3c(-c3ccc4ccc5c(-c6cccc6N(c6cccc6)c6cccc6)ccc6ccc3c4c65)c3cccc23)cc1c1ccc(N(c2cccc2)c2ccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7ccc(N(c8cccc8)c8cccc8)cc7)ccc7ccc4c5c76)c4cccc34)cc2)cc1N#Cc1ccc(-c2ccc3c(c2)c2cc(-c4ccc(C#N)cc4)ccc2c2nc4cc(-c5ccc(N(c6cccc6)c6cccc6)cc5)c(-c5ccc(N(c6cccc6)c6cccc6)cc5)cc4nc32)cc1c1ccc(N(c2cccc2)c2cccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7cccc(N(c8cccc8)c8cccc8)c7)ccc7ccc4c5c76)c4cccc34)c2)cc1CC1(C)C(=O)c2cc(-c3ccc(N4c5cccc50c5cccc54)cc3)c(-c3ccc(N4c5cccc50c5cccc54)cc3)cc2C1=O</chem>	2	0	2D-0A	865.092	18.831
TNPZ	<chem>Cc1cc(C2=NC3=NC(c4ccc(F)c(C(C)c4)=NC4=NC(c5ccc(F)c(CC1(C)c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc2c(c1)-c1cccc1C21c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc(N(c2cccc2)c2cccc2-c2c3cccc3c(-c3ccc4ccc5c(-c6cccc6N(c6cccc6)c6cccc6)ccc6ccc3c4c65)c3cccc23)cc1c1ccc(N(c2cccc2)c2ccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7ccc(N(c8cccc8)c8cccc8)cc7)ccc7ccc4c5c76)c4cccc34)cc2)cc1N#Cc1ccc(-c2ccc3c(c2)c2cc(-c4ccc(C#N)cc4)ccc2c2nc4cc(-c5ccc(N(c6cccc6)c6cccc6)cc5)c(-c5ccc(N(c6cccc6)c6cccc6)cc5)cc4nc32)cc1c1ccc(N(c2cccc2)c2cccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7cccc(N(c8cccc8)c8cccc8)c7)ccc7ccc4c5c76)c4cccc34)c2)cc1CC1(C)C(=O)c2cc(-c3ccc(N4c5cccc50c5cccc54)cc3)c(-c3ccc(N4c5cccc50c5cccc54)cc3)cc2C1=O</chem>	2	6	Multi-D/A	969.164	18.440
m-TPA-AP-TPA	<chem>Cc1cc(C2=NC3=NC(c4ccc(F)c(C(C)c4)=NC4=NC(c5ccc(F)c(CC1(C)c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc2c(c1)-c1cccc1C21c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc(N(c2cccc2)c2cccc2-c2c3cccc3c(-c3ccc4ccc5c(-c6cccc6N(c6cccc6)c6cccc6)ccc6ccc3c4c65)c3cccc23)cc1c1ccc(N(c2cccc2)c2ccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7ccc(N(c8cccc8)c8cccc8)cc7)ccc7ccc4c5c76)c4cccc34)cc2)cc1N#Cc1ccc(-c2ccc3c(c2)c2cc(-c4ccc(C#N)cc4)ccc2c2nc4cc(-c5ccc(N(c6cccc6)c6cccc6)cc5)c(-c5ccc(N(c6cccc6)c6cccc6)cc5)cc4nc32)cc1c1ccc(N(c2cccc2)c2cccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7cccc(N(c8cccc8)c8cccc8)c7)ccc7ccc4c5c76)c4cccc34)c2)cc1CC1(C)C(=O)c2cc(-c3ccc(N4c5cccc50c5cccc54)cc3)c(-c3ccc(N4c5cccc50c5cccc54)cc3)cc2C1=O</chem>	2	0	2D-0A	865.092	18.831
5,6PXZ-PIDO	<chem>Cc1cc(C2=NC3=NC(c4ccc(F)c(C(C)c4)=NC4=NC(c5ccc(F)c(CC1(C)c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc2c(c1)-c1cccc1C21c2cccc2-c2ccc(-c3ncc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)cc21c1ccc(N(c2cccc2)c2cccc2-c2c3cccc3c(-c3ccc4ccc5c(-c6cccc6N(c6cccc6)c6cccc6)ccc6ccc3c4c65)c3cccc23)cc1c1ccc(N(c2cccc2)c2ccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7ccc(N(c8cccc8)c8cccc8)cc7)ccc7ccc4c5c76)c4cccc34)cc2)cc1N#Cc1ccc(-c2ccc3c(c2)c2cc(-c4ccc(C#N)cc4)ccc2c2nc4cc(-c5ccc(N(c6cccc6)c6cccc6)cc5)c(-c5ccc(N(c6cccc6)c6cccc6)cc5)cc4nc32)cc1c1ccc(N(c2cccc2)c2cccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7cccc(N(c8cccc8)c8cccc8)c7)ccc7ccc4c5c76)c4cccc34)c2)cc1CC1(C)C(=O)c2cc(-c3ccc(N4c5cccc50c5cccc54)cc3)c(-c3ccc(N4c5cccc50c5cccc54)cc3)cc2C1=O</chem>	4	0	Multi-D/A	688.783	12.577

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
TRZ-3SO ₂	<chem>O=S(=O)(c1ccccc1)c1cccc(-c2nc(-c3cccc(S(=O)(=O)c4cccc4)c3)nc(-c3cccc(S(=O)(=O)c4cccc4)c3)n2)c1CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1cc(C(=O)c2cncnc2)cc(-n2c3ccc(C(C)(C)C)ccc32)c1</chem>	0	1	0D-1A	729.861	7.371
35tCzBPym	<chem>O=C(c1cncnc1)c1cc(-n2c3ccc3c3cccc32)cc(-n2c3ccc3c3cccc32)c1</chem>	2	1	D-A-D	739.020	13.092
35CzBPym	<chem>O=C(c1cncnc1)c1cc(-n2c3ccc3c3cccc32)ccc1-n1c2ccccc2c2cccc21CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-n2c3ccc(C(C)(C)C)ccc32)c(C(=O)c2cncnc2)c1</chem>	2	1	D-A-D	514.588	7.902
25CzBPym	<chem>O=C(c1cncnc1)c1cc(-n2c3ccc3c3cccc32)ccc1-n1c2ccccc2c2cccc21CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-n2c3ccc(C(C)(C)C)ccc32)c(C(=O)c2cncnc2)c1</chem>	2	1	D-A-D	514.588	7.902
25tCzBPym	<chem>O=C(c1cncnc1)c1cc(-n2c3ccc3c3cccc32)ccc1-n1c2ccccc2c2cccc21CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-n2c3ccc(C(C)(C)C)ccc32)c(C(=O)c2cncnc2)c1</chem>	2	1	D-A-D	739.020	13.092
BPBiPA	<chem>c(-c3c4cccc4c(-c4ccc(-n5c(-c6cccc6)nc6cccc65)cc4)c4cccc34)cc2)cc1CC1(C)c2cccc2N(c2ccc(N3c4cccc4C(C)(C)c4cccc43)c3nsnc23)c2cccc21c1ccc(N(c2cccc2)c2ccc(N(c3cccc3)c3cccc3)c3nsnc23)cc1c1ccc(N(c2cccc2)c2ccc3c(c2)c2cc(N(c4cccc4)c4cccc4)ccc2n3-c2ccc(-c3ccc4ncnc4c3)cc2)cc1</chem>	0	0	Unknown	714.872	13.339
BTZ-DMAC	<chem>CC1(C)c2cccc2N(c2ccc(N3c4cccc4C(C)(C)c4cccc43)c3nsnc23)c2cccc21c1ccc(N(c2cccc2)c2ccc(N(c3cccc3)c3cccc3)c3nsnc23)cc1c1ccc(N(c2cccc2)c2ccc3c(c2)c2cc(N(c4cccc4)c4cccc4)ccc2n3-c2ccc(-c3ccc4ncnc4c3)cc2)cc1</chem>	2	0	2D-0A	550.731	9.909
BTZ-DPA	<chem>CC1(C)c2cccc2N(c2ccc(N3c4cccc4C(C)(C)c4cccc43)c3nsnc23)c2cccc21c1ccc(N(c2cccc2)c2ccc(N(c3cccc3)c3cccc3)c3nsnc23)cc1c1ccc(N(c2cccc2)c2ccc3c(c2)c2cc(N(c4cccc4)c4cccc4)ccc2n3-c2ccc(-c3ccc4ncnc4c3)cc2)cc1</chem>	2	0	2D-0A	470.601	8.631
2F	<chem>CC1(C)c2cccc2N(c2ccc(N3c4cccc4C(C)(C)c4cccc43)c3nsnc23)c2cccc21c1ccc(N(c2cccc2)c2ccc(N(c3cccc3)c3cccc3)c3nsnc23)cc1c1ccc(N(c2cccc2)c2ccc3c(c2)c2cc(N(c4cccc4)c4cccc4)ccc2n3-c2ccc(-c3ccc4ncnc4c3)cc2)cc1</chem>	3	1	Multi-D/A	705.865	13.333

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
PFBP-2a	<chem>CC(C)(C)c1ccc2c(c1)C(C)(C)c1cc(C(C)(C)C)ccc1N2c1c(F)c(F)c(-c2c(F)c(F)c(F)c(F)c2F)c(F)c1F</chem>	1	0	1D-0A	635.614	11.310
PFBP-2b	<chem>CC(C)(C)c1ccc2c(c1)C(C)(C)c1cc(C(C)(C)C)ccc1N2c1c(F)c(F)c(-c2c(F)c(F)c(N3c4ccc(C(C)(C)C)cc4C(C)(C)c4cc(C(C)(C)C)ccc43)c(F)c2F)c(F)c1F</chem>	2	0	2D-0A	937.116	17.875
PXZ-PPO	<chem>Cc1cc2nc(N3c4cccc40c4cccc43)ccc2c(=O)o1</chem>	1	0	1D-0A	342.354	5.072
PXZ-BOO	<chem>Cc1nc2cc(N3c4cccc40c4cccc43)ccc2c(=O)o1</chem>	2	0	2D-0A	342.354	5.072
PPZTPI	<chem>CC(C)(C)c1ccc(-n2c(-c3cccc(N4c5cccc5N(c5cccc5)c5cccc54)cc3)nc3c4cccc4c4cccc4c32)cc1</chem>	2	0	2D-0A	682.871	13.549
PPZPPI	<chem>c1ccc(N2c3cccc3N(c3ccc(-c4nc5c6cccc6c6cccc6c5n4-c4cccc4)cc3)c3cccc32)c1</chem>	2	0	2D-0A	626.763	12.252
Pra-2DMAC	<chem>CC1(C)c2cccc2N(c2ccnc(N3c4cccc4C(C)(C)c4cccc43)c2)c2cccc21</chem>	0	0	Unknown	493.654	9.300
BAPCN	<chem>CC1(C)c2cccc2N(c2cccc2)c2ccc3cc(-c4ccc(C#N)cc4)cc3c21</chem>	1	2	A-D-A	436.558	8.488
BACN	<chem>CC1(C)c2cccc2N(c2cccc2)c2ccc3cc(C#N)ccc3c21</chem>	1	2	A-D-A	360.460	6.821

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
BDQC-2	<chem>c1ccc(-c2ccc(N(c3ccc(-c4cccc4)cc3)c3ccc4c5cccc5n(-c5ccc6nc(-c7cccc7)c(-c7cccc7)nc6c5)c4c3)cc2)cc1</chem>	2	1	D-A-D	766.948	14.865
NIDPA-1	<chem>c1ccc(N(c2cccc2)c2ccc3c(c2)cc2c4cccc(N(c5cccc5)c5cccc5)c4n4c5cccc5c3c24)cc1</chem>	4	0	Multi-D/A	625.775	12.929
NIDPA-2	<chem>c1ccc(N(c2cccc2)c2ccc3c(c2)c2cc(N(c4cccc4)c4cccc4)cc4c5c6cccc6ccc5n3c24)cc1</chem>	4	0	Multi-D/A	625.775	12.929
SABIQ	<chem>c1ccc(N2c3cccc3C3(c4cccc4-c4nc5cc6cccc6cc5nc43)c3cccc32)cc1</chem>	1	1	D-A	509.612	8.929
Perylene	<chem>c1cc2cccc3c4cccc5cccc(c(c1)c23)c54</chem>	0	0	Unknown	252.316	5.737
SAIP	<chem>c1ccc(N2c3cccc3C3(c4cccc4-c4ncnc43)c3cccc32)cc1</chem>	1	0	1D-0A	409.492	6.623
FAC	<chem>c1ccc2c(c1)Nc1cccc1C21c2cccc2-c2cccc21</chem>	0	0	Unknown	331.418	6.107
phenylperylene	<chem>c1ccc(-c2ccc3cccc4c5cccc6cccc(c2c34)c65)cc1</chem>	0	0	Unknown	328.414	7.404
ImPy-2	<chem>C0c1ccc2cc(-c3nc(-c4cccc4)c4cccn34)ccc2c1</chem>	0	0	Unknown	350.421	5.830
ImPy-1	<chem>c1ccc(-c2nc(-c3ccc4cccc4c3)n3cccc23)cc1</chem>	0	0	Unknown	320.395	5.822
ImPy-3	<chem>c1ccc(-c2nc(-c3ccc4ccc5cc6ccc3c4c56)n3cccc23)cc1</chem>	0	0	Unknown	394.477	7.566

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
BTaz-Ar-PXZ	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc(-c2nc3cccc3s2)cc1CC1(C)c2cccc2-c2ccc(N(c3cccc3-c3cccc3)c3cccc4c3-c3cccc3C43c4cccc4-c4cccc43)cc21</chem>	2	0	2D-0A	392.483	7.539
FSF4A	<chem>c1ccc2c(c1)nc1n(-c3ccc(-c4ccc5oc6cccc6c5c4)cn3)c3cccc3n21</chem>	1	0	1D-0A	675.875	13.473
TPA-L-BN	<chem>N#Cc1ccc(CCCc2ccc(N(c3cccc3)c3cccc3)cc2)cc1CC(C)c1ccc2occ(-c3ccc(N4c5cccc5C(C)(C)c5cc6c(cc54)oc4cccc46)cc3)c(=O)c2c1</chem>	1	2	A-D-A	388.514	7.203
IpCm-PhBzAc	<chem>c1ccc2c(c1)nc1n(-c3ccc(-c4ccc5oc6cccc6c5c4)cn3)c3cccc3n21</chem>	1	0	1D-0A	561.681	10.592
DphAn-Br	<chem>O=C1c2cccc2C(c2cccc2)(c2ccc(Br)cc2)c2cccc21</chem>	0	0	Unknown	425.325	6.376
5-DBFBI	<chem>c1ccc2c(c1)nc1n(-c3ccc(-c4ccc5oc6cccc6c5c4)cn3)c3cccc3n21</chem>	0	0	Unknown	450.501	7.393
4-DBFBI	<chem>c1ccc2c(c1)nc1n(-c3ccc(-c4ccc5oc6cccc6c5c4)ccn3)c3cccc3n21</chem>	0	0	Unknown	450.501	7.393
DMACMNPTO	<chem>Cc1cc(N2c3cccc3S(=O)(=O)c3cccc32)c(C)cc1N1c2cccc2C(C)(C)c2cccc21CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc2c(c1)-c1cc(-n3c4ccc(C(C)(C)C)cc4c4cc(C(C)(C)C)ccc43)cc1S2(=O)=OCC1(C)c2cccc2N(c2ccc(C3(c4cccc4)c4cccc4C(=O)c4cccc43)cc2)c2c1ccc1c2oc2cccc21</chem>	3	0	Multi-D/A	542.704	9.029
1CzSO	<chem>c1ccc2c(c1)nc1n(-c3ccc(-c4ccc5oc6cccc6c5c4)cn3)c3cccc3n21</chem>	2	0	2D-0A	771.083	13.884
DpAn-BzAc	<chem>c1ccc2c(c1)nc1n(-c3ccc(-c4ccc5oc6cccc6c5c4)cn3)c3cccc3n21</chem>	1	0	1D-0A	643.786	11.622

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
BzITz	<chem>N#Cc1cc(-c2nc(-c3ccccc3)n c(-c3ccccc3)n2)ccc1-n1c2c cccc2n2c3ccccc3nc12 c1ccc2c(c1)Oc1ccccc1N2c1c cc(-c2nnc(-c3ccc(N4c5cccc c50c5ccccc54)cc3)c(-c3ccc (N4c5ccccc50c5ccccc54)cc3)n2)cc1 CC(C)(C)c1ccc(N(c2ccc(-c3 cc(N)c(N)cc3C#N)cc2)c2ccc (C(C)(C)C)cc2)cc1</chem>	0	3	Multi-D/A	539.602	7.489
TPXZ-as-TAZ	<chem>c1ccc2nc3c4cccn4c4ncccc4 c3nc2c1 N#Cc1cc2nc(-c3ccccc3)c(-c 3ccc(-n4c5ccccc5c5c6c(c7c cccc7n6-c6ccccc6)c6c(c7cc ccc7n6-c6ccccc6)c54)cc3)n c2cc1C#N c1ccc(-c2nc(-n3c4ccccc4c4 ccc5c6ccccc6n(-c6ccccc6)c 5c43)nc(-n3c4ccccc4c4ccc5 c6ccccc6n(-c6ccccc6)c5c43)n2)cc1 O=[N+]([O-])c1ccc(/C(=C(/ c2ccc([N+](=O)[O-])cc2)n2 c3ccccc3c3ccccc32)n2c3ccc cc3c3ccccc32)cc1 Cc1ccc(N(c2ccc(C)cc2)c2cc c(C3(c4ccc(N(c5ccc(C)cc5) c5ccc(C)cc5)cc4)CCCC3)cc 2)cc1 c1cc(-c2ccc(-n3c4ccccc4c4 ccccc43)cc2)cc(-c2ccc3c4c cccc4c4ccccc4c3c2)c1</chem>	6	0	Multi-D/A	852.954	15.599
TPA-CN-N4-2PY	<chem>c1ccc2nc3c4cccn4c4ncccc4 c3nc2c1 N#Cc1cc2nc(-c3ccccc3)c(-c 3ccc(-n4c5ccccc5c5c6c(c7c cccc7n6-c6ccccc6)c6c(c7cc ccc7n6-c6ccccc6)c54)cc3)n c2cc1C#N c1ccc(-c2nc(-n3c4ccccc4c4 ccc5c6ccccc6n(-c6ccccc6)c 5c43)nc(-n3c4ccccc4c4ccc5 c6ccccc6n(-c6ccccc6)c5c43)n2)cc1 O=[N+]([O-])c1ccc(/C(=C(/ c2ccc([N+](=O)[O-])cc2)n2 c3ccccc3c3ccccc32)n2c3ccc cc3c3ccccc32)cc1 Cc1ccc(N(c2ccc(C)cc2)c2cc c(C3(c4ccc(N(c5ccc(C)cc5) c5ccc(C)cc5)cc4)CCCC3)cc 2)cc1 c1cc(-c2ccc(-n3c4ccccc4c4 ccccc43)cc2)cc(-c2ccc3c4c cccc4c4ccccc4c3c2)c1</chem>	1	2	A-D-A	488.679	8.454
TPA-CN-N4	<chem>c1ccc2nc3c4cccn4c4ncccc4 c3nc2c1 N#Cc1cc2nc(-c3ccccc3)c(-c 3ccc(-n4c5ccccc5c5c6c(c7c cccc7n6-c6ccccc6)c6c(c7cc ccc7n6-c6ccccc6)c54)cc3)n c2cc1C#N c1ccc(-c2nc(-n3c4ccccc4c4 ccc5c6ccccc6n(-c6ccccc6)c 5c43)nc(-n3c4ccccc4c4ccc5 c6ccccc6n(-c6ccccc6)c5c43)n2)cc1 O=[N+]([O-])c1ccc(/C(=C(/ c2ccc([N+](=O)[O-])cc2)n2 c3ccccc3c3ccccc32)n2c3ccc cc3c3ccccc32)cc1 Cc1ccc(N(c2ccc(C)cc2)c2cc c(C3(c4ccc(N(c5ccc(C)cc5) c5ccc(C)cc5)cc4)CCCC3)cc 2)cc1 c1cc(-c2ccc(-n3c4ccccc4c4 ccccc43)cc2)cc(-c2ccc3c4c cccc4c4ccccc4c3c2)c1</chem>	0	2	0D-2A	282.306	3.879
6,7-DCNQx-DICz	<chem>c1ccc2nc3c4cccn4c4ncccc4 c3nc2c1 N#Cc1cc2nc(-c3ccccc3)c(-c 3ccc(-n4c5ccccc5c5c6c(c7c cccc7n6-c6ccccc6)c6c(c7cc ccc7n6-c6ccccc6)c54)cc3)n c2cc1C#N c1ccc(-c2nc(-n3c4ccccc4c4 ccc5c6ccccc6n(-c6ccccc6)c 5c43)nc(-n3c4ccccc4c4ccc5 c6ccccc6n(-c6ccccc6)c5c43)n2)cc1 O=[N+]([O-])c1ccc(/C(=C(/ c2ccc([N+](=O)[O-])cc2)n2 c3ccccc3c3ccccc32)n2c3ccc cc3c3ccccc32)cc1 Cc1ccc(N(c2ccc(C)cc2)c2cc c(C3(c4ccc(N(c5ccc(C)cc5) c5ccc(C)cc5)cc4)CCCC3)cc 2)cc1 c1cc(-c2ccc(-n3c4ccccc4c4 ccccc43)cc2)cc(-c2ccc3c4c cccc4c4ccccc4c3c2)c1</chem>	3	5	Multi-D/A	827.951	13.998
PBICT	<chem>c1ccc2nc3c4cccn4c4ncccc4 c3nc2c1 N#Cc1cc2nc(-c3ccccc3)c(-c 3ccc(-n4c5ccccc5c5c6c(c7c cccc7n6-c6ccccc6)c6c(c7cc ccc7n6-c6ccccc6)c54)cc3)n c2cc1C#N c1ccc(-c2nc(-n3c4ccccc4c4 ccc5c6ccccc6n(-c6ccccc6)c 5c43)nc(-n3c4ccccc4c4ccc5 c6ccccc6n(-c6ccccc6)c5c43)n2)cc1 O=[N+]([O-])c1ccc(/C(=C(/ c2ccc([N+](=O)[O-])cc2)n2 c3ccccc3c3ccccc32)n2c3ccc cc3c3ccccc32)cc1 Cc1ccc(N(c2ccc(C)cc2)c2cc c(C3(c4ccc(N(c5ccc(C)cc5) c5ccc(C)cc5)cc4)CCCC3)cc 2)cc1 c1cc(-c2ccc(-n3c4ccccc4c4 ccccc43)cc2)cc(-c2ccc3c4c cccc4c4ccccc4c3c2)c1</chem>	4	1	Multi-D/A	817.956	13.927
DCBNPE	<chem>c1ccc2nc3c4cccn4c4ncccc4 c3nc2c1 N#Cc1cc2nc(-c3ccccc3)c(-c 3ccc(-n4c5ccccc5c5c6c(c7c cccc7n6-c6ccccc6)c6c(c7cc ccc7n6-c6ccccc6)c54)cc3)n c2cc1C#N c1ccc(-c2nc(-n3c4ccccc4c4 ccc5c6ccccc6n(-c6ccccc6)c 5c43)nc(-n3c4ccccc4c4ccc5 c6ccccc6n(-c6ccccc6)c5c43)n2)cc1 O=[N+]([O-])c1ccc(/C(=C(/ c2ccc([N+](=O)[O-])cc2)n2 c3ccccc3c3ccccc32)n2c3ccc cc3c3ccccc32)cc1 Cc1ccc(N(c2ccc(C)cc2)c2cc c(C3(c4ccc(N(c5ccc(C)cc5) c5ccc(C)cc5)cc4)CCCC3)cc 2)cc1 c1cc(-c2ccc(-n3c4ccccc4c4 ccccc43)cc2)cc(-c2ccc3c4c cccc4c4ccccc4c3c2)c1</chem>	2	0	2D-0A	600.634	9.641
TPA-BT-Q	<chem>c1ccc2nc3c4cccn4c4ncccc4 c3nc2c1 N#Cc1cc2nc(-c3ccccc3)c(-c 3ccc(-n4c5ccccc5c5c6c(c7c cccc7n6-c6ccccc6)c6c(c7cc ccc7n6-c6ccccc6)c54)cc3)n c2cc1C#N c1ccc(-c2nc(-n3c4ccccc4c4 ccc5c6ccccc6n(-c6ccccc6)c 5c43)nc(-n3c4ccccc4c4ccc5 c6ccccc6n(-c6ccccc6)c5c43)n2)cc1 O=[N+]([O-])c1ccc(/C(=C(/ c2ccc([N+](=O)[O-])cc2)n2 c3ccccc3c3ccccc32)n2c3ccc cc3c3ccccc32)cc1 Cc1ccc(N(c2ccc(C)cc2)c2cc c(C3(c4ccc(N(c5ccc(C)cc5) c5ccc(C)cc5)cc4)CCCC3)cc 2)cc1 c1cc(-c2ccc(-n3c4ccccc4c4 ccccc43)cc2)cc(-c2ccc3c4c cccc4c4ccccc4c3c2)c1</chem>	2	0	2D-0A	626.888	13.110
D5	<chem>c1ccc2nc3c4cccn4c4ncccc4 c3nc2c1 N#Cc1cc2nc(-c3ccccc3)c(-c 3ccc(-n4c5ccccc5c5c6c(c7c cccc7n6-c6ccccc6)c6c(c7cc ccc7n6-c6ccccc6)c54)cc3)n c2cc1C#N c1ccc(-c2nc(-n3c4ccccc4c4 ccc5c6ccccc6n(-c6ccccc6)c 5c43)nc(-n3c4ccccc4c4ccc5 c6ccccc6n(-c6ccccc6)c5c43)n2)cc1 O=[N+]([O-])c1ccc(/C(=C(/ c2ccc([N+](=O)[O-])cc2)n2 c3ccccc3c3ccccc32)n2c3ccc cc3c3ccccc32)cc1 Cc1ccc(N(c2ccc(C)cc2)c2cc c(C3(c4ccc(N(c5ccc(C)cc5) c5ccc(C)cc5)cc4)CCCC3)cc 2)cc1 c1cc(-c2ccc(-n3c4ccccc4c4 ccccc43)cc2)cc(-c2ccc3c4c cccc4c4ccccc4c3c2)c1</chem>	1	0	1D-0A	545.685	11.577

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
D6	<chem>c1cc(-c2ccc3c4ccccc4c4ccc cc4c3c2)cc(-n2c3ccccc3c3c cccc32)c1</chem>	1	0	1D-0A	469.587	9.910
PxPYM	<chem>c1ccc2c(c1)0c1ccccc1N2c1c cc(-c2cnc(-c3ccc(N4c5cccc c50c5ccccc54)cc3)nc2)cc1</chem>	4	1	Multi-D/A	594.674	10.962
D1–D6	<chem>c1cc(-c2ccc3c4ccccc4c4ccc cc4c3c2)cc(-n2c3ccccc3c3c cccc32)c1</chem>	1	0	1D-0A	469.587	9.910
DTPCZTZ	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C (C)(C)C)ccc1n2-c1ccc(-c2c cc(-c3nnc(-c4ccccc4)n3-c3 ccccc3)cc2)cc1</chem>	1	0	1D-0A	650.870	11.960
DHPZ-2BI	<chem>c1ccc(-n2c(-c3ccc(N4c5ccc cc5N(c5ccc(-c6nc7ccccc7n6 -c6ccccc6)cc5)c5ccccc54)c c3)nc3ccccc32)cc1</chem>	2	0	2D-0A	718.864	12.951
DHPZ-2BN	<chem>N#Cc1ccc(N2c3ccccc3N(c3cc c(C#N)cc3)c3ccccc32)cc1</chem>	2	4	Multi-D/A	384.442	6.683
SAF-2NP	<chem>c1ccc2c(c1)-c1ccccc1C21c2 ccccc2N(c2ccc3nc4c5cccn5 c5ncccc5c4nc3c2)c2ccccc21</chem>	1	2	A-D-A	611.708	10.026
BTPO	<chem>O=S1(=O)c2ccccc2N2c3ccccc 3S(=O)(=O)c3ccccc1c32 CC1(C)c2ccccc2N(c2ccc(-c3 cc(-c4ccc(C#N)cc4)nc(-c4c cc(C#N)cc4)c3)cc2)c2ccccc 21</chem>	3	0	Multi-D/A	369.423	3.449
PyDCN–DMAC	<chem>cc(-c4ccc(C#N)cc4)nc(-c4c cc(C#N)cc4)c3)cc2)c2ccccc 21</chem>	1	4	Multi-D/A	564.692	9.935
PyDCN	<chem>N#Cc1ccc(-c2cc(-c3ccccc3) cc(-c3ccc(C#N)cc3)n2)cc1</chem>	0	4	Multi-D/A	357.416	5.826
PXZ-OXD	<chem>c1ccc(-c2nnc(-c3ccc(N4c5c ccccc50c5ccccc54)cc3)o2)cc 1</chem>	2	0	2D-0A	403.441	6.979

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
TCz-3PA-TCz	<chem>CC(C)(C)c1ccc(-c2ccc3c(c2)cc2cc(-c4cnc(-c5ccc6c(c5)c5cc(-c7ccc(C(C)(C)C)cc7)ccc5n6-c5ccc(C(C)(C)C)cc5)cn4)ccc2n3-c2ccc(C(C)(C)C)cc2)cc1Cc1c(-c2ccccc2)cc(N2C(=O)c3cccc4c(N(c5ccc(-c6ccccc6)cc5)c5ccc6c(c5)C(C)(C)cc5ccccc5-6)ccc(c34)C2=O)cc1-c1ccccc1Cc1c(-c2ccccc2)cc(N2C(=O)c3cccc4c(-c5ccc(N(c6ccc(-c7ccccc7)cc6)c6ccc7c(c6)C(C)(C)c6ccccc6-7)cc5)ccc(c34)C2=O)cc1-c1ccccc1CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(S(=O)(=O)c2ccc(-n3c4ccc(C(C)(C)C)cc4c4cc(C(C)(C)C)ccc43)cn2)nc1O=C(c1ccc(N2c3ccccc3C3(c4ccccc4-c4ccccc43)c3cc4ccc4cc32)cc1)c1ccc(N2c3ccccc3C3(c4ccccc4-c4ccccc43)c3cc4ccccc4cc32)cc1c1ccc(N2c3ccccc3C3(c4ccccc4-c4ccccc43)c3cc4ccccc4c32)cc1CC1(C)c2cc(-c3nc(-c4ccccc4)c(-c4ccccc4)n3-c3ccccc3)ccc2-c2ccc(N(c3ccccc3)c3ccccc3)cc21</chem>	2	0	2D-0A	939.304	18.529
NAI-BiFA	<chem>c3cccc4c(N(c5ccc(-c6ccccc6)cc5)c5ccc6c(c5)C(C)(C)cc5ccccc5-6)ccc(c34)C2=O)cc1-c1ccccc1Cc1c(-c2ccccc2)cc(N2C(=O)c3cccc4c(-c5ccc(N(c6ccc(-c7ccccc7)cc6)c6ccc7c(c6)C(C)(C)c6ccccc6-7)cc5)ccc(c34)C2=O)cc1-c1ccccc1CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(S(=O)(=O)c2ccc(-n3c4ccc(C(C)(C)C)cc4c4cc(C(C)(C)C)ccc43)cn2)nc1O=C(c1ccc(N2c3ccccc3C3(c4ccccc4-c4ccccc43)c3cc4ccc4cc32)cc1)c1ccc(N2c3ccccc3C3(c4ccccc4-c4ccccc43)c3cc4ccccc4c32)cc1CC1(C)c2cc(-c3nc(-c4ccccc4)c(-c4ccccc4)n3-c3ccccc3)ccc2-c2ccc(N(c3ccccc3)c3ccccc3)cc21</chem>	1	0	1D-0A	798.986	14.726
NAI-PhBiFA	<chem>c3cccc4c(-c5ccc(N(c6ccc(-c7ccccc7)cc6)c6ccc7c(c6)C(C)(C)c6ccccc6-7)cc5)ccc(c34)C2=O)cc1-c1ccccc1CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(S(=O)(=O)c2ccc(-n3c4ccc(C(C)(C)C)cc4c4cc(C(C)(C)C)ccc43)cn2)nc1O=C(c1ccc(N2c3ccccc3C3(c4ccccc4-c4ccccc43)c3cc4ccc4cc32)cc1)c1ccc(N2c3ccccc3C3(c4ccccc4-c4ccccc43)c3cc4ccccc4c32)cc1CC1(C)c2cc(-c3nc(-c4ccccc4)c(-c4ccccc4)n3-c3ccccc3)ccc2-c2ccc(N(c3ccccc3)c3ccccc3)cc21</chem>	1	0	1D-0A	875.084	16.393
pDTCz-2DPyS	<chem>CC(C)(C)c1ccc(-c2ccc3c(c2)cc2cc(-c4cnc(-c5ccc6c(c5)c5cc(-c7ccc(C(C)(C)C)cc7)ccc5n6-c5ccc(C(C)(C)C)cc5)cn4)ccc2n3-c2ccc(C(C)(C)C)cc2)cc1Cc1c(-c2ccccc2)cc(N2C(=O)c3cccc4c(N(c5ccc(-c6ccccc6)cc5)c5ccc6c(c5)C(C)(C)cc5ccccc5-6)ccc(c34)C2=O)cc1-c1ccccc1Cc1c(-c2ccccc2)cc(N2C(=O)c3cccc4c(-c5ccc(N(c6ccc(-c7ccccc7)cc6)c6ccc7c(c6)C(C)(C)c6ccccc6-7)cc5)ccc(c34)C2=O)cc1-c1ccccc1CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(S(=O)(=O)c2ccc(-n3c4ccc(C(C)(C)C)cc4c4cc(C(C)(C)C)ccc43)cn2)nc1O=C(c1ccc(N2c3ccccc3C3(c4ccccc4-c4ccccc43)c3cc4ccc4cc32)cc1)c1ccc(N2c3ccccc3C3(c4ccccc4-c4ccccc43)c3cc4ccccc4c32)cc1CC1(C)c2cc(-c3nc(-c4ccccc4)c(-c4ccccc4)n3-c3ccccc3)ccc2-c2ccc(N(c3ccccc3)c3ccccc3)cc21</chem>	2	0	2D-0A	775.075	12.694
2SPBAC-BP	<chem>c3cccc4c(-c5ccc(N(c6ccc(-c7ccccc7)cc6)c6ccc7c(c6)C(C)(C)c6ccccc6-7)cc5)ccc(c34)C2=O)cc1-c1ccccc1CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(S(=O)(=O)c2ccc(-n3c4ccc(C(C)(C)C)cc4c4cc(C(C)(C)C)ccc43)cn2)nc1O=C(c1ccc(N2c3ccccc3C3(c4ccccc4-c4ccccc43)c3cc4ccc4cc32)cc1)c1ccc(N2c3ccccc3C3(c4ccccc4-c4ccccc43)c3cc4ccccc4c32)cc1CC1(C)c2cc(-c3nc(-c4ccccc4)c(-c4ccccc4)n3-c3ccccc3)ccc2-c2ccc(N(c3ccccc3)c3ccccc3)cc21</chem>	2	0	2D-0A	941.146	17.517
SPBAC-PH	<chem>c3cccc4c(-c5ccc(N(c6ccc(-c7ccccc7)cc6)c6ccc7c(c6)C(C)(C)c6ccccc6-7)cc5)ccc(c34)C2=O)cc1-c1ccccc1CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(S(=O)(=O)c2ccc(-n3c4ccc(C(C)(C)C)cc4c4cc(C(C)(C)C)ccc43)cn2)nc1O=C(c1ccc(N2c3ccccc3C3(c4ccccc4-c4ccccc43)c3cc4ccc4cc32)cc1)c1ccc(N2c3ccccc3C3(c4ccccc4-c4ccccc43)c3cc4ccccc4c32)cc1CC1(C)c2cc(-c3nc(-c4ccccc4)c(-c4ccccc4)n3-c3ccccc3)ccc2-c2ccc(N(c3ccccc3)c3ccccc3)cc21</chem>	1	0	1D-0A	457.576	8.986
DFPBI	<chem>CC1(C)c2cc(-c3nc(-c4ccccc4)c(-c4ccccc4)n3-c3ccccc3)ccc2-c2ccc(N(c3ccccc3)c3ccccc3)cc21</chem>	1	0	1D-0A	655.845	12.649

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
TDPA-TRZ	<chem>c1ccc(N(c2ccccc2)c2ccc(-c3nc(-c4ccc(N(c5ccccc5)c5ccccc5)cc4)nc(-c4ccc(N(c5ccccc5)c5ccccc5)cc4)n3)cc2)cc1</chem>	3	1	Multi-D/A	811.005	15.282
DMAC-11-DPPZ	<chem>CC1(C)c2ccccc2N(c2ccc3nc4c5ccccc5c5ccccc5c4nc3c2)c2ccccc21</chem>	1	2	A-D-A	487.606	9.198
PXZ-11-DPPZ	<chem>c1ccc2c(c1)Oc1ccccc1N2c1ccc2nc3c4ccccc4c4ccccc4c3nc2c1</chem>	2	2	2D-2A	461.524	8.665
PXZ-10-DPPZ	<chem>c1ccc2c(c1)Oc1ccccc1N2c1ccc2nc3c4ccccc4c4ccccc4c3nc12</chem>	2	2	2D-2A	461.524	8.665
CCC	<chem>c1ccc2c(c1)Cc1c-2ccc2c1-c1ccccc1C2</chem>	0	0	Unknown	254.332	4.829
CzmPPC	<chem>c1ccc(-c2cccc(-c3nc(-c4ccc(-c5ccccc5)c4)nc(-c4ccc(-c5ccccc5)c4)n3)c2)cc1</chem>	0	1	0D-1A	537.666	9.874
C2FLA-1	<chem>CCCC1(CCC)c2ccccc2-c2ccc(C#Cc3ccc4c(c3)C(CCC)(CCC)c3cc(C#N)ccc3-4)cc21</chem>	0	2	0D-2A	547.786	10.691
MDP3FL	<chem>Cc1cccc(N(c2ccccc2)c2ccc3c(c2)C(C)(C)c2cc(-c4ccc5c(c4)C(C)(C)c4cc(-c6ccc7c(c6)C(C)(C)c6cc(N(c8ccccc8)c8cccc(C)c8)ccc6-7)ccc4-5)ccc2-3)c1</chem>	2	0	2D-0A	941.275	19.496
C3FLA-2	<chem>CCCCC1(CCCC)c2cc(C#Cc3ccc4c(c3)C(CCC)(CCC)c3cc(C#N)ccc3-4)ccc2-c2ccc(C#Cc3ccc4c(c3)C(CCC)(CCC)c3cc(C#N)ccc3-4)cc21</chem>	0	4	Multi-D/A	873.241	16.610

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
9CzFDPEPO	<chem>O=P(c1ccccc1)(c1ccccc1)c1ccccc10c1ccc(-c2ccc(C3(c4ccc(-n5c6ccccc6c6ccccc65)cc4)c4ccccc4-c4ccccc43)cc2)cc1P(=O)(c1ccccc1)c1ccc</chem>	1	0	1D-0A	1052.163	15.885
9CzFDPEsPO	<chem>O=P(c1ccccc1)(c1ccccc1)c1ccccc10c1ccc(-c2ccc(C3(c4ccc(-n5c6ccccc6c6ccccc65)cc4)c4ccccc4-c4ccccc43)cc2)cc1</chem>	1	0	1D-0A	851.986	14.245
CTM	<chem>O=C(c1ccc(-n2c3ccccc3c3ccc32)cc1)c1ccc2sc3ccccc3c2c1</chem>	1	0	1D-0A	453.566	8.383
tCTM	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(C(=O)c2ccc3sc4ccccc4c3c2)cc1</chem>	1	0	1D-0A	565.782	10.978
PSBC	<chem>CC(C)(C)c1ccc(-n2c(-c3ccc(-c4ccc(S(=O)(=O)c5ccc(-c6ccc(-n7c8ccccc8c8ccccc87)cc6)cc5)cc4)cc3)nc3c4ccc</chem>	1	0	1D-0A	884.117	15.560
26DAcBPy	<chem>cc4c4ccccc4c32)cc1CC1(C)c2ccccc2N(c2ccc(C(=O)c3cccc(C(=O)c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)n3)cc2)c2ccccc21</chem>	2	0	2D-0A	701.870	11.762
25DAcBPy	<chem>CC1(C)c2ccccc2N(c2ccc(C(=O)c3ccc(C(=O)c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)nc3)cc2)c2ccccc21</chem>	2	0	2D-0A	701.870	11.762
26DPXZBPy	<chem>O=C(c1ccc(N2c3ccccc30c3ccc32)cc1)c1cccc(C(=O)c2ccc(N3c4ccccc40c4ccccc43)c2)n1</chem>	4	0	Multi-D/A	649.706	10.695

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
BTITrz	<chem>N#Cc1cc(-c2nc(-c3ccccc3)n</chem> <chem>c(-c3ccccc3)n2)ccc1-n1c2c</chem>	0	3	Multi-D/A	555.666	9.056
mTPA-PPI	<chem>cccc2c2c3ccccc3sc21</chem> <chem>c1ccc(N(c2ccccc2)c2ccc(-c</chem> <chem>3cccc(-c4nc5c6ccccc6c6ccc</chem> <chem>cc6c5n4-c4ccccc4)c3)cc2)c</chem> <chem>c1</chem>	1	0	1D-0A	613.764	12.136
PPI	<chem>c1ccc(-c2nc3c4ccccc4c4ccc</chem> <chem>cc4c3n2-c2ccccc2)cc1</chem> <chem>c1ccc(-c2ccc(N(c3ccc(-c4c</chem> <chem>cc5c(c4)c4ccccc4n5-c4cccc</chem> <chem>c4)cc3)c3cccc(-c4nc(-c5cc</chem> <chem>ccc5)nc(-c5ccccc5)n4)c3)c</chem> <chem>c2)cc1</chem> <chem>c1ccc(N(c2ccc(-c3ccc(-c4n</chem> <chem>c5c6ccccc6c6ccccc6c5n4-c4</chem> <chem>ccc(N(c5ccccc5)c5ccccc5)c</chem> <chem>c4)cc3)cc2)c2ccc(-c3ccc(-</chem> <chem>c4nc5c6ccccc6c6ccccc6c5n4</chem> <chem>-c4ccc(N(c5ccccc5)c5ccccc</chem> <chem>5)cc4)cc3)cc2)cc1</chem> <chem>C(=C/c1nc2c3ccccc3c3ccccc</chem> <chem>3c2n1-c1ccc(N(c2ccccc2)c2</chem> <chem>ccccc2)cc1)\c1ccc(-c2ccc(</chem> <chem>N(c3ccccc3)c3ccc(-c4ccc(/</chem> <chem>C=C/c5nc6c7ccccc7c7ccccc7</chem> <chem>c6n5-c5ccc(N(c6ccccc6)c6c</chem> <chem>ccccc6)cc5)cc4)cc3)cc2)cc1</chem> <chem>CC(C)(C)c1ccc2c(c1)c1cc(C</chem> <chem>(C)(C)C)ccc1n2-c1c(-n2c3c</chem> <chem>cccc3c3ccccc32)cc(-n2c3cc</chem> <chem>ccc3c3ccccc32)c(-n2c3ccc(</chem> <chem>C(C)(C)C)cc3c3cc(C(C)(C)C</chem> <chem>)ccc32)c1C#N</chem>	0	0	Unknown	370.455	6.999
CzTPA-m-Trz	<chem>c1ccc(-c2nc3c4ccccc4c4ccc</chem> <chem>cc4c3n2-c2ccccc2)cc1</chem> <chem>c1ccc(-c2ccc(N(c3ccc(-c4c</chem> <chem>cc5c(c4)c4ccccc4n5-c4cccc</chem> <chem>c4)cc3)c3cccc(-c4nc(-c5cc</chem> <chem>ccc5)nc(-c5ccccc5)n4)c3)c</chem> <chem>c2)cc1</chem> <chem>c1ccc(N(c2ccc(-c3ccc(-c4n</chem> <chem>c5c6ccccc6c6ccccc6c5n4-c4</chem> <chem>ccc(N(c5ccccc5)c5ccccc5)c</chem> <chem>c4)cc3)cc2)c2ccc(-c3ccc(-</chem> <chem>c4nc5c6ccccc6c6ccccc6c5n4</chem> <chem>-c4ccc(N(c5ccccc5)c5ccccc</chem> <chem>5)cc4)cc3)cc2)cc1</chem> <chem>C(=C/c1nc2c3ccccc3c3ccccc</chem> <chem>3c2n1-c1ccc(N(c2ccccc2)c2</chem> <chem>ccccc2)cc1)\c1ccc(-c2ccc(</chem> <chem>N(c3ccccc3)c3ccc(-c4ccc(/</chem> <chem>C=C/c5nc6c7ccccc7c7ccccc7</chem> <chem>c6n5-c5ccc(N(c6ccccc6)c6c</chem> <chem>ccccc6)cc5)cc4)cc3)cc2)cc1</chem> <chem>CC(C)(C)c1ccc2c(c1)c1cc(C</chem> <chem>(C)(C)C)ccc1n2-c1c(-n2c3c</chem> <chem>cccc3c3ccccc32)cc(-n2c3cc</chem> <chem>ccc3c3ccccc32)c(-n2c3ccc(</chem> <chem>C(C)(C)C)cc3c3cc(C(C)(C)C</chem> <chem>)ccc32)c1C#N</chem>	2	1	D-A-D	793.974	14.773
TPA-2PPITPA	<chem>c1ccc(-c2nc3c4ccccc4c4ccc</chem> <chem>cc4c3n2-c2ccccc2)cc1</chem> <chem>c1ccc(-c2ccc(N(c3ccc(-c4c</chem> <chem>cc5c(c4)c4ccccc4n5-c4cccc</chem> <chem>c4)cc3)c3cccc(-c4nc(-c5cc</chem> <chem>ccc5)nc(-c5ccccc5)n4)c3)c</chem> <chem>c2)cc1</chem> <chem>c1ccc(N(c2ccc(-c3ccc(-c4n</chem> <chem>c5c6ccccc6c6ccccc6c5n4-c4</chem> <chem>ccc(N(c5ccccc5)c5ccccc5)c</chem> <chem>c4)cc3)cc2)c2ccc(-c3ccc(-</chem> <chem>c4nc5c6ccccc6c6ccccc6c5n4</chem> <chem>-c4ccc(N(c5ccccc5)c5ccccc</chem> <chem>5)cc4)cc3)cc2)cc1</chem> <chem>C(=C/c1nc2c3ccccc3c3ccccc</chem> <chem>3c2n1-c1ccc(N(c2ccccc2)c2</chem> <chem>ccccc2)cc1)\c1ccc(-c2ccc(</chem> <chem>N(c3ccccc3)c3ccc(-c4ccc(/</chem> <chem>C=C/c5nc6c7ccccc7c7ccccc7</chem> <chem>c6n5-c5ccc(N(c6ccccc6)c6c</chem> <chem>ccccc6)cc5)cc4)cc3)cc2)cc1</chem> <chem>CC(C)(C)c1ccc2c(c1)c1cc(C</chem> <chem>(C)(C)C)ccc1n2-c1c(-n2c3c</chem> <chem>cccc3c3ccccc32)cc(-n2c3cc</chem> <chem>ccc3c3ccccc32)c(-n2c3ccc(</chem> <chem>C(C)(C)C)cc3c3cc(C(C)(C)C</chem> <chem>)ccc32)c1C#N</chem>	3	0	Multi-D/A	1316.625	26.055
TPA-2SPPITPA	<chem>c1ccc(-c2nc3c4ccccc4c4ccc</chem> <chem>cc4c3n2-c2ccccc2)cc1</chem> <chem>c1ccc(-c2ccc(N(c3ccc(-c4c</chem> <chem>cc5c(c4)c4ccccc4n5-c4cccc</chem> <chem>c4)cc3)c3cccc(-c4nc(-c5cc</chem> <chem>ccc5)nc(-c5ccccc5)n4)c3)c</chem> <chem>c2)cc1</chem> <chem>c1ccc(N(c2ccc(-c3ccc(-c4n</chem> <chem>c5c6ccccc6c6ccccc6c5n4-c4</chem> <chem>ccc(N(c5ccccc5)c5ccccc5)c</chem> <chem>c4)cc3)cc2)c2ccc(-c3ccc(-</chem> <chem>c4nc5c6ccccc6c6ccccc6c5n4</chem> <chem>-c4ccc(N(c5ccccc5)c5ccccc</chem> <chem>5)cc4)cc3)cc2)cc1</chem> <chem>C(=C/c1nc2c3ccccc3c3ccccc</chem> <chem>3c2n1-c1ccc(N(c2ccccc2)c2</chem> <chem>ccccc2)cc1)\c1ccc(-c2ccc(</chem> <chem>N(c3ccccc3)c3ccc(-c4ccc(/</chem> <chem>C=C/c5nc6c7ccccc7c7ccccc7</chem> <chem>c6n5-c5ccc(N(c6ccccc6)c6c</chem> <chem>ccccc6)cc5)cc4)cc3)cc2)cc1</chem> <chem>CC(C)(C)c1ccc2c(c1)c1cc(C</chem> <chem>(C)(C)C)ccc1n2-c1c(-n2c3c</chem> <chem>cccc3c3ccccc32)cc(-n2c3cc</chem> <chem>ccc3c3ccccc32)c(-n2c3ccc(</chem> <chem>C(C)(C)C)cc3c3cc(C(C)(C)C</chem> <chem>)ccc32)c1C#N</chem>	3	0	Multi-D/A	1368.701	27.061
2Cz2tCzBn	<chem>c1ccc(-c2nc3c4ccccc4c4ccc</chem> <chem>cc4c3n2-c2ccccc2)cc1</chem> <chem>c1ccc(-c2ccc(N(c3ccc(-c4c</chem> <chem>cc5c(c4)c4ccccc4n5-c4cccc</chem> <chem>c4)cc3)c3cccc(-c4nc(-c5cc</chem> <chem>ccc5)nc(-c5ccccc5)n4)c3)c</chem> <chem>c2)cc1</chem> <chem>c1ccc(N(c2ccc(-c3ccc(-c4n</chem> <chem>c5c6ccccc6c6ccccc6c5n4-c4</chem> <chem>ccc(N(c5ccccc5)c5ccccc5)c</chem> <chem>c4)cc3)cc2)c2ccc(-c3ccc(-</chem> <chem>c4nc5c6ccccc6c6ccccc6c5n4</chem> <chem>-c4ccc(N(c5ccccc5)c5ccccc</chem> <chem>5)cc4)cc3)cc2)cc1</chem> <chem>C(=C/c1nc2c3ccccc3c3ccccc</chem> <chem>3c2n1-c1ccc(N(c2ccccc2)c2</chem> <chem>ccccc2)cc1)\c1ccc(-c2ccc(</chem> <chem>N(c3ccccc3)c3ccc(-c4ccc(/</chem> <chem>C=C/c5nc6c7ccccc7c7ccccc7</chem> <chem>c6n5-c5ccc(N(c6ccccc6)c6c</chem> <chem>ccccc6)cc5)cc4)cc3)cc2)cc1</chem> <chem>CC(C)(C)c1ccc2c(c1)c1cc(C</chem> <chem>(C)(C)C)ccc1n2-c1c(-n2c3c</chem> <chem>cccc3c3ccccc32)cc(-n2c3cc</chem> <chem>ccc3c3ccccc32)c(-n2c3ccc(</chem> <chem>C(C)(C)C)cc3c3cc(C(C)(C)C</chem> <chem>)ccc32)c1C#N</chem>	4	2	Multi-D/A	988.336	19.137

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
2tCz2CzBn	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c(-n2c3cccc3c3cccc32)c(C#N)c1-n1c2ccc cc2c2cccc21</chem>	4	2	Multi-D/A	988.336	19.137
PmPmP	<chem>c1cncc(-c2nc(-c3ccc(N4c5cccc50c5cccc54)cc3)cc(-c3cncnc3)n2)c1</chem>	2	2	2D-2A	492.542	7.238
PLQYs	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1cc2ccc3c c(-n4c5ccc(C(C)(C)C)cc5c5 cc(C(C)(C)C)ccc54)cc4ccc(c1)c2c34</chem>	2	0	2D-0A	757.078	15.968
PXZ-ND	<chem>c1ccc2c(c1)0c1cccc1N2c1c cc2ccc(N3c4cccc40c4cccc 43)nc2n1</chem>	2	0	2D-0A	492.538	8.781
CPPyC	<chem>c1cc(-c2cncc(-n3c4cccc4c 4cccc43)c2)cc(-n2c3cccc 3c3cccc32)c1</chem>	2	0	2D-0A	485.590	8.943
DMAC-ND	<chem>CC1(C)c2cccc2N(c2ccc3ccc (N4c5cccc5C(C)(C)c5cccc 54)nc3n2)c2cccc21</chem>	0	0	Unknown	544.702	9.848
PrDPhAc	<chem>c1ccc(C2(c3cccc3)c3cccc 3N(c3cncc(N4c5cccc5C(c5c cccc5)(c5cccc5)c5cccc54)n3)c3cccc32)cc1</chem>	0	0	Unknown	742.926	12.808
2Mi2SB	<chem>Cc1nc2cccc2n1-c1cc(S(=O) (=O)c2ccc(C(C)(C)C)cc2)c(- n2c(C)nc3cccc32)cc1S(=O) (=O)c1ccc(C(C)(C)C)cc1</chem>	0	0	Unknown	730.956	9.242
BFDMAc-NAI	<chem>CC(C)(C)c1ccc(N2C(=O)c3cc cc4c(N5c6cccc6C(C)(C)c6c cc7c(oc8cccc87)c65)ccc(c 34)C2=O)cc1</chem>	1	0	1D-0A	626.756	10.946

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
HPIPXZ	<chem>c1ccc(N(c2ccc(-c3ccc(N(c4cccc4)c4cccc5cccc45)cc3)cc2)c2cccc3cccc23)cc1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1cc(C(=O)c2cccn2)cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1cc(C(=O)c2cccn2)cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1cc(C(=O)c2cccn2)cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1</chem>	2	0	2D-0A	588.754	12.600
3BPy-mDTC	<chem>c1ccc(N(c2ccc(-c3ccc(N(c4cccc4)c4cccc5cccc45)cc3)cc2)c2cccc3cccc23)cc1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1cc(C(=O)c2cccn2)cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1cc(C(=O)c2cccn2)cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1</chem>	2	0	2D-0A	738.032	13.697
2BPy-mDTC	<chem>c1ccc(N(c2ccc(-c3ccc(N(c4cccc4)c4cccc5cccc45)cc3)cc2)c2cccc3cccc23)cc1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1cc(C(=O)c2cccn2)cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1cc(C(=O)c2cccn2)cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1</chem>	2	0	2D-0A	738.032	13.697
BP-mDTC	<chem>c1ccc(N(c2ccc(-c3ccc(N(c4cccc4)c4cccc5cccc45)cc3)cc2)c2cccc3cccc23)cc1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1cc(C(=O)c2cccn2)cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1cc(C(=O)c2cccn2)cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1</chem>	2	0	2D-0A	737.044	14.302
Cz2ICz	<chem>c1ccc2c(c1)c1cccc1n2-c1cc2c(c1)c1cccc3c4cc(-n5c6cccc6c6cccc65)ccc4n2c13CC1(C)c2cccc2N(c2ccc3nc4c5cc(C#N)ccc5c5ccc(C#N)cc5c4nc3c2)c2cccc21c1ccc2c(c1)c1cccc1n2-c1cc2c(c1)c1cccc3c4cccc4n2</chem>	4	0	Multi-D/A	571.683	11.031
Ac-BPCN	<chem>c1ccc2c(c1)c1cccc1n2-c1cc2c(c1)c1cccc3c4cc(-n5c6cccc6c6cccc65)ccc4n2c13CC1(C)c2cccc2N(c2ccc3nc4c5cc(C#N)ccc5c5ccc(C#N)cc5c4nc3c2)c2cccc21c1ccc2c(c1)c1cccc1n2-c1cc2c(c1)c1cccc3c4cccc4n2</chem>	1	6	Multi-D/A	537.626	8.942
5CzICz	<chem>c1ccc2c(c1)c1cccc1n2-c1cc2c(c1)c1cccc3c4cccc4n2</chem>	3	0	Multi-D/A	406.488	7.934
DCPDAPM	<chem>c1ccc2c(c1)c1cccc1n2-c1cc2c(c1)c1cccc3c4cccc4n2</chem>	3	0	Multi-D/A	719.888	13.221
SPFS-PXZ	<chem>c1ccc2c(c1)c1cccc1n2-c1cc2c(c1)c1cccc3c4cccc4n2</chem>	4	0	Multi-D/A	742.856	12.347

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
DPS-PXZ	<chem>O=S(=O)(c1ccc(N2c3ccccc3Occcc32)cc1)c1ccc(N2c3ccccc3Occcc32)cc1</chem>	4	0	Multi-D/A	580.665	9.670
TX-CzPh	<chem>O=c1c2ccccc2sc2ccc(-c3ccc4c(c3)c3ccccc3n4-c3ccccc3)cc12</chem>	1	0	1D-0A	453.566	8.179
bpipH	<chem>BrC1CCCC(-c2nc3c4cccnC4C4NCCCC4C3[nH]2)n1</chem>	0	0	Unknown	376.217	4.484
Tri-PXZ-PCN	<chem>N#Cc1c(-c2ccc(N3c4ccccc4Occcc43)cc2)nc(-c2ccc(N3c4ccccc4Occcc43)cc2)c(C#N)c1-c1ccc(N2c3ccccc3Occcc32)cc1</chem>	6	2	Multi-D/A	900.998	16.552
ACID-BPPZ	<chem>CC1(C)c2ccccc2N(c2ccc3c(c2)c2ccccc2c2nc4c5cccnC5C5NCCCC5C4nc32)c2cc3c4ccccc4n(-c4ccccc4)c3cc21</chem>	2	2	2D-2A	754.897	13.392
C-BPyIA	<chem>c1ccc(-n2c(-c3ccc(-c4c5ccccc5c(-c5ccc(-c6nc7c8cccc9ccc%10cccc(c%10c98)c7n6-c6ccccc6)cc5)c5ccccc45)cc3)nc3c4cccc5ccc6cccc(c6c54)c32)cc1</chem>	0	0	Unknown	963.156	19.134
N-BPyIA	<chem>c1ccc(-c2nc3c4cccc5ccc6ccc(c6c54)c3n2-c2ccc(-c3c4ccccc4c(-c4ccc(-n5c(-c6ccccc6)nc6c7cccc8ccc9cccc(c9c87)c65)cc4)c4ccccc34)cc2)cc1</chem>	0	0	Unknown	963.156	19.134
2-TA-AN	<chem>c1ccc2c(c1)Sc1ccc(-c3c4cccc4cc4ccccc34)cc1S2</chem>	0	0	Unknown	392.548	8.276

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
ANOP-MeFAC	<chem>Cc1ccc2c(c1)C1(c3ccccc3-c3ccccc31)c1cc(C)ccc1N2c1ccc2c3c(cccc13)-c1nc3nonc3nc1-2</chem>	1	0	1D-0A	603.685	9.577
ANOP-DMAC	<chem>CC1(C)c2ccccc2N(c2ccc3c4c(cccc24)-c2nc4nonc4nc2-3)</chem>	1	0	1D-0A	453.505	6.923
4PyCNICz	<chem>c2ccccc21N#Cc1c(-c2ccccc2)nc(-c2ccc2)nc1-c1ccc(-n2c3ccccc3c3ccc4c(c5ccccc5n4-c4ccc4)c32)cc1</chem>	2	2	2D-2A	663.784	11.543
2PyCNBCz	<chem>N#Cc1c(-c2ccccc2)nc(-c2ccc(-n3c4ccccc4c4cc(-c5ccc6c(c5)c5ccccc5n6-c5ccccc5)ccc43)cc2)nc1-c1ccccc1</chem>	2	2	2D-2A	739.882	13.210
CzA-PPO	<chem>Cc1ccc(N(c2ccc(C)cc2)c2c3ccccc3c(-c3ccc(P(=O)(c4ccc4)c4ccc(-n5c6ccccc6c6ccccc65)cc4)cc3)c3ccccc23)cc1</chem>	2	0	2D-0A	814.969	14.483
2PyCNICz	<chem>N#Cc1c(-c2ccccc2)nc(-c2ccc(-n3c4ccccc4c4ccc5c(c6ccc6n5-c5ccccc5)c43)cc2)nc1-c1ccccc1</chem>	2	2	2D-2A	663.784	11.543
TAA-PPO	<chem>Cc1ccc(N(c2ccc(C)cc2)c2c3ccccc3c(-c3ccc(P(=O)(c4ccc4)c4cccc4)cc3)c3ccccc23)cc1</chem>	1	0	1D-0A	649.774	11.386
pSFIAc2	<chem>Cc1ccc2c(c1)c1c3c4c(c5c1n2-c1ccc(C(C)(C)C)cc1C51c2ccccc2-c2ccccc21)c1cc(C)cc1n4-c1ccc(C(C)(C)C)cc1C31c2ccccc2-c2ccccc21</chem>	2	0	2D-0A	873.156	16.446

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
TBN-TPA	<chem>CC(C)(C)c1ccc(N2c3ccc(C(C)(C)C)cc3B3c4cc(C(C)(C)C)ccc4N(c4ccc(C(C)(C)C)cc4)c4cc(-n5c6ccc(C(C)(C)C)cc6c6cc(C(C)(C)C)ccc65)cc2c43)cc1</chem>	3	0	Multi-D/A	922.167	16.651
CzBPDCb	<chem>c1cc(-c2cccc(-n3c4cccc4c4ncccc43)c2)cc(-n2c3cccc3c3cccc32)c1</chem>	1	0	1D-0A	485.590	8.943
[2,1-b]IF	<chem>C1=CC2(C=C3C1=CC1=C3C3(C4=c5cccc5=CC4=C1)c1cccc1Sc1cccc13)c1cccc1Sc1cccc12</chem>	0	0	Unknown	618.826	9.206
cyclohexanone	<chem>BrCCCCCCC1(CCCCCCBr)c2cc(Br)ccc2-c2ccc(Br)cc21</chem>	0	0	Unknown	650.131	10.169
thio-ether	<chem>C1=CC2(C=C3C1=CC1=C3C3(C4=c5cccc5=CC4=C1)c1cccc1Sc1cccc13)c1cccc1Sc1cccc12</chem>	0	0	Unknown	618.826	9.206
ICz-2	<chem>c1ccc(-c2cccnc2-c2cc(-c3c[nH]nn3)cc(-c3cccn3)n2)c1</chem>	0	0	Unknown	376.423	4.658
DDCzTrz	<chem>c1ccc(-c2nc(-c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)nc(-c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)n2)cc1</chem>	4	1	Multi-D/A	970.152	17.261
B-O-dpa	<chem>CC(C)(C)c1cc2c3c(c1)N(c1cccc1)c1cccc1B3c1cccc10</chem>	1	0	1D-0A	401.318	5.389
B-dpa-SpiroAC	<chem>CC(C)(C)c1cc2c3c(c1)N1c4cccc4C4(c5cccc5-c5cccc54)c4cccc(c41)B3c1cccc1N2c1cccc1</chem>	2	0	2D-0A	638.623	9.743

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
DCz-OTP	<chem>c1cc(-c2ccccc2-c2cccc(-n3c4ccccc4c4ccccc43)c2)cc(-n2c3ccccc3c3ccccc32)c1CC(C)(C)c1cc2c3c(c1)-n1c4cccc4c4cccc(c41)B3c1cccc</chem>	2	0	2D-0A	560.700	11.215
B-dpa-Cz	<chem>c1N2c1cccc1CC(C)(C)c1ccc2c(c1)c1cc(C)(C)(C)ccc1n2-c1ccc(N2c3ccccc3C3(c4ccccc4-c4cc5c(cc43)C3(c4ccccc4-5)c4cccc4N(c4ccc(-c5nc(-c6ccccc6)nc(C6CCCCC6)n5)cc4)c4cccc43)c3ccccc32)cc1O=C(c1cccc1)c1ccc2nc3c4c</chem>	2	0	2D-0A	474.416	6.694
c-D2T	<chem>cc(N5c6ccccc6Oc6ccccc65)c4c4cc(N5c6ccccc6Oc6ccccc65)ccc4c3nc2c1N#Cc1ccc(-n2c(-c3ccc(-n4c5ccccc5c5ccccc54)cc3)nc3c4ccccc4c4ccccc4c32)cc1N#Cc1ccc(-n2c(-c3ccc(-c4c</chem>	3	1	Multi-D/A	1251.635	23.238
DPXZ-DPPM	<chem>cc(-n5c6ccccc6c6ccccc65)c4)cc3)nc3c4ccccc4c4ccccc4c32)cc1N#Cc1ccc(-c2cc(-c3ccccc3)c(-c3ccc(N(c4ccccc4)c4cccc4)cc3)cc2-c2ccccc2)cc1Cc1nc(C)c(C#N)c(-c2ccc(N3c4ccccc4C(C)(C)c4ccccc43)cc2)c1C#NN#Cc1c(-n2c(-c3ccccc3)nc3ccccc32)cc(-n2c(-c3ccccc3)nc3ccccc32)cc1-n1c(-c2cccc2)nc2cccc21</chem>	4	2	Multi-D/A	746.826	13.471
CBPMCn	<chem>cc(-n5c6ccccc6c6ccccc65)c4)cc3)nc3c4ccccc4c4ccccc4c32)cc1N#Cc1ccc(-n2c(-c3ccc(-c4c</chem>	1	2	A-D-A	560.660	9.968
CPBPMCn	<chem>cc(-n5c6ccccc6c6ccccc65)c4)cc3)nc3c4ccccc4c4ccccc4c32)cc1N#Cc1ccc(-c2cc(-c3ccccc3)c(-c3ccc(N(c4ccccc4)c4cccc4)cc3)cc2-c2ccccc2)cc1Cc1nc(C)c(C#N)c(-c2ccc(N3c4ccccc4C(C)(C)c4ccccc43)cc2)c1C#NN#Cc1c(-n2c(-c3ccccc3)nc3ccccc32)cc(-n2c(-c3ccccc3)nc3ccccc32)cc1-n1c(-c2cccc2)nc2cccc21</chem>	1	2	A-D-A	636.758	11.635
TPB-AC	<chem>cc(-n5c6ccccc6c6ccccc65)c4)cc3)nc3c4ccccc4c4ccccc4c32)cc1N#Cc1ccc(-c2cc(-c3ccccc3)c(-c3ccc(N(c4ccccc4)c4cccc4)cc3)cc2-c2ccccc2)cc1Cc1nc(C)c(C#N)c(-c2ccc(N3c4ccccc4C(C)(C)c4ccccc43)cc2)c1C#NN#Cc1c(-n2c(-c3ccccc3)nc3ccccc32)cc(-n2c(-c3ccccc3)nc3ccccc32)cc1-n1c(-c2cccc2)nc2cccc21</chem>	1	2	A-D-A	574.727	11.696
Me-DMAC	<chem>cc(-n5c6ccccc6c6ccccc65)c4)cc3)nc3c4ccccc4c4ccccc4c32)cc1N#Cc1ccc(-c2cc(-c3ccccc3)c(-c3ccc(N(c4ccccc4)c4cccc4)cc3)cc2-c2ccccc2)cc1Cc1nc(C)c(C#N)c(-c2ccc(N3c4ccccc4C(C)(C)c4ccccc43)cc2)c1C#NN#Cc1c(-n2c(-c3ccccc3)nc3ccccc32)cc(-n2c(-c3ccccc3)nc3ccccc32)cc1-n1c(-c2cccc2)nc2cccc21</chem>	1	2	A-D-A	440.550	7.218
iTPBI-CN	<chem>cc(-n5c6ccccc6c6ccccc65)c4)cc3)nc3c4ccccc4c4ccccc4c32)cc1N#Cc1ccc(-c2cc(-c3ccccc3)c(-c3ccc(N(c4ccccc4)c4cccc4)cc3)cc2-c2ccccc2)cc1Cc1nc(C)c(C#N)c(-c2ccc(N3c4ccccc4C(C)(C)c4ccccc43)cc2)c1C#NN#Cc1c(-n2c(-c3ccccc3)nc3ccccc32)cc(-n2c(-c3ccccc3)nc3ccccc32)cc1-n1c(-c2cccc2)nc2cccc21</chem>	0	2	0D-2A	679.787	10.576

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
TCP	<chem>c1ccc2c(c1)c1ccccc1n2-c1cc(-n2c3ccccc3c3ccccc32)cc(-n2c3ccccc3c3ccccc32)c1Cc1cc(-n2c3ccccc3c3ccccc32)ccc1-c1ccc(-n2c3ccccc3c3ccccc3cc(C(=O)c4ccc(N5c6ccccc6Cc6ccccc65)cc4)ccc32)cc1Cc1cc(-n2c3ccccc3c3ccccc32)ccc1-c1ccc(-n2c3ccccc3cc(C(=O)c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)ccc32)cc1C</chem>	3	0	Multi-D/A	573.699	10.978
CDBP-BP-PXZ	<chem>c1ccc(-c2ccc3c(c2)c2ccccc2n3-c2ccc(-c3cc(-c4ccccc4)nc(-c4ccccc4)n3)cc2)cc1c1ccc(-c2cc(-c3ccc(-n4c5ccccc5c5cc6c(cc54)C4(c5cccc5-c5ccccc54)c4ccccc4-6)cc3)nc(-c3ccccc3)n2)cc1</chem>	4	0	Multi-D/A	797.958	14.971
CDBP-BP-DMAC	<chem>c1ccc(-c2ccc3c(c2)c2ccccc2n3-c2ccc(-c3cc(-c4ccccc4)nc(-c4ccccc4)n3)cc2)cc1c1ccc(-c2cc(-c3ccc(-n4c5ccccc5c5cc6c(cc54)C4(c5cccc5-c5ccccc54)c4ccccc4-6)cc3)nc(-c3ccccc3)n2)cc1</chem>	3	0	Multi-D/A	824.040	15.505
3CzPhpPM	<chem>c1ccc(-c2ccc3c(c2)c2ccccc2n3-c2ccc(-c3cc(-c4ccccc4)nc(-c4ccccc4)n3)cc2)cc1c1ccc(-c2cc(-c3ccc(-n4c5ccccc5c5cc6c(cc54)C4(c5cccc5-c5ccccc54)c4ccccc4-6)cc3)nc(-c3ccccc3)n2)cc1</chem>	1	1	D-A	549.677	10.242
SFI23pPM	<chem>c1ccc(-c2ccc3c(c2)c2ccccc2n3-c2ccc(-c3cc(-c4ccccc4)nc(-c4ccccc4)n3)cc2)cc1c1ccc(-c2cc(-c3ccc(-n4c5ccccc5c5cc6c(cc54)C4(c5cccc5-c5ccccc54)c4ccccc4-6)cc3)nc(-c3ccccc3)n2)cc1</chem>	1	1	D-A	711.868	12.918
3PCzPFP	<chem>c1ccc(-c2ccc3c(c2)c2ccccc2n3-c2ccc(-c3cc(-c4ccccc4)nc(-c4ccccc4)n3)cc2)cc1c1ccc(-c2cc(-c3ccc(-n4c5ccccc5c5cc6c(cc54)C4(c5cccc5-c5ccccc54)c4ccccc4-6)cc3)nc(-c3ccccc3)n2)cc1</chem>	1	0	1D-0A	411.464	7.140
SFI34pPM	<chem>c1ccc(-c2ccc3c(c2)c2ccccc2n3-c2ccc(-c3cc(-c4ccccc4)nc(-c4ccccc4)n3)cc2)cc1c1ccc(-c2cc(-c3ccc(-n4c5ccccc5c5cc6c(cc54)C4(c5cccc5-c5ccccc54)c4ccccc4-6)cc3)nc(-c3ccccc3)n2)cc1</chem>	1	1	D-A	711.868	12.918
SBA-2DPS	<chem>c1ccc(-c2ccc3c(c2)c2ccccc2n3-c2ccc(-c3cc(-c4ccccc4)nc(-c4ccccc4)n3)cc2)cc1c1ccc(-c2cc(-c3ccc(-n4c5ccccc5c5cc6c(cc54)C4(c5cccc5-c5ccccc54)c4ccccc4-6)cc3)nc(-c3ccccc3)n2)cc1</chem>	2	0	2D-0A	778.955	11.301
2DMAC-BP-F	<chem>c1ccc(-c2ccc3c(c2)c2ccccc2n3-c2ccc(-c3cc(-c4ccccc4)nc(-c4ccccc4)n3)cc2)cc1c1ccc(-c2cc(-c3ccc(-n4c5ccccc5c5cc6c(cc54)C4(c5cccc5-c5ccccc54)c4ccccc4-6)cc3)nc(-c3ccccc3)n2)cc1</chem>	2	2	2D-2A	712.872	13.447

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
DCA	<chem>N#Cc1cc(C#N)cc(-c2c3ccccc3c(-c3cc(C#N)cc(C#N)c3)c3ccccc23)c1</chem>	0	8	Multi-D/A	430.470	6.814
DPFPhe	<chem>CCCCCN1C2=CCC(=CC3c4cc(-c5ccccc5)ccc4-c4ccc(-c5ccccc5)cc43)C=C20c2ccccc21</chem>	0	0	Unknown	597.802	11.710
DFPFPhe	<chem>CCCCCN1C2=CCC(=CC3c4cc(-c5ccc(F)cc5)ccc4-c4ccc(-c5ccc(F)cc5)cc43)C=C20c2ccccc21</chem>	0	0	Unknown	633.782	11.988
3ai	<chem>CCCCc1cc2c(s1)-c1sc(CCCC)cc1P2(=O)c1ccc(-n2c3ccccc3c3ccccc32)cc1</chem>	1	0	1D-0A	565.744	9.212
DNFPh	<chem>CCCCCN1C2=CCC(=CC3c4cc(-c5cccc6ccccc56)ccc4-c4ccc(-c5cccc6ccccc56)cc43)C=C2Sc2ccccc21</chem>	0	0	Unknown	713.990	14.730
pCzAnBzt	<chem>N#Cc1ccc(-c2c3ccccc3c(-c3ccc(-n4c5ccccc5c5ccccc54)cc3)c3ccccc23)cc1</chem>	1	2	A-D-A	520.635	10.296
DPE-DPXZ	<chem>C(#Cc1ccc(N2c3ccccc30c3ccccc32)cc1)c1ccc(N2c3ccccc30c3ccccc32)cc1</chem>	4	0	Multi-D/A	540.622	10.237
3CzAnBzt	<chem>N#Cc1ccc(-c2c3ccccc3c(-c3ccc4c(c3)c3ccccc3n4-c3ccc(cc3)c3ccccc23)cc1</chem>	1	2	A-D-A	520.635	10.296
TABD-COOH	<chem>O=C(0)c1ccc(/C(=C\C=C(\c2ccccc2)c2ccc(C(=O)O)cc2)c2ccccc2)cc1</chem>	0	0	Unknown	446.502	6.647
ADN	<chem>c1ccc2cc(-c3c4ccccc4c(-c4ccc5ccccc5c4)c4ccccc34)cc2c1</chem>	0	0	Unknown	430.550	9.633

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
Et3SiH	<chem>FC1=C(F)C(c2c(F)c(F)c(F)c(F)c2F)(c2c(F)c(F)c(F)c(F)c2F)C(c2c(F)c(F)c(F)c(F)c2F)B1c1c(F)c(F)c(F)c(F)c1F</chem>	0	0	Unknown	932.132	11.081
DPE-DDMAc	<chem>CC1(C)c2cccc2N(c2ccc(C#Cc3ccc(N4c5cccc5C(C)(C)c5cccc54)cc3)cc2)c2cccc21</chem>	2	0	2D-0A	592.786	11.305
mCz-TAn-CN	<chem>CC(C)(C)c1ccc2c(-c3cccc(-n4c5cccc5c5cccc54)c3)c3cc(C(C)(C)C)ccc3c(-c3ccc(C#N)cc3)c2c1</chem>	1	2	A-D-A	632.851	12.891
indenoacridine	<chem>c1ccc2[nH]c3ccc4c5cccc5c4c3cc2c1</chem>	1	0	1D-0A	267.331	5.628
MPAc	<chem>Cc1ccc2c(c1)C(c1cccc1)(c1cccc1)c1cc(C)ccc1N2</chem>	0	0	Unknown	361.488	6.743
TBAc	<chem>CC(C)(C)C1=CC=CC(C(C)(C)C)(C2(C3(C(C)(C)C)C=CC=C(C(C)(C)C)C3)c3cccc3N(c3ccc3)c3cccc32)C1</chem>	1	0	1D-0A	637.996	13.686
poCz-SO	<chem>O=P(c1cccc1)(c1cccc1)c1ccc2[nH]c3ccc(P(=O)(c4ccc4)c4cccc4)cc3c2c1</chem>	1	0	1D-0A	567.565	6.600
H4TCPE	<chem>O=C(O)c1ccc(CC(c2ccc(C(=O)O)cc2)(c2ccc(C(=O)O)cc2)c2ccc(C(=O)O)cc2)cc1</chem>	0	0	Unknown	510.498	5.057
CbBPCb	<chem>c1cc(-c2cccc(-n3c4cccc4c4ccnc43)c2)cc(-n2c3cccc3c3ccnc32)c1</chem>	0	0	Unknown	486.578	8.338
TMOE	<chem>COc1ccc(C(=C(c2ccc(OC)cc2)c2ccc(OC)cc2)cc1</chem>	0	0	Unknown	452.550	6.728

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
FAPbCl ₃	<chem>c1ccc(-c2ccc(-c3ccccc3)c(-c3ccccc3)c2-c2ccccc2)cc1</chem>	0	0	Unknown	382.506	8.355
DBT-SADF	<chem>O=S1(=O)c2ccccc2-c2cc(N3c4ccccc4C4(c5ccccc5-c5cccc54)c4ccccc43)ccc21</chem>	1	0	1D-0A	545.663	8.646
mPXTMP	<chem>Cc1c(-c2ccc3c(c2)Oc2ccccc2S3)cccc1-c1ccc2c(c1)Oc1cccc1S2</chem>	0	0	Unknown	488.633	9.843
APDC-tPh	<chem>N#Cc1nc2c(nc1C#N)-c1ccc(-c3cc(-c4ccccc4)cc(-c4cccc4)c3)c3cccc-2c13</chem>	0	2	0D-2A	482.546	8.022
POZDBPHZ	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc2c(ccc3nc4ccc5cc(N6c7cccc7Oc7cccc76)ccc5c4nc32)c1</chem>	4	2	Multi-D/A	642.718	12.240
CPTBF	<chem>CC(C)(C)c1ccc2c(c1)C(c1ccc(-n3c4ccccc4c4ccccc43)cc1)(c1ccc(-n3c4ccccc4c4cccc43)cc1)c1cc(C(C)(C)C)cc1-2</chem>	2	0	2D-0A	761.025	14.839
trimesitylboron	<chem>Cc1cc(C)c(B(c2c(C)cc(C)cc2C)c2c(C)cc(C)cc2C)c(C)c1</chem>	0	0	Unknown	368.373	4.979
CMP	<chem>c1ccc(C(c2ccccc2)(c2ccccc2)c2ccccc2)cc1</chem>	0	0	Unknown	320.435	6.069
CH ₃ OH-CHCl ₃	<chem>O=P(c1ccccc1)(c1ccccc1)c1cccc(-c2ccc3ccc4cccc5ccc2c3c45)c1</chem>	0	0	Unknown	478.531	7.890
s-triazine	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc([Si](c2ccccc2)(c2ccccc2)c2ccc(B(O)O)cc2)cc1</chem>	1	0	1D-0A	657.739	6.436

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
[BMIM][PF6]	<chem>CCCCCCC1(CCCCCC)c2ccccc2-c2ccc(Br)cc21</chem>	0	0	Unknown	413.443	8.656
1SBFN	<chem>c1ccc2c(c1)-c1ccccc1C21c2ccccc2-c2ccc(-c3cccc4cccc34)cc21</chem>	0	0	Unknown	442.561	8.850
2SBFN	<chem>c1ccc2c(c1)-c1ccccc1C21c2ccccc2-c2ccc(-c3ccc4cccc4c3)cc21</chem>	0	0	Unknown	442.561	8.850
F2PA	<chem>CC1(C)c2ccccc2-c2ccc(N(c3cccc3)c3ccc4c(c3)C(C)(C)c3cccc3-4)cc21</chem>	1	0	1D-0A	477.651	9.769
TSBFB	<chem>c1ccc2c(c1)-c1ccccc1C21c2ccccc2-c2ccc(-c3cc(-c4ccc5c(c4)C4(c6cccc6-c6cccc64)c4cccc4-5)cc(-c4ccc5c(c4)C4(c6cccc6-c6cccc64)c4cccc4-5)c3)cc21</chem>	0	0	Unknown	1021.275	19.718
DPPP	<chem>c1ccc(-c2cc(-c3cccc3)cc(-c3cccc3-c3ccc4ccc5c(-c6cccc6-c6cc(-c7cccc7)cc(-c7cccc7)c6)ccc6ccc3c4c65)c2)cc1</chem>	0	0	Unknown	811.040	17.920
TPA-NZP	<chem>c1ccc(-c2c3cccc3c(-c3ccc(N(c4cccc4)c4cccc4)cc3)c3nsnc23)cc1</chem>	1	0	1D-0A	505.646	9.648
mTPA-PPI	<chem>c1ccc(N(c2cccc2)c2ccc(-c3cccc(-c4nc5c6cccc6c6ccc6c5n4-c4cccc4)c3)cc2)c1</chem>	1	0	1D-0A	613.764	12.136
BSBFB	<chem>c1cc(-c2ccc3c(c2)C2(c4cccc4-c4cccc42)c2cccc2-3)cc(-c2ccc3c(c2)C2(c4cccc4-c4cccc42)c2cccc2-3)c1</chem>	0	0	Unknown	706.888	13.708

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
BN-DBC-Ph2	<chem>C1=CB2C(=C3C(=C4C=C(c5ccc</chem> <chem>cc5)C=CN4c4ccc(-c5ccccc5)</chem> <chem>cc43)c3ccccc32)C=C1</chem> <chem>c1ccc(-c2nc(-c3ccccc3)nc(</chem> <chem>-c3cc(-n4c5ccccc5c5ncccc5</chem> <chem>4)c(-n4c5ccccc5c5ncccc54)</chem> <chem>c(-n4c5ccccc5c5ncccc54)c3</chem> <chem>)n2)cc1</chem> <chem>O=c1c2ccccc2oc2ccc(-c3cc(</chem> <chem>-n4c5ccccc5c5ccccc54)cc(-</chem> <chem>n4c5ccccc5c5ccccc54)c3)cc</chem> <chem>12</chem>	0	0	Unknown	481.407	7.869
TCbTrz	<chem>CC1(C)OB(c2cc(-n3c4ccccc4</chem> <chem>c4ccccc43)cc(-n3c4ccccc4c</chem> <chem>4ccccc43)c2)OC1(C)C</chem> <chem>O=S(=O)(c1ccc(N2c3ccccc3C</chem> <chem>=Cc3ccccc32)cc1)c1ccc(N2c</chem> <chem>3ccccc3C=Cc3ccccc32)cc1</chem> <chem>CC1(C)c2ccccc2N(c2c(F)c(N</chem> <chem>3c4ccccc4C(C)(C)c4ccccc43</chem> <chem>)c(C#N)c(N3c4ccccc4C(C)(C</chem> <chem>)c4ccccc43)c2C#N)c2ccccc2</chem> <chem>1</chem>	0	1	0D-1A	807.921	12.349
Xm-mCP	<chem>CC1(C)OB(c2cc(-n3c4ccccc4</chem> <chem>c4ccccc43)cc(-n3c4ccccc4c</chem> <chem>4ccccc43)c2)OC1(C)C</chem> <chem>O=S(=O)(c1ccc(N2c3ccccc3C</chem> <chem>=Cc3ccccc32)cc1)c1ccc(N2c</chem> <chem>3ccccc3C=Cc3ccccc32)cc1</chem> <chem>CC1(C)c2ccccc2N(c2c(F)c(N</chem> <chem>3c4ccccc4C(C)(C)c4ccccc43</chem> <chem>)c(C#N)c(N3c4ccccc4C(C)(C</chem> <chem>)c4ccccc43)c2C#N)c2ccccc2</chem> <chem>1</chem>	2	0	2D-0A	602.693	10.807
Xp-mCP	<chem>CC1(C)OB(c2cc(-n3c4ccccc4</chem> <chem>c4ccccc43)cc(-n3c4ccccc4c</chem> <chem>4ccccc43)c2)OC1(C)C</chem> <chem>O=S(=O)(c1ccc(N2c3ccccc3C</chem> <chem>=Cc3ccccc32)cc1)c1ccc(N2c</chem> <chem>3ccccc3C=Cc3ccccc32)cc1</chem> <chem>CC1(C)c2ccccc2N(c2c(F)c(N</chem> <chem>3c4ccccc4C(C)(C)c4ccccc43</chem> <chem>)c(C#N)c(N3c4ccccc4C(C)(C</chem> <chem>)c4ccccc43)c2C#N)c2ccccc2</chem> <chem>1</chem>	2	0	2D-0A	534.468	8.180
ethylacetate	<chem>CC1(C)OB(c2cc(-n3c4ccccc4</chem> <chem>c4ccccc43)cc(-n3c4ccccc4c</chem> <chem>4ccccc43)c2)OC1(C)C</chem> <chem>O=S(=O)(c1ccc(N2c3ccccc3C</chem> <chem>=Cc3ccccc32)cc1)c1ccc(N2c</chem> <chem>3ccccc3C=Cc3ccccc32)cc1</chem> <chem>CC1(C)c2ccccc2N(c2c(F)c(N</chem> <chem>3c4ccccc4C(C)(C)c4ccccc43</chem> <chem>)c(C#N)c(N3c4ccccc4C(C)(C</chem> <chem>)c4ccccc43)c2C#N)c2ccccc2</chem> <chem>1</chem>	2	0	2D-0A	600.743	10.427
3DMACIPN	<chem>CC1(C)OB(c2cc(-n3c4ccccc4</chem> <chem>c4ccccc43)cc(-n3c4ccccc4c</chem> <chem>4ccccc43)c2)OC1(C)C</chem> <chem>O=S(=O)(c1ccc(N2c3ccccc3C</chem> <chem>=Cc3ccccc32)cc1)c1ccc(N2c</chem> <chem>3ccccc3C=Cc3ccccc32)cc1</chem> <chem>CC1(C)c2ccccc2N(c2c(F)c(N</chem> <chem>3c4ccccc4C(C)(C)c4ccccc43</chem> <chem>)c(C#N)c(N3c4ccccc4C(C)(C</chem> <chem>)c4ccccc43)c2C#N)c2ccccc2</chem> <chem>1</chem>	3	4	Multi-D/A	767.952	13.896
ICZ	<chem>CCCCC1c2[nH]c3ccccc3c2cc</chem> <chem>2[nH]c3ccccc3c12</chem> <chem>C(=Cc1ccc(N(c2ccccc2)c2cc</chem> <chem>ccc2)cc1)c1ccc(-c2ccc(C=C</chem> <chem>c3ccc(N(c4ccccc4)c4ccccc4</chem> <chem>)cc3)cc2)cc1</chem> <chem>C(=C(c1ccccc1)c1ccccc1)c1</chem> <chem>ccc(-c2ccc(C=C(c3ccccc3)c</chem> <chem>3ccccc3)cc2)cc1</chem>	2	0	2D-0A	326.443	6.688
BDAVBi	<chem>CCCCC1c2[nH]c3ccccc3c2cc</chem> <chem>2[nH]c3ccccc3c12</chem> <chem>C(=Cc1ccc(N(c2ccccc2)c2cc</chem> <chem>ccc2)cc1)c1ccc(-c2ccc(C=C</chem> <chem>c3ccc(N(c4ccccc4)c4ccccc4</chem> <chem>)cc3)cc2)cc1</chem> <chem>C(=C(c1ccccc1)c1ccccc1)c1</chem> <chem>ccc(-c2ccc(C=C(c3ccccc3)c</chem> <chem>3ccccc3)cc2)cc1</chem>	2	0	2D-0A	692.906	14.634
DPVBi	<chem>CCCCC1c2[nH]c3ccccc3c2cc</chem> <chem>2[nH]c3ccccc3c12</chem> <chem>C(=Cc1ccc(N(c2ccccc2)c2cc</chem> <chem>ccc2)cc1)c1ccc(-c2ccc(C=C</chem> <chem>c3ccc(N(c4ccccc4)c4ccccc4</chem> <chem>)cc3)cc2)cc1</chem> <chem>C(=C(c1ccccc1)c1ccccc1)c1</chem> <chem>ccc(-c2ccc(C=C(c3ccccc3)c</chem> <chem>3ccccc3)cc2)cc1</chem>	0	0	Unknown	510.680	10.531

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
CBTD2	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-c2ccc(-c3ccc(-n4c5ccc(C(C)(C)C)cc5c5cc(C(C)(C)C)ccc54)cc3)c3nsnc23)cc1</chem>	2	0	2D-0A	843.197	16.409
MeOCzPCN	<chem>COC1ccc2c(c1)c1cc(OC)ccc1n2-c1ncnc(-n2c3ccc(OC)cc3c3cc(OC)ccc32)c1C#N</chem>	2	2	2D-2A	555.594	6.577
Na-DMAC	<chem>CC1(C)c2cccc2N(c2ccc(-c3c(C#N)c(-c4cccc5cccc45)n c(-c4cccc5cccc45)c3C#N)c c2)c2cccc21</chem>	1	2	A-D-A	664.812	12.241
3Py-DMAC	<chem>CC1(C)c2cccc2N(c2ccc(-c3c(C#N)c(-c4cccn4)nc(-c4ccn4)c3C#N)cc2)c2cccc21</chem>	1	2	A-D-A	566.668	8.725
6tBPA	<chem>CC(C)(C)c1ccc(N(c2ccc(C(C)(C)C)cc2)c2c3ccc(C(C)(C)C)cc3c3ccc(C(C)(C)C)cc3)c3cc(C(C)(C)C)cc23)cc1</chem>	2	0	2D-0A	849.304	18.718
C-O	<chem>COC1ccc2c(c1)c1cccc1n2-c1cc2c(cc1-n1c3cccc3c3ccc cc31)Cc1cccc102</chem>	2	0	2D-0A	542.638	9.586
IDFL-2DPA	<chem>CC1(C)c2cccc2-c2c1cc1c3cc(N(c4cccc4)c4cccc4)ccc3n3c4ccc(N(c5cccc5)c5ccc cc5)cc4c2c13</chem>	4	0	Multi-D/A	691.878	14.083
ACRF	<chem>N#Cc1ccc2c(c1)C1(c3cc(C#N)ccc3-2)c2cccc2N(c2cccc c2)c2cccc21</chem>	1	4	Multi-D/A	457.536	7.576
TPSi-F	<chem>c1ccc(C2(c3ccc([Si](c4ccc cc4)(c4cccc4)c4cccc4)cc3)c3cccc3-c3cccc32)cc1</chem>	0	0	Unknown	576.815	7.427

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
PAB	<chem>c1ccc(N(c2ccccc2)c2cc3c4c(c2)N(c2ccccc2)c2ccccc2B4c2ccccc2N3c2ccccc2)cc1CC(C)(C)c1ccc(N2c3ccc(C(C)(C)C)cc3B3c4cc(C(C)(C)C)ccc4N(c4ccc(C(C)(C)C)cc4)c4cc(N(c5ccccc5)c5ccccc5)cc2c43)cc1</chem>	3	0	Multi-D/A	587.535	9.239
2tPAB	<chem>c1ccc(N(c2ccccc2)c2ccc(-c3ccc(-c4ccc(N(c5ccccc5)c5ccccc5)cc4)c4nc5c(nc34)c3sccc3c3ccsc35)cc2)cc1</chem>	3	0	Multi-D/A	811.967	14.429
pDTPA-DTPZ	<chem>c1ccc(N(c2ccccc2)c2ccc(-c3ccc(-c4ccc(N(c5ccccc5)c5ccccc5)cc4)c4nc5c(nc34)c3sccc3c3ccsc35)cc2)cc1</chem>	2	2	2D-2A	779.006	15.486
oDTPA-DTPZ	<chem>c1ccc(N(c2ccccc2)c2ccc(-c3ccc4nc5c(nc4cc3-c3ccc(N(c4ccccc4)c4ccccc4)cc3)c3sc</chem>	2	2	2D-2A	779.006	15.486
oTPAPO-DTPZ	<chem>cc3c3ccsc35)cc2)cc1O=P(c1ccccc1)(c1ccccc1)c1cc2nc3c(nc2cc1-c1ccc(N(c2ccccc2)c2ccccc2)cc1)c1scc</chem>	1	2	A-D-A	735.874	11.989
BPZPP	<chem>c1c1ccsc13c1ccc(-c2cc3nc4c5ccc(N6c7ccccc70c7ccccc76)nc5c5nc(N6c7ccccc70c7ccccc76)ccc5</chem>	2	2	2D-2A	796.890	14.364
BIPOC _z	<chem>c4nc3cc2-c2ccccc2)cc1c1ccc(-c2nc(-c3ccc(OC(=O)Cc4ccccc5c4[nH]c4ccccc45)cc3)n(-c3ccccc3)c2-c2ccccc2)</chem>	1	0	1D-0A	609.773	10.909
tDPAC-BN	<chem>cc1CC(C)(C)c1ccc(N2c3ccc(C(C)(C)C)cc3B3c4cc(C(C)(C)C)ccc4N(c4ccc(C(C)(C)C)cc4)c4cc(N5c6ccccc6C(c6ccccc6)(c6ccccc6)c6ccccc65)cc2c43)cc1</chem>	3	0	Multi-D/A	976.174	17.125

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
tDMAC-BN	<chem>CC(C)(C)c1ccc(N(c2ccc(C(C)(C)C)cc2)c2cc3c4c(c2)N2c5cccc5C(C)(C)c5cccc(c52)B4c2cc(C(C)(C)C)ccc2N3c2ccc(C(C)(C)C)cc2)cc1c1ccc(-n2c3cccc3c3c2ccc2c4cccc4n(-c4ccc(-c5nnc(-c6ccc(-n7c8cccc8c8ccc9c(c%10cccc%10n9-c9cccc9)c87)cc6)o5)cc4)c23)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4)cc3C(=O)c3ccc(-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4-c4cccc4)cc3C(=O)c3ccc(-c4cccc4-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3cccc3C(=O)c3cccc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1c1ccc(N(c2cccc2)c2ccc(-c3nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)n3)cc2)cc1Nc1ccc2c(c1)Oc1cc(N)cc3c1B2c1ccc(Nc2cccc4cc5ccc6cc7cc8cccc8cc7cc6c5cc24)cc103Cc1ccc(C)c(N(c2ccc3ccc4c(N(c5cc(C)ccc5C)c5cccc6c5oc5cccc56)ccc5ccc2c3c54)c2cccc3c2oc2cccc23)c1</chem>	3	0	Multi-D/A	852.032	15.068
2ICz-OXD	<chem>CC(C)(C)c1ccc(N(c2ccc(C(C)(C)C)cc2)c2cc3c4c(c2)N2c5cccc5C(C)(C)c5cccc(c52)B4c2cc(C(C)(C)C)ccc2N3c2ccc(C(C)(C)C)cc2)cc1c1ccc(-n2c3cccc3c3c2ccc2c4cccc4n(-c4ccc(-c5nnc(-c6ccc(-n7c8cccc8c8ccc9c(c%10cccc%10n9-c9cccc9)c87)cc6)o5)cc4)c23)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4)cc3C(=O)c3ccc(-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4-c4cccc4)cc3C(=O)c3ccc(-c4cccc4-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3cccc3C(=O)c3cccc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1c1ccc(N(c2cccc2)c2ccc(-c3nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)n3)cc2)cc1Nc1ccc2c(c1)Oc1cc(N)cc3c1B2c1ccc(Nc2cccc4cc5ccc6cc7cc8cccc8cc7cc6c5cc24)cc103Cc1ccc(C)c(N(c2ccc3ccc4c(N(c5cc(C)ccc5C)c5cccc6c5oc5cccc56)ccc5ccc2c3c54)c2cccc3c2oc2cccc23)c1</chem>	4	0	Multi-D/A	883.027	15.792
SIR-B	<chem>CC(C)(C)c1ccc(N(c2ccc(C(C)(C)C)cc2)c2cc3c4c(c2)N2c5cccc5C(C)(C)c5cccc(c52)B4c2cc(C(C)(C)C)ccc2N3c2ccc(C(C)(C)C)cc2)cc1c1ccc(-n2c3cccc3c3c2ccc2c4cccc4n(-c4ccc(-c5nnc(-c6ccc(-n7c8cccc8c8ccc9c(c%10cccc%10n9-c9cccc9)c87)cc6)o5)cc4)c23)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4)cc3C(=O)c3ccc(-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4-c4cccc4)cc3C(=O)c3ccc(-c4cccc4-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3cccc3C(=O)c3cccc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1c1ccc(N(c2cccc2)c2ccc(-c3nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)n3)cc2)cc1Nc1ccc2c(c1)Oc1cc(N)cc3c1B2c1ccc(Nc2cccc4cc5ccc6cc7cc8cccc8cc7cc6c5cc24)cc103Cc1ccc(C)c(N(c2ccc3ccc4c(N(c5cc(C)ccc5C)c5cccc6c5oc5cccc56)ccc5ccc2c3c54)c2cccc3c2oc2cccc23)c1</chem>	1	0	1D-0A	671.884	12.645
SIR-C	<chem>CC(C)(C)c1ccc(N(c2ccc(C(C)(C)C)cc2)c2cc3c4c(c2)N2c5cccc5C(C)(C)c5cccc(c52)B4c2cc(C(C)(C)C)ccc2N3c2ccc(C(C)(C)C)cc2)cc1c1ccc(-n2c3cccc3c3c2ccc2c4cccc4n(-c4ccc(-c5nnc(-c6ccc(-n7c8cccc8c8ccc9c(c%10cccc%10n9-c9cccc9)c87)cc6)o5)cc4)c23)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4)cc3C(=O)c3ccc(-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4-c4cccc4)cc3C(=O)c3ccc(-c4cccc4-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3cccc3C(=O)c3cccc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1c1ccc(N(c2cccc2)c2ccc(-c3nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)n3)cc2)cc1Nc1ccc2c(c1)Oc1cc(N)cc3c1B2c1ccc(Nc2cccc4cc5ccc6cc7cc8cccc8cc7cc6c5cc24)cc103Cc1ccc(C)c(N(c2ccc3ccc4c(N(c5cc(C)ccc5C)c5cccc6c5oc5cccc56)ccc5ccc2c3c54)c2cccc3c2oc2cccc23)c1</chem>	1	0	1D-0A	824.080	15.979
SIR-A	<chem>CC(C)(C)c1ccc(N(c2ccc(C(C)(C)C)cc2)c2cc3c4c(c2)N2c5cccc5C(C)(C)c5cccc(c52)B4c2cc(C(C)(C)C)ccc2N3c2ccc(C(C)(C)C)cc2)cc1c1ccc(-n2c3cccc3c3c2ccc2c4cccc4n(-c4ccc(-c5nnc(-c6ccc(-n7c8cccc8c8ccc9c(c%10cccc%10n9-c9cccc9)c87)cc6)o5)cc4)c23)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4)cc3C(=O)c3ccc(-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4-c4cccc4)cc3C(=O)c3ccc(-c4cccc4-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3cccc3C(=O)c3cccc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1c1ccc(N(c2cccc2)c2ccc(-c3nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)n3)cc2)cc1Nc1ccc2c(c1)Oc1cc(N)cc3c1B2c1ccc(Nc2cccc4cc5ccc6cc7cc8cccc8cc7cc6c5cc24)cc103Cc1ccc(C)c(N(c2ccc3ccc4c(N(c5cc(C)ccc5C)c5cccc6c5oc5cccc56)ccc5ccc2c3c54)c2cccc3c2oc2cccc23)c1</chem>	1	0	1D-0A	519.688	9.311
TRZ3DPA	<chem>CC(C)(C)c1ccc(N(c2ccc(C(C)(C)C)cc2)c2cc3c4c(c2)N2c5cccc5C(C)(C)c5cccc(c52)B4c2cc(C(C)(C)C)ccc2N3c2ccc(C(C)(C)C)cc2)cc1c1ccc(-n2c3cccc3c3c2ccc2c4cccc4n(-c4ccc(-c5nnc(-c6ccc(-n7c8cccc8c8ccc9c(c%10cccc%10n9-c9cccc9)c87)cc6)o5)cc4)c23)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4)cc3C(=O)c3ccc(-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4-c4cccc4)cc3C(=O)c3ccc(-c4cccc4-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3cccc3C(=O)c3cccc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1c1ccc(N(c2cccc2)c2ccc(-c3nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)n3)cc2)cc1Nc1ccc2c(c1)Oc1cc(N)cc3c1B2c1ccc(Nc2cccc4cc5ccc6cc7cc8cccc8cc7cc6c5cc24)cc103Cc1ccc(C)c(N(c2ccc3ccc4c(N(c5cc(C)ccc5C)c5cccc6c5oc5cccc56)ccc5ccc2c3c54)c2cccc3c2oc2cccc23)c1</chem>	3	1	Multi-D/A	811.005	15.282
BO3N	<chem>CC(C)(C)c1ccc(N(c2ccc(C(C)(C)C)cc2)c2cc3c4c(c2)N2c5cccc5C(C)(C)c5cccc(c52)B4c2cc(C(C)(C)C)ccc2N3c2ccc(C(C)(C)C)cc2)cc1c1ccc(-n2c3cccc3c3c2ccc2c4cccc4n(-c4ccc(-c5nnc(-c6ccc(-n7c8cccc8c8ccc9c(c%10cccc%10n9-c9cccc9)c87)cc6)o5)cc4)c23)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4)cc3C(=O)c3ccc(-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4-c4cccc4)cc3C(=O)c3ccc(-c4cccc4-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3cccc3C(=O)c3cccc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1c1ccc(N(c2cccc2)c2ccc(-c3nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)n3)cc2)cc1Nc1ccc2c(c1)Oc1cc(N)cc3c1B2c1ccc(Nc2cccc4cc5ccc6cc7cc8cccc8cc7cc6c5cc24)cc103Cc1ccc(C)c(N(c2ccc3ccc4c(N(c5cc(C)ccc5C)c5cccc6c5oc5cccc56)ccc5ccc2c3c54)c2cccc3c2oc2cccc23)c1</chem>	0	0	Unknown	641.539	9.088
BPPyA	<chem>CC(C)(C)c1ccc(N(c2ccc(C(C)(C)C)cc2)c2cc3c4c(c2)N2c5cccc5C(C)(C)c5cccc(c52)B4c2cc(C(C)(C)C)ccc2N3c2ccc(C(C)(C)C)cc2)cc1c1ccc(-n2c3cccc3c3c2ccc2c4cccc4n(-c4ccc(-c5nnc(-c6ccc(-n7c8cccc8c8ccc9c(c%10cccc%10n9-c9cccc9)c87)cc6)o5)cc4)c23)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4)cc3C(=O)c3ccc(-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3ccc(-c4cccc4-c4cccc4)cc3C(=O)c3ccc(-c4cccc4-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1Cc1ccc2c(c1)C1(c3cccc3C(=O)c3cccc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1c1ccc(N(c2cccc2)c2ccc(-c3nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)n3)cc2)cc1Nc1ccc2c(c1)Oc1cc(N)cc3c1B2c1ccc(Nc2cccc4cc5ccc6cc7cc8cccc8cc7cc6c5cc24)cc103Cc1ccc(C)c(N(c2ccc3ccc4c(N(c5cc(C)ccc5C)c5cccc6c5oc5cccc56)ccc5ccc2c3c54)c2cccc3c2oc2cccc23)c1</chem>	2	0	2D-0A	772.948	16.556

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
SiCN2Cz	<chem>N#Cc1cc(-c2nc(-n3c4cccc4c4cccc43)nc(-n3c4cccc4c4cccc43)n2)cc([Si](c2cccc2)(c2cccc2)c2cccc2)c1c1ccc(-c2c3cccc3c(-c3ccc(-n4c5cccc5c5cccc54)cc3)c3cccc23)cc1</chem>	2	3	Multi-D/A	770.972	8.982
PhPC	<chem>C0c1cc(/C=C/c2ccc(-n3cnc4cccc43)cc2)c(OC)cc1/C=C/c1ccc(-n2cnc3cccc32)cc1</chem>	1	0	1D-0A	495.625	10.424
1MOBI	<chem>BrC1cc(/C=C/c2ccc(-n3cnc4cccc43)cc2)c(Br)cc1/C=C/c1ccc(-n2cnc3cccc32)cc1</chem>	0	0	Unknown	574.684	8.722
3BrBI	<chem>CC1(C)c2cccc2-c2cccc(-c3ccc4sc5ccc(-c6cccc7c6C(C)(C)c6cccc6-7)cc5c(=O)c4c3)c21</chem>	0	0	Unknown	672.424	10.230
DMBFTX	<chem>N#Cc1ccc(-c2c3cccc3c(-c3ccc4c(c3)C3(c5cccc5-c5ccc53)c3cccc3-4)c3cccc23)cc1</chem>	0	2	0D-2A	593.729	11.542
DMF-AnCN	<chem>CC1(C)c2cccc2-c2ccc(-c3c4cccc4c(-c4ccc(C#N)cc4)c4cccc34)cc21</chem>	0	2	0D-2A	471.603	9.505
DPF-AnCN	<chem>N#Cc1ccc(-c2c3cccc3c(-c3ccc4c(c3)C(c3cccc3)(c3ccc3)c3cccc3-4)c3cccc23)cc1</chem>	0	2	0D-2A	595.745	11.562
DPA	<chem>c1ccc(-c2c3cccc3c(-c3ccc3)c3cccc23)cc1</chem>	0	0	Unknown	330.430	7.327

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
TPA-DAP-TPA	<chem>c1ccc(N(c2ccccc2)c2ccc(-c3c4ccccc4c(-c4ccc5ccc6c(-c7c8ccccc8c(-c8ccc(N(c9ccccc9)c9ccccc9)cc8)c8ccccc78)ccc7ccc4c5c76)c4ccccc34)cc2)cc1</chem>	2	0	2D-0A	1041.310	22.804
AcDbp	<chem>CC1(C)c2ccccc2N(c2ccc3c4ccc(N5c6ccccc6C(C)(C)c6ccc65)cc4c4nc5ccccc5nc4c3c2)c2ccccc21</chem>	2	2	2D-2A	694.882	13.308
4-Ac	<chem>CC(C)(C)c1ccc2c(c1)C(C)(C)c1cc(C(C)(C)C)ccc1N2c1ccc(C(F)(F)F)cc1</chem>	1	0	1D-0A	465.603	9.409
czpzpy	<chem>c1cc(-c2cccc3c2[nH]c2cccc23)nc(-n2cccn2)c1</chem>	1	0	1D-0A	310.360	4.569
NyDPO	<chem>c1ccc2c(c1)Oc1ccccc1N2c1ccc(-c2ccc3nc(-c4ccc(N5c6ccccc6Oc6ccccc65)cc4)ccc3n2)cc1</chem>	4	0	Multi-D/A	644.734	12.115
1MPA	<chem>Cc1nc(N2c3ccccc3C(c3ccccc3)(c3ccccc3)c3ccccc32)cc(N2c3ccccc3C(c3ccccc3)(c3ccccc3)c3ccccc32)n1</chem>	0	1	0D-1A	756.953	13.117
LEt	<chem>CCc1ccccc1P(c1ccccc1CC)c1ccccc1P(c1ccccc1CC)c1ccccc1CC</chem>	0	0	Unknown	558.686	7.453
DC-TC	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(C(=O)C(=O)c2ccc(-n3c4ccc(C(C)(C)C)cc4c4cc(C(C)(C)C)ccc43)cc2)cc1</chem>	2	0	2D-0A	765.054	14.136
DC-ACR	<chem>CC1(C)c2ccccc2N(c2ccc(C(=O)C(=O)c3ccc(N4c5ccccc5C(C)(C)c5ccccc54)cc3)cc2)c2ccccc21</chem>	2	0	2D-0A	624.784	10.970

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
1G-F1	<chem>c1ccc(P(c2cccc2)c2cscc2P(c2cccc2)c2cccc2)cc1</chem>	0	0	Unknown	452.499	5.265
P-Cu-C	<chem>c1ccc(P(c2cccc2)c2cccc2-c2cccc2P(c2cccc2)c2cccc2)cc1</chem>	0	0	Unknown	522.568	6.870
BPCP-2CPC	<chem>N#Cc1cc(-n2c3cccc3c3cc(-c4ccc(C5(c6ccc(-n7c8cccc8c8cccc87)cc6)CCCC5)cc4)ccc32)c(-n2c3cccc3c3cc(-c4ccc(C5(c6ccc(-n7c8cccc8c8cccc87)cc6)CCCC5)cc4)ccc32)cc1C#N</chem>	4	4	Multi-D/A	1 257.598	23.653
2H-Qz	<chem>c1ccc(-c2cc(-c3ccccn3)cc(-c3nc(-c4ccc(N5c6cccc60c6cccc65)cc4)c4cccc4n3)c2)nc1</chem>	2	1	D-A-D	617.712	10.663
PPINCN-Cz	<chem>N#Cc1ccc(-n2c(-c3ccc4ccc5cccc6ccc3c4c56)nc3c4ccc(-c5ccc(-n6c7cccc7c7cccc76)cc5)cc4c4cc(-c5ccc(-n6c7cccc7c7cccc76)cc5)ccc4c32)c2cccc12</chem>	2	2	2D-2A	1 052.253	20.449
PPINCN	<chem>N#Cc1ccc(-n2c(-c3ccc4ccc5cccc6ccc3c4c56)nc3c4cccc4c4cccc4c32)c2cccc12</chem>	0	2	0D-2A	569.667	10.921
BDAPM	<chem>O=C(c1ccc(N2c3cccc3[Si](c3cccc3)(c3cccc3)c3cccc32)cc1)c1ccc(N2c3cccc3[Si](c3cccc3)(c3cccc3)c3cccc32)cc1</chem>	2	0	2D-0A	877.208	9.239
12AcCz-PM	<chem>CC1(C)c2cccc2N(c2ccc(-c3cc(-c4cccc4)nc(-c4cccc4)n3)cc2)c2c1ccc1c2cccc2n1-c1cccc1</chem>	2	1	D-A-D	680.855	12.684

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
23AcCz-PM	<chem>CC1(C)c2cccc2N(c2ccc(-c3cc(-c4cccc4)nc(-c4cccc4)n3)cc2)c2cc3c(cc21)c1ccc</chem>	2	1	D-A-D	680.855	12.684
CDDPI	<chem>cc1n3-c1cccc1c1ccc2c(-n3c4cccc4c4cccc43)ccc(-c3nc4c5cccc5c5cccc5c4n3-c3ccc4c(c3)OCCO4)c2c1</chem>	1	0	1D-0A	643.746	11.020
DDBP	<chem>c1ccc(-c2nc(-c3ccc(-c4ccc(N(c5cccc5)c5cccc5)cc4)c4cccc34)n(-c3ccc4c(c3)OCCO4)c2-c2cccc2)cc1</chem>	1	0	1D-0A	723.876	12.934
CCDD	<chem>O=C(c1cccc1)c1ccc(-n2c3ccc(N(c4cccc4)c4cccc4)cc3cc(N(c4cccc4)c4cccc4)ccc32)cc1</chem>	3	0	Multi-D/A	681.839	12.954
DPNC	<chem>COc1ccc(N(c2ccc(OC)cc2)c2ccc3c(c2)c2cc(N(c4ccc(OC)cc4)c4ccc(OC)cc4)ccc2n3-c2cccc3cccc23)cc1</chem>	3	0	Multi-D/A	747.895	12.911
LiPr	<chem>CC(C)c1cccc1P(c1cccc1C(C)C)c1cccc1P(c1cccc1C(C)C)c1cccc1C(C)C</chem>	0	0	Unknown	614.794	9.697
Spiro-CN	<chem>Cc1ccc(N(c2ccc(C)cc2)c2ccc3c(c2)C2(c4cc(C#N)ccc4-c4ccc(C#N)cc42)c2cc(N(c4ccc(C)cc4)c4ccc(C)cc4)ccc2-3)cc1</chem>	2	4	Multi-D/A	756.953	13.947
TMDBP	<chem>CC1(C)c2cccc2Nc2c1ccc1c2C(C)(C)c2cccc2N1</chem>	0	0	Unknown	340.470	6.452
ACRSA	<chem>O=C1c2cccc2C2(c3cccc31)c1cccc1N(c1cccc1)c1cccc12</chem>	1	0	1D-0A	435.526	7.397

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
CAPI	<chem>c1ccc2c(-n3c(-c4ccc(-c5c6cccc6c(-n6c7cccc7c7cccc76)c6cccc56)cc4)nc4c5ccc5c5cccc5c43)cccc2c1</chem>	1	0	1D-0A	761.928	15.223
PhCzSPOTz	<chem>O=P(c1cccc1)(c1cccc1)c1nc(-c2cccc2)nc(-n2c3cccc3c3cccc32)n1</chem>	1	1	D-A	522.548	6.275
CCAPI	<chem>N#Cc1ccc(-n2c(-c3ccc(-c4c5cccc5c(-n5c6cccc6c6cccc65)c5cccc45)cc3)nc3c4cccc4c4cccc4c32)c2cccc1</chem>	1	2	A-D-A	786.938	15.094
2CzCbPy	<chem>c1cc(-n2c3cccc3c3cccc32)nc(-n2c3ccc(-n4c5cccc5c5cccc54)cc3c3ncccc32)c1</chem>	2	0	2D-0A	575.675	9.768
CzCbPy	<chem>c1ccc(-n2c3ccc(-n4c5cccc5c5cccc54)cc3c3ncccc32)n</chem>	1	0	1D-0A	410.480	6.671
DCE	<chem>c1CC(C)CCC[C@H](C)[C@H]1CC[C@H]2[C@H]3CC=C4C[C@H](O)CC[C@]4(C)[C@H]3CC[C@]12</chem>	0	0	Unknown	386.664	7.389
PICFOCz	<chem>FC(F)(F)c1cccc(-n2c(-c3cc(OCn4c5cccc5c5cccc54)cc3)nc3c4cccc4c4cccc4c32)c1</chem>	1	0	1D-0A	675.754	11.985
dpmt	<chem>Cc1sc(C)c(P(c2cccc2)c2cccc2)c1P(c1cccc1)c1cccc1</chem>	0	0	Unknown	480.553	5.881
BICFOCz	<chem>FC(F)(F)c1cccc(-n2c(-c3cc(OCn4c5cccc5c5cccc54)cc3)nc(-c3cccc3)c2-c2cccc2)c1</chem>	1	0	1D-0A	677.770	11.859

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
19a	<chem>c1ccc2cc(-c3ccc4cc5cc(-c6ccc7ccccc7c6)ccc5cc4c3)cc</chem>	0	0	Unknown	430.550	9.633
BTPBFCz	<chem>c2c1c1ccc(-c2nc(-c3ccccc3-n3c4ccccc4c4c5oc6ccccc6c5ccc43)c3sc4ccccc4c3n2)cc1</chem>	1	1	D-A	593.711	11.175
Ph-OBNA	<chem>c1ccc(-n2c3ccccc3c3c4c5c(cc32)0c2ccccc2B5c2ccccc204)cc1</chem>	1	0	1D-0A	435.291	5.511
HPVAnt	<chem>CCCCCc1ccc(C=Cc2ccc3cc4ccccc4cc3c2)cc1</chem>	0	0	Unknown	364.532	8.286
BTPBFCz-D3	<chem>c1ccc(-c2nc(-c3ccc(-n4c5ccccc5c5ccc6c(c7ccccc7n6-c6ccccc6)c54)cc3-n3c4ccccc4c4c5oc6ccccc6c5ccc43)c3sc4ccccc4c3n2)cc1</chem>	3	1	Multi-D/A	924.101	17.369
ICBNTrz4	<chem>CC1(C)c2ccccc2-c2cc3c4cccc4n(-c4ccc(-c5nc(-c6cccc6)nc(-c6ccccc6)n5)cc4C#N)c3cc21</chem>	1	3	Multi-D/A	615.740	10.148
1PCTRz	<chem>c1ccc(-c2nc(-c3ccccc3)nc(-c3ccc(-n4c5ccccc5c5cc(-n6c7ccccc7c7ccccc76)ccc54)c(-c4ccccc4)c3)n2)cc1</chem>	2	1	D-A-D	715.860	12.734
DMACN-B	<chem>CC1cc(C)c(N(c2ccc3c(c2)C(c2ccccc2)(c2ccccc2)c2cccc4c2N3c2ccccc2C4(C)C)c2c(C)cc(C)cc2C)c(C)c1</chem>	2	0	2D-0A	700.970	13.812
CzSO-1CO	<chem>CC1(OCOC(=O)c2ccccc2)CC=Cc2c1c1ccccc1n2-c1ccc(S(=O)(=O)c2ccccc2)cc1</chem>	0	0	Unknown	549.648	6.927
PPO21	<chem>0=P(c1ccccc1)(c1ccccc1)c1ccc(-n2c3ccccc3c3cc(P(=O)(c4ccccc4)c4ccccc4)ccc32)cc1</chem>	1	0	1D-0A	643.663	8.063

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
OXD-7	<chem>CC(C)(C)c1ccc(-c2nnc(-c3ccc(-c4nnc(-c5ccc(C(C)(C)C)cc5)o4)c3)o2)cc1</chem>	0	0	Unknown	478.596	7.716
TCZ1	<chem>CCCCC(CC)Cn1c2ccc(-n3c4ccc4c4cccc43)cc2c2cc(-n3c4cccc4c4cccc43)ccc21</chem>	3	0	Multi-D/A	609.817	12.205
APPT	<chem>c1cc2c3c(cccc3c1)-c1nc3c4cccnc4c4ncccc4c3nc1-2</chem>	0	1	0D-1A	356.388	5.527
ANQDC	<chem>N#Cc1cc2nc3c(nc2cc1C#N)-c1cccc2cccc-3c12</chem>	0	5	Multi-D/A	304.312	4.174
ECDPTT	<chem>CCn1c2cccc2c2cc(C(=C(c3cccc3)c3cccc3)c3ccc(-c4cc5c(c4)S(=O)(=O)c4cccc4S5(=O)=O)cc3)ccc21</chem>	1	0	1D-0A	727.907	10.468
1CM	<chem>CCCCCCCCn1c2cccc2c2cc(Nc3nc(OC)nc(OC)n3)ccc21</chem>	1	1	D-A	433.556	6.101
NPE-AcDPS	<chem>CC(c1cccc1)N1c2cc(N3c4cccc4C(C)(C)c4cccc43)ccc2S(=O)(=O)c2ccc(N3c4cccc4C(C)(C)c4cccc43)cc21</chem>	3	0	Multi-D/A	749.980	12.950
DMAC-Cz	<chem>CC1(C)c2cccc2N(c2ccc3[nH]c4ccc(N5c6cccc6C(C)(C)c6cccc65)cc4c3c2)c2cccc21</chem>	3	0	Multi-D/A	581.763	11.539
VBC-PTZ	<chem>CCN(CC)c1ccc2c(c1)oc(=O)c1c(=O)oc3cc(N4c5cccc5Sc5cccc54)ccc3c12</chem>	2	0	2D-0A	532.621	7.833
t-BuPCz	<chem>CC(C)(C)c1ccc(-c2ccc3[nH]c4ccc(-c5ccc(C(C)(C)C)cc5)cc4c3c2)cc1</chem>	1	0	1D-0A	431.623	9.250

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
4BPy-mDTC	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1cc(C(=O)c2ccncc2)cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1Cc1ccc(N(c2ccc(C)cc2)c2cc(C(C3(c4ccc(N(c5ccc(C)cc5)c5ccc(C)cc5)cc4)CCCC3)cc2)cc1Fc1cc2nc(-c3ccc(N4c5ccccc50c5ccccc54)cc3)c(-c3ccc(N4c5ccccc50c5ccccc54)cc3)nc2cc1FC(#Cc1ccc(N2c3ccccc30c3ccc32)cc1)c1ccc(N2c3ccccc30c3ccccc32)cc1CCCCC0c1ccc2c3ccc(OC(C)C)cc3n(-c3ccnc(-c4ccc(F)c(C#N)c4F)c3)c2c1</chem>	2	0	2D-0A	738.032	13.697
DC-TC	<chem>c(C3(c4ccc(N(c5ccc(C)cc5)c5ccc(C)cc5)cc4)CCCC3)cc2)cc1Fc1cc2nc(-c3ccc(N4c5ccccc50c5ccccc54)cc3)c(-c3ccc(N4c5ccccc50c5ccccc54)cc3)nc2cc1FC(#Cc1ccc(N2c3ccccc30c3ccc32)cc1)c1ccc(N2c3ccccc30c3ccccc32)cc1CCCCC0c1ccc2c3ccc(OC(C)C)cc3n(-c3ccnc(-c4ccc(F)c(C#N)c4F)c3)c2c1</chem>	2	0	2D-0A	626.888	13.110
DFDBQPXZ	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc2[nH]c3ccc(-n4c5ccc(C(C)(C)C)c5c5cc(C(C)(C)C)ccc54)cc3c2c1c1cc(-c2cccc(-c3cccc(-n4c5ccccc5c5ccccc54)c3)n2)cc(-n2c3ccccc3c3ccccc32)c1c1cc(-c2cccc(-n3c4ccccc4c4ccccc43)c2)cc(-n2c3ccccc3c3ccccc32)c1Cc1sc2ccccc2c1C1=C(c2c(C)sc3ccccc23)C(F)(F)C(F)(F)C1(F)F</chem>	4	1	Multi-D/A	680.714	12.393
DPE	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc2[nH]c3ccc(-n4c5ccc(C(C)(C)C)c5c5cc(C(C)(C)C)ccc54)cc3c2c1c1cc(-c2cccc(-c3cccc(-n4c5ccccc5c5ccccc54)c3)n2)cc(-n2c3ccccc3c3ccccc32)c1c1cc(-c2cccc(-n3c4ccccc4c4ccccc43)c2)cc(-n2c3ccccc3c3ccccc32)c1Cc1sc2ccccc2c1C1=C(c2c(C)sc3ccccc23)C(F)(F)C(F)(F)C1(F)F</chem>	4	0	Multi-D/A	540.622	10.237
CBZ	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc2[nH]c3ccc(-n4c5ccc(C(C)(C)C)c5c5cc(C(C)(C)C)ccc54)cc3c2c1c1cc(-c2cccc(-c3cccc(-n4c5ccccc5c5ccccc54)c3)n2)cc(-n2c3ccccc3c3ccccc32)c1c1cc(-c2cccc(-n3c4ccccc4c4ccccc43)c2)cc(-n2c3ccccc3c3ccccc32)c1Cc1sc2ccccc2c1C1=C(c2c(C)sc3ccccc23)C(F)(F)C(F)(F)C1(F)F</chem>	1	2	A-D-A	581.707	9.914
tBuCz2	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc2[nH]c3ccc(-n4c5ccc(C(C)(C)C)c5c5cc(C(C)(C)C)ccc54)cc3c2c1c1cc(-c2cccc(-c3cccc(-n4c5ccccc5c5ccccc54)c3)n2)cc(-n2c3ccccc3c3ccccc32)c1c1cc(-c2cccc(-n3c4ccccc4c4ccccc43)c2)cc(-n2c3ccccc3c3ccccc32)c1Cc1sc2ccccc2c1C1=C(c2c(C)sc3ccccc23)C(F)(F)C(F)(F)C1(F)F</chem>	3	0	Multi-D/A	722.033	14.705
26DCzPPy	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc2[nH]c3ccc(-n4c5ccc(C(C)(C)C)c5c5cc(C(C)(C)C)ccc54)cc3c2c1c1cc(-c2cccc(-c3cccc(-n4c5ccccc5c5ccccc54)c3)n2)cc(-n2c3ccccc3c3ccccc32)c1c1cc(-c2cccc(-n3c4ccccc4c4ccccc43)c2)cc(-n2c3ccccc3c3ccccc32)c1Cc1sc2ccccc2c1C1=C(c2c(C)sc3ccccc23)C(F)(F)C(F)(F)C1(F)F</chem>	2	0	2D-0A	561.688	10.610
mCBP	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc2[nH]c3ccc(-n4c5ccc(C(C)(C)C)c5c5cc(C(C)(C)C)ccc54)cc3c2c1c1cc(-c2cccc(-c3cccc(-n4c5ccccc5c5ccccc54)c3)n2)cc(-n2c3ccccc3c3ccccc32)c1c1cc(-c2cccc(-n3c4ccccc4c4ccccc43)c2)cc(-n2c3ccccc3c3ccccc32)c1Cc1sc2ccccc2c1C1=C(c2c(C)sc3ccccc23)C(F)(F)C(F)(F)C1(F)F</chem>	2	0	2D-0A	484.602	9.548
benzoxazole	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc2[nH]c3ccc(-n4c5ccc(C(C)(C)C)c5c5cc(C(C)(C)C)ccc54)cc3c2c1c1cc(-c2cccc(-c3cccc(-n4c5ccccc5c5ccccc54)c3)n2)cc(-n2c3ccccc3c3ccccc32)c1c1cc(-c2cccc(-n3c4ccccc4c4ccccc43)c2)cc(-n2c3ccccc3c3ccccc32)c1Cc1sc2ccccc2c1C1=C(c2c(C)sc3ccccc23)C(F)(F)C(F)(F)C1(F)F</chem>	0	0	Unknown	468.487	8.563

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
dbG2TAZ	<chem>c1ccc2c(c1)ccc1[nH]c3ccc4cccc4c3c12C=Cc1ccc(C0c2cccc3c2c2ccc</chem>	1	0	1D-0A	267.331	5.628
V-TPA	<chem>cc2n3-c2ccc(N(c3ccc(-n4c5cccc5c5cccc54)cc3)c3ccc(-n4c5cccc5c5cccc54)cc3)cc2)cc1O=C(c1cccc1)c1ccc(-n2c3c</chem>	4	0	Multi-D/A	873.072	16.670
CCDC	<chem>cc(-n4c5cccc5c5cccc54)c3c3cc(-n4c5cccc5c5cccc54)ccc32)cc1O=C(C=C(O)c1ccc(N2c3cccc</chem>	3	0	Multi-D/A	677.807	12.209
PXZPDO	<chem>30c3cccc32)cc1)c1ccc(N2c3cccc30c3cccc32)cc1CC1(C)c2cccc2N(c2ccc(C(=</chem>	4	0	Multi-D/A	586.647	10.619
DMAC-BP	<chem>O)c3ccc(N4c5cccc5C(C)(C)c5cccc54)cc3)cc2)c2cccc21c1ccc2c(c1)-c1cccc1C21c2</chem>	2	0	2D-0A	596.774	11.136
SBTF	<chem>c(ccc3cc(-c4ccc5cccc5c4)ccc23)-c2ccc(-c3cc4cccc4c4cccc34)c3cccc1c23N#CC(C#N)=C(c1cccc(N(c2cc</chem>	0	0	Unknown	718.899	15.130
3-DPM	<chem>ccc2)c2cccc2)c1)c1cccc(N(c2cccc2)c2cccc2)c1N#CC(C#N)=C(c1ccc(N(c2ccc</chem>	2	2	2D-2A	564.692	10.475
4-DPM	<chem>cc2)c2cccc2)cc1)c1ccc(N(c2cccc2)c2cccc2)cc1c1ccc(-c2ccc(-n3c4cccc4c</chem>	2	2	2D-2A	564.692	10.475
2CzPy	<chem>4cccc43)cc2-n2c3cccc3c3cccc32)nc1</chem>	2	0	2D-0A	485.590	8.943

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
2PBCzPy	<chem>c1ccc(-n2c3cccc3c3cc(-c4ccc5c(c4)c4cccc4n5-c4ccc(-c5cccn5)c(-n5c6cccc6c6cc(-c7ccc8c(c7)c7cccc7n8-c7cccc7)ccc65)c4)ccc32)cc1</chem>	4	0	Multi-D/A	968.176	18.471
2BFCzPy	<chem>c1ccc(-c2ccc(-n3c4cccc4c4c5oc6cccc6c5ccc43)cc2-n2c3cccc3c3c4oc5cccc5c4ccc32)nc1</chem>	2	0	2D-0A	665.752	12.742
CNTPA-CZ	<chem>N#Cc1ccc(N(c2cccc2)c2cc(-n3c4cccc4c4cccc43)cc(-n3c4cccc4c4cccc43)c2)cc1</chem>	3	2	Multi-D/A	600.725	11.222
CNTPA-PXZ	<chem>N#Cc1ccc(N(c2cccc2)c2cc(N3c4cccc40c4cccc43)cc(N3c4cccc40c4cccc43)c2)cc1</chem>	5	2	Multi-D/A	632.723	12.179
BMZ-TZ	<chem>c1cc(-c2nc3cccc3[nH]2)cc(-c2nnnc(-c3cccc(-c4nc5cccc5[nH]4)c3)c2-c2cccc(-c3nc4cccc4[nH]3)c2)c1</chem>	0	0	Unknown	657.741	9.502
TPAPm	<chem>N#Cc1c(-c2ccc(N(c3cccc3)c3cccc3)cc2)ncnc1-c1ccc(N(c2cccc2)c2cccc2)cc1</chem>	2	2	2D-2A	591.718	10.622
CzPm	<chem>N#Cc1c(-c2ccc(-n3c4cccc4c4cccc43)cc2)ncnc1-c1ccc(-n2c3cccc3c3cccc32)cc1</chem>	2	2	2D-2A	587.686	9.876
TCPZ	<chem>C1=C(c2cccc(-n3c4cccc4c4cccc43)c2)NN(c2cccc(-n3c4cccc4c4cccc43)c2)N=C1c1cccc(-n2c3cccc3c3cccc32)c1</chem>	3	0	Multi-D/A	806.973	13.748

Continued on next page

Table S1 – Continued from previous page

Molecule Id	SMILES	Total donors	Total acceptors	Architecture	Molecular weight	logp
TrisCz-Trz	<chem>c1ccc(-n2c3ccccc3c3ccccc32)c(-c2nc(-c3ccccc3-n3c4ccccc4c4ccccc43)nc(-c3cccc3-n3c4ccccc4c4ccccc43)n2)c1</chem>	3	1	Multi-D/A	804.957	14.164
PTZ-TAZ	<chem>c1ccc2c(c1)Oc1ccccc1N2c1ccc(-c2nnc(-c3ccc(N4c5ccccc5O4)cc3)o2)cc1CC1(C)c2ccccc2N(c2ccc3c(c2)C(C)(C)c2cc(N4c5ccccc5C(C)(C)c5ccccc54)ccc2S3(=O)=O)c2ccccc21c1ccc([Si](c2ccccc2)(c2ccc2)c2cccc([Si](c3ccccc3)(c3ccccc3)c3ccccc3)c2)cc1</chem>	4	0	Multi-D/A	584.635	10.555
DDMA-TXO2	<chem>c1ccc2c(c1)Oc1ccccc1N2c1ccc(-c2nnc(-c3ccc(N4c5ccccc5O4)cc3)o2)cc1CC1(C)c2ccccc2N(c2ccc3c(c2)C(C)(C)c2cc(N4c5ccccc5C(C)(C)c5ccccc54)ccc2S3(=O)=O)c2ccccc21c1ccc([Si](c2ccccc2)(c2ccc2)c2cccc([Si](c3ccccc3)(c3ccccc3)c3ccccc3)c2)cc1</chem>	2	0	2D-0A	672.894	11.377
UGH3	<chem>c1ccc2c(c1)Oc1ccccc1N2c1ccc(-c2nnc(-c3ccc(N4c5ccccc5O4)cc3)o2)cc1CC1(C)c2ccccc2N(c2ccc3c(c2)C(C)(C)c2cc(N4c5ccccc5C(C)(C)c5ccccc54)ccc2S3(=O)=O)c2ccccc21c1ccc([Si](c2ccccc2)(c2ccc2)c2cccc([Si](c3ccccc3)(c3ccccc3)c3ccccc3)c2)cc1</chem>	0	0	Unknown	594.906	4.441

1

Experimental Validation Data

Table S2: Experimental validation benchmark set (379 molecules). Complete dataset comparing calculated (gas-phase screening) and experimental ΔE_{ST} values across diverse TADF architectures.

Molecule	Arch.	Exp. ΔE_{ST} (eV)	Calc. ΔE_{ST} (eV)	Error (eV)	f	Overlap	Ref.
Diphenylsulfane	0D-1A	0.280	0.094	-0.186	0.001	0.479	S1
CzmPPC	0D-1A	0.280	0.103	-0.177	0.001	0.472	S1
TCbTrz	0D-1A	0.270	0.220	-0.050	0.381	0.133	S2
APPT	0D-1A	0.070	0.186	+0.116	0.011	0.340	S3
PX-TRZ	0D-1A	0.056	0.242	+0.186	0.031	0.256	S4
DMIC-TRZ	0D-1A	0.056	0.519	+0.463	0.231	0.778	S5
D-A-D	0D-2A	0.242	0.028	-0.214	0.007	0.544	S6
phenazasiline	0D-2A	0.242	0.028	-0.214	0.007	0.545	S6
TPA-CN-N4	0D-2A	0.190	0.024	-0.166	0.007	0.559	S7
DTRZSFX	0D-2A	0.310	0.356	+0.046	2.021	0.444	S8
mTPA-PPI	1D-0A	0.760	0.258	-0.502	0.535	0.250	S9
mTPA-PPI	1D-0A	0.760	0.260	-0.500	0.556	0.251	S9

Continued on next page

Table S2 – Continued from previous page

Molecule	Arch.	Exp. ΔE_{ST} (eV)	Calc. ΔE_{ST} (eV)	Error (eV)	f	Overlap	Ref.
DPACPhTPI	1D-0A	0.720	0.255	-0.465	0.472	0.130	S10
PTPC	1D-0A	0.636	0.244	-0.391	0.473	0.187	S11
mP2MPC	1D-0A	0.650	0.284	-0.366	0.306	0.045	S12
9PhCzFDBFSPO	1D-0A	0.690	0.352	-0.338	0.218	0.006	S13
9PhCzFDBFDPO	1D-0A	0.630	0.354	-0.276	0.207	0.007	S13
BFDMAc-NAI	1D-0A	0.235	0.104	-0.131	0.091	0.270	S14
TX-CzPh	1D-0A	0.670	0.551	-0.119	0.170	0.288	S15
PXZ-PPO	1D-0A	0.600	0.510	-0.090	0.610	0.568	S16
dbG2TAZ	1D-0A	0.530	0.447	-0.083	0.040	0.871	S17
5-Cz-Sac	1D-0A	0.380	0.341	-0.039	0.264	0.365	S18
ACRSA	1D-0A	0.035	0.002	-0.033	0.000	0.177	S19
Cz-SO	1D-0A	0.410	0.395	-0.015	0.224	0.391	S20
PhCO-DMAC	1D-0A	0.090	0.076	-0.014	0.134	0.196	S21
6AcBIQ	1D-0A	0.100	0.086	-0.014	0.079	0.257	S22
TS-1	1D-0A	0.410	0.396	-0.014	0.224	0.391	S20
C3F7	1D-0A	0.083	0.078	-0.006	0.046	0.338	S23
CzoB	1D-0A	0.083	0.079	-0.005	0.047	0.340	S23
SIR-A	1D-0A	0.004	0.000	-0.004	0.000	0.157	S24
ANOP-DMAC	1D-0A	0.060	0.079	+0.019	0.188	0.209	S25
TPADSO2	1D-0A	0.380	0.405	+0.025	1.154	0.290	S26
MCZ-P2-DTM	1D-0A	0.230	0.270	+0.040	1.072	0.341	S27
PT-BZ-DMAC	1D-0A	0.008	0.068	+0.059	0.131	0.166	S28
ANOP-MeFAC	1D-0A	0.050	0.112	+0.062	0.280	0.242	S25
IpCm-PhBzAc	1D-0A	0.350	0.508	+0.158	1.088	0.044	S29
NA-TNA	1D-0A	0.240	0.404	+0.164	2.199	0.294	S30
PCTXO	1D-0A	0.040	0.206	+0.166	0.000	0.136	S31
TXO-TPA	1D-0A	0.080	0.253	+0.173	0.002	0.143	S31
CTM	1D-0A	0.110	0.284	+0.174	0.790	0.324	S32
BuPCzBCO	1D-0A	0.185	0.370	+0.185	0.159	0.645	S33
4-SCZ	1D-0A	0.240	0.446	+0.206	0.185	0.748	S34
DpAn-BzAc	1D-0A	0.167	0.377	+0.211	0.179	0.033	S35
9ArFDBFxPO	1D-0A	0.150	0.364	+0.214	0.200	0.013	S13
9CzFDBFDPO	1D-0A	0.150	0.376	+0.226	0.212	0.012	S13
NAI-PhBiFA	1D-0A	0.120	0.370	+0.250	1.378	0.262	S36
t-BuPCz	1D-0A	0.080	0.333	+0.253	0.043	0.632	S37
Ph-OBNA	1D-0A	0.075	0.336	+0.261	0.172	0.544	S38
FSF4A	1D-0A	0.022	0.286	+0.264	0.288	0.559	S39
ECDPTT	1D-0A	0.080	0.345	+0.265	0.561	0.200	S40
PXZ-QL	1D-0A	0.100	0.370	+0.270	0.334	0.474	S41
9CzFDBFSPO	1D-0A	0.120	0.396	+0.276	0.201	0.023	S13
poCz-SO	1D-0A	0.110	0.389	+0.279	0.069	0.672	S42
czpzpy	1D-0A	0.130	0.420	+0.290	0.371	0.523	S43
ACRXTN	1D-0A	0.173	0.478	+0.305	0.906	0.494	S44
DPACpBr	1D-0A	0.011	0.335	+0.324	0.273	0.525	S45
PPO21	1D-0A	0.025	0.377	+0.352	0.214	0.533	S46
Yad	1D-0A	0.020	0.387	+0.367	0.286	0.214	S47
DMAC-AQ	1D-0A	0.063	0.436	+0.373	0.314	0.477	S48
SPBP-DPAC	1D-0A	0.030	0.496	+0.466	0.857	0.486	S49
4-Ac	1D-0A	0.080	0.559	+0.479	0.624	0.702	S50
PS-BZ-DMAC	1D-0A	0.004	0.556	+0.551	1.150	0.444	S28
3ai	1D-0A	0.180	0.806	+0.626	0.797	0.397	S51
2BFCzPy	2D-0A	0.650	0.265	-0.385	0.669	0.411	S52
2CzPy	2D-0A	0.710	0.333	-0.377	0.172	0.441	S52
DC-TC	2D-0A	0.140	0.005	-0.135	0.001	0.298	S53
DPE-DDMAc	2D-0A	0.310	0.175	-0.135	0.156	0.197	S54
DiCz-Sac	2D-0A	0.353	0.240	-0.113	0.175	0.432	S18
BTaz-Ar-PXZ	2D-0A	0.300	0.194	-0.106	0.464	0.319	S55
VBC-PTZ	2D-0A	0.530	0.434	-0.096	1.225	0.433	S56
PTZ-NAI	2D-0A	0.087	0.000	-0.086	0.000	0.094	S57
BFCZPZ1	2D-0A	0.348	0.268	-0.080	0.959	0.633	S58

Continued on next page

Table S2 – Continued from previous page

Molecule	Arch.	Exp. ΔE_{ST} (eV)	Calc. ΔE_{ST} (eV)	Error (eV)	f	Overlap	Ref.
BFCZPZ3	2D-0A	0.340	0.265	-0.075	0.653	0.629	S58
pDTCz-2DPyS	2D-0A	0.300	0.227	-0.073	0.171	0.456	S59
BFCZPZ2	2D-0A	0.323	0.269	-0.055	1.804	0.543	S58
mAP	2D-0A	0.650	0.605	-0.045	0.008	0.235	S60
BTCZPZ1	2D-0A	0.270	0.246	-0.024	0.873	0.518	S58
DBCP	2D-0A	0.081	0.063	-0.018	0.070	0.214	S61
DMAC-FO	2D-0A	0.140	0.125	-0.015	0.113	0.284	S62
DC-ACR	2D-0A	0.010	0.003	-0.007	0.000	0.246	S53
DMAC-BP	2D-0A	0.070	0.064	-0.006	0.240	0.211	S63
SPAC-FO	2D-0A	0.130	0.140	+0.010	0.132	0.294	S62
3TCPM	2D-0A	0.250	0.262	+0.012	1.529	0.116	S64
TCB	2D-0A	0.250	0.263	+0.013	1.659	0.113	S64
2BPy-mDTC	2D-0A	0.065	0.080	+0.015	0.035	0.208	S65
TDBA-TPDICz	2D-0A	0.410	0.430	+0.020	0.391	0.205	S66
TRZ-pIC	2D-0A	0.410	0.430	+0.020	0.391	0.208	S66
2DN	2D-0A	0.070	0.092	+0.022	0.098	0.282	S67
SXAC-BP-SXAC	2D-0A	0.061	0.083	+0.022	0.332	0.214	S68
4BPy-mDTC	2D-0A	0.130	0.154	+0.024	0.115	0.214	S65
MPPA	2D-0A	0.172	0.196	+0.024	0.019	0.285	S69
STAC-BP-STAC	2D-0A	0.052	0.086	+0.034	0.320	0.225	S68
3BPy-mDTC	2D-0A	0.120	0.155	+0.035	0.091	0.197	S65
25DAcBPy	2D-0A	0.018	0.058	+0.040	0.146	0.176	S70
SFAC-BP-SFAC	2D-0A	0.036	0.077	+0.041	0.313	0.215	S68
PhQLPXZ	2D-0A	0.090	0.137	+0.047	0.192	0.288	S71
BTD-PXZ	2D-0A	0.130	0.186	+0.056	0.140	0.348	S72
pSFIAc2	2D-0A	0.310	0.367	+0.057	0.421	0.662	S73
1CzSO	2D-0A	0.260	0.320	+0.060	0.293	0.324	S74
BTZ-DMAC	2D-0A	0.065	0.127	+0.062	0.192	0.326	S75
DHPZ-2BI	2D-0A	0.127	0.212	+0.085	0.000	0.273	S76
PFBP-2b	2D-0A	0.060	0.152	+0.092	0.635	0.413	S77
m-DTPACO	2D-0A	0.210	0.317	+0.107	0.032	0.355	S78
PPZPPI	2D-0A	0.110	0.220	+0.110	0.000	0.234	S79
PPZTPI	2D-0A	0.110	0.223	+0.113	0.000	0.233	S79
t-BuCz2BP	2D-0A	0.215	0.337	+0.122	1.121	0.357	S80
BP-mDTC	2D-0A	0.070	0.193	+0.123	0.166	0.230	S65
OPDPO	2D-0A	0.020	0.145	+0.125	0.019	0.138	S81
CCDMB	2D-0A	0.181	0.325	+0.144	0.551	0.210	S82
PXZ-BOO	2D-0A	0.060	0.241	+0.181	0.254	0.368	S16
2SPBAC-BP	2D-0A	0.220	0.401	+0.181	0.186	0.167	S83
26DAcBPy	2D-0A	0.006	0.195	+0.189	0.305	0.126	S70
PXZ-NAI	2D-0A	0.060	0.265	+0.205	0.194	0.367	S57
F2AcBO	2D-0A	0.075	0.285	+0.210	0.255	0.123	S84
PXZ-ND	2D-0A	0.090	0.303	+0.213	0.891	0.542	S85
PSPP	2D-0A	0.077	0.319	+0.243	0.015	0.144	S86
Me2ACBO	2D-0A	0.050	0.314	+0.264	0.468	0.215	S84
B-dpa-SpiroAC	2D-0A	0.140	0.406	+0.266	0.578	0.568	S87
DMAC-DPS	2D-0A	0.173	0.442	+0.268	0.093	0.191	S88
BBCz	2D-0A	0.128	0.405	+0.277	0.001	0.730	S89
BDAPM	2D-0A	0.060	0.339	+0.279	1.193	0.535	S90
t-DABNA	2D-0A	0.160	0.448	+0.288	0.467	0.586	S91
B-dpa-Cz	2D-0A	0.150	0.440	+0.290	0.614	0.582	S87
DC-TC	2D-0A	0.150	0.465	+0.315	0.379	0.670	S92
TPA-BT-Q	2D-0A	0.150	0.467	+0.317	0.345	0.664	S92
o-Ac2BBP	2D-0A	0.010	0.362	+0.352	0.259	0.226	S93
IDAC-MCO	2D-0A	0.090	0.447	+0.357	0.127	0.290	S94
SBA-2DPS	2D-0A	0.090	0.456	+0.366	0.085	0.237	S95
p-DTPACO	2D-0A	0.050	0.418	+0.368	1.859	0.314	S78
TDBA-SBA	2D-0A	0.010	0.405	+0.395	0.282	0.029	S96
DDMA-TXO2	2D-0A	0.140	0.541	+0.401	0.108	0.277	S97
BPZPP	2D-2A	0.200	0.133	-0.067	0.178	0.553	S98

Continued on next page

Table S2 – Continued from previous page

Molecule	Arch.	Exp. ΔE_{ST} (eV)	Calc. ΔE_{ST} (eV)	Error (eV)	f	Overlap	Ref.
2PCzBN-tPh	2D-2A	0.275	0.221	-0.054	0.617	0.327	S99
DDPhCz-DPPZ	2D-2A	0.160	0.123	-0.037	0.331	0.568	S100
DCz-DPPZ	2D-2A	0.190	0.156	-0.034	0.703	0.562	S100
CzBN	2D-2A	0.300	0.298	-0.002	0.254	0.349	S101
2PCzBN	2D-2A	0.305	0.304	-0.001	0.725	0.343	S99
α -2DMAC-DBP	2D-2A	0.025	0.055	+0.030	0.099	0.437	S102
pDPBPZ-PXZ	2D-2A	0.060	0.091	+0.031	0.413	0.240	S103
PXZ-10-DPPZ	2D-2A	0.036	0.092	+0.056	0.091	0.413	S104
m2Cz2Trz	2D-2A	0.090	0.151	+0.061	0.210	0.490	S105
BTrztCz	2D-2A	0.020	0.086	+0.066	0.073	0.113	S106
3-DPM	2D-2A	0.099	0.169	+0.070	0.031	0.218	S107
PXZ-11-DPPZ	2D-2A	0.051	0.121	+0.070	0.260	0.331	S104
36DTPAFM	2D-2A	0.120	0.207	+0.087	0.129	0.274	S108
CzPm	2D-2A	0.180	0.287	+0.107	1.329	0.296	S109
2DMAC-BP-F	2D-2A	0.152	0.263	+0.111	0.895	0.545	S110
TPAPm	2D-2A	0.320	0.450	+0.130	2.000	0.402	S111
BPCN-2Cz	2D-2A	0.220	0.356	+0.136	0.198	0.172	S112
APDC-DTPA	2D-2A	0.140	0.280	+0.140	1.503	0.277	S113
BMIM	2D-2A	0.140	0.280	+0.140	1.502	0.276	S113
oDTPA-DTPZ	2D-2A	0.040	0.224	+0.184	0.041	0.411	S114
ACID-BPPZ	2D-2A	0.015	0.204	+0.189	0.127	0.166	S115
pDTPA-DTPZ	2D-2A	0.040	0.236	+0.196	0.563	0.370	S114
VBN0	2D-2A	0.110	0.355	+0.245	0.496	0.320	S116
4-DPM	2D-2A	0.143	0.448	+0.305	1.200	0.569	S107
MeOCzPCN	2D-2A	0.010	0.468	+0.458	0.542	0.470	S117
DMAC-11-DPPZ	A-D-A	0.116	0.056	-0.060	0.141	0.248	S102
PO-TPA	A-D-A	0.250	0.193	-0.057	0.582	0.395	S118
IABTCN	A-D-A	0.400	0.355	-0.045	1.413	0.367	S119
α -DMAC-DBP	A-D-A	0.070	0.052	-0.017	0.060	0.408	S102
BPCN-Cz2Ph	A-D-A	0.310	0.304	-0.006	0.149	0.240	S112
3TPAFM	A-D-A	0.160	0.159	-0.001	0.088	0.298	S108
3Py-DMAC	A-D-A	0.040	0.065	+0.025	0.255	0.054	S120
DMAC-PCN	A-D-A	0.095	0.134	+0.039	0.215	0.045	S121
SAF-2NP	A-D-A	0.040	0.082	+0.042	0.224	0.299	S122
Me-DMAC	A-D-A	0.010	0.112	+0.102	0.184	0.052	S123
Me-MOC	A-D-A	0.250	0.380	+0.130	0.415	0.111	S124
CBZ	A-D-A	0.003	0.159	+0.156	0.162	0.043	S125
oTPAPO-DTPZ	A-D-A	0.060	0.222	+0.162	0.049	0.223	S114
Me-DOC	A-D-A	0.240	0.429	+0.189	0.525	0.082	S124
Na-DMAC	A-D-A	0.075	0.270	+0.195	0.141	0.035	S120
FICz	D-A	0.510	0.195	-0.315	0.152	0.450	S126
SFCz	D-A	0.480	0.201	-0.279	0.213	0.445	S126
SFI23pPM	D-A	0.330	0.233	-0.097	0.377	0.480	S127
3CzPhpPM	D-A	0.330	0.241	-0.089	0.450	0.477	S127
DMAC-TRZ	D-A	0.075	0.048	-0.027	0.126	0.191	S128
SpiroAC-TRZ	D-A	0.072	0.068	-0.004	0.207	0.211	S129
BDBTCz-Trz	D-A	0.200	0.232	+0.032	0.673	0.284	S130
BDBFCz-Trz	D-A	0.200	0.236	+0.036	0.734	0.321	S130
CzDCbTrz	D-A	0.135	0.176	+0.041	0.360	0.165	S131
BTPBFCz	D-A	0.090	0.266	+0.176	0.147	0.221	S132
1CM	D-A	0.290	0.507	+0.217	0.083	0.701	S133
2BrCPT	D-A	0.040	0.270	+0.230	0.490	0.465	S134
24PQ-Cz	D-A-D	0.600	0.253	-0.347	0.052	0.473	S135
BDQPCz	D-A-D	0.495	0.152	-0.343	0.183	0.584	S136
AcCYP	D-A-D	0.200	0.072	-0.128	0.205	0.112	S137
12AcCz-PM	D-A-D	0.390	0.266	-0.124	0.924	0.475	S138
tCbz-mPYRs	D-A-D	0.415	0.299	-0.116	0.638	0.445	S139
TPA-2	D-A-D	0.145	0.042	-0.103	0.005	0.432	S140
1PXZP	D-A-D	0.260	0.165	-0.095	0.373	0.290	S141
IQ-oTPA	D-A-D	0.070	-0.000	-0.070	0.002	0.140	S142

Continued on next page

Table S2 – Continued from previous page

Molecule	Arch.	Exp. ΔE_{ST} (eV)	Calc. ΔE_{ST} (eV)	Error (eV)	f	Overlap	Ref.
BDQ-tBuCz	D-A-D	0.180	0.159	-0.021	0.322	0.595	S136
BDQDMAC	D-A-D	0.075	0.065	-0.010	0.129	0.453	S136
25CzBPym	D-A-D	0.055	0.083	+0.028	0.066	0.259	S143
25tCzBPym	D-A-D	0.046	0.085	+0.038	0.061	0.261	S143
SBDBQ-PXZ	D-A-D	0.070	0.109	+0.039	0.208	0.320	S144
BDQC-2	D-A-D	0.120	0.165	+0.045	0.275	0.114	S145
DCzTrz	D-A-D	0.190	0.241	+0.051	0.273	0.269	S2
35CzBPym	D-A-D	0.074	0.149	+0.076	0.064	0.199	S143
2H-Qz	D-A-D	0.137	0.214	+0.077	0.016	0.208	S146
FDQPXZ	D-A-D	0.045	0.128	+0.083	0.232	0.340	S71
AcCN	D-A-D	0.070	0.161	+0.091	0.266	0.104	S147
35tCzBPym	D-A-D	0.055	0.148	+0.093	0.063	0.183	S143
1PCTRz	D-A-D	0.160	0.257	+0.097	0.569	0.162	S148
pDPAPQ-PXZ	D-A-D	0.060	0.178	+0.118	0.295	0.346	S103
MPA	D-A-D	0.140	0.260	+0.120	0.016	0.162	S149
23AcCz-PM	D-A-D	0.060	0.200	+0.140	0.171	0.137	S138
2CM	D-A-D	0.310	0.474	+0.164	0.087	0.039	S133
APPT-PXZ	D-A-D	0.010	0.183	+0.173	0.297	0.346	S3
MBF	D-A-D	0.010	0.220	+0.210	0.843	0.302	S149
Cbz-PYR	D-A-D	0.060	0.280	+0.220	0.738	0.501	S150
IQ-Se	Multi-D/A	1.180	0.332	-0.848	0.077	0.559	S151
V-TPA	Multi-D/A	0.621	0.280	-0.341	1.076	0.042	S152
DBP-3TPA	Multi-D/A	0.390	0.092	-0.298	0.007	0.543	S153
DCzBNPh-1	Multi-D/A	0.500	0.256	-0.244	0.100	0.317	S154
NIDPA-1	Multi-D/A	0.600	0.388	-0.212	0.195	0.546	S155
IDFL-2DPA	Multi-D/A	0.450	0.245	-0.205	0.305	0.455	S156
2PBCzPy	Multi-D/A	0.530	0.337	-0.193	0.144	0.327	S52
PXZPDO	Multi-D/A	0.383	0.250	-0.134	0.017	0.158	S157
DQP	Multi-D/A	0.147	0.019	-0.128	0.000	0.575	S129
ACRFLCN	Multi-D/A	0.100	0.000	-0.100	0.000	0.136	S158
ACRF	Multi-D/A	0.100	0.001	-0.099	0.000	0.137	S158
TNPZ	Multi-D/A	0.170	0.082	-0.088	0.007	0.300	S159
3-AcBPtCN	Multi-D/A	0.134	0.064	-0.070	0.268	0.262	S160
DCPDAPM	Multi-D/A	0.130	0.070	-0.060	0.130	0.183	S161
β -DI-DBP	Multi-D/A	0.160	0.107	-0.053	0.016	0.563	S102
3-ACBPmCN	Multi-D/A	0.091	0.047	-0.044	0.010	0.171	S160
6,7-DCNQx-DICz	Multi-D/A	0.046	0.009	-0.037	0.013	0.052	S162
CZ9CZPZ	Multi-D/A	0.315	0.286	-0.029	1.574	0.276	S58
5,6PXZ-PIDO	Multi-D/A	0.235	0.212	-0.023	0.071	0.139	S163
2CzPN	Multi-D/A	0.266	0.243	-0.023	0.205	0.460	S164
3DMAC-BP-Br	Multi-D/A	0.040	0.020	-0.020	0.082	0.228	S165
Py-TPA	Multi-D/A	0.140	0.125	-0.015	0.042	0.238	S166
BTPO	Multi-D/A	0.433	0.436	+0.002	0.220	0.547	S167
TPXZ-as-TAZ	Multi-D/A	0.093	0.095	+0.002	0.439	0.118	S168
TPXZ-as-TAZ	Multi-D/A	0.093	0.098	+0.005	0.681	0.122	S168
BPCP-2CPC	Multi-D/A	0.166	0.174	+0.008	0.150	0.015	S169
2F	Multi-D/A	0.203	0.215	+0.012	0.112	0.121	S170
TATP-BP	Multi-D/A	0.129	0.145	+0.016	0.163	0.124	S171
ANQDC-PSTA	Multi-D/A	0.080	0.097	+0.017	0.240	0.200	S172
DV-3CzCN	Multi-D/A	0.230	0.248	+0.018	0.610	0.249	S173
DICzTrz	Multi-D/A	0.150	0.168	+0.018	0.103	0.187	S174
α -2CbPN	Multi-D/A	0.270	0.293	+0.023	0.253	0.221	S131
PTZ-2PTO	Multi-D/A	0.270	0.293	+0.023	0.033	0.175	S175
4CzIPN	Multi-D/A	0.123	0.151	+0.028	0.151	0.325	S176
TPA-DCPP	Multi-D/A	0.145	0.175	+0.030	0.901	0.257	S129
TCzTrz	Multi-D/A	0.150	0.180	+0.030	0.353	0.305	S131
3-AcBPdCN	Multi-D/A	0.180	0.211	+0.031	0.637	0.518	S160
c-D2T	Multi-D/A	0.088	0.124	+0.035	0.290	0.044	S177
SBF-PXZ	Multi-D/A	0.260	0.305	+0.045	0.038	0.390	S178
3CzBN	Multi-D/A	0.283	0.329	+0.046	0.317	0.367	S101

Continued on next page

Table S2 – Continued from previous page

Molecule	Arch.	Exp. ΔE_{ST} (eV)	Calc. ΔE_{ST} (eV)	Error (eV)	f	Overlap	Ref.
DBBPZ-DPXZ	Multi-D/A	0.050	0.104	+0.054	0.692	0.321	S179
PTZ-TAZ	Multi-D/A	0.144	0.199	+0.055	0.861	0.337	S180
2PXZ-OXD	Multi-D/A	0.144	0.203	+0.059	0.867	0.339	S180
DPIBPZ-DPXZ	Multi-D/A	0.050	0.110	+0.060	0.691	0.346	S179
2Cz2tCzBn	Multi-D/A	0.130	0.191	+0.061	0.438	0.368	S181
BzITz	Multi-D/A	0.185	0.251	+0.066	0.941	0.377	S182
2tCz2CzBn	Multi-D/A	0.130	0.199	+0.069	0.395	0.357	S181
2PS-2FPh	Multi-D/A	0.198	0.273	+0.075	0.032	0.091	S183
DBQ-3PXZ	Multi-D/A	0.030	0.106	+0.076	0.228	0.135	S144
TrisCz-Trz	Multi-D/A	0.057	0.136	+0.079	0.077	0.290	S184
DPXZ-DPPM	Multi-D/A	0.023	0.105	+0.082	0.593	0.325	S185
2PXZP	Multi-D/A	0.134	0.218	+0.084	0.455	0.273	S141
CCDC	Multi-D/A	0.222	0.307	+0.085	0.261	0.192	S186
TRZ3DPA	Multi-D/A	0.282	0.367	+0.085	1.556	0.456	S187
TDPA-TRZ	Multi-D/A	0.282	0.369	+0.087	1.557	0.441	S187
mDPBPZ-DPXZ	Multi-D/A	0.020	0.107	+0.087	0.693	0.330	S179
PTZ-DCPP	Multi-D/A	0.187	0.278	+0.091	0.383	0.527	S188
DPE	Multi-D/A	0.190	0.284	+0.094	0.005	0.362	S54
DPE-DPXZ	Multi-D/A	0.190	0.285	+0.095	0.002	0.363	S54
CZ-TRZ	Multi-D/A	0.150	0.245	+0.095	0.579	0.158	S189
t3Cz-SO	Multi-D/A	0.260	0.356	+0.096	0.207	0.240	S20
PxCYP	Multi-D/A	0.135	0.232	+0.097	0.441	0.160	S137
Fene	Multi-D/A	0.025	0.130	+0.105	0.076	0.295	S47
BTPBFCz-D3	Multi-D/A	0.120	0.226	+0.106	0.074	0.174	S132
DHPZ-2BN	Multi-D/A	0.100	0.209	+0.109	0.000	0.349	S76
m-CZ-DPS-DMAC	Multi-D/A	0.210	0.320	+0.110	0.393	0.152	S190
12BTCzTPN	Multi-D/A	0.083	0.196	+0.113	0.360	0.358	S191
NyDPO	Multi-D/A	0.033	0.149	+0.116	1.071	0.234	S192
ICBNTrz4	Multi-D/A	0.085	0.204	+0.119	0.855	0.329	S193
BTITrz	Multi-D/A	0.102	0.222	+0.119	0.748	0.311	S194
3DPA3CN	Multi-D/A	0.103	0.223	+0.120	1.412	0.265	S195
SiTCNCz	Multi-D/A	0.130	0.250	+0.120	0.418	0.424	S196
2PN	Multi-D/A	0.010	0.132	+0.122	0.093	0.332	S67
Tri-PXZ-PCN	Multi-D/A	0.050	0.173	+0.123	0.612	0.078	S197
CNPP-TPA	Multi-D/A	0.100	0.223	+0.123	0.632	0.301	S198
CN-QP	Multi-D/A	0.030	0.156	+0.126	0.034	0.150	S199
DVCz-2CzCN	Multi-D/A	0.160	0.287	+0.127	0.260	0.034	S200
oPTBC	Multi-D/A	0.007	0.134	+0.127	0.094	0.309	S201
δ -2CbPN	Multi-D/A	0.130	0.261	+0.131	0.236	0.154	S131
CDBP-BP-DMAC	Multi-D/A	0.039	0.183	+0.144	0.193	0.157	S202
BTCz3CzTPN	Multi-D/A	0.036	0.195	+0.159	0.460	0.142	S203
PxPYM	Multi-D/A	0.010	0.169	+0.159	0.715	0.300	S204
4CzTPN	Multi-D/A	0.047	0.210	+0.163	0.422	0.388	S205
3DMAC-BP-CN	Multi-D/A	0.020	0.185	+0.165	0.019	0.239	S165
26DPXZBPY	Multi-D/A	0.035	0.201	+0.166	0.004	0.290	S70
DFDBQPXZ	Multi-D/A	0.000	0.169	+0.169	0.012	0.157	S206
pCNBCzoCF3	Multi-D/A	0.100	0.279	+0.178	0.774	0.383	S207
CNTPA-CZ	Multi-D/A	0.350	0.528	+0.178	0.657	0.371	S208
2CzTPEPCz	Multi-D/A	0.130	0.310	+0.180	1.519	0.533	S209
Ac-BPCN	Multi-D/A	0.130	0.313	+0.183	0.469	0.498	S210
PSPBP	Multi-D/A	0.110	0.294	+0.184	0.021	0.176	S86
AI-4Cz	Multi-D/A	0.020	0.205	+0.185	0.051	0.367	S211
PBICT	Multi-D/A	0.028	0.213	+0.185	0.377	0.282	S162
DMeCzIPN	Multi-D/A	0.091	0.279	+0.188	0.176	0.372	S212
2MeCz-TRZ	Multi-D/A	0.050	0.242	+0.192	0.620	0.161	S189
MeCz-TRZ	Multi-D/A	0.060	0.253	+0.193	0.512	0.183	S189
4F-m- ν -DABNA	Multi-D/A	0.070	0.264	+0.194	1.012	0.249	S213
CNTPA-PXZ	Multi-D/A	0.130	0.327	+0.197	0.012	0.103	S208
DMAC-Cz-TTR	Multi-D/A	0.066	0.271	+0.205	0.350	0.038	S214
DPS-PXZ	Multi-D/A	0.081	0.288	+0.207	0.039	0.380	S215

Continued on next page

Table S2 – Continued from previous page

Molecule	Arch.	Exp. ΔE_{ST} (eV)	Calc. ΔE_{ST} (eV)	Error (eV)	f	Overlap	Ref.
DCzIPN	Multi-D/A	0.143	0.359	+0.215	0.272	0.545	S212
D-BP-PXZ	Multi-D/A	0.030	0.246	+0.216	0.048	0.322	S93
ANQDC	Multi-D/A	0.020	0.237	+0.217	0.006	0.511	S172
m- ν -DABNA	Multi-D/A	0.070	0.289	+0.219	0.898	0.530	S213
Fens	Multi-D/A	0.020	0.240	+0.220	0.022	0.155	S47
PTZ-mPYR	Multi-D/A	0.018	0.239	+0.221	0.062	0.059	S216
4F- ν -DABNA	Multi-D/A	0.050	0.271	+0.221	0.931	0.272	S213
BDPCC	Multi-D/A	0.110	0.333	+0.223	0.348	0.547	S164
PxCN	Multi-D/A	0.037	0.261	+0.224	0.004	0.179	S147
CDBP-BP-PXZ	Multi-D/A	0.060	0.288	+0.227	0.017	0.257	S202
mPTBC	Multi-D/A	0.006	0.234	+0.228	0.284	0.309	S201
OCzBN	Multi-D/A	0.140	0.370	+0.230	0.603	0.340	S217
BCZ-DPS-AD	Multi-D/A	0.097	0.328	+0.232	0.158	0.181	S218
DTPTCzDP-CN	Multi-D/A	0.030	0.264	+0.234	0.588	0.303	S219
11-AcBPdCN	Multi-D/A	0.066	0.301	+0.235	0.463	0.533	S160
PyDCN-DMAC	Multi-D/A	0.010	0.246	+0.236	0.037	0.078	S220
PCDMB	Multi-D/A	0.087	0.323	+0.236	0.015	0.076	S82
CZ-DPS-DMAC	Multi-D/A	0.070	0.308	+0.238	0.253	0.326	S221
2Cz-DMAC-BTB	Multi-D/A	0.020	0.266	+0.246	0.080	0.143	S222
2Cz-DMAC-TTR	Multi-D/A	0.047	0.293	+0.246	0.431	0.156	S222
mCP-BP-DMAC	Multi-D/A	0.016	0.287	+0.271	0.017	0.254	S223
TPD4PA	Multi-D/A	0.050	0.324	+0.274	0.271	0.356	S224
o-Px2BBP	Multi-D/A	0.015	0.302	+0.287	0.345	0.200	S93
Spiro-CN	Multi-D/A	0.001	0.289	+0.288	1.537	0.053	S225
DACIPN	Multi-D/A	0.035	0.343	+0.308	0.194	0.274	S226
SPFS-PXZ	Multi-D/A	0.015	0.329	+0.314	0.014	0.101	S215
PXZ2PTO	Multi-D/A	0.023	0.338	+0.315	0.015	0.339	S227
tBuCz2	Multi-D/A	0.080	0.397	+0.317	0.049	0.398	S228
2Cz-DMAC-TXO	Multi-D/A	0.036	0.368	+0.332	0.021	0.110	S222
DMACMNPTO	Multi-D/A	0.205	0.538	+0.333	0.043	0.087	S229
TCz1	Multi-D/A	0.058	0.401	+0.343	0.056	0.451	S226
TCZ1	Multi-D/A	0.058	0.402	+0.344	0.059	0.450	S226
TCPZ	Multi-D/A	0.140	0.534	+0.394	0.414	0.771	S230
DMAC-Cz	Multi-D/A	0.001	0.420	+0.419	0.063	0.142	S214
CN-TPA	Multi-D/A	0.080	0.502	+0.422	0.592	0.634	S231
NPE-AcDPS	Multi-D/A	0.030	0.490	+0.460	0.091	0.140	S232
TXO-P-Si	Unknown	0.522	-0.005	-0.528	0.000	0.079	S46
PyIAAnTPh	Unknown	1.140	0.722	-0.418	0.780	0.497	S233
PIAnTPh	Unknown	1.140	0.777	-0.363	0.862	0.872	S233
N-B	Unknown	0.456	0.175	-0.281	0.184	0.551	S234
E-o-DHMPA	Unknown	1.069	0.791	-0.278	0.386	0.931	S235
Z-o-DHMPA	Unknown	1.069	0.791	-0.278	0.385	0.931	S235
m-DHMPA	Unknown	1.019	0.781	-0.238	0.435	0.936	S235
p-DHMPA	Unknown	1.004	0.780	-0.224	0.453	0.936	S235
UGH3	Unknown	0.170	-0.001	-0.171	0.000	0.610	S97
BMZ-TZ	Unknown	0.018	-0.010	-0.028	0.009	0.474	S236
BTZPP	Unknown	0.300	0.307	+0.007	2.874	0.249	S237
NyDPAC	Unknown	0.152	0.166	+0.014	0.035	0.031	S238
mPXTMP	Unknown	0.270	0.305	+0.035	0.001	0.653	S239
Pra-2DMAC	Unknown	0.440	0.507	+0.067	0.086	0.264	S60
DHID-DBS	Unknown	0.330	0.445	+0.115	0.374	0.262	S240
trimesitylboron	Unknown	0.014	0.221	+0.208	0.175	0.638	S241
DBT-BZ-DMAC	Unknown	0.080	0.314	+0.234	0.022	0.216	S129
benzoxazole	Unknown	0.180	0.429	+0.249	0.309	0.630	S242
DMBFTX	Unknown	0.280	0.544	+0.264	0.206	0.588	S243
DMAC-ND	Unknown	0.060	0.354	+0.294	0.952	0.467	S85
SBTF	Unknown	0.200	0.502	+0.302	1.478	0.909	S244
2Mi2SB	Unknown	0.030	0.346	+0.316	0.129	0.283	S245
DHID-DPS	Unknown	0.180	0.548	+0.368	0.192	0.461	S240
PrDPhAc	Unknown	0.075	0.472	+0.397	0.572	0.644	S246

Continued on next page

Table S2 – Continued from previous page

Molecule	Arch.	Exp. ΔE_{ST} (eV)	Calc. ΔE_{ST} (eV)	Error (eV)	f	Overlap	Ref.
TMDBP	Unknown	0.100	0.520	+0.420	0.158	0.740	S247
FAC	Unknown	0.090	0.517	+0.427	0.059	0.173	S248
SpiroAc	Unknown	0.090	0.520	+0.430	0.049	0.170	S248
P4CzCN-BCz	Unknown	0.170	0.601	+0.431	0.001	0.099	S249
BO3N	Unknown	0.298	0.735	+0.437	0.104	0.693	S187
TPSi-F	Unknown	0.100	0.629	+0.529	0.343	0.846	S250
OXD-7	Unknown	0.000	0.573	+0.573	2.008	0.235	S251

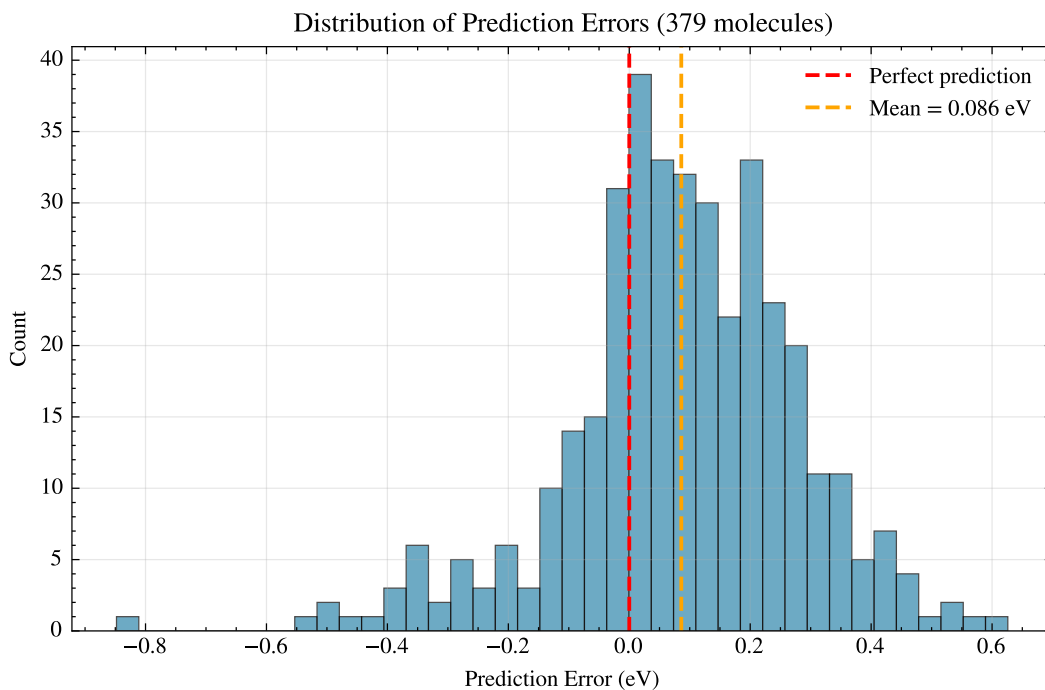


Figure S1: Distribution of prediction errors for the 379-molecule validation dataset. The histogram shows that most molecules have errors within ± 0.2 eV, with a mean error of $+0.086$ eV (systematic overestimation). The distribution is approximately normal with some outliers at larger absolute errors.

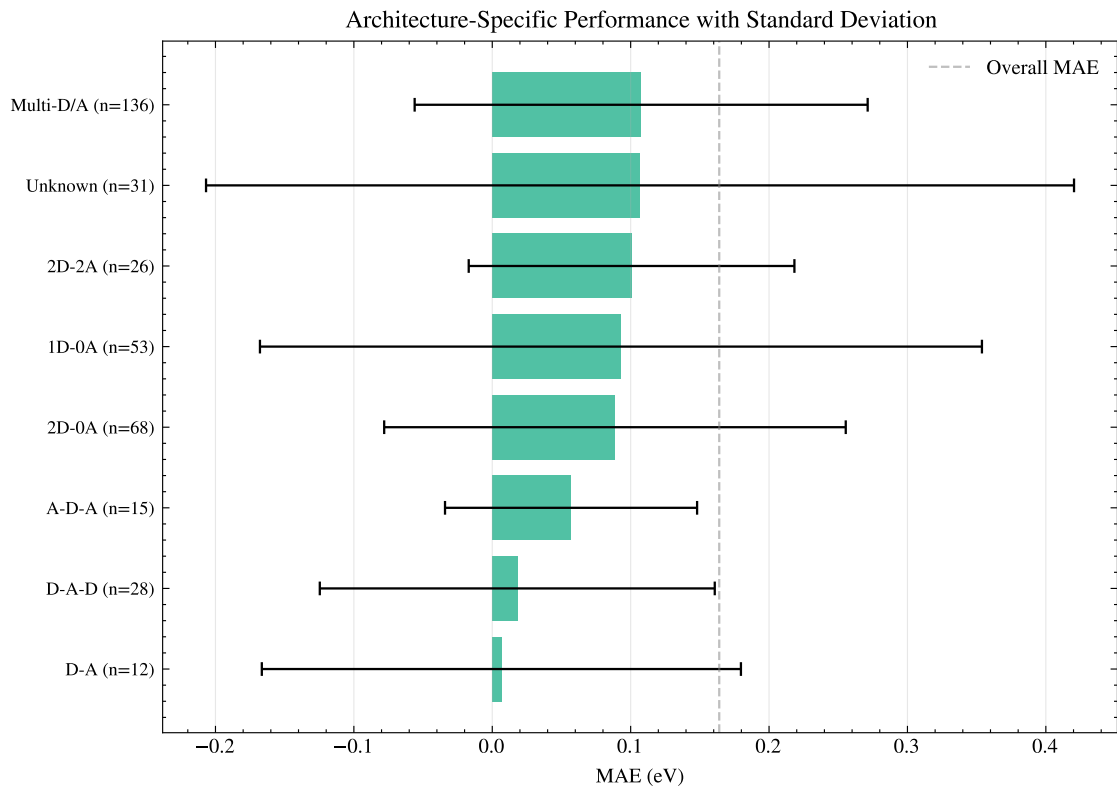


Figure S2: Architecture-specific performance with standard deviations. Error bars represent the variability in prediction accuracy within each architecture class. A-D-A and D-A-D architectures show the lowest mean absolute errors and smallest variability, while 1D-0A systems exhibit larger errors and greater variability, reflecting the challenges in predicting charge-transfer states for these motifs.

Outlier Analysis

To understand the limitations of our computational approach, we analyzed the 59 molecules (15.6% of the dataset) with prediction errors exceeding 0.3 eV. Figure S3 shows the distribution of these outliers and their characteristic descriptors.

Error Patterns

The outliers exhibit two distinct error patterns:

- **Overestimation** (Calc > Exp): 43 molecules (72.9% of outliers), predominantly with very small experimental ΔE_{ST} values (<0.1 eV). The mean experimental value for this group is 0.04 eV, while calculations predict 0.41 eV, suggesting systematic overestimation for near-zero singlet-triplet gaps.
- **Underestimation** (Calc < Exp): 16 molecules (27.1% of outliers), primarily with large experimental ΔE_{ST} values (>0.5 eV). The mean experimental value is 0.72 eV versus calculated 0.32 eV, indicating difficulty in predicting molecules with large singlet-triplet splittings.

Descriptor Analysis

Outliers show distinct characteristics compared to well-predicted molecules:

- **HOMO-LUMO overlap**: Outliers exhibit significantly higher overlap (0.394 ± 0.238) compared to well-predicted molecules (0.312 ± 0.177). High overlap (>0.5) is present in 28.8% of outliers versus only 13.4% of well-predicted molecules, suggesting that strong orbital delocalization challenges the computational approach.

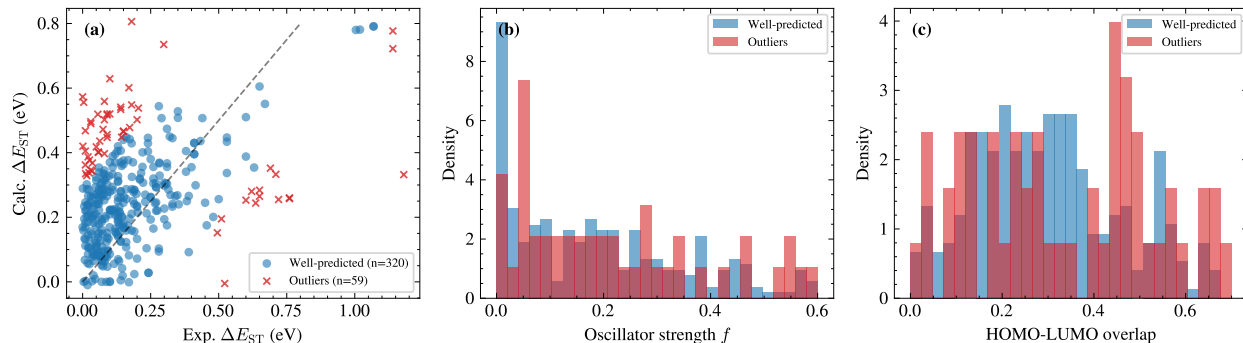


Figure S3: Outlier analysis of the 379-molecule validation set. (a) Scatter plot highlighting outliers (red crosses, $|\text{error}| > 0.3$ eV) versus well-predicted molecules (blue circles). (b) Distribution of oscillator strength f for outliers and well-predicted molecules. (c) Distribution of HOMO-LUMO overlap for both groups. Outliers show systematically higher overlap values, indicating stronger charge-transfer character.

- **Oscillator strength:** Outliers show slightly elevated f values (0.409 ± 0.447) compared to well-predicted molecules (0.366 ± 0.432), with 15.3% having $f > 0.8$ indicating strong π - π^* character.
- **Experimental range:** Nearly half of outliers (49.2%) have experimental $\Delta E_{\text{ST}} < 0.1$ eV, representing the challenging regime where thermal energy ($k_{\text{B}}T \approx 0.026$ eV at 300 K) becomes comparable to the singlet-triplet gap.

Architecture-Specific Patterns

Outliers are not uniformly distributed across molecular architectures:

- **1D-0A** (15 outliers, 25.4%): Single-donor architectures show moderate systematic overestimation (mean error = +0.085 eV), with relatively low HOMO-LUMO overlap (0.343).
- **Unknown** (14 outliers, 23.7%): Molecules with unclassified architectures exhibit the largest systematic overestimation (mean error = +0.237 eV) and highest overlap (0.479), suggesting complex electronic structures not captured by standard D-A classifications.
- **Multi-D/A** (14 outliers, 23.7%): Complex multi-component systems show significant overestimation (mean error = +0.222 eV) despite low oscillator strength (0.197), indicating challenges in treating multiple charge-transfer pathways.
- **2D-0A** (10 outliers, 16.9%): Dual-donor architectures display consistent overestimation (mean error = +0.211 eV) with moderate descriptors.

Extreme Cases

Table S3 lists the most challenging predictions. The worst case (IQ-Se) shows an underestimation of 0.85 eV for a molecule with experimental $\Delta E_{\text{ST}} = 1.18$ eV, far outside the typical TADF range. Conversely, molecules like 3ai and PS-BZ-DMAC with near-zero experimental gaps are overestimated by >0.5 eV, likely due to environmental effects (solvent, solid-state packing) not captured in gas-phase calculations.

Implications

This analysis reveals that the computational approach performs best for molecules with:

- Moderate ΔE_{ST} values (0.1–0.5 eV)
- Low-to-moderate HOMO-LUMO overlap (<0.5)
- Well-defined D-A architectures

Conversely, predictions are less reliable for molecules with near-zero experimental gaps, high orbital overlap, or complex multi-component architectures. These limitations likely stem from the gas-phase approximation, which neglects environmental effects (solvent stabilization, solid-state packing) that can significantly modulate singlet-triplet gaps in real TADF systems.

Table S3: Top 10 most challenging predictions from the 379-molecule validation set.

Molecule	Architecture	Exp (eV)	Calc (eV)	Error (eV)	f	Overlap
IQ-Se	Multi-D/A	1.180	0.332	−0.848	0.077	0.559
3ai	1D-0A	0.180	0.806	+0.626	0.797	0.397
OXD-7	Unknown	0.000	0.573	+0.573	2.008	0.235
PS-BZ-DMAC	1D-0A	0.004	0.556	+0.551	1.150	0.444
TPSi-F	Unknown	0.100	0.629	+0.529	0.343	0.846
TXO-P-Si	Unknown	0.522	−0.005	−0.528	0.000	0.079
mTPA-PPI	1D-0A	0.760	0.258	−0.502	0.535	0.250
mTPA-PPI	1D-0A	0.760	0.260	−0.500	0.556	0.251
4-Ac	1D-0A	0.080	0.559	+0.479	0.624	0.702
SPBP-DPAC	1D-0A	0.030	0.496	+0.466	0.857	0.486

Multi-Reference TADF Validation Data

Table S4: Validation of xTB-based screening for MR-TADF molecules: Comparison with TD-DFT/CAM-B3LYP/def2-TZVP calculations. While absolute values show systematic deviations, the relative ordering of molecules by ΔE_{ST} is preserved, demonstrating the validity of the screening protocol for candidate prioritization.

	sTD-DFT-xTB (Screening)			TD-DFT (Validation)			
Molecule	S_1 (eV)	T_1 (eV)	ΔE_{ST} (eV)	S_1 (eV)	T_1 (eV)	ΔE_{ST} (eV)	Ranking
phenazasiline ^a	3.23	3.12	0.04	3.72	2.29	1.43	1 / 1 ^b
24PQ-Cz ^a	3.77	3.27	0.27	4.07	2.63	1.43	2 / 2 ^b
MAE ^c	0.40	0.73	1.28	—	—	—	—
MRE ^d (%)	10.3	29.0	89.3	—	—	—	—

^a MR-TADF molecules characterized by B-N doping and multiple resonance structures.

^b Ranking by ΔE_{ST} : xTB ranking / TD-DFT ranking. Identical rankings confirm trend preservation.

^c Mean Absolute Error: $MAE = \frac{1}{N} \sum_{i=1}^N |xTB_i - TD-DFT_i|$

^d Mean Relative Error: $MRE = \frac{1}{N} \sum_{i=1}^N \frac{|xTB_i - TD-DFT_i|}{TD-DFT_i} \times 100\%$

Note: While absolute ΔE_{ST} values show large deviations (typical for single-reference methods on multi-reference systems), the relative ordering is preserved. This confirms the validity of xTB-based screening for identifying promising MR-TADF candidates, consistent with literature precedent showing that approximate methods capture qualitative trends despite quantitative limitations. ^{S252,S253}

Spin-Orbit Coupling Validation Data

Table S5: Spin-orbit coupling matrix elements for representative TADF emitters. T_1-S_1 and T_1-S_0 SOC values calculated using TD-DFT/TDA with the SOMF(1X) operator (ω B97X-D4/def2-TZVP, ORCA 6.1.0). All values in cm^{-1} .

Molecule	T_1-S_0 SOC	T_1-S_1 SOC	ΔE_{ST} (eV)	Overlap	Architecture
DMAC-BP	12.079	14.025	0.051	0.603	D-A
α -DMAC-DBP	0.792	2.476	0.051	0.603	D-A
24PQ-Cz	0.941	4.145	0.062	0.603	MR-TADF
PBTPA	3.710	0.095	0.270	0.061	2D-0A
PXZ-10-DPPZ	0.828	1.681	0.265	0.061	2D-0A
phenazasiline	0.391	11.328	0.259	0.061	Heterocycle
FTAT-FBO	1.371	0.650	0.259	0.603	D-A
2CzPN	0.462	0.580	0.259	0.603	Multi-D/A
DMAC-DPS	0.218	0.636	0.051	0.603	D-A
DMAC-TRZ	1.210	0.400	0.051	0.603	D-A
PXZ-OXD	1.021	0.381	0.265	0.061	2D-0A
TPA-APy	0.283	0.350	0.259	0.603	D-A
SpiroAC-TRZ	0.970	0.341	0.062	0.603	Spiro D-A
2BrCPT	0.262	0.270	0.272	0.603	D-A
IQ-oTPA	41.159	0.136	0.259	0.603	D-A
BACN	0.185	0.058	0.486	0.603	A-D-A
PtOEP	81.760	3.021	0.259	0.603	Pt complex

Statistical Summary (organic molecules, excluding PtOEP): $n = 16$; T_1-S_1 SOC median = 0.490 cm^{-1} ; range = $0.058\text{--}14.025 \text{ cm}^{-1}$.

Weak correlations with structural descriptors: HOMO-LUMO overlap ($r = 0.037$, $p = 0.89$), ΔE_{ST} ($r = 0.067$, $p = 0.80$).

PtOEP shows enhanced SOC due to heavy atom effect (Pt).

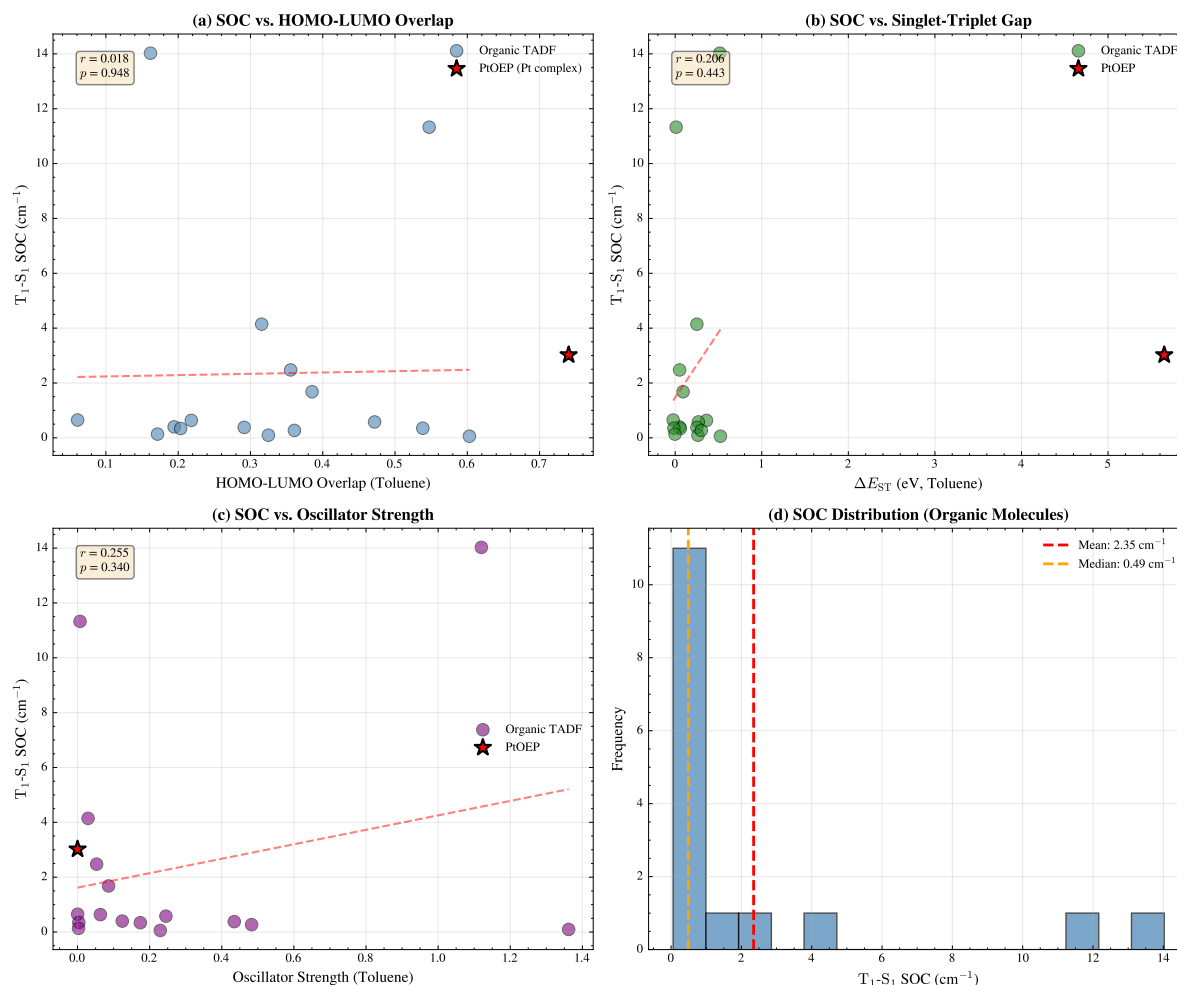


Figure S4: Correlation analysis of spin-orbit coupling with structural descriptors. (a) SOC vs. HOMO-LUMO overlap, (b) SOC vs. singlet-triplet gap, (c) SOC vs. oscillator strength, (d) SOC distribution for organic molecules. Weak correlations ($r < 0.2$) indicate that SOC depends on complex orbital angular momentum factors beyond simple geometric descriptors. PtOEP (red star) shows enhanced SOC due to heavy atom effect.

Adiabatic ΔE_{ST} Validation

Computational Methodology

To validate the vertical approximation employed in our high-throughput screening, we performed TD-DFT geometry optimizations of S_1 and T_1 excited states for 6 representative molecules using the ω B97X-D4 functional^{S254} with the def2-TZVP basis set^{S255} and CPCM solvation model for toluene.^{S256}

Choice of Functional

The ω B97X-D4 functional was selected based on several considerations:

Range-separated hybrid character: ω B97X-D4 is a range-separated hybrid functional with 100% Hartree-Fock exchange at long range, making it particularly well-suited for describing charge-transfer (CT) excited states characteristic of TADF emitters.^{S257}

Dispersion correction: The D4 dispersion correction^{S258} represents the latest generation of Grimme's dispersion schemes, providing improved treatment of non-covalent interactions important in π -stacked and sterically hindered TADF architectures.

Literature validation: The ω B97X-D functional family has been extensively validated for TADF emitters,^{S259,S260} with ω B97X-D4 offering improved performance through enhanced dispersion treatment while maintaining the reliable CT state description of its predecessor.

Comparison with xTB Adiabatic Estimates

Table S6 compares xTB adiabatic ΔE_{ST} estimates (from both sTDA and sTD-DFT methods) with TD-DFT adiabatic values obtained from excited-state geometry optimizations. Figure S5 shows the correlation between xTB and TD-DFT adiabatic energies.

Table S6: Comparison of xTB adiabatic ΔE_{ST} estimates with TD-DFT adiabatic values from excited-state geometry optimizations. All values in eV.

Molecule	xTB sTDA	xTB sTD-DFT	TD-DFT Adiabatic
DMAC-TRZ	0.049	0.051	1.018
SpiroAC-TRZ	0.060	0.062	1.000
2CzPN	0.268	0.259	1.046
PBTPA	0.263	0.270	1.233
2BrCPT	0.302	0.272	1.041
BACN	0.523	0.486	1.171

Statistics:

sTDA: $R^2 = 0.395$, Spearman $\rho = 0.543$ ($p = 0.266$)

sTD-DFT: $R^2 = 0.454$, Spearman $\rho = 0.657$ ($p = 0.156$)

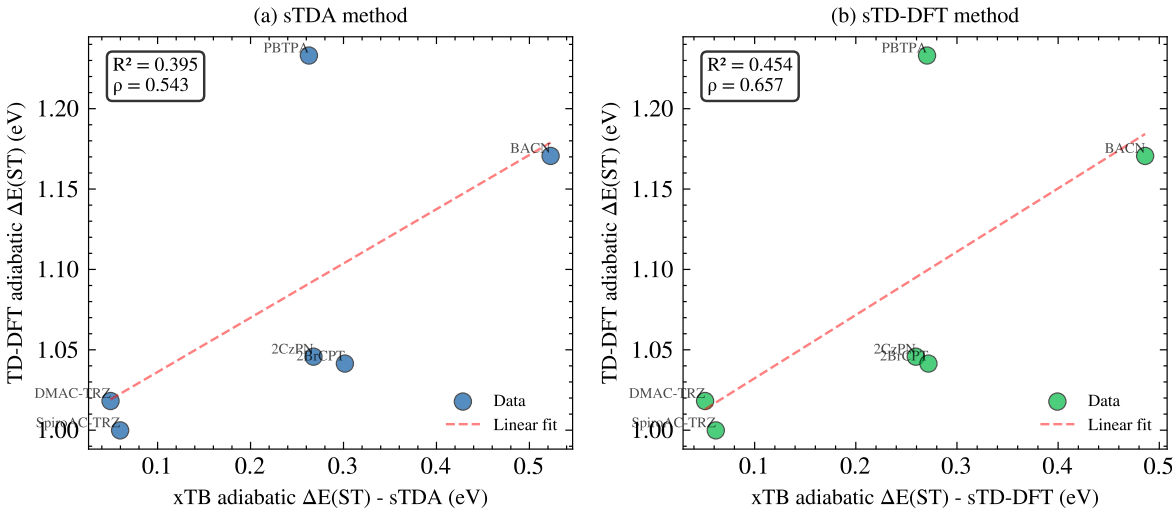


Figure S5: Correlation between xTB adiabatic ΔE_{ST} estimates and TD-DFT adiabatic values from excited-state geometry optimizations. (a) sTDA method shows moderate correlation ($R^2 = 0.40$, $\rho = 0.54$). (b) sTD-DFT method shows improved correlation ($R^2 = 0.45$, $\rho = 0.66$) due to better treatment of electron correlation. Both methods preserve molecular rankings, validating the use of xTB for high-throughput screening. Red dashed lines show linear fits.

Systematic Comparison with Experiment

To assess the accuracy of our TD-DFT approach, we compared adiabatic ΔE_{ST} values with available experimental data (Table S7). For the three molecules with experimental measurements, we observe systematic overestimation by approximately 0.9–1.0 eV. This discrepancy is consistent with known limitations of TD-DFT for strongly CT-dominated excited states in TADF systems,^{S261,S262} where the single-reference nature of TD-DFT struggles to fully capture the multi-configurational character of near-degenerate singlet-triplet states.

Table S7: Comparison of TD-DFT adiabatic ΔE_{ST} with experimental values. All values in eV.

Molecule	Experimental	TD-DFT	Difference
DMAC-TRZ	0.11 ^{S263}	1.018	+0.91
SpiroAC-TRZ	0.11 ^{S264}	1.000	+0.89
2CzPN	0.08 ^{S265}	1.046	+0.97
Mean overestimation: 0.92 ± 0.04 eV			

Despite this systematic offset, the **relative ordering of molecules is preserved**, which is the critical requirement for validating our high-throughput screening protocol. The consistent overestimation across all molecules suggests that ω B97X-D4 provides a reliable framework for comparing relative ΔE_{ST} values, even if absolute predictions require higher-level methods such as SCS-CC2 or STEOM-DLPNO-CCSD.^{S266,S267}

Conclusions from Adiabatic Validation

The adiabatic validation demonstrates that:

1. xTB adiabatic estimates show moderate correlation with TD-DFT adiabatic values ($R^2 = 0.40\text{--}0.45$, $\rho = 0.54\text{--}0.66$).
2. sTD-DFT performs slightly better than sTDA, as expected from improved electron correlation treatment.
3. Molecular rankings are preserved despite systematic underestimation by xTB.
4. TD-DFT systematically overestimates experimental ΔE_{ST} by ~ 1 eV, a known limitation for CT states.
5. The vertical approximation employed in our screening is validated for identifying promising TADF candidates through relative comparisons.

References

- (S1) Chen, Y.-K.; Jayakumar, J.; Ko, C.-L.; Hung, W.-Y.; Wu, T.-L.; Cheng, C.-H. Increase the molecular length and donor strength to boost horizontal dipole orientation for high-efficiency OLEDs. *Journal of Materials Chemistry C* **2022**, *10*, 9241–9248.
- (S2) Youn, K. M.; Ahn, D. H.; Kim, G. H.; Karthik, D.; Lee, J. Y.; Kwon, J. H. Blue thermally activated delayed fluorescence emitters with a δ -pyridoindole donor moiety. *New Journal of Chemistry* **2018**, *42*, 5532–5539.
- (S3) Yang, W.; Ning, W.; Gong, S.; Yang, C. Deep-red thermally activated delayed fluorescence emitters based on a phenanthroline-containing planar acceptor. *Dyes and Pigments* **2021**, *192*, 109474.
- (S4) Qu, C.; Xia, G.; Xu, Y.; Zhu, Y.; Liang, J.; Zhang, H.; Wang, J.; Zhang, Z.; Wang, Y. Boron-containing D–A–A type TADF materials with tiny singlet–triplet energy splittings and high photoluminescence quantum yields for highly efficient OLEDs with low efficiency roll-offs. *Journal of Materials Chemistry C* **2020**, *8*, 3846–3854.
- (S5) Chen, K.; Zhang, F.; Li, A.; Zhang, R.; Sheng, R.; Duan, Y.; Zhao, Y.; Chen, P. High efficiency, ultra-low roll-offs in orange phosphorescent organic light-emitting devices using a novel exciplex system. *Organic Electronics* **2022**, *106*, 106536.

- (S6) Hosono, T.; Decarli, N. O.; Crocomo, P. Z.; Goya, T.; de Sousa, L. E.; Tohnai, N.; Minakata, S.; de Silva, P.; Data, P.; Takeda, Y. The regioisomeric effect on the excited-state fate leading to room-temperature phosphorescence or thermally activated delayed fluorescence in a dibenzophenazine-cored donor–acceptor–donor system. *Journal of Materials Chemistry C* **2022**, *10*, 4905–4913.
- (S7) He, J.-L.; Tang, Y.; Zhang, K.; Zhao, Y.; Lin, Y.-C.; Hsu, C.-K.; Chen, C.-H.; Chiu, T.-L.; Lee, J.-H.; Wang, C.-K.; Wu, C.-C.; Fan, J. An extended π -backbone for highly efficient near-infrared thermally activated delayed fluorescence with enhanced horizontal molecular orientation. *Materials Horizons* **2022**, *9*, 772–779.
- (S8) Cao, H.-T.; Wan, J.; Li, B.; Zhang, H.; Xie, L.-H.; Sun, C.; Feng, Q.-Y.; Yu, W.-J.; Huang, W. Highly efficient exciplex-emission from spiro[fluorene-9,9'-xanthene] derivatives. *Dyes and Pigments* **2021**, *185*, 108894.
- (S9) Liu, H.; Bai, Q.; Yao, L.; Zhang, H.; Xu, H.; Zhang, S.; Li, W.; Gao, Y.; Li, J.; Lu, P.; Wang, H.; Yang, B.; Ma, Y. Highly efficient near ultraviolet organic light-emitting diode based on a meta-linked donor–acceptor molecule. *Chemical Science* **2015**, *6*, 3797–3804.
- (S10) Huang, Z.; Wang, B.; Zhang, Q.; Xiang, S.; Lv, X.; Ma, L.; Yang, B.; Gao, Y.; Wang, L. Highly twisted bipolar emitter for efficient nondoped deep-blue electroluminescence. *Dyes and Pigments* **2017**, *140*, 328–336.
- (S11) Ying, S.; Lv, J.; Li, Y.; Huo, Y.; Liu, Y.; Ma, D.; Peng, L.; Yan, S. A large-scale deep-blue tetraphenylbenzene-bridged hybridized local and charge transfer fluorophore exhibiting small efficiency roll-off and low amplified spontaneous emission threshold. *Materials Chemistry Frontiers* **2022**, *6*, 2085–2094.
- (S12) Peng, L.; Lv, J.; Xiao, S.; Huo, Y.; Liu, Y.; Ma, D.; Ying, S.; Yan, S. High-performance non-doped near ultraviolet OLEDs with the EQE $\sim$ 6% and CIEy $\sim$ 0.03 from high-lying reverse intersystem crossing. *Chemical Engineering Journal* **2022**, *450*, 138339.
- (S13) Zhang, Z.; Ding, D.; Wei, Y.; Zhang, J.; Han, C.; Xu, H. Excited-state engineering of universal ambipolar hosts for highly efficient blue phosphorescence and thermally activated delayed fluorescence organic light-emitting diodes. *Chemical Engineering Journal* **2020**, *382*, 122485.
- (S14) Chen, T.; Lu, C.-H.; Huang, C.-W.; Zeng, X.; Gao, J.; Chen, Z.; Xiang, Y.; Zeng, W.; Huang, Z.; Gong, S.; Wu, C.-C.; Yang, C. Tuning the emissive characteristics of TADF emitters by fusing heterocycles with acridine as donors: highly efficient orange to red organic light-emitting diodes with EQE over 20. *Chemical Engineering Journal* **2020**, *382*, 122485.
- (S15) Wang, S.; Shu, H.; Han, X.; Wu, X.; Tong, H.; Wang, L. A highly efficient purely organic room-temperature phosphorescence film based on a selenium-containing emitter for sensitive oxygen detection. *Journal of Materials Chemistry C* **2021**, *9*, 9907–9913.
- (S16) Chen, J.-X.; Xiao, Y.-F.; Wang, K.; Fan, X.-C.; Cao, C.; Chen, W.-C.; Zhang, X.; Shi, Y.-Z.; Yu, J.; Geng, F.-X.; Zhang, X.-H.; Lee, C.-S. Origin of thermally activated delayed fluorescence in a donor–acceptor type emitter with an optimized nearly planar geometry. *Journal of Materials Chemistry C* **2020**, *8*, 13263–13269.
- (S17) Albrecht, K.; Matsuoka, K.; Yokoyama, D.; Sakai, Y.; Nakayama, A.; Fujita, K.; Yamamoto, K. Thermally activated delayed fluorescence OLEDs with fully solution processed organic layers exhibiting nearly 10% efficiency. *Chemical Communications* **2017**, *53*, 2439–2442.
- (S18) Xiao, Z.; Li, N.; Yang, W.; Huang, Z.; Cao, X.; Huang, T.; Chen, Z.; Yang, C. Saccharin-derived multifunctional emitters featuring concurrently room temperature phosphorescence, thermally activated delayed fluorescence and aggregation-induced enhanced emission. *Chemical Engineering Journal* **2021**, *419*, 129628.
- (S19) Nasu, K.; Nakagawa, T.; Nomura, H.; Lin, C.-J.; Cheng, C.-H.; Tseng, M.-R.; Yasuda, T.; Adachi, C. A highly luminescent spiro-anthracenone-based organic light-emitting diode exhibiting thermally activated delayed fluorescence. *Chemical Communications* **2013**, *49*, 10385–10387.
- (S20) Liu, Y.; Qiu, S.; Yu, J.; Ban, X.; Pan, J.; Gao, K.; Zhu, A.; Zhang, D.; Zhang, T.; Tong, Z. Strategy to improve the efficiency of solution-processed phosphorescent organic light-emitting devices by modified TADF host with tert-butyl carbazole. *Tetrahedron* **2021**, *81*, 131869.
- (S21) Jing, Y.-Y.; Tao, X.-D.; Yang, M.-X.; Chen, X.-L.; Lu, C.-Z. Triptycene-imbedded thermally activated delayed fluorescence emitters with excellent film morphologies for applications in efficient nondoped and doped organic light-emitting devices. *Chemical Engineering Journal* **2021**, *413*, 127418.
- (S22) Yun, J. H.; Lee, J. Y. Benzoisoquinoline-1,3-dione acceptor based red thermally activated delayed fluorescent emitters. *Dyes and Pigments* **2017**, *144*, 212–217.
- (S23) Kumar, A.; Oh, J.; Kim, J.; Jung, J.; Lee, M. H. Facile color tuning of thermally activated delayed fluorescence by substituted ortho-carbazole-appended triarylboron emitters. *Dyes and Pigments* **2019**, *168*, 273–280.

- (S24) Pei, R.; Lou, J.; Li, G.; Liu, H.; Yin, X.; Zhou, C.; Wang, Z.; Yang, C. Modulating LUMO extension of Spiro-junction TADF emitters for efficient OLEDs with relieved efficiency Roll-Off. *Chemical Engineering Journal* **2022**, *437*, 135222.
- (S25) Gong, X.; Xiang, Y.; Ning, W.; Zhan, L.; Gong, S.; Xie, G.; Yang, C. A heterocycle fusing strategy for simple construction of efficient solution-processable pure-red thermally activated delayed fluorescence emitters. *Journal of Materials Chemistry C* **2022**, *10*, 15981–15988.
- (S26) Sun, K.; Sun, Y.; Jiang, W.; Huang, S.; Tian, W.; Sun, Y. Highly efficient and color tunable thermally activated delayed fluorescent emitters and their applications for the solution-processed OLEDs. *Dyes and Pigments* **2017**, *139*, 326–333.
- (S27) Ma, F.; Cheng, Y.; Zhang, X.; Gu, X.; Zheng, Y.; Hasrat, K.; Qi, Z. Enhancing performance for blue TADF emitters by introducing intramolecular CH $\cdots$ N hydrogen bonding between donor and acceptor. *Dyes and Pigments* **2019**, *166*, 245–253.
- (S28) Ma, M.; Li, J.; Liu, D.; Li, D.; Dong, R.; Mei, Y. Low efficiency roll-off thermally activated delayed fluorescence emitters for non-doped OLEDs: Substitution effect of thioether and sulfone groups. *Dyes and Pigments* **2021**, *194*, 109649.
- (S29) Chen, M.; Yang, J.; Ye, Z.; Wang, S.; Tang, Z.; Chen, G.; Zheng, Y.; Shi, Y.; Wei, B.; Wong, W.-Y. Extremely low-efficiency roll-off of phosphorescent organic light-emitting diodes at high brightness based on acridine heterocyclic derivatives. *Journal of Materials Chemistry C* **2018**, *6*, 9713–9722.
- (S30) Zheng, X.; Peng, Q.; Lin, J.; Wang, Y.; Zhou, J.; Jiao, Y.; Bai, Y.; Huang, Y.; Li, F.; Liu, X.; Pu, X.; Lu, Z. Simultaneous harvesting of triplet excitons in OLEDs by both guest and host materials with an intramolecular charge-transfer feature via triplet–triplet annihilation. *Journal of Materials Chemistry C* **2015**, *3*, 6970–6978.
- (S31) Khammultri, P.; Chasing, P.; Chitpakdee, C.; Namuangruk, S.; Sudyoasuk, T.; Promarak, V. Red to orange thermally activated delayed fluorescence polymers based on 2-(4-(diphenylamino)-phenyl)-9H-thioxanthen-9-one-10,10-dioxide for efficient solution-processed OLEDs. *RSC Advances* **2021**, *11*, 24794–24806.
- (S32) Tan, J.-H.; Chen, W.-C.; Ni, S.-F.; Qiu, Z.; Zhan, Y.; Yang, Z.; Xiong, J.; Cao, C.; Huo, Y.; Lee, C.-S. Aggregation-state engineering and emission switching in D–A–D' AIEgens featuring dual emission, MCL and white electroluminescence. *Journal of Materials Chemistry C* **2020**, *8*, 8061–8068.
- (S33) Shin, K.; Lee, E.; Lee, T.; Lee, Y. H.; Kim, D. H.; Kim, C.; Jung, J.; Jung, B. J.; Lee, M. H. Efficient TADF from carbon-carbon bonded donor-acceptor molecules based on boron-carbonyl hybrid acceptor. *Dyes and Pigments* **2023**, *209*, 110937.
- (S34) Zhu, X.-D.; Tian, Q.-S.; Zheng, Q.; Wang, Y.-K.; Yuan, Y.; Li, Y.; Jiang, Z.-Q.; Liao, L.-S. Deep-blue thermally activated delayed fluorescence materials with high glass transition temperature. *Journal of Luminescence* **2019**, *206*, 146–153.
- (S35) Li, S.; Sun, T.; Niu, Y.; Tian, Y.; Yan, L.; Shi, W.; Yang, X.; Wei, B. Thermochromic properties of acridine heterocyclic derivatives with a donor–acceptor configuration. *New Journal of Chemistry* **2022**, *46*, 18815–18823.
- (S36) Chen, S.; Zeng, P.; Wang, W.; Wang, X.; Wu, Y.; Lin, P.; Peng, Z. Naphthalimide–arylamine derivatives with aggregation induced delayed fluorescence for realizing efficient green to red electroluminescence. *Journal of Materials Chemistry C* **2019**, *7*, 2886–2897.
- (S37) Zhou, X.; Xiang, Y.; Gong, S.; Chen, Z.; Ni, F.; Xie, G.; Yang, C. Simple construction of deep-red hexaazatrinaphthylene-based thermally activated delayed fluorescence emitters for efficient solution-processed OLEDs with a peak at 692 nm. *Chemical Communications* **2019**, *55*, 14190–14193.
- (S38) Park, E. Y.; Park, J. H.; Kim, Y.-H.; Suh, M. C. The effect of molecular aggregation of thermally activated delayed fluorescence sensitizers for hyperfluorescence in organic light-emitting diodes. *Journal of Materials Chemistry C* **2022**, *10*, 4705–4716.
- (S39) Song, J.; Zhang, F.; Yang, L.; Chen, K.; Li, A.; Sheng, R.; Duan, Y.; Chen, P. Highly efficient, ultralow turn-on voltage red and white organic light-emitting devices based on a novel exciplex host. *Materials Advances* **2021**, *2*, 3677–3684.
- (S40) Dong, X.; Wang, S.; Gui, C.; Shi, H.; Cheng, F.; Tang, B. Z. Synthesis, aggregation-induced emission and thermally activated delayed fluorescence properties of two new compounds based on phenylethene, carbazole and 9,9',10,10'-tetraoxidethianthrene. *Tetrahedron* **2018**, *74*, 497–505.
- (S41) Shen, Y.-F.; Li, M.; Zhao, W.-L.; Wang, Y.-F.; Lu, H.-Y.; Chen, C.-F. Quinoline-based TADF emitters exhibiting aggregation-induced emission for efficient non-doped organic light-emitting diodes. *Materials Chemistry Frontiers* **2021**, *5*, 834–842.
- (S42) Godumala, M.; Choi, S.; Cho, M. J.; Choi, D. H. Recent breakthroughs in thermally activated delayed fluorescence organic light emitting diodes containing non-doped emitting layers. *Journal of Materials Chemistry C* **2019**, *7*, 2172–2198.

- (S43) Chen, X.-L.; Lin, C.-S.; Wu, X.-Y.; Yu, R.; Teng, T.; Zhang, Q.-K.; Zhang, Q.; Yang, W.-B.; Lu, C.-Z. Highly efficient cuprous complexes with thermally activated delayed fluorescence and simplified solution process OLEDs using the ligand as host. *Journal of Materials Chemistry C* **2015**, *3*, 1187–1195.
- (S44) Wei, Z.; Zuo, T.; Jiang, S.; Qi, F.; Yang, M.; Meng, L.; Lu, C.-Z. Theoretically elucidating high photoluminescence performance of dimethylacridan-based blue-color thermally activated delayed fluorescent materials. *New Journal of Chemistry* **2022**, *46*, 3464–3471.
- (S45) Lee, Y. H.; Lee, D.; Lee, T.; Lee, J.; Jung, J.; Yoo, S.; Lee, M. H. Impact of boryl acceptors in para-acridine-appended triarylboron emitters on blue thermally activated delayed fluorescence OLEDs. *Dyes and Pigments* **2021**, *188*, 109224.
- (S46) Wei, X.; Hu, T.; Li, Z.; Liu, Y.; Hu, X.; Gao, H.; Liu, G.; Wang, P.; Yi, Y.; Wang, Y. Rational strategy of exciplex-type thermally activated delayed fluorescent (TADF) emitters: Stacking of donor and acceptor units of the intramolecular TADF molecule. *Chemical Engineering Journal* **2022**, *433*, 133546.
- (S47) Zhang, L.; Wang, Y.-F.; Li, M.; Gao, Q.-Y.; Chen, C.-F. Quinoline-based aggregation-induced delayed fluorescence materials for highly efficient non-doped organic light-emitting diodes. *Chinese Chemical Letters* **2021**, *32*, 740–744.
- (S48) Huang, B.; Jiang, W.; Yang, L.; Zhang, X.; Li, Y.; Chen, W.-C.; Gu, N. The effect of electron donor and acceptor conformations on thermally activated delayed fluorescence. *Organic Electronics* **2022**, *111*, 106657.
- (S49) Wu, L.; Wang, K.; Wang, C.; Fan, X.-C.; Shi, Y.-Z.; Zhang, X.; Zhang, S.-L.; Ye, J.; Zheng, C.-J.; Li, Y.-Q.; Yu, J.; Ou, X.-M.; Zhang, X.-H. Using fluorene to lock electronically active moieties in thermally activated delayed fluorescence emitters for high-performance non-doped organic light-emitting diodes with suppressed roll-off. *Chemical Science* **2021**, *12*, 1495–1502.
- (S50) Keruckiene, R.; Guzauskas, M.; Narbutaitis, E.; Tsiko, U.; Volyniuk, D.; Lee, P.-H.; Chen, C.-H.; Chiu, T.-L.; Lin, C.-F.; Lee, J.-H.; Grazulevicius, J. Exciplex-forming derivatives of 2,7-di-tert-butyl-9,9-dimethylacridan and benzotrifluoride for efficient OLEDs. *Organic Electronics* **2020**, *78*, 105576.
- (S51) Puntischer, H.; Kautny, P.; Stöger, B.; Tissot, A.; Hametner, C.; Hagemann, H. R.; Fröhlich, J.; Baumgartner, T.; Lumpi, D. Structure–property studies of P-triarylamine-substituted dithieno[3,2-b:2',3'-d]phospholes. *RSC Advances* **2015**, *5*, 93797–93807.
- (S52) Bian, C.; Wang, Q.; Ran, Q.; Liu, X.-Y.; Fan, J.; Liao, L.-S. New carbazole-based bipolar hosts for efficient blue phosphorescent organic light-emitting diodes. *Organic Electronics* **2018**, *52*, 138–145.
- (S53) Chen, D.; Cai, X.; Li, X.-L.; He, Z.; Cai, C.; Chen, D.; Su, S.-J. Efficient solution-processed red all-fluorescent organic light-emitting diodes employing thermally activated delayed fluorescence materials as assistant hosts: molecular design strategy and exciton dynamic analysis. *Journal of Materials Chemistry C* **2017**, *5*, 5223–5231.
- (S54) He, Z.; Cai, X.; Wang, Z.; Li, Y.; Xu, Z.; Liu, K.; Chen, D.; Su, S.-J. Sky-blue thermally activated delayed fluorescence material employing a diphenylethyne acceptor for organic light-emitting diodes. *Journal of Materials Chemistry C* **2018**, *6*, 36–42.
- (S55) Paredis, S.; Cardeynals, T.; Deckers, J.; Danos, A.; Vanderzande, D.; Monkman, A. P.; Champagne, B.; Maes, W. Bridge control of photophysical properties in benzothiazole-phenoxazine emitters – from thermally activated delayed fluorescence to room temperature phosphorescence. *Journal of Materials Chemistry C* **2022**, *10*, 4775–4784.
- (S56) Su, R.; Zhao, Y.; Liu, S.; Li, P.; Ma, J.; Zhang, D.; Han, M.; Yu, T. Synthesis and luminescence properties of functionalized V-shaped bis-coumarin derivatives. *New Journal of Chemistry* **2022**, *46*, 6793–6803.
- (S57) Wang, B.; Zheng, Y.; Wang, T.; Ma, D.; Wang, Q. 1,8-Naphthalimide-based hybrids for efficient red thermally activated delayed fluorescence organic light-emitting diodes. *Organic Electronics* **2021**, *88*, 106012.
- (S58) Cai, M.; Auffray, M.; Zhang, D.; Zhang, Y.; Nagata, R.; Lin, Z.; Tang, X.; Chan, C.-Y.; Lee, Y.-T.; Huang, T.; Song, X.; Tsuchiya, Y.; Adachi, C.; Duan, L. Enhancing spin-orbital coupling in deep-blue/blue TADF emitters by minimizing the distance from the heteroatoms in donors to acceptors. *Chemical Engineering Journal* **2021**, *420*, 127591.
- (S59) Rajamalli, P.; Chen, D.; Li, W.; Samuel, I. D. W.; Cordes, D. B.; Slawin, A. M. Z.; Zysman-Colman, E. Enhanced thermally activated delayed fluorescence through bridge modification in sulfone-based emitters employed in deep blue organic light-emitting diodes. *Journal of Materials Chemistry C* **2019**, *7*, 6664–6671.
- (S60) Zhang, N.; Zheng, C.; Chen, Z.; Zhao, J.; Zhang, M.; Yang, H.; He, Z.; Du, X.; Tao, S. Improving the efficiency of exciplex based OLEDs by controlling the different configurations of the donor. *Journal of Materials Chemistry C* **2021**, *9*, 600–608.

- (S61) Chen, J.-X.; Wang, H.; Zhang, X.; Xiao, Y.-F.; Wang, K.; Zhou, L.; Shi, Y.-Z.; Yu, J.; Lee, C.-S.; Zhang, X.-H. Using fullerene fragments as acceptors to construct thermally activated delayed fluorescence emitters for high-efficiency organic light-emitting diodes. *Chemical Engineering Journal* **2022**, *435*, 134731.
- (S62) Yu, Y.-J.; Tang, X.; Ge, H.-T.; Yuan, Y.; Jiang, Z.-Q.; Liao, L.-S. Fluorenone-based thermally activated delayed fluorescence materials for orange-red emission. *Organic Electronics* **2019**, *73*, 240–246.
- (S63) Shakeel, U.; Singh, J. Study of processes of reverse intersystem crossing (RISC) and thermally activated delayed fluorescence (TADF) in organic light emitting diodes (OLEDs). *Organic Electronics* **2018**, *59*, 121–124.
- (S64) Ma, F.; Ji, H.; Zhang, D.; Xue, K.; Zhang, P.; Qi, Z.; Zhu, H. Adjusting the photophysical properties of AIE-active TADF emitters from through-bond to through-space charge transfer for high-performance solution-processed OLEDs. *Dyes and Pigments* **2021**, *188*, 109208.
- (S65) Rajamalli, P.; Thangaraji, V.; Senthilkumar, N.; Ren-Wu, C.-C.; Lin, H.-W.; Cheng, C.-H. Thermally activated delayed fluorescence emitters with a m,m-di-tert-butyl-carbazolyl benzoylpyridine core achieving extremely high blue electroluminescence efficiencies. *Journal of Materials Chemistry C* **2017**, *5*, 2919–2926.
- (S66) Maeng, J. H.; Ahn, D. H.; Lee, H.; Jung, Y. H.; Karthik, D.; Lee, J. Y.; Kwon, J. H. Rigid indolocarbazole donor moiety for highly efficient thermally activated delayed fluorescent device. *Dyes and Pigments* **2020**, *180*, 108485.
- (S67) Lin, C.; Wu, Z.; Liu, J.; Deng, W.-T.; Zhuang, Y.; Xuan, T.; Xue, J.; Zhang, L.; Wei, G.; Xie, R.-J. Extremely low efficiency roll-off in vacuum- and solution-processed deep-red/near-infrared OLEDs based on 1,8-naphthalimide TADF emitters. *Journal of Luminescence* **2022**, *243*, 118683.
- (S68) Huang, R.; Chen, H.; Liu, H.; Zhuang, Z.; Wang, J.; Yu, M.; Yang, D.; Ma, D.; Zhao, Z.; Zhong Tang, B. Creating efficient delayed fluorescence luminogens with acridine-based spiro donors to improve horizontal dipole orientation for high-performance OLEDs. *Chemical Engineering Journal* **2022**, *435*, 134934.
- (S69) Sharma, S.; Pal, A. K. Recent advances of NIR-TADF ($\lambda_{\text{maxPL/EL}} > 700$ nm) emitters and their applications in OLEDs. *Journal of Materials Chemistry C* **2022**, *10*, 15681–15707.
- (S70) Pandidurai, J.; Jayakumar, J.; Senthilkumar, N.; Cheng, C.-H. Effects of intramolecular hydrogen bonding on the conformation and luminescence properties of dibenzoylpyridine-based thermally activated delayed fluorescence materials. *Journal of Materials Chemistry C* **2019**, *7*, 13104–13110.
- (S71) Liu, T.; Xie, G. Exploiting new feasibility of a phenylquinoline unit for establishing efficient green thermally activated delayed fluorescent emitter with short delayed fluorescent lifetime. *Organic Electronics* **2022**, *106*, 106518.
- (S72) Ma, Y.-P.; Wang, G.; Zhao, M.; Shi, Z.-F.; Miao, Y.; Cao, X.-P.; Zhang, H.-L. A red thermally activated delayed fluorescence emitter based on benzo[c][1,2,5]thiadiazole. *Dyes and Pigments* **2023**, *212*, 111084.
- (S73) Meng, G.; Zhang, D.; Wei, J.; Zhang, Y.; Huang, T.; Liu, Z.; Yin, C.; Hong, X.; Wang, X.; Zeng, X.; Yang, D.; Ma, D.; Li, G.; Duan, L. Highly efficient and stable deep-blue OLEDs based on narrowband emitters featuring an orthogonal spiro-configured indolo[3,2,1-de]acridine structure. *Chemical Science* **2022**, *13*, 5622–5630.
- (S74) Li, Y.; Chen, T.; Huang, M.; Gu, Y.; Gong, S.; Xie, G.; Yang, C. Tuning the twist angle of thermally activated delayed fluorescence molecules via a dendronization strategy: high-efficiency solution-processed non-doped OLEDs. *Journal of Materials Chemistry C* **2017**, *5*, 3480–3487.
- (S75) Ni, F.; Wu, Z.; Zhu, Z.; Chen, T.; Wu, K.; Zhong, C.; An, K.; Wei, D.; Ma, D.; Yang, C. Teaching an old acceptor new tricks: rationally employing 2,1,3-benzothiadiazole as input to design a highly efficient red thermally activated delayed fluorescence emitter. *Journal of Materials Chemistry C* **2017**, *5*, 1363–1368.
- (S76) Lee, J.; Shizu, K.; Tanaka, H.; Nakanotani, H.; Yasuda, T.; Kaji, H.; Adachi, C. Controlled emission colors and singlet–triplet energy gaps of dihydrophenazine-based thermally activated delayed fluorescence emitters. *Journal of Materials Chemistry C* **2015**, *3*, 2175–2181.
- (S77) Hladka, I.; Volyniuk, D.; Bezvikonnyi, O.; Kinzhybalo, V.; Bednarchuk, T. J.; Danyliv, Y.; Lytvyn, R.; Lazauskas, A.; Grazulevicius, J. V. Polymorphism of derivatives of tert-butyl substituted acridan and perfluorobiphenyl as sky-blue OLED emitters exhibiting aggregation induced thermally activated delayed fluorescence. *Journal of Materials Chemistry C* **2018**, *6*, 13179–13189.
- (S78) Hu, J.; Zhang, X.; Zhang, D.; Cao, X.; Jiang, T.; Zhang, X.; Tao, Y. Linkage modes on phthaloyl/triphenylamine hybrid compounds: Multi-functional AIE luminogens, non-doped emitters and organic hosts for highly efficient solution-processed delayed fluorescence OLEDs. *Dyes and Pigments* **2017**, *137*, 480–489.

- (S79) Huang, Z.; Xiang, S.; Zhang, Q.; Lv, X.; Ye, S.; Guo, R.; Wang, L. Highly efficient green organic light emitting diodes with phenanthroimidazole-based thermally activated delayed fluorescence emitters. *Journal of Materials Chemistry C* **2018**, *6*, 2379–2386.
- (S80) Huang, B.; Ban, X.; Sun, K.; Ma, Z.; Mei, Y.; Jiang, W.; Lin, B.; Sun, Y. Thermally activated delayed fluorescence materials based on benzophenone derivative as emitter for efficient solution-processed non-doped green OLED. *Dyes and Pigments* **2016**, *133*, 380–386.
- (S81) Chen, X.; Yang, Z.; Xie, Z.; Zhao, J.; Yang, Z.; Zhang, Y.; Aldred, M. P.; Chi, Z. An efficient yellow thermally activated delayed fluorescence emitter with universal applications in both doped and non-doped organic light-emitting diodes. *Materials Chemistry Frontiers* **2018**, *2*, 1017–1023.
- (S82) Yang, D.; Kim, J.-M.; Huh, J.-S.; Kim, J.-J.; Hong, J.-I. The effect of the electron-donor ability on the OLED efficiency of twisted donor-acceptor type emitters. *Organic Electronics* **2021**, *95*, 106187.
- (S83) Song, X.; Liu, Z.; Lu, M.; Zou, S.; Guo, F.; Gao, S.; Zhao, Z.; Tang, B. Z.; Zhang, Y. Efficient thermally activated delayed fluorescent emitter based on a spiro-type benzo[b]acridine donor and a benzophenone acceptor. *Journal of Materials Chemistry C* **2022**, *10*, 17550–17556.
- (S84) Chen, Y.; Li, N.; Huang, Z.; Xie, G.; Yang, C. Molecular engineering by π -linkers enables delayed fluorescence emitters for high-efficiency sky-blue solution-processed OLEDs. *Chemical Engineering Journal* **2022**, *430*, 133078.
- (S85) Chen, C.; Lu, H.-Y.; Wang, Y.-F.; Li, M.; Shen, Y.-F.; Chen, C.-F. Naphthyridine-based thermally activated delayed fluorescence emitters for multi-color organic light-emitting diodes with low efficiency roll-off. *Journal of Materials Chemistry C* **2019**, *7*, 4673–4680.
- (S86) Wu, P.; Xie, F.-m.; Wei, H.-x.; Li, Y.-Q.; Dai, G.-l.; Wang, Y.; Tang, J.-X.; Zhao, X. Thermally activated delayed fluorescent emitters based on 3-(phenylsulfonyl)pyridine. *Chemical Physics Letters* **2021**, *771*, 138474.
- (S87) Park, J.; Moon, J.; Lim, J.; Woo, J.; Yoon, S. S.; Lee, J. Y. Fine-tuned asymmetric blue multiple resonance thermally activated delayed fluorescence emitters with high efficiency and narrow emission band. *Journal of Materials Chemistry C* **2022**, *10*, 12300–12306.
- (S88) Luo, J.; Gong, S.; Gu, Y.; Chen, T.; Li, Y.; Zhong, C.; Xie, G.; Yang, C. Multi-carbazole encapsulation as a simple strategy for the construction of solution-processed, non-doped thermally activated delayed fluorescence emitters. *Journal of Materials Chemistry C* **2016**, *4*, 2442–2446.
- (S89) Zheng, X.; Cao, F.; Wang, C.; Tsuboi, T.; Zhu, Y.; Ai, Q.; Deng, C.; Wang, D.; Su, L.; Liu, Z.; Zhang, Q. Expanding the hole delocalization range in excited molecules for stable organic light-emitting diodes employing thermally activated delayed fluorescence. *Journal of Materials Chemistry C* **2020**, *8*, 10021–10030.
- (S90) Sun, J. W.; Baek, J. Y.; Kim, K.-H.; Huh, J.-S.; Kwon, S.-K.; Kim, Y.-H.; Kim, J.-J. Azasilene-based thermally activated delayed fluorescence emitters for blue organic light emitting diodes. *Journal of Materials Chemistry C* **2017**, *5*, 1027–1032.
- (S91) Kim, J.; Chung, W.; Kim, J.; Lee, J. Concentration quenching-resistant multiresonance thermally activated delayed fluorescence emitters. *Materials Today Energy* **2021**, *21*, 100792.
- (S92) Lee, J.; Aizawa, N.; Yasuda, T. Molecular engineering of phosphacycle-based thermally activated delayed fluorescence materials for deep-blue OLEDs. *Journal of Materials Chemistry C* **2018**, *6*, 3578–3583.
- (S93) He, Z.; Li, J.; Liu, D.; Mei, Y.; Wan, H. Performance enhancement in thermally activated delayed fluorescence emitters by multi-fold layout. *Dyes and Pigments* **2023**, *212*, 111104.
- (S94) Chen, J.-X.; Tao, W.-W.; Wang, K.; Zheng, C.-J.; Liu, W.; Li, X.; Ou, X.-M.; Zhang, X.-H. Highly efficient thermally activated delayed fluorescence emitters based on novel Indolo[2,3-b]acridine electron-donor. *Organic Electronics* **2018**, *57*, 327–334.
- (S95) Zeng, X.; Pan, K.-C.; Lee, W.-K.; Gong, S.; Ni, F.; Xiao, X.; Zeng, W.; Xiang, Y.; Zhan, L.; Zhang, Y.; Wu, C.-C.; Yang, C. High-efficiency pure blue thermally activated delayed fluorescence emitters with a preferentially horizontal emitting dipole orientation via a spiro-linked double D–A molecular architecture. *Journal of Materials Chemistry C* **2019**, *7*, 10851–10859.
- (S96) Hua, T.; Liu, Y.-C.; Huang, C.-W.; Li, N.; Zhou, C.; Huang, Z.; Cao, X.; Wu, C.-C.; Yang, C. High-efficiency and low roll-off deep-blue OLEDs enabled by thermally activated delayed fluorescence emitter with preferred horizontal dipole orientation. *Chemical Engineering Journal* **2022**, *433*, 133598.

- (S97) Ward, J. S.; Danos, A.; Stachelek, P.; Fox, M. A.; Batsanov, A. S.; Monkman, A. P.; Bryce, M. R. Exploiting trifluoromethyl substituents for tuning orbital character of singlet and triplet states to increase the rate of thermally activated delayed fluorescence. *Materials Chemistry Frontiers* **2020**, *4*, 3602–3615.
- (S98) Chen, J.-X.; Wang, H.; Zhou, L.; Wang, K.; Yu, J.; Zhang, X.-H. Structure-property investigation of two red TADF isomers with different D-A conjugation for superior exciton utilization. *Dyes and Pigments* **2023**, *208*, 110801.
- (S99) Hong, X.; Zhang, D.; Yin, C.; Wang, Q.; Zhang, Y.; Huang, T.; Wei, J.; Zeng, X.; Meng, G.; Wang, X.; Li, G.; Yang, D.; Ma, D.; Duan, L. TADF molecules with π -extended acceptors for simplified high-efficiency blue and white organic light-emitting diodes. *Chem* **2022**, *8*, 1705–1719.
- (S100) Zhang, Y.; Zhang, J.; Shi, C.; Sun, N.; Wang, Q. Dipyrido[3,2-a:2',3'-c]phenazine acceptor based thermally activated delayed fluorescence emitters. *Dyes and Pigments* **2022**, *206*, 110634.
- (S101) Zhang, D.; Cai, M.; Bin, Z.; Zhang, Y.; Zhang, D.; Duan, L. Highly efficient blue thermally activated delayed fluorescent OLEDs with record-low driving voltages utilizing high triplet energy hosts with small singlet–triplet splittings. *Chemical Science* **2016**, *7*, 3355–3363.
- (S102) Maeng, J. H.; Braveenth, R.; Jung, Y. H.; Hwang, S. J.; Lee, H.; Min, H. L.; Kim, J. Y.; Han, C. W.; Kwon, J. H. Efficiency enhancement in orange red thermally activated delayed fluorescence OLEDs by using a rigid di-indolocarbazole donor moiety. *Dyes and Pigments* **2021**, *194*, 109580.
- (S103) Zhou, L.; Wang, H.; Shi, Y.-Z.; Fan, X.-C.; Chen, J.-X.; Wang, K.; Yu, J.; Zhang, X.-H. Controlling the conjugation extension inside acceptors for enhancing reverse intersystem crossing of red thermally activated delayed fluorescence emitters. *Chemical Engineering Journal* **2022**, *440*, 135775.
- (S104) Zhou, C.; Chen, W.-C.; Liu, H.; Cao, X.; Li, N.; Zhang, Y.; Lee, C.-S.; Yang, C. Isomerization enhanced quantum yield of dibenzo[a,c]phenazine-based thermally activated delayed fluorescence emitters for highly efficient orange OLEDs. *Journal of Materials Chemistry C* **2020**, *8*, 9639–9645.
- (S105) Lee, D. R.; Han, S. H.; Lee, C. W.; Lee, J. Y. Bis(diphenyltriazine) as a new acceptor of efficient thermally activated delayed fluorescent emitters. *Dyes and Pigments* **2018**, *151*, 75–80.
- (S106) Lee, H. L.; Han, S. H.; Hong, W. P.; Song, O.-K.; Lee, J. Y. π linker mediated coupling of two emitting units for improved efficiency in thermally activated delayed fluorescent emitters. *Dyes and Pigments* **2019**, *162*, 36–42.
- (S107) Sohn, S.; Hyun Koh, B.; Baek, J. Y.; Chan Byun, H.; Lee, J. H.; Shin, D.-S.; Ahn, H.; Lee, H.-K.; Hwang, J.; Jung, S.; Kim, Y.-H. Synthesis and characterization of diphenylamine derivative containing malononitrile for thermally activated delayed fluorescent emitter. *Dyes and Pigments* **2017**, *140*, 14–21.
- (S108) Li, Q.; Xu, J.; Tan, S.; Dai, Y.; Xue, J.; Qiao, J. Approaching 800 nm near-infrared thermally activated delay fluorescence materials via simple malononitrile condensation strategy. *Organic Electronics* **2022**, *110*, 106645.
- (S109) Leitonas, K.; Guzauskas, M.; Tsiko, U.; Simokaitiene, J.; Volyniuk, D.; Grazulevicius, J. V. White vertical organic permeable-base light-emitting transistors obtained by mixing of blue exciton and orange interface exciplex emissions. *Journal of Materials Chemistry C* **2022**, *10*, 9786–9793.
- (S110) Hong, G.; Si, C.; Gupta, A. K.; Bizzarri, C.; Nieger, M.; Samuel, I. D. W.; Zysman-Colman, E.; Bräse, S. Fluorinated dibenzo[a, c]-phenazine-based green to red thermally activated delayed fluorescent OLED emitters. *Journal of Materials Chemistry C* **2022**, *10*, 4757–4766.
- (S111) Tsiko, U.; Volyniuk, D.; Andruleviciene, V.; Leitonas, K.; Sych, G.; Bezikonnyi, O.; Jasinskas, V.; Gulbinas, V.; Stakhira, P.; Grazulevicius, J. Triphenylamino or 9-phenyl carbazolyl-substituted pyrimidine-5-carbonitriles as bipolar emitters and hosts with triplet harvesting abilities. *Materials Today Chemistry* **2022**, *25*, 100955.
- (S112) Zhang, Y.-L.; Ran, Q.; Wang, Q.; Tian, Q.-S.; Kong, F.-C.; Fan, J.; Liao, L.-S. High-performance sky-blue phosphorescent organic light-emitting diodes employing wide-bandgap bipolar host materials with thermally activated delayed fluorescence characteristics. *Organic Electronics* **2020**, *81*, 105660.
- (S113) Chen, Y.; Wang, Y.-X.; Lu, C.-W.; Su, H.-C. Deep-red light-emitting electrochemical cells based on phosphor-sensitized thermally activated delayed fluorescence. *Journal of Materials Chemistry C* **2022**, *10*, 11211–11219.
- (S114) Qu, C.; Wang, H.; Man, Y.; Li, Z.; Ma, P.; Chang, P.; Li, X.; Han, C.; Pang, Y.; Xu, H. Asymmetric intramolecular charge transfer enables highly efficient red thermally activated delayed fluorescent emitters. *Chemical Engineering Journal* **2023**, *457*, 141061.

- (S115) Chen, J.-X.; Tao, W.-W.; Xiao, Y.-F.; Tian, S.; Chen, W.-C.; Wang, K.; Yu, J.; Geng, F.-X.; Zhang, X.-H.; Lee, C.-S. Isomeric thermally activated delayed fluorescence emitters based on indolo[2,3-b]acridine electron-donor: a compromising optimization for efficient orange-red organic light-emitting diodes. *Journal of Materials Chemistry C* **2019**, *7*, 2898–2904.
- (S116) Sun, K.; Tian, W.; Ge, C.; Gu, F.; Zhou, Y.; Wang, W.; Cai, Z.; Jiang, W.; Sun, Y. Creation of efficient solution-processed OLEDs via a strategy of the host-guest system constructing with two small cross-linkable TADF molecules. *Organic Electronics* **2022**, *101*, 106417.
- (S117) Tsiko, U.; Bezvikonnyi, O.; Sych, G.; Keruckiene, R.; Volyniuk, D.; Simokaitiene, J.; Danyliv, I.; Danyliv, Y.; Bucinskas, A.; Tan, X.; Grazulevicius, J. V. Multifunctional derivatives of pyrimidine-5-carbonitrile and differently substituted carbazoles for doping-free sky-blue OLEDs and luminescent sensors of oxygen. *Journal of Advanced Research* **2021**, *33*, 41–51.
- (S118) Liang, J.-X.; Pan, Z.-H.; Zhang, K.; Yang, D.; Tai, J.-W.; Wang, C.-K.; Fung, M.-K.; Ma, D.; Fan, J. A facile method to achieve red thermally activated delayed fluorescence emitters with EQE over 30% aspect ratio engineering. *Chemical Engineering Journal* **2023**, *457*, 141074.
- (S119) Kim, C. Y.; Yoon, J.; Choi, S.; Jeong, C. H.; Kim, H. J.; Jeong, J.-E.; Kim, S. K.; Lee, H.; Woo, H. Y.; Kwon, J. H.; Cho, M. J.; Choi, D. H. Blue-emitting dendritic molecule with dual functionality as host and dopant for solution-processed white OLEDs with red-emitting material. *Synthetic Metals* **2019**, *258*, 116198.
- (S120) Liu, H.; Li, J.; Chen, W.-C.; Chen, Z.; Liu, Z.; Zhan, Q.; Cao, X.; Lee, C.-S.; Yang, C. Modulating the acceptor structure of dicyanopyridine based TADF emitters: Nearly 30% efficiency roll-off in OLED. *Chemical Engineering Journal* **2020**, *401*, 126107.
- (S121) Li, N.; Chai, D.; Chen, Z.; Zhou, C.; Ni, F.; Huang, Z.; Cao, X.; Xie, G.; Li, K.; Yang, C. Molecular engineering by If-Bond spacer enables solution-processable host materials for TADF emitter towards high-performance OLEDs. *Chemical Engineering Journal* **2020**, *396*, 125276.
- (S122) Zhang, H.-Y.; Yang, H.-Y.; Zhang, M.; Lin, H.; Tao, S.-L.; Zheng, C.-J.; Zhang, X.-H. A novel orange-red thermally activated delayed fluorescence emitter with high molecular rigidity and planarity realizing 32.5% quantum efficiency in organic light-emitting diodes. *Materials Horizons* **2022**, *9*, 2425–2432.
- (S123) Li, J.; Chen, W.-C.; Liu, H.; Chen, Z.; Chai, D.; Lee, C.-S.; Yang, C. Double-twist pyridine-carbonitrile derivatives yielding excellent thermally activated delayed fluorescence emitters for high-performance OLEDs. *Journal of Materials Chemistry C* **2020**, *8*, 602–606.
- (S124) Liu, H.; Li, J.; Li, G.; Zhang, B.; Zhan, Q.; Liu, Z.; Zhou, C.; Li, K.; Wang, Z.; Yang, C. A simple strategy to achieve efficient thermally activated delayed fluorescent emitters via enhancing electron donating ability of donors. *Dyes and Pigments* **2020**, *180*, 108521.
- (S125) Benjamin, H.; Zheng, Y.; Kozhevnikov, V. N.; Siddle, J. S.; O'Driscoll, L. J.; Fox, M. A.; Batsanov, A. S.; Griffiths, G. C.; Dias, F. B.; Monkman, A. P.; Bryce, M. R. Unusual dual-emissive heteroleptic iridium complexes incorporating TADF cyclometalating ligands. *Dalton Transactions* **2020**, *49*, 2190–2208.
- (S126) Zhen, Y.; Zhang, F.; Liu, H.; Yan, Y.; Li, X.; Wang, S. Impact of peripheral groups on pyrimidine acceptor-based HLCT materials for efficient deep blue OLED devices. *Journal of Materials Chemistry C* **2022**, *10*, 9953–9960.
- (S127) Lv, X.; Sun, S.; Zhang, Q.; Ye, S.; Liu, W.; Wang, Y.; Guo, R.; Wang, L. A strategy to construct multifunctional TADF materials for deep blue and high efficiency yellow fluorescent devices. *Journal of Materials Chemistry C* **2020**, *8*, 4818–4826.
- (S128) Sun, H.; Tan, X.; Sang, S.; Liu, Q.; Sun, P.; Zhang, J.; Hang, X.-C.; Chen, F.; Chen, Z.-K. A novel design strategy for deeper blue and more stable thermally activated delayed fluorescent emitters. *Organic Electronics* **2020**, *78*, 105610.
- (S129) Barman, D.; Narang, K.; Gogoi, R.; Barman, D.; Iyer, P. K. Exceptional class of thermally activated delayed fluorescent emitters that display pure blue, near-IR, circularly polarized luminescence and multifunctional behaviour for highly efficient and stable OLEDs. *Journal of Materials Chemistry C* **2022**, *10*, 8536–8583.
- (S130) Tao, W.-W.; Wang, K.; Chen, J.-X.; Shi, Y.-Z.; Liu, W.; Zheng, C.-J.; Li, Y.-Q.; Yu, J.; Ou, X.-M.; Zhang, X.-H. Dibenzofuran/dibenzothiophene as the secondary electron-donors for highly efficient blue thermally activated delayed fluorescence emitters. *Journal of Materials Chemistry C* **2019**, *7*, 4475–4483.
- (S131) Kim, G. H.; Lampande, R.; Im, J. B.; Lee, J. M.; Lee, J. Y.; Kwon, J. H. Controlling the exciton lifetime of blue thermally activated delayed fluorescence emitters using a heteroatom-containing pyridoindole donor moiety. *Materials Horizons* **2017**, *4*, 619–624.
- (S132) Lee, H. L.; Chung, W. J.; Lee, J. Y. Selective efficiency boosting in thermally activated delayed fluorescence emitters by a secondary donor. *Chemical Engineering Journal* **2021**, *408*, 127293.

- (S133) Zassowski, P.; Ledwon, P.; Kurowska, A.; Herman, A. P.; Jarosz, T.; Lapkowski, M.; Cherpak, V.; Stakhira, P.; Peculyte, L.; Volyniuk, D.; Grazulevicius, J. V. Efficient synthesis and structural effects of ambipolar carbazole derivatives. *Synthetic Metals* **2017**, *223*, 1–11.
- (S134) Huang, T.; Song, X.; Cai, M.; Zhang, D.; Duan, L. Improving reverse intersystem crossing in exciplex-forming hosts by introducing heavy atom effect. *Materials Today Energy* **2021**, *21*, 100705.
- (S135) Li, B.; Song, X.; Jiang, X.; Li, Z.; Guo, F.; Wang, Y.; Zhao, L.; Zhang, Y. Stable deep blue organic light emitting diodes with CIE of y < 0.10 based on quinazoline and carbazole units. *Chinese Chemical Letters* **2020**, *31*, 1188–1192.
- (S136) Yu, L.; Wu, Z.; Zhong, C.; Xie, G.; Wu, K.; Ma, D.; Yang, C. Tuning the emission from local excited-state to charge-transfer state transition in quinoxaline-based butterfly-shaped molecules: Efficient orange OLEDs based on thermally activated delayed fluorescence emitter. *Dyes and Pigments* **2017**, *141*, 325–332.
- (S137) Choi, S. H.; Lee, C. H.; Adachi, C.; Lee, S. Y. Highly effective nicotinonitrile-derivatives-based thermally activated delayed fluorescence emitter with asymmetric molecular architecture for high-performance organic light-emitting diodes. *Dyes and Pigments* **2020**, *172*, 107849.
- (S138) Zhang, Q.; Sun, S.; Chung, W. J.; Yoon, S. J.; Wang, Y.; Guo, R.; Ye, S.; Lee, J. Y.; Wang, L. Highly efficient TADF OLEDs with low efficiency roll-off based on novel acridine–carbazole hybrid donor-substituted pyrimidine derivatives. *Journal of Materials Chemistry C* **2019**, *7*, 12248–12255.
- (S139) Serevičius, T.; Skaisgiris, R.; Fiodorova, I.; Steckis, V.; Dodonova, J.; Banevičius, D.; Kazlauskas, K.; Juršėnas, S.; Tumkevičius, S. Achieving efficient deep-blue TADF in carbazole-pyrimidine compounds. *Organic Electronics* **2020**, *82*, 105723.
- (S140) Zhao, Y.; Fu, C.; Fu, L.; Liu, Y.; Lu, Z.; Pu, X. Data-driven machine learning models for quick prediction of thermal stability properties of OLED materials. *Materials Today Chemistry* **2021**, *22*, 100625.
- (S141) Wang, Z.; Yao, J.; Zhan, L.; Gong, S.; Ma, D.; Yang, C. Purine-based thermally activated delayed fluorescence emitters for efficient organic light-emitting diodes. *Dyes and Pigments* **2020**, *180*, 108437.
- (S142) Xie, W.; Li, M.; Peng, X.; Qiu, W.; Gan, Y.; Chen, Z.; He, Y.; Li, W.; Liu, K.; Wang, L.; Gu, Q.; Su, S.-J. “On-off” switch between red thermally activated delayed fluorescence and conventional fluorescence by isomeric regulation. *Chemical Engineering Journal* **2021**, *425*, 131510.
- (S143) Pandidurai, J.; Jayakumar, J.; Chen, Y.-K.; Hsieh, C.-M.; Cheng, C.-H. Constitutional isomers of carbazole–benzoyl-pyrimidine-based thermally activated delayed fluorescence emitters for efficient OLEDs. *Journal of Materials Chemistry C* **2021**, *9*, 15900–15909.
- (S144) Yu, L.; Wu, Z.; Xie, G.; Zeng, W.; Ma, D.; Yang, C. Molecular design to regulate the photophysical properties of multifunctional TADF emitters towards high-performance TADF-based OLEDs with EQEs up to 22.4% efficiency roll-offs. *Chemical Science* **2018**, *9*, 1385–1391.
- (S145) Kim, D. H.; Inada, K.; Zhao, L.; Komino, T.; Matsumoto, N.; Ribierre, J. C.; Adachi, C. Organic light emitting diodes with horizontally oriented thermally activated delayed fluorescence emitters. *Journal of Materials Chemistry C* **2017**, *5*, 1216–1223.
- (S146) Li, P.; Xiang, Y.; Gong, S.; Lee, W.-K.; Huang, Y.-H.; Wang, C.-Y.; Yang, C.; Wu, C.-C. Quinazoline-based thermally activated delayed fluorescence emitters for high-performance organic light-emitting diodes with external quantum efficiencies about 28%. *Journal of Materials Chemistry C* **2021**, *9*, 12633–12641.
- (S147) Choi, S. H.; Lee, C. H.; Adachi, C.; Lee, S. Y. Molecular design of highly effective thermally activated delayed fluorescence emitters based on ortho-substituted donor-acceptor-donor pyridinecarbonitrile derivatives and their application for high-performance OLEDs. *Dyes and Pigments* **2019**, *171*, 107775.
- (S148) Lee, K. H.; Jeon, S. O.; Chung, Y. S.; Numata, M.; Lee, H.; Lee, E. K.; Kwon, E. S.; Sim, M.; Choi, H.; Lee, J. Y. An excited state managing molecular design platform of blue thermally activated delayed fluorescence emitters by π -linker engineering. *Journal of Materials Chemistry C* **2020**, *8*, 1736–1745.
- (S149) Zhan, L.; Ning, W.; Gong, S.; Xie, G.; Yang, C. Difluoroboron locking tactic enhances photo- and electroluminescence of TADF emitter. *Dyes and Pigments* **2021**, *192*, 109392.
- (S150) Achelle, S.; Hodée, M.; Massue, J.; Fihey, A.; Katan, C. Diazine-based thermally activated delayed fluorescence chromophores. *Dyes and Pigments* **2022**, *200*, 110157.
- (S151) Zhang, X.; Wang, D.; Lei, Y.; Liu, M.; Cai, Z.; Wu, H.; Shen, G.; Huang, X.; Dong, Y. Selenium atoms induce organic doped systems to produce pure phosphorescence emission. *Chemical Communications* **2022**, *58*, 1179–1182.

- (S152) Ban, X.; Qiu, S.; Cao, Q.; Zhang, K.; Zhou, T.; Xu, H.; Pei, M.; Ge, F.; Tong, Z.; Jiang, W. Exciplex polymer with strong AIE for constructing fully-solution-processed organic light-emitting diodes with 100-fold efficiency improvement compared to physically blended exciplex. *Polymer Chemistry* **2022**, *13*, 6500–6511.
- (S153) Sun, K.; Cai, Z.; Jiang, J.; Tian, W.; Guo, W.; Shao, J.; Jiang, W.; Sun, Y. Multi-substituted dibenzo[a,c]phenazine derivatives as solution-processable thermally activated delayed fluorescence materials for orange-red organic light-emitting diodes. *Dyes and Pigments* **2020**, *173*, 107957.
- (S154) Yan, X.; Cheng, Z.; Yang, T.; Li, C.; Wang, Y. Carbazole-benzonitrile based organic semiconductors: Synthesis, characterization and electroluminescent property. *Organic Electronics* **2022**, *102*, 106445.
- (S155) Patil, V. V.; Lee, K. H.; Lee, J. Y. Isomeric fused benzocarbazole as a chromophore for blue fluorescent organic light-emitting diodes. *Journal of Materials Chemistry C* **2020**, *8*, 8320–8327.
- (S156) Patil, V. V.; Lee, K. H.; Lee, J. Y. 11,11-Dimethyl-11H-indeno[1,2-b]indolo[1,2,3-jk]carbazole: A rigid chromophore with novel amalgamation strategy for long lifetime blue fluorescent organic light-emitting diodes. *Chemical Engineering Journal* **2020**, *395*, 125125.
- (S157) Zhao, K.; Omar, O. H.; Nematiram, T.; Padula, D.; Troisi, A. Novel thermally activated delayed fluorescence materials by high-throughput virtual screening: going beyond donor-acceptor design. *Journal of Materials Chemistry C* **2021**, *9*, 3324–3333.
- (S158) Zhu, X.-D.; Peng, C.-C.; Kong, F.-C.; Yang, S.-Y.; Li, H.-C.; Kumar, S.; Wang, T.-T.; Jiang, Z.-Q.; Liao, L.-S. Acceptor modulation for improving a spiro-type thermally activated delayed fluorescence emitter. *Journal of Materials Chemistry C* **2020**, *8*, 8579–8584.
- (S159) Liu, Y.; Yang, J.; Mao, Z.; Wang, Y.; Zhao, J.; Su, S.-J.; Chi, Z. Isomeric thermally activated delayed fluorescence emitters for highly efficient organic light-emitting diodes. *Chemical Science* **2023**, *14*, 1551–1556.
- (S160) Kothavale, S.; Lim, J.; Yeob Lee, J. Rational design of CN substituted dibenzo[a,c]phenazine acceptor for color tuning of thermally activated delayed fluorescent emitters. *Chemical Engineering Journal* **2022**, *431*, 134216.
- (S161) Zhao, Y.; Wang, W.; Gui, C.; Fang, L.; Zhang, X.; Wang, S.; Chen, S.; Shi, H.; Tang, B. Z. Thermally activated delayed fluorescence material with aggregation-induced emission properties for highly efficient organic light-emitting diodes. *Journal of Materials Chemistry C* **2018**, *6*, 2873–2881.
- (S162) Kothavale, S.; Chung, W. J.; Lee, J. Y. High efficiency and long lifetime orange-red thermally activated delayed fluorescent organic light emitting diodes by donor and acceptor engineering. *Journal of Materials Chemistry C* **2021**, *9*, 528–536.
- (S163) Xiang, Y.; Zhu, Z.-L.; Xie, D.; Gong, S.; Wu, K.; Xie, G.; Lee, C.-S.; Yang, C. Revealing the new potential of an indandione unit for constructing efficient yellow thermally activated delayed fluorescence emitters with short emissive lifetimes. *Journal of Materials Chemistry C* **2018**, *6*, 7111–7118.
- (S164) Kumar, S.; Tourneur, P.; Adsetts, J. R.; Wong, M. Y.; Rajamalli, P.; Chen, D.; Lazzaroni, R.; Viville, P.; Cordes, D. B.; Slawin, A. M. Z.; Olivier, Y.; Cornil, J.; Ding, Z.; Zysman-Colman, E. Photoluminescence and electrochemiluminescence of thermally activated delayed fluorescence (TADF) emitters containing diphenylphosphine chalcogenide-substituted carbazole donors. *Journal of Materials Chemistry C* **2022**, *10*, 4646–4667.
- (S165) Huang, F.-X.; Li, H.-Z.; Xie, F.-M.; Zeng, X.-Y.; Li, Y.-Q.; Hu, Y.-Y.; Tang, J.-X.; Zhao, X. Efficient orange-red thermally activated delayed fluorescence material containing a cyano group. *Dyes and Pigments* **2021**, *195*, 109731.
- (S166) Tai, J.-W.; Tang, Y.; Zhang, K.; Yang, C.-Z.; Pan, Z.-H.; Lin, Y.-C.; Shih, Y.-W.; Chen, C.-H.; Chiu, T.-L.; Lee, J.-H.; Wang, C.-K.; Wu, C.-C.; Fan, J. 13.2 *Chemical Engineering Journal* **2023**, *452*, 139534.
- (S167) Ma, Z.; Yang, Z.; Mu, L.; Deng, L.; Chen, L.; Wang, B.; Qiao, X.; Hu, D.; Yang, B.; Ma, D.; Peng, J.; Ma, Y. Converting molecular luminescence to ultralong room-temperature phosphorescence via the excited state modulation of sulfone-containing heteroaromatics. *Chemical Science* **2021**, *12*, 14808–14814.
- (S168) Zhang, J.; Yuan, H.; Feng, S.; Wen, K.; Guo, X. Theoretical studies on electroluminescent mechanism of a series of thermally activated delayed fluorescence emitters possessing asymmetric-triazine-cored triads. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy* **2018**, *202*, 102–106.
- (S169) Kim, H. J.; Kim, S. K.; Godumala, M.; Yoon, J.; Kim, C. Y.; Jeong, J.-E.; Woo, H. Y.; Kwon, J. H.; Cho, M. J.; Choi, D. H. Novel molecular triad exhibiting aggregation-induced emission and thermally activated fluorescence for efficient non-doped organic light-emitting diodes. *Chemical Communications* **2019**, *55*, 9475–9478.

- (S170) Wang, L.; Li, T.; Feng, P.; Song, Y. Theoretical tuning of the singlet–triplet energy gap to achieve efficient long-wavelength thermally activated delayed fluorescence emitters: the impact of substituents. *Physical Chemistry Chemical Physics* **2017**, *19*, 21639–21647.
- (S171) Chen, Y.; Wang, S.; Wu, X.; Xu, Y.; Li, H.; Liu, Y.; Tong, H.; Wang, L. Triazatruxene-based small molecules with thermally activated delayed fluorescence, aggregation-induced emission and mechanochromic luminescence properties for solution-processable nondoped OLEDs. *Journal of Materials Chemistry C* **2018**, *6*, 12503–12508.
- (S172) Gong, X.; Lu, C.-H.; Lee, W.-K.; Li, P.; Huang, Y.-H.; Chen, Z.; Zhan, L.; Wu, C.-C.; Gong, S.; Yang, C. High-efficiency red thermally activated delayed fluorescence emitters based on benzothiophene-fused spiro-acridine donor. *Chemical Engineering Journal* **2021**, *405*, 126663.
- (S173) Sun, K.; Wu, J.; Zhu, L.; Liu, H.; Zhou, Y.; Tian, W.; Cai, Z.; Jiang, W.; Sun, Y. Highly efficient blue all-solution-processed organic light-emitting diodes based on the strategy of constructing a thermally cross-linkable TADF dendrimer. *Dyes and Pigments* **2022**, *198*, 109967.
- (S174) Konidena, R. K.; Lee, K. H.; Lee, J. Y. Molecular design featuring carbazole-decorated 15H-diindolo[2,3-b:1',2',3'-lm]carbazole for improved efficiency and lifetime of thermally activated delayed fluorescence emitters. *Journal of Materials Chemistry C* **2020**, *8*, 2491–2499.
- (S175) Xiang, S.; Guo, R.; Huang, Z.; Lv, X.; Sun, S.; Chen, H.; Zhang, Q.; Wang, L. Highly efficient yellow nondoped thermally activated delayed fluorescence OLEDs by utilizing energy transfer between dual conformations based on phenothiazine derivatives. *Dyes and Pigments* **2019**, *170*, 107636.
- (S176) Zhang, J.; Xu, X.; Yao, C.; Peng, J.; Jia, M.; Li, L. A novel ternary organic microwire radial heterojunction with high photoconductivity. *Journal of Materials Chemistry C* **2016**, *4*, 4505–4511.
- (S177) Peng, H.; Lou, J.; Li, G.; Zhou, C.; Wang, Z.; Liu, H. Efficient thermally activated delayed fluorescence emitters based on a parallelly aligned bi-spiro-acridine donor. *Journal of Materials Chemistry C* **2022**, *10*, 5813–5820.
- (S178) Jesuraj, P. J.; Somasundaram, S.; Kamaraj, E.; Hafeez, H.; Lee, C.; Kim, D.; Won, S. H.; Shin, S. T.; Song, M.; Kim, C.-S.; Park, S.; Ryu, S. Y. Intramolecular charge transfer-based spirobifluorene-coupled heteroaromatic moieties as efficient hole transport layer and host in phosphorescent organic light-emitting diodes. *Organic Electronics* **2020**, *85*, 105825.
- (S179) Wang, H.; Zhou, L.; Shi, Y.-Z.; Fan, X.-C.; Chen, J.-X.; Wang, K.; Yu, J.; Zhang, X.-H. A facile strategy for enhancing reverse intersystem crossing of red thermally activated delayed fluorescence emitters. *Chemical Engineering Journal* **2022**, *433*, 134423.
- (S180) Xie, F.-M.; Zhou, J.-X.; Li, Y.-Q.; Tang, J.-X. Effects of the relative position and number of donors and acceptors on the properties of TADF materials. *Journal of Materials Chemistry C* **2020**, *8*, 9476–9494.
- (S181) Xie, F.-M.; An, Z.-D.; Xie, M.; Li, Y.-Q.; Zhang, G.-H.; Zou, S.-J.; Chen, L.; Chen, J.-D.; Cheng, T.; Tang, J.-X. tert-Butyl substituted hetero-donor TADF compounds for efficient solution-processed non-doped blue OLEDs. *Journal of Materials Chemistry C* **2020**, *8*, 5769–5776.
- (S182) Raikwar, M. M.; Kim, S. C.; Lee, J. Y. Highly efficient thermally activated delayed fluorescence emitter based on the 5H-benzo[d]benzo[4,5]imidazo[1,2-a]imidazole donor. *Materials Chemistry Frontiers* **2022**, *6*, 3382–3390.
- (S183) Skhirtladze, L.; Leitonas, K.; Bucinskas, A.; Woon, K. L.; Volyniuk, D.; Keruckienė, R.; Mahmoudi, M.; Lapkowski, M.; Ariffin, A.; Grazulevicius, J. V. Turn on of room temperature phosphorescence of donor-acceptor-donor type compounds via transformation of excited states by rigid hosts for oxygen sensing. *Sensors and Actuators B: Chemical* **2023**, *380*, 133295.
- (S184) Cha, J.-R.; Lee, C. W.; Gong, M.-S. Effect of increasing electron donor units for high-efficiency blue thermally activated delayed fluorescence. *Dyes and Pigments* **2017**, *140*, 399–406.
- (S185) Liang, J.; Li, C.; Cui, Y.; Li, Z.; Wang, J.; Wang, Y. Rational design of efficient orange-red to red thermally activated delayed fluorescence emitters for OLEDs with external quantum efficiency of up to 26.0% efficiency roll-off. *Journal of Materials Chemistry C* **2020**, *8*, 1614–1622.
- (S186) Zhao, H.; Wang, Z.; Cai, X.; Liu, K.; He, Z.; Liu, X.; Cao, Y.; Su, S.-J. Highly efficient thermally activated delayed fluorescence materials with reduced efficiency roll-off and low on-set voltages. *Materials Chemistry Frontiers* **2017**, *1*, 2039–2046.
- (S187) Gan, Y.; Peng, X.; Qiu, W.; Wang, L.; Li, D.; Xie, W.; Liu, D.; Li, M.; Lin, J.; Su, S.-J. Multiple charge transfer disk-like emitters with fast fluorescence radiation rate and high horizontal dipole orientation for pure blue organic light-emitting diodes. *Chemical Engineering Journal* **2022**, *430*, 133030.

- (S188) Wang, B.; Qiao, X.; Yang, Z.; Wang, Y.; Liu, S.; Ma, D.; Wang, Q. Realizing efficient red thermally activated delayed fluorescence organic light-emitting diodes using phenoxazine/phenothiazine-phenanthrene hybrids. *Organic Electronics* **2018**, *59*, 32–38.
- (S189) Kumar, S.; Ma, Y.-Y.; Khan, A.; Yuan, Y.; Yang, S.-Y.; Jiang, Z.-Q.; Fung, M.-K.; Liao, L.-S. Structurally controlled singlet-triplet splitting for blue star-shaped thermally activated delayed fluorescence emitters incorporating the tricarbazoles-triazine motifs. *Organic Electronics* **2020**, *84*, 105783.
- (S190) Dong, X.; Li, R.; Zheng, Y.; Huo, J.; Cao, Y.; Shi, H. Synthesis, photoluminescence and electroluminescence properties of a new blue emitter with aggregation-induced emission and thermally activated delayed fluorescence characteristics. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy* **2023**, *291*, 122344.
- (S191) Shin, D. J.; Hwang, S. J.; Lim, J.; Jeon, C. Y.; Lee, J. Y.; Kwon, J. H. Reverse intersystem crossing accelerating assistant dopant for high efficiency and long lifetime in red hyperfluorescence organic light-emitting diodes. *Chemical Engineering Journal* **2022**, *446*, 137181.
- (S192) Lee, Y.; Hong, J.-I. High-efficiency thermally activated delayed fluorescence emitters via a high horizontal dipole ratio and controlled dual emission. *Journal of Materials Chemistry C* **2020**, *8*, 8012–8017.
- (S193) Lee, H. J.; Lee, H. L.; Han, S. H.; Lee, J. Y. Novel secondary acceptor based molecular design for superb lifetime in thermally activated delayed fluorescent organic light-emitting diodes through high bond energy and fast up-conversion. *Chemical Engineering Journal* **2022**, *427*, 130988.
- (S194) Konidena, R. K.; Lee, K. H.; Lee, J. Y. 6H-Benzo[4,5]thieno[2,3-b]indole as a novel donor for efficient thermally activated delayed fluorescence emitters with EQEs over 20%, 13912–13919.
- (S195) Taneda, M.; Shizu, K.; Tanaka, H.; Adachi, C. High efficiency thermally activated delayed fluorescence based on 1,3,5-tris(4-(diphenylamino)phenyl)-2,4,6-tricyanobenzene. *Chemical Communications* **2015**, *51*, 5028–5031.
- (S196) Yun, J. H.; Kim, J.-M.; Chung, W. J.; Lim, J.; Lee, J. Y.; Lee, Y.; Choo, C. A novel electroplex host with dual triplet exciton up-converting channels suppressing triplet exciton induced degradation mechanisms in blue organic light-emitting diodes. *Journal of Materials Chemistry C* **2021**, *9*, 15242–15250.
- (S197) Chen, Z.; Wu, Z.; Ni, F.; Zhong, C.; Zeng, W.; Wei, D.; An, K.; Ma, D.; Yang, C. Emitters with a pyridine-3,5-dicarbonitrile core and short delayed fluorescence lifetimes of about 1.5 μ s: orange-red TADF-based OLEDs with very slow efficiency roll-offs at high luminance. *Journal of Materials Chemistry C* **2018**, *6*, 6543–6548.
- (S198) Wang, H.; Zhao, B.; Qu, C.; Duan, C.; Li, Z.; Ma, P.; Chang, P.; Han, C.; Xu, H. 2,3-Dicyanopyrazino phenanthroline enhanced charge transfer for efficient near-infrared thermally activated delayed fluorescent diodes. *Chemical Engineering Journal* **2022**, *436*, 135080.
- (S199) Huang, T.; Liu, D.; Li, D.; Jiang, W.; Jiang, J. Novel yellow thermally activated delayed fluorescence emitters for highly efficient full-TADF WOLEDs with low driving voltages and remarkable color stability. *New Journal of Chemistry* **2019**, *43*, 13339–13348.
- (S200) Sun, K.; Tian, W.; Gao, H.; Bi, C.; Yao, J.; Wang, Z.; Cai, Z.; Jiang, W. Creation of a thermally cross-linkable encapsulated TADF molecule for highly efficient solution-processed hybrid white OLEDs. *Organic Electronics* **2022**, *102*, 106442.
- (S201) Wang, D.-Q.; Zhang, M.; Wang, K.; Zheng, C.-J.; Shi, Y.-Z.; Chen, J.-X.; Lin, H.; Tao, S.-L.; Zhang, X.-H. Fine-tuning the emissions of highly efficient thermally activated delayed fluorescence emitters with different linking positions of electron-deficient substituent groups. *Dyes and Pigments* **2017**, *143*, 62–70.
- (S202) Chen, H.; Liu, H.; Xiong, Y.; He, J.; Zhao, Z.; Tang, B. Z. New aggregation-induced delayed fluorescent materials for efficient OLEDs with high stabilities of emission color and efficiency. *Materials Chemistry Frontiers* **2022**, *6*, 924–932.
- (S203) Shin, D. J.; Kim, S. C.; Lee, J. Y. A reverse intersystem crossing managing assistant dopant for high external quantum efficiency red organic light-emitting diodes. *Journal of Materials Chemistry C* **2022**, *10*, 4821–4830.
- (S204) Lee, C. H.; Choi, S. H.; Oh, S. J.; Lee, J. H.; Shim, J. W.; Adachi, C.; Lee, S. Y. Highly effective organic light-emitting diodes containing thermally activated delayed fluorescence emitters with horizontal molecular orientation. *RSC Advances* **2020**, *10*, 42897–42902.
- (S205) Yokoyama, M.; Inada, K.; Tsuchiya, Y.; Nakanotani, H.; Adachi, C. Trifluoromethane modification of thermally activated delayed fluorescence molecules for high-efficiency blue organic light-emitting diodes. *Chemical Communications* **2018**, *54*, 8261–8264.

- (S206) Yu, L.; Wu, Z.; Xie, G.; Zhong, C.; Zhu, Z.; Ma, D.; Yang, C. An efficient exciton harvest route for high-performance OLEDs based on aggregation-induced delayed fluorescence. *Chemical Communications* **2018**, *54*, 1379–1382.
- (S207) Grybauskaitė-Kaminskienė, G.; Ivaniuk, K.; Bagdziunas, G.; Turyk, P.; Stakhira, P.; Baryshnikov, G.; Volyniuk, D.; Cherpak, V.; Minaev, B.; Hotra, Z.; Ågren, H.; Grazulevicius, J. V. Contribution of TADF and exciplex emission for efficient “warm-white” OLEDs. *Journal of Materials Chemistry C* **2018**, *6*, 1543–1550.
- (S208) Ran, Q.; Zhang, Y.-L.; Hua, X.; Fung, M.-K.; Liao, L.-S.; Fan, J. Modulation of p-type units in tripodal bipolar hosts towards highly efficient red phosphorescent OLEDs. *Dyes and Pigments* **2019**, *162*, 632–639.
- (S209) Zhao, J.; Yang, Z.; Chen, X.; Xie, Z.; Liu, T.; Chi, Z.; Yang, Z.; Zhang, Y.; Aldred, M. P.; Chi, Z. Efficient triplet harvesting in fluorescence–TADF hybrid warm-white organic light-emitting diodes with a fully non-doped device configuration. *Journal of Materials Chemistry C* **2018**, *6*, 4257–4264.
- (S210) Kothavale, S.; Chung, W. J.; Lee, J. Y. Color tuning of dibenzo[a,c]phenazine-2,7-dicarbonitrile-derived thermally activated delayed fluorescence emitters from yellow to deep-red. *Journal of Materials Chemistry C* **2020**, *8*, 7059–7066.
- (S211) Liu, X.; Wang, Y.-F.; Li, M.; Zhu, Y.; Chen, C.-F. Aromatic-imide-based TADF material as emitter for efficient yellow and white organic light-emitting diodes. *Organic Electronics* **2021**, *88*, 106017.
- (S212) Kreiza, G.; Berenis, D.; Banevičius, D.; Juršėnas, S.; Javorskis, T.; Orentas, E.; Kazlauskas, K. High efficiency and extremely low roll-off solution- and vacuum-processed OLEDs based on isophthalonitrile blue TADF emitter. *Chemical Engineering Journal* **2021**, *412*, 128574.
- (S213) Rayappa Naveen, K.; Lee, H.; Braveenth, R.; Joon Yang, K.; Jae Hwang, S.; Hyuk Kwon, J. Deep blue diboron embedded multi-resonance thermally activated delayed fluorescence emitters for narrowband organic light emitting diodes. *Chemical Engineering Journal* **2022**, *432*, 134381.
- (S214) Zhang, M.; Dai, G.-L.; Zheng, C.-J.; Wang, K.; Shi, Y.-Z.; Fan, X.-C.; Lin, H.; Tao, S.-L.; Zhang, X.-H. Novel D-DaC²-A structure thermally activated delayed fluorescence emitters realizing over 20 efficiencies in both evaporation- and solution-processed organic light-emitting diodes. *Organic Electronics* **2021**, *99*, 106312.
- (S215) Fan, X.-C.; Wang, K.; Zheng, C.-J.; Dai, G.-L.; Shi, Y.-Z.; Li, Y.-Q.; Yu, J.; Ou, X.-M.; Zhang, X.-H. Thermally activated delayed fluorescence emitters with low concentration sensitivity for highly efficient organic light emitting devices. *Journal of Materials Chemistry C* **2019**, *7*, 8923–8928.
- (S216) Serevičius, T.; Skaisgiris, R.; Dodonova, J.; Kazlauskas, K.; Juršėnas, S.; Tumkevičius, S. Minimization of solid-state conformational disorder in donor–acceptor TADF compounds. *Physical Chemistry Chemical Physics* **2020**, *22*, 265–272.
- (S217) Ban, X.; Chen, F.; Liu, Y.; Pan, J.; Zhu, A.; Jiang, W.; Sun, Y. Design of efficient thermally activated delayed fluorescence blue host for high performance solution-processed hybrid white organic light emitting diodes. *Chemical Science* **2019**, *10*, 3054–3064.
- (S218) Huo, J.; Xiao, S.; Wu, Y.; Li, M.; Tong, H.; Shi, H.; Ma, D.; Tang, B. Z. Molecular engineering of blue diphenylsulfone-based emitter with aggregation-enhanced emission and thermally activated delayed fluorescence characteristics: impairing intermolecular electron-exchange interactions using steric hindrance. *Chemical Engineering Journal* **2023**, *452*, 138957.
- (S219) Shi, C.; Liu, D.; Li, J.; He, Z.; Song, K.; Liu, B.; Wu, Q.; Xu, M. tert-Butyltriazine-Diphenylaminocarbazole based TADF materials: Ĩ€-Bridge modification for enhanced kRISC and efficiency stability. *Dyes and Pigments* **2022**, *204*, 110430.
- (S220) Dong, R.; Liu, D.; Li, J.; Ma, M.; Mei, Y.; Li, D.; Jiang, J. Acceptor modulation for blue and yellow TADF materials and fabrication of all-TADF white OLED. *Materials Chemistry Frontiers* **2022**, *6*, 40–51.
- (S221) Zheng, Y.; Huo, J.; Xiao, S.; Shi, H.; Ma, D.; Tang, B. Z. Synthesis, photoluminescence and electroluminescence properties of a new blue emitter containing carbazole, acridine and diphenyl sulfone units. *Organic Electronics* **2022**, *101*, 106411.
- (S222) Zhang, M.; Dai, G.; Zheng, C.; Wang, K.; Shi, Y.; Fan, X.; Lin, H.; Tao, S.; Zhang, X. New electron-donating segment to develop thermally activated delayed fluorescence emitters for efficient solution-processed non-doped organic light-emitting diodes. *Chinese Chemical Letters* **2022**, *33*, 1110–1115.
- (S223) Wu, X.; Zeng, J.; Peng, X.; Liu, H.; Tang, B. Z.; Zhao, Z. Robust sky-blue aggregation-induced delayed fluorescence materials for high-performance top-emitting OLEDs and single emissive layer white OLEDs. *Chemical Engineering Journal* **2023**, *451*, 138919.

- (S224) Naveen, K. R.; Lee, H.; Seung, L. H.; Jung, Y. H.; Keshavananda Prabhu, C.; Muruganantham, S.; Kwon, J. H. Modular design for constructing narrowband deep-blue multiresonant thermally activated delayed fluorescent emitters for efficient organic light emitting diodes. *Chemical Engineering Journal* **2023**, *451*, 138498.
- (S225) Nakagawa, T.; Ku, S.-Y.; Wong, K.-T.; Adachi, C. Electroluminescence based on thermally activated delayed fluorescence generated by a spirobifluorene donor-acceptor structure. *Chemical Communications* **2012**, *48*, 9580.
- (S226) Skuodis, E.; Bezvikonnyi, O.; Tomkeviciene, A.; Volyniuk, D.; Mimaite, V.; Lazauskas, A.; Bucinskas, A.; Keruckiene, R.; Sini, G.; Grazulevicius, J. V. Aggregation, thermal annealing, and hosting effects on performances of an acridan-based TADF emitter. *Organic Electronics* **2018**, *63*, 29–40.
- (S227) Xiang, S.; Huang, Z.; Sun, S.; Lv, X.; Fan, L.; Ye, S.; Chen, H.; Guo, R.; Wang, L. Highly efficient non-doped OLEDs using aggregation-induced delayed fluorescence materials based on 10-phenyl-10H-phenothiazine 5,5-dioxide derivatives. *Journal of Materials Chemistry C* **2018**, *6*, 11436–11443.
- (S228) Shi, Y.-Z.; Wu, H.; Wang, K.; Yu, J.; Ou, X.-M.; Zhang, X.-H. Recent progress in thermally activated delayed fluorescence emitters for nondoped organic light-emitting diodes. *Chemical Science* **2022**, *13*, 3625–3651.
- (S229) Guo, R.; Wang, Y.; Huang, Z.; Zhang, Q.; Xiang, S.; Ye, S.; Liu, W.; Wang, L. Phenothiazine dioxide-containing derivatives as efficient hosts for blue, green and yellow thermally activated delayed fluorescence OLEDs. *Journal of Materials Chemistry C* **2020**, *8*, 3705–3714.
- (S230) Zhang, D.; Duan, L.; Zhang, D.; Qiao, J.; Dong, G.; Wang, L.; Qiu, Y. Extremely low driving voltage electrophosphorescent green organic light-emitting diodes based on a host material with small singlet–triplet exchange energy without p- or n-doping layer. *Organic Electronics* **2013**, *14*, 260–266.
- (S231) He, J.-L.; Kong, F.-C.; Sun, B.; Wang, X.-J.; Tian, Q.-S.; Fan, J.; Liao, L.-S. Highly efficient deep-red TADF organic light-emitting diodes via increasing the acceptor strength of fused polycyclic aromatics. *Chemical Engineering Journal* **2021**, *424*, 130470.
- (S232) Huang, Z.; Huang, C.-W.; Tang, Y.-K.; Xiao, Z.; Li, N.; Hua, T.; Cao, X.; Zhou, C.; Wu, C.-C.; Yang, C. Chiral thermally activated delayed fluorescence emitters for circularly polarized luminescence and efficient deep blue OLEDs. *Dyes and Pigments* **2022**, *197*, 109860.
- (S233) Liu, F.; Cheng, Z.; Wan, L.; Gao, L.; Yan, Z.; Hu, D.; Ying, L.; Lu, P.; Ma, Y. Anthracene-based emitters for highly efficient deep blue organic light-emitting diodes with narrow emission spectrum. *Chemical Engineering Journal* **2021**, *426*, 131351.
- (S234) Kitamoto, Y.; Namikawa, T.; Suzuki, T.; Miyata, Y.; Kita, H.; Sato, T.; Oi, S. Dimesitylarylborane-based luminescent emitters exhibiting highly-efficient thermally activated delayed fluorescence for organic light-emitting diodes. *Organic Electronics* **2016**, *34*, 208–217.
- (S235) Yu, X.; Fan, C.; Dai, G.; Wang, X.; Ye, C.; Tao, X. 9,10-Di(hydroxymethylphenyl)anthracenes: Highly efficient triplet annihilators with small singlet-triplet energy gap ($\bar{\Gamma}E$) and planar configuration. *Dyes and Pigments* **2020**, *176*, 108166.
- (S236) Lu, T.; Jiang, W.; Sun, K.; Tian, W.; Sun, Y. Reduced efficiency roll-off and enhanced excitation confinement in exciplex-type host: Electron transport materials based on benzimidazole units. *Dyes and Pigments* **2018**, *151*, 35–44.
- (S237) Zhou, Y.; Zhang, M.; Ye, J.; Liu, H.; Wang, K.; Yuan, Y.; Du, Y.-Q.; Zhang, C.; Zheng, C.-J.; Zhang, X.-H. Efficient solution-processed red organic light-emitting diode based on an electron-donating building block of pyrrolo[3,2-b]pyrrole. *Organic Electronics* **2019**, *65*, 110–115.
- (S238) Lee, Y.; Woo, S.-J.; Kim, J.-J.; Hong, J.-I. Linear-shaped thermally activated delayed fluorescence emitter using 1,5-naphthyridine as an electron acceptor for efficient light extraction. *Organic Electronics* **2020**, *78*, 105600.
- (S239) Zhang, M.; Zheng, C.-J.; Lin, H.; Tao, S.-L. Thermally activated delayed fluorescence exciplex emitters for high-performance organic light-emitting diodes. *Materials Horizons* **2021**, *8*, 401–425.
- (S240) Tongsuk, S.; Malatong, R.; Unjarern, T.; Wongkaew, C.; Surawatanawong, P.; Sudyoadsuk, T.; Promarak, V.; Ruangsapapichat, N. Enhancement of performance of OLEDs using double indolo[3,2-b]indole electron-donors based emitter. *Journal of Luminescence* **2021**, *238*, 118287.
- (S241) Godumala, M.; Choi, S.; Cho, M. J.; Choi, D. H. Thermally activated delayed fluorescence blue dopants and hosts: from the design strategy to organic light-emitting diode applications. *Journal of Materials Chemistry C* **2016**, *4*, 11355–11381.
- (S242) Nosova, E. V.; Lipunova, G. N.; Zyryanov, G. V.; Charushin, V. N.; Chupakhin, O. N. Functionalized 1,3,5-triazine derivatives as components for photo- and electroluminescent materials. *Organic Chemistry Frontiers* **2022**, *9*, 6646–6683.

- (S243) Wang, H.; Lv, X.; Meng, L.; Wei, X.; Wang, Y.; Wang, P. Novel thioxanthone host material with thermally activated delayed fluorescence for reduced efficiency roll-off of phosphorescent OLEDs. *Chinese Chemical Letters* **2018**, *29*, 471–474.
- (S244) Cha, S. J.; Han, N. S.; Song, J. K.; Park, S.-R.; Jeon, Y. M.; Suh, M. C. Efficient deep blue fluorescent emitter showing high external quantum efficiency. *Dyes and Pigments* **2015**, *120*, 200–207.
- (S245) Matsui, Y.; Yokoyama, Y.; Ogaki, T.; Ishiharaguchi, K.; Niwa, A.; Ohta, E.; Saigo, M.; Miyata, K.; Onda, K.; Naito, H.; Ikeda, H. Aggregation-induced emission active thermally-activated delayed fluorescence materials possessing N-heterocycle and sulfonyl groups. *Journal of Materials Chemistry C* **2022**, *10*, 4607–4613.
- (S246) Liu, X.-Y.; Liang, F.; Yuan, Y.; Cui, L.-S.; Jiang, Z.-Q.; Liao, L.-S. An effective host material with thermally activated delayed fluorescence formed by confined conjugation for red phosphorescent organic light-emitting diodes. *Chemical Communications* **2016**, *52*, 8149–8151.
- (S247) Chai, D.; Zeng, X.; Su, X.; Zhou, C.; Zou, Y.; Zhong, C.; Gong, S.; Yang, C. A simple and effective strategy to lock the quasi-equatorial conformation of acridine by H–H repulsion for highly efficient thermally activated delayed fluorescence emitters. *Chemical Communications* **2020**, *56*, 2308–2311.
- (S248) Zhu, Y.; Chen, Z.; Ying, A.; Gong, S.; Wang, T.; Yang, C. Nematic liquid crystals induce and amplify the circularly polarized luminescence of chiral TADF emitters. *Journal of Materials Chemistry C* **2022**, *10*, 5065–5069.
- (S249) Zhou, T.; Zhang, K.; Cao, Q.; Xu, H.; Ban, X.; Zhu, P.; Li, Q.; Shi, L.; Ge, F.; Jiang, W. Benzonitrile-based AIE polymer host with a simple synthesis process for high-efficiency solution-processable green and blue TADF organic light emitting diodes. *Journal of Materials Chemistry C* **2022**, *10*, 2109–2120.
- (S250) Méhes, G.; Goushi, K.; Potscavage, W. J.; Adachi, C. Influence of host matrix on thermally-activated delayed fluorescence: Effects on emission lifetime, photoluminescence quantum yield, and device performance. *Organic Electronics* **2014**, *15*, 2027–2037.
- (S251) Cao, H.-T.; Hong, C.-S.; Ye, D.-Q.; Liu, L.-H.; Xie, L.-H.; Chen, S.-F.; Sun, C.; Wang, S.-S.; Zhang, H.-M.; Huang, W. Tetracyano-substituted spiro[fluorene-9,9'-xanthene] as electron acceptor for exciplex thermally activated delayed fluorescence. *Journal of Molecular Structure* **2019**, *1196*, 132–138.
- (S252) Pershin, A.; Hall, D.; Lemaire, V.; Sancho-García, J.-C.; Muccioli, L.; Zysman-Colman, E.; Beljonne, D.; Olivier, Y. Highly emissive excitons with reduced exchange energy in thermally activated delayed fluorescent molecules. *Nat. Commun.* **2019**, *10*, 597.
- (S253) Olivier, Y.; Moral, M.; Muccioli, L.; Sancho-García, J.-C. Computational Design of Thermally Activated Delayed Fluorescence Materials: The Challenges Ahead. *The Journal of Physical Chemistry Letters* **2018**, *9*, 6149–6163.
- (S254) Najibi, A.; Goerigk, L. The Nonlocal Kernel in van der Waals Density Functionals as an Additive Correction: An Extensive Analysis with Special Emphasis on the B97M-V and ω B97M-V Approaches. *J. Chem. Theory Comput.* **2018**, *14*, 5725–5738.
- (S255) Weigend, F.; Ahlrichs, R. Balanced basis sets of split valence, triple zeta valence and quadruple zeta valence quality for H to Rn: Design and assessment of accuracy. *Phys. Chem. Chem. Phys.* **2005**, *7*, 3297.
- (S256) Barone, V.; Cossi, M. Quantum Calculation of Molecular Energies and Energy Gradients in Solution by a Conductor Solvent Model. *J. Phys. Chem. A* **1998**, *102*, 1995–2001.
- (S257) Penfold, T. J.; Gindensperger, E.; Daniel, C.; Marian, C. M. Structure-property relationships in donor-acceptor TADF emitters. *Chemical Reviews* **2018**, *118*, 6975–7025.
- (S258) Caldeweyher, E.; Ehlert, S.; Hansen, A.; Neugebauer, H.; Spicher, S.; Bannwarth, C.; Grimme, S. A generally applicable atomic-charge dependent London dispersion correction. *The Journal of chemical physics* **2019**, *150*.
- (S259) Sun, H.; Zhong, C.; Brédas, J.-L. Reliable Prediction with Tuned Range-Separated Functionals of the Singlet-Triplet Gap in Organic Emitters for Thermally Activated Delayed Fluorescence. *Journal of Chemical Theory and Computation* **2015**, *11*, 3851–3858.
- (S260) Moral, M.; Muccioli, L.; Son, W.-J.; Olivier, Y.; Sancho-García, J. C. Theoretical Rationalization of the Singlet-Triplet Gap in OLEDs Materials: Impact of Charge-Transfer Character. *Journal of Chemical Theory and Computation* **2015**, *11*, 168–177.
- (S261) Penfold, T. J.; Gindensperger, E.; Daniel, C.; Marian, C. M. Spin-Vibronic Mechanism for Intersystem Crossing. *Chem. Rev.* **2018**, *118*, 6975–7025.

- (S262) Olivier, Y.; Moral, M.; Muccioli, L.; Sancho-García, J.-C. Dynamic Nature of Excited States of Donor-Acceptor TADF Materials for OLEDs: How Theory Can Reveal Structure-Property Relationships. *Journal of Materials Chemistry C* **2017**, *5*, 5718–5729.
- (S263) Uoyama, H.; Goushi, K.; Shizu, K.; Nomura, H.; Adachi, C. Efficient blue organic light-emitting diodes employing thermally activated delayed fluorescence. *Nature* **2012**, *492*, 234–238.
- (S264) Lee, D.; Ma, X.; Jung, J. H.; Chung, E. W.; Lee, Y.; Yook, K. S.; Lee, J. Y. Blue organic light-emitting diodes: current status, challenges, and future outlook. *J. Mater. Chem. C* **2013**, *1*, 2666–2671.
- (S265) Kaji, H.; Suzuki, H.; Fukushima, T.; Shizu, K.; Suzuki, K.; Kubo, S.; Komino, T.; Oiwa, H.; Suzuki, F.; Wakamiya, A.; Murata, Y.; Adachi, C. Purely organic electroluminescent material realizing 100% conversion from electricity to light. *Nature Communications* **2015**, *6*, 8476.
- (S266) Samanta, P. K.; Kim, D.; Coropceanu, V.; Brédas, J.-L. Up-Conversion Intersystem Crossing Rates in Organic Emitters for Thermally Activated Delayed Fluorescence: Impact of the Nature of Singlet vs Triplet Excited States. *Journal of the American Chemical Society* **2017**, *139*, 4042–4051, PMID: 28244314.
- (S267) Gibson, J.; Monkman, A. P.; Penfold, T. J. The Importance of Vibronic Coupling for Efficient Reverse Intersystem Crossing in Thermally Activated Delayed Fluorescence Molecules. *ChemPhysChem* **2016**, *17*, 2956–2961.